MedMasters Collaborative

Cardiovascular System

Active recall and clinical reasoning workbook

NPLEX Part I · by Dr. Sharilyn Rennie

Every competency in this volume has two brain dumps. The first asks you to rebuild the foundation from memory. The second asks you to use it on something new. Work without your notes, rate your confidence, then check the answer key and write down what you missed.

Answers are in the separate answer key so nothing on these pages gives itself away.

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Cardiovascular System (CARD) · NPLEX-P1-CARD-01

CV-1 Cardiovascular Embryology

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-01-01-C01				
NPLEX-P1-CARD-01-01-C02				
NPLEX-P1-CARD-01-01-C03				
NPLEX-P1-CARD-01-01-C04				
NPLEX-P1-CARD-01-01-C05				
NPLEX-P1-CARD-01-01-C06				
NPLEX-P1-CARD-01-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C01-BD1

Draw the heart tube and name what each segment becomes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-BD1 Date: _____

Without looking at your notes, draw the straight heart tube and label its five segments from caudal to cranial, then write beside each segment the main adult structure it becomes. Mark which segments come from the primary heart field and which come from the secondary heart field. Above the drawing write the sequence that builds the tube, from cardiac progenitors leaving the primitive streak to a tube that beats, and put the day or week beside each step. Name the three layers of the tube wall and write what each one gives rise to, including the tissue that supplies the epicardium and the coronary vessels. Finish by writing which parts of the atrial and ventricular walls end up smooth and which end up trabeculated, and explain that difference from the segment each wall came from.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C01-BD2

A child with chronic sinusitis and an unusual chest film

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-BD2 Date: _____

A child has had a wet cough and recurrent ear and sinus infections since the newborn period, and the chest radiograph shows the cardiac apex on the right. Write which organelle is defective and which protein inside it is at fault, then explain step by step why that defect leaves the side of the heart unpredictable rather than always reversed. Write the proportion of these patients you would expect to have fully mirrored organs, the proportion with normal organ position, and the proportion with mixed sidedness, and name the triad that goes with the mirrored form. Next write why the same defect causes the cough, the ear disease, infertility in men, and a raised risk of ectopic pregnancy. Finish by writing which of these children carry a high risk of complex heart defects and which usually have a structurally normal heart, and state what the limb lead and precordial tracings look like when the whole body is mirrored.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-01-01-C01-RT01

Rebuild the left and right signalling sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-RT01 Date: _____

Rebuild from memory the five-step sequence that runs from beating cilia at the primitive node to a heart tube that loops to the right, one step per line and in order. Beside the first step write where those cilia sit, how they are tilted, and which motor protein drives them. Beside the signalling steps write which genes switch on in left lateral plate mesoderm and which gene forms the midline barrier, and write what that barrier is for. Beside the last step write which organs take their sidedness from this signal. Underneath, write what happens to each of the five steps when the fluid flow at the node is absent, and draw the looping heart as it ends up, marking where the atria sit relative to the ventricles and which chamber ends up on the left.

Rebuild from memory: Left and right signalling that directs looping

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C02-BD1

Draw atrial septation in four panels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C02 Activity: NPLEX-P1-CARD-01-01-C02-BD1 Date: _____

Draw four panels of the developing atrial septum, from the first septum growing down to the sealed adult septum. In each panel label every septum and every opening present at that moment, and write the week the panel shows. Make clear in your drawing which opening is a hole inside the first septum and which opening is the gap left by the second septum, and write one sentence on how the two are different. Mark which septum is thin and which is thick, and which side of the heart each one sits on. Then write how the offset openings make a valve that passes blood one way only, state which way blood crosses before birth and why, and write what presses the valve shut after birth. Finish by naming the adult structure left behind and writing which part of it is the first septum and which part is the second.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C02-BD2

An unsealed flap against a true hole

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C02 Activity: NPLEX-P1-CARD-01-01-C02-BD2 Date: _____

Two adults have an opening between the atria. One has a flap that never fused and the other has a genuine deficiency of septal tissue at the floor of the oval fossa. For each person write whether tissue is missing, which way blood crosses and why, what has to happen before any blood crosses at all, and what the usual consequence is. For the one with the true deficiency, write which chambers and vessels carry the extra volume and which chamber is spared, then write the murmur with its timing and where you hear it, explain why the flow through the hole itself is nearly silent, and write what the added rumble in diastole tells you about shunt size. Write what the limb lead axis and the shape of the complex in V1 look like, and explain why that pattern follows from the chamber that is loaded. Write out in four or five steps how decades of high pulmonary flow can reverse the shunt and produce blue skin. Finish by explaining why the second heart sound is both widely split and unchanging with breathing here, and why a block in the right bundle gives a wide split that still moves with breathing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C03-BD1

What the endocardial cushions build

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C03 Activity: NPLEX-P1-CARD-01-01-C03-BD1 Date: _____

Without your notes, draw the atrioventricular canal before and after the cushions fuse, and write where the canal sits before it shifts. List every structure the cushions go on to build, one per line, and beside each one write the week it is finished. Write the cell biology in order: which tissue signals, which signal families it uses, which cells change their character, what that change is called, and where those cells end up. Then write how the leaflets, the cords and the papillary muscles are carved out of cushion tissue, how many leaflets each orifice ends up with, and what tissue replaces the muscle inside the cords. Separately, write how the muscular part of the ventricular septum forms, how the membranous part closes, and name the three tissues that have to meet there. Finish by writing which part of the ventricular septum most defects sit in and explain that from the way it closes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C03-BD2

Why one cushion failure damages several structures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C03 Activity: NPLEX-P1-CARD-01-01-C03-BD2 Date: _____

An infant with trisomy 21 is in heart failure at three months with a loud murmur, and the tracing shows a leftward and superior limb lead axis with a prolonged interval from the start of the P wave to the start of the complex. Write which developmental step failed, then list every structure that is abnormal because of that one failure. Write why this infant is in failure at three months while a child with a hole at the oval fossa of similar size can stay well for decades, using the pressure as well as the flow in your answer. Explain the limb lead axis and the lengthened interval from where the conducting tissue has been pushed and what has happened to the left anterior fascicle, and contrast that axis with the axis you expect at the oval fossa. Then write the difference between the complete and the partial form of this lesion in terms of the openings present, the valve leaflets, and the valve leak. Finish by writing what goes wrong with the leaflets in Ebstein anomaly and what blood has to do instead when the right atrioventricular orifice never opens.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-01-01-C03-CP01

Three openings between the atria compared

Fill in every cell from memory, then check it against the notes and correct anything you missed.

Feature	Patent foramen ovale	Secundum atrial septal defect	Complete atrioventricular septal defect
Developmental step that failed			
Is septal tissue actually missing			
Where the opening sits			
State of the atrioventricular valves			
Direction of flow and what makes it move			
Limb lead axis on the tracing			
Age at which it usually causes trouble, and why			
Chromosomal or single gene association from the notes			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C04-BD1

Draw the spiral outflow septum and the three defects

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C04 Activity: NPLEX-P1-CARD-01-01-C04-BD1 Date: _____

Draw the single outflow tract in cross section at two levels and show the ridges that divide it, naming which ridge is superior and which is inferior and which way each one grows. Mark how far the septum twists and write what that twist achieves for the final position of each great vessel, including which vessel ends up anterior and which ends up posterior and which ventricle each one leaves. Write where the cells that build this septum come from, which arches they travel through, and the other structures in the neck they help to form. Then draw three more outflow tracts: one where the ridges never form, one where the septum comes down straight instead of spiralling, and one where the septum is pushed forward and upward. Beside each drawing name the defect and list its anatomical features. Finish by writing why displacing the outlet septum also leaves a hole in the ventricular septum.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C04-BD2

Two blue newborns from two different failures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C04 Activity: NPLEX-P1-CARD-01-01-C04-BD2 Date: _____

One newborn is deeply blue within hours of birth and gets worse on day two, with the great vessels lying one in front of the other on the film. A second child is pink for the first year and then has spells of deep blue colour during crying, with a heart on the film that has a raised apex and a scooped out upper left border. For each child name the defect, name the step of outflow division that failed, and write where the blood is going. For the first child write why the two circuits cannot supply each other, name the three places mixing can happen, and write what changed on day two. For the second child write why the colour depends on the balance between the resistance at the right ventricular outflow and the resistance in the body, and explain why mild obstruction leaves the child pink. Write where the murmur comes from and what happens to its loudness during a spell, and explain that from the flow. Finish by writing the two blood findings that follow years of low oxygen, and name the deletion syndrome that links outflow defects to a small thymus, low calcium and facial changes, with the gene involved and the reason the defect happens.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-01-01-C04-PR01

Predict what happens during a spell and during squatting

A two-year-old with pulmonary outflow narrowing, a large malalignment hole in the ventricular septum, an overriding aorta and a thick right ventricle begins crying hard and turns deep blue. Predict each item below, write the direction of the change, and give the reason in one line from pressures, resistances or flow.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the resistance in the systemic circulation while the child cries		
What happens to the resistance at the right ventricular outflow compared with the systemic resistance		
Which way blood crosses the hole in the ventricular septum, and what happens to the amount crossing		
What happens to the blood flow reaching the lungs and to the arterial oxygen saturation		
What happens to the loudness of the murmur, and from which part of the heart that murmur arises		
What the falling oxygen and rising acid in the blood do to the systemic resistance next, and what that does to the spell		
What happens to each of the items above when the child squats and folds the hips and knees		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C05-BD1

Draw the fetal circulation with its saturations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-BD1 Date: _____

Draw the fetal circulation from the placenta back to the placenta. Label the umbilical vein, the three shunts, both atria and ventricles, the pulmonary trunk, the ascending and descending aorta, and the umbilical arteries, and mark the direction of flow at every point. Write the approximate oxygen saturation in the umbilical vein and in the umbilical arteries, and write the partial pressure of oxygen in the umbilical vein. Mark the ridge in the right atrium that aims the incoming stream at the atrial opening and write what it does. Then write the four reasons the resistance in the fetal lungs is high and the one reason the resistance in the body is low, and write which pressure is higher, the pulmonary trunk or the descending aorta. Finish by writing which ventricle handles the larger share of the combined output and what that explains about the thickness of the two ventricular walls at birth.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C05-BD2

Why the fetal brain is well supplied at a low oxygen tension

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-BD2 Date: _____

The fetus lives with an arterial oxygen tension far below an adult value and still oxygenates its tissues. Write the four reasons that works, and for the haemoglobin reason name the two chains in fetal haemoglobin, the exact amino acid substitution responsible, the molecule that no longer binds, and the two half saturation pressures so the difference is on the page as numbers. Write what the double effect of carbon dioxide at the placenta does to the affinity on each side of the barrier and which way oxygen therefore moves. Next write out the route by which the best oxygenated blood reaches the heart muscle and the brain, naming the branches that leave the aorta before the mixing point, and write what reaches the lower body instead and why. Then predict and explain what happens to each ventricle and to the lower body if the vessel joining the pulmonary trunk to the descending aorta narrows in mid gestation, and what happens to the growth of the left heart if the atrial opening narrows instead. Finish by writing why embryonic chambers grow to the size they do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-01-01-C05-RT02

Rebuild the path of blood through the fetus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-RT02 Date: _____

Rebuild from memory the seven-step path of blood through the fetal circulation, one step per line and in order from the umbilical vein to the umbilical arteries. Beside each step write the saturation or the pressure relationship that explains why blood goes that way rather than the other way. Mark the two steps where a shunt is being used and write what each shunt bypasses. Beside the step through the liver, write what regulates how much blood takes the bypass and what that protects the heart from. Underneath the list, write in one line which great vein's blood ends up in the left heart and the brain and which great vein's blood ends up in the lower body, and name the vessel the umbilical arteries branch from.

Rebuild from memory: Path of blood through the fetal circulation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C06-BD1

Rebuild the changes at birth and the remnants table

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C06 Activity: NPLEX-P1-CARD-01-01-C06-BD1 Date: _____

Write the six-step sequence that turns a parallel circulation into a circulation in series, one step per line and in order, starting with the first breaths. Beside the breathing step write the three reasons the resistance in the lungs falls and name the two substances released by oxygen and shear stress. Beside the cord clamping step write what happens to the resistance in the body and to the venous return from below. Then write which pressure change closes the atrial opening, how long functional closure takes, and how long full anatomical fusion takes. Write the mechanism that closes the vessel between the pulmonary trunk and the aorta in two parts: the oxygen part, naming the channel that closes, the change in membrane voltage and the channel that then opens, and the prostaglandin part, naming the substance, where it came from, where it is now broken down, and the second messenger it worked through. Below that, build the whole table of fetal vessels and shunts with one row each and columns for the route and the reason for flow and for the adult remnant and its location, and include the cord structure that was never a vessel.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C06-BD2

A vessel that stays open and the lesions that depend on it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C06 Activity: NPLEX-P1-CARD-01-01-C06-BD2 Date: _____

Write why the vessel between the pulmonary trunk and the aorta stays open most often in babies born early, naming the two responses of immature ductal muscle, and name the infection and the environmental exposure that also make it likely. Then write the murmur it produces: its quality, its timing across the cardiac cycle, where you hear it, and why the sound continues rather than stopping. Write what happens to the pulse pressure and why, and which side of the heart carries the extra volume. Next write the three groups of lesions that become dangerous the moment this vessel closes, and for each one write whether the baby depends on it for mixing, for lung blood flow, or for body blood flow. Then explain how a large opening of this kind can end up making the toes blue and clubbed while the fingers stay pink, using the point at which the vessel joins the aorta in your answer, and write why a reversed shunt between the ventricles or between the atria colours fingers and toes alike. Finish by writing what is happening in a newborn whose lung resistance never falls despite normal heart anatomy, and where blood crosses in that baby.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-01-01-C06-CA01

A newborn who collapses on the third day

A term newborn girl was well at delivery and fed normally for the first two days. On the third day she becomes pale, mottled and breathless. Temperature is 36.5 degrees Celsius (97.7 degrees Fahrenheit), pulse is 184 per minute, respirations are 68 per minute, and blood pressure in the right arm is 74 over 48 mm Hg. The femoral pulses cannot be felt and the liver edge is 3 cm below the costal margin. Oxygen saturation measured on the right hand is 96 percent and on the right foot is 82 percent. The chest radiograph shows a large heart and congested lung fields. Umbilical venous blood drawn at delivery had been reported as normal, and the newborn examination on day one, including femoral pulses, had been recorded as normal.

1. Recognize. Name the fetal structure that has just changed state and state what was supplying the lower body until the third day.

2. Explain. Explain from the fetal circulation why this baby was well on day one and why the signs appeared on day three, including why the liver enlarged and why the breathing is fast.

3. Differentiate. Write how this picture differs from a newborn who is blue in the first hours with parallel circuits, and from a newborn whose lung resistance never fell, naming the findings that separate the three.

4. Investigate. Write which measurements and which parts of the examination you would use to confirm where the obstruction sits, and what value or finding each one would give in this baby.

5. Decide. State which two signals normally close that structure and which of them was lost first in this baby, and state what the lower body circulation now depends on.

6. Justify. Justify the difference between the saturation at the right hand and the saturation at the right foot from the point at which that fetal vessel joins the aorta and the branches that leave the arch before it.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C07-BD1

Build the arch artery table and draw both nerve loops

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C07 Activity: NPLEX-P1-CARD-01-01-C07-BD1 Date: _____

Draw the pharyngeal arch arteries running between the aortic sac and the paired dorsal aortae, number the arches, and write whether all of them are ever present at the same time. Then build a table from memory with one row for each embryonic source and a column for the adult derivative, covering the outflow trunk, the aortic sac on each side, the first and second arch arteries, the third, the fourth on the left and on the right, the fifth, the sixth on both sides, the left seventh intersegmental artery and the left dorsal aorta. Mark on your drawing the two segments of dorsal aorta that disappear. Underneath, draw both recurrent laryngeal nerves. Write which arch each nerve first hooked under, which vessel each ends up under in the adult, and why the two differ. Write which branch of the vagus supplies the fourth arch and which muscle it serves, and which muscles the recurrent nerve serves.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C07-BD2

Four remodelling errors and a hoarse voice

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C07 Activity: NPLEX-P1-CARD-01-01-C07-BD2 Date: _____

For each of the four situations below write which arch artery or dorsal aortic segment persisted or regressed abnormally, what the resulting anatomy is, and what the person notices: the right subclavian artery arising as the last branch of the arch and running behind the oesophagus; a vascular ring around the trachea and oesophagus; an interrupted arch between the left common carotid and the left subclavian; and a right laryngeal nerve that does not loop at all. For the interrupted arch write what the lower body depends on and explain why this one has the strongest link with the deletion syndrome in the notes. Then write which single muscle the recurrent nerve supplies that opens the vocal fold, where the fold comes to rest when the nerve is injured, and what the voice sounds like. Finish with the vulnerability of each nerve: list the structures along the left nerve's course in the chest and the conditions along it that cause hoarseness, name the dilated chamber and vessel that can trap it, list where the right nerve is at risk and from what, and name the operation that puts both at risk.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-01-01-C02-Q01

An atrial opening that seals too early

Serial ultrasound examinations of a fetus at 24 weeks of gestation show that the flap valve between the atria has sealed against the thicker right sided septum and no flow crosses it. Follow up studies over the next 8 weeks show a progressively small left atrium and a small, thick walled left ventricle, with a right atrium and right ventricle that are larger than expected. Which of the following best explains the appearance of the left side of this heart?

A. Oxygenated blood is diverted into the pulmonary arteries, so the pulmonary vascular bed carries the volume instead

B. Pressure in the left atrium rises above pressure in the right atrium, so the left ventricle ejects against a greater load

C. Most left sided filling normally arrives across that opening, and chambers grow in proportion to the volume that passes through them

D. The endocardial cushions fail to fuse once flow through the atrioventricular canal is reduced, so the mitral orifice does not form

Why did you pick it?

Board-style question NPLEX-P1-CARD-01-01-C07-Q02

Hoarseness with a dilated aortic arch

A 68-year-old man comes to the physician because of 4 months of progressive hoarseness. Laryngoscopy shows that the left vocal fold lies near the midline and does not move on attempted abduction. Computed tomography of the chest shows a 6 cm aneurysm of the aortic arch. The nerve responsible for his hoarseness is held in the chest by a remnant of which of the following embryonic structures?

A. The distal part of the left sixth pharyngeal arch artery

B. The left fourth pharyngeal arch artery

C. The left seventh intersegmental artery

D. The left horn of the aortic sac

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-02

CV-2 Histology of the Heart and Blood Vessels

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-02-01-C01				
NPLEX-P1-CARD-02-01-C02				
NPLEX-P1-CARD-02-01-C03				
NPLEX-P1-CARD-02-01-C04				
NPLEX-P1-CARD-02-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C01-BD1

Draw the heart wall and the fibrous skeleton

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-BD1 Date: _____

Without looking at your notes, draw a cross section of the left ventricular wall from the cavity outward. Label the endocardium with its subendocardial layer, the myocardium, and the epicardium, and write beside each layer the tissue it is made of and what runs inside it. Write the vessel layer each heart layer corresponds to. Give the normal thickness of the atrial wall, the right ventricular wall, and the left ventricular wall. Then draw the heart in cross section at the level of the valves and lay out the fibrous skeleton: the four fibrous rings, the two fibrous trigones, and the membranous septum. Mark the thickest part, give its other name, and write which structure passes through it and where that structure runs next. Finish by writing the three layers of a valve leaflet with what each layer does, and name the layer that expands in myxomatous mitral valve disease.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C01-BD2

Wall stress, hypertrophy, and two failures of the skeleton

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-BD2 Date: _____

Write the relationship between wall stress, chamber pressure, chamber radius, and wall thickness. Using it, work three cases and show your numbers: a ventricle with a radius of 2.5 cm and a wall of 1.0 cm at a systolic pressure of 120 mmHg; the same ventricle when systolic pressure rises to 200 mmHg; and a ventricle with a radius of 3.5 cm and a wall of 0.8 cm at 120 mmHg. For each case write what the number means for the work each myocyte has to do. Then write how the wall remodels in pressure overload and in volume overload, where the new sarcomeres go in each, what happens to cavity size, and what the nuclei of hypertrophied myocytes look like under the microscope. Write one cost of thickening the wall. Finish with two failures of the fibrous skeleton: write what happens electrically when a strand of myocardium crosses a fibrous ring, including the three tracing findings that follow, and write why an abscess of the aortic valve ring in infective endocarditis can produce new conduction block.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C01-RT01

Rebuild why the subendocardium is injured first

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C01 Activity: NPLEX-P1-CARD-02-01-C01-RT01 Date: _____

Rebuild the six-step sequence that explains why the inner third of the ventricular wall is the first layer to become ischemic, one step per line and in order. Beside the step about systole, draw a small cross section of the wall and mark where intramural pressure is highest and which vessels are compressed. Beside the step about demand, write why wall stress is highest in the inner layers, using the relationship you already know. Then write the three situations named in the notes that tip supply below demand, write during which part of the cardiac cycle the inner wall is perfused, and write what the resulting infarct is called and why it can happen without an artery being completely blocked.

Rebuild from memory: Why the subendocardium is injured first

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-02-01-C01-CA01

Chest tightness on exertion in a man with a thick left ventricle

A 71-year-old man comes to the physician because of 6 months of tightness in the chest that comes on when he walks uphill and settles within a few minutes of stopping. Temperature is 36.8 C (98.2 F), pulse is 76/min, respirations are 16/min, and blood pressure is 128/80 mm Hg. A harsh systolic murmur is heard over the right upper chest and the carotid upstroke is slow and low in volume. Echocardiography shows a left ventricular wall thickness of 1.8 cm with a cavity radius of 2.4 cm, a peak systolic pressure in the left ventricle of 210 mm Hg, and vigorous contraction. Coronary angiography shows smooth epicardial coronary arteries with no narrowing.

1. Recognize. Name the pattern of hypertrophy present and the loading problem that produced it, and state where the new sarcomeres have been laid down.

2. Explain. Calculate wall stress for a radius of 2.4 cm at 210 mm Hg with a wall of 1.0 cm and again with a wall of 1.8 cm, then explain from the two numbers what the thickening achieved.

3. Differentiate. Write how this man's chest tightness differs in mechanism from chest tightness caused by a plaque narrowing an epicardial coronary artery, and name the finding here that separates the two.

4. Investigate. Write which layer of the ventricular wall is at greatest risk and which three features of a stained section of the wall you would look at to support that.

5. Decide. State the layer in which necrosis would appear first if his coronary perfusion fell during exertion, and name the type of infarct that results.

6. Justify. Justify that answer from how the coronary arteries enter the wall, what systole does to the inner vessels, and how wall stress is distributed from endocardium to epicardium.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C02-BD1

Draw a cardiac myocyte and its intercalated disc

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C02 Activity: NPLEX-P1-CARD-02-01-C02-BD1 Date: _____

Without looking at your notes, draw a working ventricular myocyte joined to its neighbour. Give its width and length in micrometres and label the nuclei, the perinuclear zone and what it contains, the mitochondria with the fraction of cell volume they occupy, and the capillary beside the cell. Write which fuel and which pathway supply nearly all of its ATP, how long contraction survives complete ischemia, and after how long injury becomes irreversible. Then draw the intercalated disc enlarged, marking the transverse portion and the lateral portion and which junctions sit in each. Build a three column table with the headings junction, structure, and function, and fill a row for the fascia adherens, the desmosome, and the gap junction, naming the proteins in each. Finish by drawing the T tubule and sarcoplasmic reticulum arrangement, naming it and stating where in the sarcomere it sits.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C02-BD2

When the junctions fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C02 Activity: NPLEX-P1-CARD-02-01-C02-BD2 Date: _____

Three hearts have a single junctional defect each. In the first, ventricular connexin 43 is greatly reduced but the desmosomes and fascia adherens are normal. In the second, a mutation in plakophilin 2 weakens the desmosomes. In the third, gap junctions are normal in number but have moved from the ends of the cells to the lateral borders, as happens at the edge of an old infarct. For each heart write whether the cells stay attached, what happens to the speed and direction of electrical spread, and what rhythm problem follows and why. For the second heart name the disease, write which chamber it affects most, what replaces the lost myocytes, and name the recessive form along with its hair and skin findings and the reason the skin is involved. Finish by writing why an adult heart cannot replace the myocytes it loses and what it does instead in overload and after an infarct.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-02-01-C02-CP01

Cardiac muscle against skeletal muscle

Fill in every cell of the table from memory, then open the notes and correct what you missed. In the last column write the demand on the heart that each cardiac feature serves.

Feature	Cardiac muscle	Skeletal muscle	What the cardiac version achieves
Cell shape and position of nuclei			
Coupling between cells			
T tubule size, position, and number of cisternae paired with it			
What triggers sarcoplasmic reticulum calcium release			
Whether extracellular calcium is needed for contraction			
Whether tetanus is possible			
How force is graded			
Capacity to regenerate after injury			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-02-01-C02-Q01

Sudden death in a young athlete with a fibrofatty right ventricle

A 19-year-old man collapses during a football match and is resuscitated from ventricular tachycardia. His father died suddenly at the age of 32. Examination shows woolly hair and thickened skin on the palms and soles. Cardiac imaging shows a dilated right ventricle with areas of the wall replaced by fat, and a biopsy specimen shows myocytes pulled apart from one another with fibrofatty tissue between them. Which of the following structures is most likely to be defective in this patient?

A. Connexon channels of the gap junction

B. Cadherin to actin linkage of the fascia adherens

C. Plakoglobin of the desmosome

D. Ryanodine receptor 2 of the sarcoplasmic reticulum

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C03-BD1

Purkinje fibers, working myocytes, and nodal cells

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-BD1 Date: _____

Without looking at your notes, write where Purkinje fibers lie in the ventricular wall, what they are continuous with, and how a stained section tells you a cell is one: its size, the colour and content of its cytoplasm, the stain it reacts with, where its myofibrils sit, and what surrounds its nucleus. Write the three features that make it the fastest conducting cell in the heart and give its conduction velocity. Then build a table from memory with one column each for the Purkinje fiber, the working ventricular myocyte, and the nodal cell, and rows for size and staining, myofibrils, gap junctions, conduction velocity, and main role. Finish by writing the rate at which Purkinje fibers pace, what that rhythm looks like on a tracing, and the unrelated cell with the same name that is a common matching trap.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C03-BD2

Read two sections and follow the consequences

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-BD2 Date: _____

A section taken just beneath the endothelium of the left ventricle shows large pale cells that stain strongly with periodic acid Schiff, have few myofibrils at the edge of the cell, and have a clear halo around the nucleus. A section of the right atrial wall shows striated cells with small membrane bound dense granules beside the nucleus and Golgi apparatus. Name each cell and write what it does. For the first, write why its stored glycogen matters when the inner wall becomes ischemic, and what the surviving cells at the inner edge of an infarct can start. For the second, name the hormone in the granules, write the stimulus that releases it, and write its receptor, its second messenger, and its four actions on the kidney and the vessels. Then write which chamber makes the related peptide that is measured in blood, why that one is not stored in prominent granules, and what its plasma level does in heart failure and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C03-RT01

Rebuild the atrial stretch feedback loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C03 Activity: NPLEX-P1-CARD-02-01-C03-RT01 Date: _____

Rebuild the five-step loop that runs from a rise in blood volume back to a fall in atrial stretch, one step per line and in order, then draw it as a closed loop with arrows. Beside the release step write which cells release the hormone, where in the cell it was stored, and which atrium holds the most of these granules. Beside the receptor step write the receptor name and the enzyme domain it carries, and write which second messenger that produces. Beside the kidney step write what happens to each of the two glomerular arterioles and what that does to filtration. Then write in one line what kind of feedback this loop is and how you can tell from the last step.

Rebuild from memory: Atrial stretch and natriuretic peptide feedback

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C04-BD1

Draw the wall of every vessel class

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C04 Activity: NPLEX-P1-CARD-02-01-C04-BD1 Date: _____

Without looking at your notes, draw cross sections of an elastic artery, a muscular artery, an arteriole, a capillary, a postcapillary venule, a medium vein, and a large vein, side by side and roughly to scale. Label the three layers on each where they exist, and write beside each drawing the feature that identifies it down the microscope and its main job. Give the number of elastic lamellae in the aortic media, the number of smooth muscle layers in a muscular artery media and in an arteriole, the diameter of a capillary, and the diameter below which a vessel counts as an arteriole. Mark where the internal elastic lamina is most distinct and where the vasa vasorum run, and write what the vasa vasorum are for and which vessel type has more of them. Finish by writing where valves are found and the three places they are absent, and what fraction of blood volume the veins hold.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C04-BD2

Which layer is damaged, and what the pressures do

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C04 Activity: NPLEX-P1-CARD-02-01-C04-BD2 Date: _____

For each of these, name the vessel layer or component that is damaged and write how the damage explains the finding: an older patient with a high systolic pressure and a low diastolic pressure; a tall young patient with a defect in fibrillin 1 who has a dilated aortic root and tearing chest pain; a patient with tertiary syphilis who has an aneurysm of the ascending aorta and an intima described as tree bark; a patient whose muscular arteries show pipestem calcification on an X ray with no narrowing of the lumen; and a patient with long standing hypertension and diabetes whose renal arterioles are thickened by pink material. Then write the relationship between resistance and vessel radius, and work out roughly how much resistance rises when an arteriole narrows by 20%. Write the pressure at the start of the arterioles and the pressure at their end, and explain why most of the pressure drop happens there. Finish by writing what the elastic recoil of the aorta contributes in diastole and what a loss of elastin does to systolic pressure, diastolic pressure, and pulse pressure.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-02-01-C04-Q02

Capillary type and what can leave the vessel

A researcher examines electron micrographs of capillaries from four tissues. One micrograph shows a wide lumen, large gaps between endothelial cells, and a basal lamina that is present in some places and absent in others. Which of the following is most likely to pass through the wall of the vessel in this micrograph?

A. Whole red and white blood cells

B. Water and dissolved gases only

C. Small peptide hormones but not albumin

D. Nothing larger than glucose, because tight junctions seal the wall

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-02-01-C05-BD1

Build the endothelial product table and the nitric oxide pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-BD1 Date: _____

Without looking at your notes, build a three column table with the headings product, mechanism, and effect, and fill a row for nitric oxide, prostacyclin, endothelin-1, von Willebrand factor, P-selectin, thrombomodulin, heparan sulfate, and tissue plasminogen activator. Then write the nitric oxide pathway in full: name the enzyme, its substrate and cofactors, both products, and the four things nitric oxide does besides relaxing smooth muscle. Write what destroys nitric oxide and how long it survives, and what that means for how far it acts. Draw the storage granule that holds von Willebrand factor, give its shape and name, list what is in it and in its membrane, and write how quickly it is released and by which four stimuli. Finish by writing the two jobs von Willebrand factor does, the enzyme that trims it, and the disease caused when that enzyme is missing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-02-01-C05-BD2

Two vessels, one drug, and the same receptor

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-BD2 Date: _____

Acetylcholine is applied to two arterial rings. The first has intact endothelium and relaxes. The second has had its endothelium rubbed away and contracts. Write the receptor acetylcholine binds in each ring, the cell it acts on, the signalling route it uses, and why the two rings move in opposite directions. Then write what happens when acetylcholine is infused into a human coronary artery that has atherosclerosis, and what that response reveals. Next write what rising shear stress does to an artery and through which molecule. Finish with aspirin: write which enzyme it modifies and how, why the platelet cannot recover the enzyme while the endothelial cell can, and what that difference does to the balance between thromboxane A2 and prostacyclin at a low dose.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-02-01-C05-RT01

Rebuild the nitric oxide relaxation pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-2 Histology of the Heart and Blood Vessels
Competency: NPLEX-P1-CARD-02-01-C05 Activity: NPLEX-P1-CARD-02-01-C05-RT01 Date: _____

Rebuild the six-step pathway from a stimulus at the endothelial surface to a dilated vessel, one step per line and in order. Beside the first step write the three stimuli given in the notes. Beside the enzyme step write what activates the enzyme. Beside the cyclase step write which cyclase it is, which nucleotide it makes, and which kinase that nucleotide activates. Beside the last two steps write what happens to calcium inside the smooth muscle cell and which enzyme acting on myosin is switched on. Underneath, write what the same signal does to platelets, and write which other hormone in this chapter raises the same second messenger and through which receptor.

Rebuild from memory: Nitric oxide and vascular relaxation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-02-01-C05-PR01

Predict what happens to an endothelium under oxidative stress

A 58-year-old man has smoked for 30 years and has type 2 diabetes, untreated hypertension, and a high level of oxidized LDL. Predict each item below, write the direction of the change, and give the molecule or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the amount of superoxide in the endothelial cell		
What happens to the amount of nitric oxide available to the smooth muscle		
What happens to the enzyme that makes nitric oxide once tetrahydrobiopterin runs low, and what it starts producing instead		
What happens to endothelin-1 and what that does to vessel diameter		
What happens to VCAM-1, ICAM-1, and the selectins on the endothelial surface, and what that does to monocytes		
What happens to tissue factor, plasminogen activator inhibitor 1, and thrombomodulin, and what that does to the surface of the vessel		
What this vessel does when acetylcholine is infused into it, compared with a healthy vessel		
Which parts of the arterial tree the resulting lesions appear in first, and what about the flow there explains it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-03

CV-3 Heart, Great Vessels, and Pericardium

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-03-01-C01				
NPLEX-P1-CARD-03-01-C02				
NPLEX-P1-CARD-03-01-C03				
NPLEX-P1-CARD-03-01-C04				
NPLEX-P1-CARD-03-01-C05				
NPLEX-P1-CARD-03-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C01-BD1

Draw the heart in the chest and label what forms each edge

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C01 Activity: NPLEX-P1-CARD-03-01-C01-BD1 Date: _____

Without looking at your notes, draw the front of the chest with the sternum, the costal cartilages and the intercostal spaces, then draw the outline of the heart on it. Mark the four corners of that outline by naming the exact cartilage or intercostal space each one sits at, and mark the apex with the space it lies behind, its relation to the midclavicular line and its distance from the midline. Then write one line for each of these and name the chamber or chambers that form it: anterior surface, inferior surface, base, apex, right border, left border, inferior border. Add which compartment of the mediastinum holds the heart and which plane divides the mediastinum into superior and inferior parts. Finish by writing how the heart is rotated in the chest, how much of it lies left of the median plane, and which chamber therefore faces forward.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C01-BD2

Read the chest film and the apex beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C01 Activity: NPLEX-P1-CARD-03-01-C01-BD2 Date: _____

Draw a posteroanterior chest film outline and label the structures that form the right contour from top to bottom and the left contour from top to bottom. Write why two of the four chambers form no edge on this view and which view shows them instead. Next, write how the cardiothoracic ratio is measured and the value above which it is abnormal. A woman with long standing mitral regurgitation has an apex beat in the sixth intercostal space lateral to the midclavicular line and a cardiothoracic ratio of 0.58 on a posteroanterior film taken in full inspiration. Write which chamber has enlarged, which border and which contour it has changed, and why the apex beat has moved. Then a man has a ratio of 0.56 on a portable anteroposterior film taken during a shallow breath. Write the two reasons that film can overstate the size of his heart and what you would ask for before calling the heart enlarged.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C02-BD1

Lay out the chambers from front to back with their neighbors

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-BD1 Date: _____

Draw the chest in cross section at the level of the heart and lay the four chambers out in their front to back order, writing the order out in words underneath. Mark what lies immediately in front of the most anterior chamber and name the bony and cartilaginous structures it sits behind. Mark what lies immediately behind the most posterior chamber, then add the vessel behind and to its left and the two airway structures just above it. Write one line saying what separates that chamber from the esophagus. Finish by naming the embryological event that put the right ventricle at the front and the left atrium at the back.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C02-BD2

Trace each finding of left atrial enlargement to a neighbor

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-BD2 Date: _____

A 42-year-old woman with past rheumatic fever has breathlessness on exertion, difficulty swallowing solid food and a hoarse voice. Her chest film shows a straight left heart border, a second curved edge inside the right heart border and a raised left main bronchus. For each of those six findings write the structure involved, why the left atrium reaches it rather than some other structure, and the direction the atrium had to expand to reach it. Write why the atrium cannot enlarge forwards. Then write which valve lesion her history points to and trace in three steps how that lesion raises left atrial pressure and then causes breathlessness. Finish by writing why a probe placed in the esophagus sees the left atrium and its auricle better than a probe on the chest wall, and what is most often looked for there in a patient with atrial fibrillation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C02-RT01

Rebuild the chain from a narrowed mitral valve to hoarseness

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-RT01 Date: _____

Rebuild the six step sequence that runs from a narrowed mitral valve to a hoarse voice, one step per line and in order. Beside the step where the atrium dilates, write the two directions it can expand and the one direction it cannot, with the reason. Beside the esophagus step, write the symptom it produces and what the barium filled esophagus looks like. Beside the bronchus step, write what happens to the angle at the carina. Beside the final step, name the nerve, the structure it hooks under and the second vessel whose dilation helps stretch it. Underneath, write the two findings this same enlargement produces on a posteroanterior film and the rhythm and complication that follow from slow flow in the dilated atrium.

Rebuild from memory: How mitral stenosis produces symptoms outside the heart

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C03-BD1

Build the pericardium layer by layer

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-BD1 Date: _____

Draw the heart in the pericardium in a side view and label the fibrous pericardium, the parietal serous layer, the pericardial cavity and the visceral serous layer. Beside the fibrous layer write what it is made of, the three places it is attached or blended, and two consequences of the fact that it does not stretch over short periods. Beside the cavity write the normal volume of fluid in it and what that fluid does. Write one or two sentences explaining why the heart is described as lying outside the serous cavity even though the sac surrounds it. Then name the two sleeves in which the serous layer reflects, naming the vessels in each, and mark on your drawing where each of the two sinuses lies. Finish by naming the artery that supplies most of the pericardium and the artery that supplies the epicardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C03-BD2

Separate pericardial pain from ischemic pain

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-BD2 Date: _____

A 34-year-old man has sharp chest pain that is worse when he breathes in deeply and when he lies flat, eases when he sits forward, and is felt along the ridge of his left trapezius. Write the nerve that carries this pain, the spinal segments it enters, which pericardial layers are pain sensitive and which are not, and the nerves that bring shoulder skin into those same segments. Then explain in three or four sentences why the pain is felt at the shoulder when nothing connects the pericardium to the shoulder. Next, write the nerves and the spinal segments that carry pain from ischemic myocardium, where that pain is felt and where it spreads. Build a short side by side list comparing the two pains on character, site, spread, effect of breathing and effect of posture. Finish by writing the added sound heard when inflamed serous surfaces rub, the number of components it can have and the cardiac events those components match.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C03-RT01

Rebuild the referral of pericardial pain to the shoulder

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-RT01 Date: _____

Rebuild the six step sequence that runs from inflammation of the parietal pericardium to pain felt at the trapezius ridge, one step per line and in order. Beside the nerve step write the nerve, the spinal segments it enters and where it runs in relation to the fibrous pericardium, the mediastinal pleura and the root of the lung. Beside the step where the signals meet, write the nerves that bring the shoulder skin to those segments. Underneath, write in one sentence what referral depends on, then write what else irritating the diaphragm would be felt in the same place and why.

Rebuild from memory: Referral of pericardial pain to the shoulder

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-03-01-C03-CP01

Transverse against oblique pericardial sinus

Fill in every cell of the table from memory, then open the notes chapter and correct any cell you missed.

Feature	Transverse sinus	Oblique sinus
Where it lies, named by the structures in front of and behind it		
Which reflection forms each boundary		
Shape, and whether it is open or blind		
How it formed in the embryo		
Why it matters in surgery or in a patient		
Which structure just outside the pericardium lies closest to it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C04-BD1

Build the great vessel table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-BD1 Date: _____

On blank paper build a table with one row for each of these: superior vena cava, inferior vena cava, pulmonary veins, pulmonary trunk, ascending aorta, aortic arch. Use the columns where it begins, where it ends and its most important relation, and fill every cell. Add the three branches of the arch in their order from right to left, and write which vein crosses in front of them. Then write which of these vessels lie inside the pericardium and which do not. Underneath, draw the arch from the left side and label the ligamentum arteriosum with the two points it joins, the aortic isthmus with the two landmarks that bound it, and the aortopulmonary window. Finish by writing what the ligamentum arteriosum was in fetal life, which arch artery it came from and what that structure carried.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C04-BD2

Three patients with vessels pressing on neighbors

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-BD2 Date: _____

Patient one is a 68-year-old smoker with a hoarse, breathy voice and a mass sitting between the underside of the aortic arch and the left pulmonary artery. Patient two has swelling of the face, neck and both arms with distended veins over the chest wall and mediastinal nodes on imaging. Patient three has difficulty swallowing and an artery that arises last from the arch and runs behind the esophagus. For each patient write the vessel or space involved, the structure being compressed or stretched, and why that structure and not another one is affected in that position. For patient one, name the nerve, the muscles it supplies and the one laryngeal muscle it does not supply, and write why a mass in the same position on the right would be far less likely to change the voice. For patient two, write the route the blood takes once the vessel is blocked. For patient three, name the variation and write what is different about the course of the laryngeal nerve on that side.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C04-RT01

Rebuild why the two recurrent laryngeal nerves loop at different levels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-RT01 Date: _____

Rebuild the six step sequence that explains why the two recurrent laryngeal nerves end up hooked at different levels, one step per line and in order. Beside the first step name the arch arteries both nerves start under. Beside the right sided steps write which arch arteries regress and which artery the right nerve ends up under, naming its arch of origin. Beside the left sided steps write what the left sixth arch artery becomes before birth and after birth, and the structure the left nerve stays hooked under. Then draw the arch from the left with both nerves on it, mark where each one hooks, and mark the groove each one ascends in. Finish by listing four diseases or procedures that can injure the left nerve in the chest and write in one line why its course makes it vulnerable.

Rebuild from memory: Why the two recurrent laryngeal nerves loop at different levels

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C05-BD1

Build the four chambers from the inside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C05 Activity: NPLEX-P1-CARD-03-01-C05-BD1 Date: _____

On blank paper build a table with one row for each chamber and the columns smooth part, rough part and distinctive internal features, and fill every cell. Then write one line for each of these saying exactly where it sits and what it marks or does: crista terminalis, fossa ovalis with its floor and its rim, opening of the coronary sinus, moderator band, supraventricular crest, aortic vestibule. Write how many papillary muscles each ventricle has, where their chordae attach in each ventricle, and what the papillary muscles do during systole. Add which conduction structures sit at the upper end of the crista terminalis and just above the coronary sinus opening, and which conduction structure the moderator band carries. Finish by writing in one or two sentences why atrial walls are smooth in some places and rough in others, and why the left atrium is almost entirely smooth with four openings.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C05-BD2

Wall thickness, wall stress and two loaded ventricles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C05 Activity: NPLEX-P1-CARD-03-01-C05-BD2 Date: _____

Write the normal wall thickness of each ventricle, the peak systolic pressure each one generates and the relationship between the two. Write the law of Laplace as an equation with each term named, and write what wall stress largely determines in the myocardium. Then take two patients. The first has had high aortic pressure for years. The second has had a large valve leak for years. For each one write whether the load is pressure or volume, whether sarcomeres are added in parallel or in series, what happens to wall thickness and to cavity radius, and what happens to wall stress, showing which term of the equation drives your answer. Next, write why both ventricles are the same thickness at birth, what changes afterwards and why, then explain why a large pulmonary embolism can fail the right ventricle while the same ventricle copes easily with a big volume load. Finish with the patient who has a leg vein clot, a probe patent foramen ovale and a pulmonary embolism, and an hour later develops weakness on one side: write in four steps how a clot from the leg reached a brain artery and what holds that opening shut in normal life.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-03-01-C05-PR01

Predict what happens when right heart pressure rises above left

A 56-year-old woman with a probe patent foramen ovale and a deep vein thrombosis in the right calf has a large pulmonary embolism. Predict each item below, write the direction of the change, and give the structure, pressure or law that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to pulmonary vascular resistance and to right ventricular systolic pressure		
What happens to right ventricular wall stress, naming the term of the law of Laplace that changes		
What happens to the pressure gradient across the fossa ovalis and to the flap of septum primum		
What happens to the direction of any flow through the fossa ovalis		
What happens to a fragment of thrombus arriving in the right atrium, and which filter it now escapes		
What happens to the position of the interventricular septum and to left ventricular filling		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C06-BD1

Blunt and penetrating injury, structure by structure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-BD1 Date: _____

Write in one sentence each how blunt trauma injures and how penetrating trauma injures, and say which structures each one damages most. Then draw the arch and descending aorta and mark the site where deceleration tears it, naming the segment and the two landmarks that bound it. Beside the drawing write why the tear happens there, naming what holds the descending aorta and what leaves the arch mobile, and write which layer can contain the leak and what that contained leak is called. List the chest film findings of this injury and the test the diagnosis actually rests on. Next write which chamber a sternal blow bruises most often, the marker it releases, the commonest rhythm disturbance and the conduction block that fits the anterior course of the right bundle. Write what a precordial blow landing on the upstroke of the T wave can trigger and the name for that. Finish by writing the order of chambers injured by stab wounds, and what decides whether blood collects in the pericardial sac or drains into the pleural cavity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C06-BD2

Explain every sign of tamponade from a stiff sac

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-BD2 Date: _____

A man is stabbed just left of the sternum. His pulse is 128 per minute, his blood pressure is 82 over 64 mm Hg, his neck veins are distended and his heart sounds are muffled. Breath sounds are equal on both sides and the trachea is midline. Write which chamber the blade most likely entered and why that chamber and not another. Then write how much blood in the sac can cause this when it enters over minutes, and explain from the shape of the pericardial pressure and volume curve why a slowly collecting effusion can reach a liter with little pressure rise. Explain each of the following in one or two sentences from rising pressure in a stiff closed sac: why the right chambers are affected before the left, why filling and stroke volume fall, why venous pressure rises, why diastolic pressures in all four chambers become equal, why the heart sounds are muffled and why the pulse is fast. Then write why systolic pressure falls by more than 10 mm Hg on inspiration, naming what the septum does and why it has nowhere else to go. Write what happens to the jugular y descent, and write why the neck veins can be flat in a patient who has also lost a lot of blood. Finish by writing the two findings on examination that would point to a tension pneumothorax instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C06-RT01

Rebuild the downward spiral of acute tamponade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-RT01 Date: _____

Rebuild the six step loop of acute tamponade, one step per line and in order, then draw it as a closed circle with arrows so each step feeds the next. Beside the first step write the volume of rapidly entering blood that can do this and the property of the fibrous sac that allows it. Beside the compression step write which chambers are squeezed first and in which part of the cardiac cycle. Beside the coronary step write the two pressures whose difference gives coronary perfusion pressure and what each one is doing here. Underneath the loop write the rhythm the spiral ends in if the blood is not removed, and write in one line what breaks the loop.

Rebuild from memory: The downward spiral of acute tamponade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-03-01-C06-CP01

Tamponade against tension pneumothorax after chest trauma

Fill in every cell of the table from memory, then open the notes chapter and correct any cell you missed.

Finding	Cardiac tamponade	Tension pneumothorax
Why cardiac output falls		
Neck veins		
Breath sounds and percussion note		
Position of the trachea		
Heart sounds		
What makes the neck veins flat in either condition		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-03-01-C06-CA01

Shock with distended neck veins after a stab wound

A 27-year-old man is brought to the emergency department 20 minutes after being stabbed with a short blade. The wound is in the fourth intercostal space 2 cm to the left of the sternal edge. Temperature is 36.8 °C (98.2 °F), pulse is 132 per minute, respirations are 26 per minute and blood pressure is 80 over 66 mm Hg. The neck veins are distended at 45 degrees, the heart sounds are muffled and the precordium is quiet. Breath sounds are equal on both sides and the percussion note is normal in both lungs. The trachea is midline. Systolic pressure falls by 18 mm Hg during quiet inspiration. The jugular pulse shows a brisk inward movement with the x descent and almost no y descent. The electrocardiogram shows sinus tachycardia with normal voltages and no alternans.

1. Recognize. Name the chamber the blade most likely entered and the condition that has developed in the pericardial sac, and name the three findings in this patient that together make up the classic triad.

2. Explain. Explain why a volume of blood this small produces shock, and explain why filling of the right chambers fails before the left.

3. Differentiate. Write how this picture differs from a tension pneumothorax and from blood loss alone, naming the findings in the vignette that separate them.

4. Investigate. Write which measurements and which findings on the chest film and on bedside imaging you would look for, and what each one would show in this patient.

5. Decide. State which chamber pressures you expect to be equal in diastole, and state what the inspiratory fall in systolic pressure and the blunted y descent each tell you about filling.

6. Justify. Justify your answer from the structure of the pericardium and from the way the two ventricles share a fixed space, and explain why a slow effusion of the same volume would not do this.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-03-01-C01-Q01

Which chamber forms the apex beat

A 54-year-old woman is examined because of 6 months of breathlessness on exertion. A forceful, sustained impulse is felt in the left sixth intercostal space 3 cm lateral to the midclavicular line. A posteroanterior chest radiograph taken in full inspiration shows a cardiothoracic ratio of 0.59 with the lower left contour displaced downward and to the left. Enlargement of which of the following chambers best accounts for these findings?

A. Right atrium

B. Right ventricle

C. Left atrium

D. Left ventricle

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C02-Q02

Dysphagia and a widened carina

A 45-year-old woman with a history of rheumatic fever has 4 months of difficulty swallowing solid food. A barium swallow shows an indentation on the anterior wall of the esophagus at the level of the carina, and a chest radiograph shows elevation of the left main bronchus with a widened carinal angle. Which of the following best explains these findings?

A. An enlarged left atrium expanding backward and upward against the esophagus and left main bronchus

B. A dilated right ventricle filling the retrosternal space and displacing the mediastinum

C. An aneurysm of the descending thoracic aorta indenting the esophagus from behind

D. An aberrant right subclavian artery passing behind the esophagus

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C04-Q03

Hoarseness from a mediastinal mass

A 68-year-old man who has smoked for 40 years has 2 months of a hoarse, breathy voice. Computed tomography shows a 4 cm mass in the space bounded above by the inferior surface of the aortic arch and below by the left pulmonary artery. Injury to a nerve at which of the following sites best explains his voice change?

A. The right recurrent laryngeal nerve where it hooks under the right subclavian artery

B. The left phrenic nerve where it crosses the aortic arch

C. The left recurrent laryngeal nerve where it hooks under the arch at the ligamentum arteriosum

D. The left vagus nerve where it lies behind the root of the left lung

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C06-Q04

Where deceleration tears the aorta

A 31-year-old man is brought to the emergency department after a head-on collision at high speed. He is hypotensive. A chest radiograph shows a widened mediastinum, loss of the normal aortic knob contour, deviation of the trachea and nasogastric tube to the right, and fractures of the left first and second ribs. Computed tomographic angiography shows an aortic injury. At which of the following sites is the tear most likely to be found, and for which reason?

A. The ascending aorta, because it lies inside the pericardium and is compressed against the sternum

B. The aortic isthmus, because the mobile arch shears against the descending aorta held to the spine

C. The aortic root at the valve, because the aortic and mitral valves are in fibrous continuity

D. The aorta at the diaphragmatic hiatus, because the aortic opening is the narrowest point

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-04

CV-4 Heart Valves and the Cardiac Cycle

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-04-01-C01				
NPLEX-P1-CARD-04-01-C02				
NPLEX-P1-CARD-04-01-C03				
NPLEX-P1-CARD-04-01-C04				
NPLEX-P1-CARD-04-01-C05				
NPLEX-P1-CARD-04-01-C06				
NPLEX-P1-CARD-04-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C01-BD1

Draw the four valves and where each is heard

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C01 Activity: NPLEX-P1-CARD-04-01-C01-BD1 Date: _____

Without looking at your notes, draw the four heart valves in position and build a table beside the drawing with one row per valve. For each valve write the chambers or vessels it separates, the number and names of its cusps, what holds it closed, and the chest location where it is best heard with the intercostal space. Then write two sentences on what the fibrous skeleton does for the valves and two sentences on why a valve cusp moves only when the pressure across it changes. Finish by writing where the murmur of aortic stenosis radiates and where the murmur of mitral regurgitation radiates, and give the anatomical reason for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C01-BD2

Why an inferior infarct tears one papillary muscle and not the other

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C01 Activity: NPLEX-P1-CARD-04-01-C01-BD2 Date: _____

A man has an occlusion of his posterior descending artery and five days later develops sudden breathlessness with a new loud murmur at the apex. Write which papillary muscle failed and why that one and not the other, naming the blood supply of each left ventricular papillary muscle. Then write in order what happens from the moment the muscle gives way: what the mitral cusps do during systole, where the blood goes, what the left atrial pressure does, and what appears in the lungs and in the forward stroke volume. Write why this is tolerated badly compared with mitral regurgitation that has built up over years. Finish by writing two other ways the same loss of cusp support occurs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-04-01-C01-CA01

Sudden breathlessness five days after an inferior infarction

A 66-year-old man was admitted five days ago with 4 hours of chest pain and ST elevation in leads II, III, and aVF. He improved, and this evening he becomes acutely breathless and cannot lie flat. Temperature is 37.1 degrees C (98.8 degrees F), pulse is 112/min and regular, respirations are 30/min, and blood pressure is 86/60 mm Hg. A new holosystolic murmur is heard at the apex and radiates to the left axilla. Crackles are present through both lung fields. The jugular venous pulsation is raised. A pressure tracing from the pulmonary artery wedge position shows a small a wave and a very large wave in late systole, with a mean wedge pressure of 30 mm Hg. The electrocardiogram is unchanged from this morning.

1. Recognize. Name the structural complication, the valve involved, and the artery whose occlusion led to it.

2. Explain. Explain the mechanism from the blood supply of the affected structure through to the murmur, the wedge tracing, and the pulmonary crackles.

3. Differentiate. Write how this presentation differs from a ventricular septal rupture and from free wall rupture with tamponade, naming the findings on examination and on the wedge tracing that separate them.

4. Investigate. Write which measurements on the wedge and jugular tracings you would look at to support your answer, and what each one would show here.

5. Decide. State why the blood pressure is low even though the total volume the ventricle ejects each beat has risen.

6. Justify. Justify that answer from where the regurgitant volume goes, from the compliance of an atrium that has had no time to dilate, and from what has happened to the forward stroke volume.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C02-BD1

Draw the Wiggers diagram and mark every valve event

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-BD1 Date: _____

On one set of axes with time across the bottom, draw left atrial pressure, left ventricular pressure, and aortic pressure, and underneath them draw left ventricular volume, the ECG, and the heart sounds. Mark the four valve events on the tracing and write beside each one which two pressures have just crossed and in which direction. Label isovolumetric contraction, rapid ejection, reduced ejection, isovolumetric relaxation, rapid filling, diastasis, and atrial systole. Write the normal numbers on the drawing: aortic diastolic and peak pressures, end-diastolic and end-systolic volumes, stroke volume, ejection fraction, pulmonary artery pressure, and the length of one cycle with its systolic and diastolic parts at 75 beats per minute. Finish by marking where S1 and S2 fall and writing what is vibrating to make each one.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C02-BD2

Name the phase from four numbers, and what a fast rate costs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-BD2 Date: _____

Three tracings are sampled at one instant each. Tracing A: left atrial pressure 9 mm Hg, left ventricular pressure 45 mm Hg, aortic pressure 84 mm Hg, left ventricular volume 120 mL. Tracing B: left atrial pressure 9 mm Hg, left ventricular pressure 45 mm Hg, aortic pressure 84 mm Hg, left ventricular volume 50 mL. Tracing C: left atrial pressure 12 mm Hg, left ventricular pressure 6 mm Hg, aortic pressure 78 mm Hg, left ventricular volume 70 mL. For each tracing write the state of the mitral valve and the aortic valve, name the phase, and show the reasoning as a comparison of two pressures at a time. Write which single measurement separates tracing A from tracing B and why pressure cannot do it. Then the heart rate rises from 75 to 160 beats per minute. Write what happens to the length of systole, the length of diastole, the part of diastole that is lost first, and ventricular filling, and explain why left ventricular coronary flow suffers. Finish by writing why the pulmonary valve opens before the aortic valve and closes after it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-04-01-C02-RT01

Rebuild the valve events of one left-sided cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C02 Activity: NPLEX-P1-CARD-04-01-C02-RT01 Date: _____

Rebuild the nine-step sequence of valve events in one left-sided cardiac cycle from atrial systole through to diastasis, one step per line and in order. Beside each step that is a valve opening or closing, write the two pressures being compared and the approximate pressure at which it happens. Mark the two steps during which every valve is closed and write what the ventricular volume is doing in each. Mark the step that produces S1 and the step that produces S2, and beside the aortic closure step write what makes the dicrotic notch on the aortic curve. Finish by writing what makes S1 loud and what makes it soft, with the reason in each case.

Rebuild from memory: Valve events in one left-sided cardiac cycle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C03-BD1

Draw the normal pressure-volume loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C03 Activity: NPLEX-P1-CARD-04-01-C03-BD1 Date: _____

Draw the left ventricular pressure-volume loop with pressure on the vertical axis and volume on the horizontal axis. Mark the direction the loop is traced. Label all four corners with the valve event that happens there and write the pressure and volume at each one. Label the two vertical sides and the top and bottom edges with the phase each represents. Then mark on the drawing what the width of the loop measures and what the enclosed area measures, and write beside the bottom edge and the top left corner which relationship each one follows and what changes its slope. Finish by writing the normal end-diastolic volume, end-systolic volume, stroke volume, and ejection fraction range.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C03-BD2

Name the lesion from the loop

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C03 Activity: NPLEX-P1-CARD-04-01-C03-BD2 Date: _____

Three loops are described. Loop 1: end-diastolic volume 190 mL, end-systolic volume 65 mL, peak pressure 115 mm Hg, and the left side of the loop slopes to the right so volume rises while pressure is falling after ejection. Loop 2: end-diastolic volume 125 mL, end-systolic volume 75 mL, peak pressure 215 mm Hg with aortic pressure 115/70 mm Hg, and a steep bottom edge with an end-diastolic pressure of 22 mm Hg. Loop 3: end-diastolic volume 200 mL, end-systolic volume 70 mL, peak pressure 105 mm Hg, and volume starts falling at the bottom right corner while pressure is only about 15 mm Hg. For each loop calculate the stroke volume, state whether the load is a pressure load or a volume load, name the lesion, and write the one feature of the loop that settles it. For each lesion write which isovolumetric phase is no longer isovolumetric and why, which pattern of hypertrophy follows, and what has happened to the forward stroke volume compared with the total. Finish by writing what the loop of significant mitral stenosis looks like and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-04-01-C03-CP01

Pressure-volume loop in three valve lesions

Fill in every cell from memory, then open the notes and correct any cell you missed.

Feature	Aortic stenosis	Aortic regurgitation	Mitral regurgitation
Pressure load or volume load			
Peak left ventricular pressure, and how it compares with aortic pressure			
End-diastolic volume			
End-systolic volume			
Total stroke volume, and what happens to the forward volume			
Which isovolumetric phase is no longer isovolumetric, and what the volume does during it			
Pattern of chronic hypertrophy			
Shape and position of the loop			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-04-01-C03-Q01

A loop that loses volume before the aortic valve opens

A pressure-volume loop recorded from the left ventricle of a 58-year-old woman shows an end-diastolic volume of 195 mL, an end-systolic volume of 70 mL, and a peak ventricular pressure of 108 mm Hg. Volume begins to fall at the bottom right corner of the loop while ventricular pressure is still only 16 mm Hg, so the right side of the loop slopes to the left rather than running vertically. Her measured forward cardiac output is below normal. Which of the following best explains the shape of this loop?

A. A fixed obstruction to left ventricular outflow raising the pressure needed to eject

B. Reduced left ventricular compliance with a steep end-diastolic pressure-volume relationship

C. Blood escaping into the left atrium from the onset of ventricular contraction

D. Blood returning from the aorta into the ventricle during relaxation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C04-BD1

Draw the jugular venous pulse against the heart sounds

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C04 Activity: NPLEX-P1-CARD-04-01-C04-BD1 Date: _____

Draw one cycle of the jugular venous pulse and underneath it draw the ECG and mark S1 and S2. Label the a wave, the c wave, the x descent, the v wave, and the y descent. Beside each label write the mechanical event in the right atrium or ventricle that produces it and its timing relative to the P wave, S1, or S2. Then write how the jugular pulse is told apart from the carotid pulse, listing four differences. Finish by writing how the height of the pulsation is used to estimate right atrial pressure, including the distance from the sternal angle to the right atrium, the angle at which the patient sits, and the height above the sternal angle that suggests raised pressure.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C04-BD2

Read four abnormal neck tracings

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C04 Activity: NPLEX-P1-CARD-04-01-C04-BD2 Date: _____

Four patients have abnormal jugular findings. Patient 1 has a pulse of 36 beats per minute and occasional very large pulsations that do not appear with every beat. Patient 2 has a narrow complex tachycardia with large pulsations that appear with every beat. Patient 3 has an irregularly irregular pulse and no a wave at all. Patient 4 has a large systolic wave with no x descent, a steep y descent, and a pulsating liver. For each patient name the abnormal wave or missing wave, write the mechanism in terms of what the right atrium is contracting or filling against, and write the rhythm or lesion responsible. For patient 1 explain why the large pulsations come only on some beats and for patient 2 explain why they come on every beat. For patient 4 write what has merged with what and why the pressure rises right through systole, then write which tracing shows the same picture in acute mitral regurgitation. Finish by writing what a large a wave that is not a cannon wave tells you and give two causes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-04-01-C04-Q02

Intermittent giant neck pulsations

A 77-year-old woman comes to the emergency department after two episodes of collapse. Pulse is 40/min and regular, and blood pressure is 118/72 mm Hg. Inspection of the neck shows occasional very large venous pulsations that are not palpable and do not occur with every beat. The electrocardiogram shows P waves at a rate of 92/min and QRS complexes at a rate of 40/min with no fixed relationship between them. Which of the following best explains the large venous pulsations?

A. Right atrial contraction against a closed tricuspid valve on beats where the timing coincides

B. Regurgitation of blood from the right ventricle into the right atrium through an incompetent tricuspid valve

C. Obstruction of right atrial emptying through a stenotic tricuspid valve

D. Transmitted pulsation from the adjacent carotid artery during ventricular ejection

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C05-BD1

Build the S2 splitting table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-BD1 Date: _____

On blank paper build a table of the four patterns of S2 splitting with one row each for physiologic, wide, fixed, and paradoxical. Use the columns order of the two components, what is heard in expiration, what is heard in inspiration, the mechanism, and the classic causes, and fill every cell. Above the table write which component of S2 is loud and heard widely, which is soft, where splitting is listened for, and how far apart the components must be for a split to be audible. Beside the physiologic row, draw the timing of the two components in expiration and in inspiration. Finish by writing the two questions you ask at the bedside to place any S2 into one of these four rows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C05-BD2

Four patients with a split S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-BD2 Date: _____

Four patients have an abnormal second sound. Patient 1 has a split heard in expiration that widens further on inspiration, and the ECG shows an rSR prime in V1 with a QRS of 0.14 s. Patient 2 has a wide split that does not change at all through the respiratory cycle. Patient 3 has a split heard in expiration that becomes single on inspiration, and an ejection murmur at the right second intercostal space with a soft A2. Patient 4 has a split heard in expiration that widens on inspiration, with a holosystolic murmur at the apex radiating to the axilla. For each patient name the pattern, write which component is early or late and why, and write whether the respiratory mechanism is still working. For patient 2 explain from the behaviour of the two atria why breathing no longer changes the split. For patient 3 explain why inspiration narrows rather than widens it. For patient 4 explain which component has moved and what gave the left ventricle a second route out. Finish by writing the two mechanisms, one electrical and one mechanical, that delay a component, and give an example of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-04-01-C05-RT01

Rebuild the sequence that widens the split on inspiration

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C05 Activity: NPLEX-P1-CARD-04-01-C05-RT01 Date: _____

Rebuild the seven-step sequence that runs from the start of inspiration to a wider split of S2, one step per line and in order. Mark which steps act on the right heart and which act on the left. Beside the step about pulmonary vessels, write what happens to pulmonary vascular impedance and what that does to the length of right ventricular ejection. Beside the step about left ventricular filling, write why filling dips and what that does to the timing of A2. Underneath, draw A2 and P2 on a timeline for expiration and for inspiration with the interval written in seconds, and write in which age group this is easiest to hear.

Rebuild from memory: Physiologic splitting of S2 during inspiration

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C06-BD1

S3 and S4 side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C06 Activity: NPLEX-P1-CARD-04-01-C06-BD1 Date: _____

Write out S3 and S4 side by side on one page. For each sound give its timing in diastole, the interval from S2 where the notes gave one, the mechanical event that produces it, what is vibrating, where and with which part of the stethoscope it is heard on the left and on the right, and what makes a right-sided one louder. Then for each sound list when it is a normal finding and when it means disease, with the conditions from the notes. Write what happens to each sound in atrial fibrillation and give the reason. Finish by writing why significant mitral stenosis prevents an S3.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C06-BD2

Decide which murmur is just flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C06 Activity: NPLEX-P1-CARD-04-01-C06-BD2 Date: _____

Write the physical reason a normal valve can produce a murmur, naming the quantity that predicts turbulence and the three things that raise or lower it. Then take four people: a woman at 30 weeks of pregnancy, a man with a haemoglobin of 68 g/L, a child of six with a thin chest, and a woman with a temperature of 39.4 degrees C (103 degrees F). For each one write which part of that quantity has changed and in which direction, and what happens to the murmur when the cause is removed. Next list the features that let you call a murmur functional: grade, timing, location, what S2 does, and what extra sounds are present. Then list the findings that argue for structural disease instead, and name the one continuous murmur that is still innocent along with the two bedside moves that abolish it. Finish by writing the three murmurs a child commonly has and where each is heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-04-01-C07-BD1

Build the murmur timing table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C07 Activity: NPLEX-P1-CARD-04-01-C07-BD1 Date: _____

Build a table of murmurs from memory with one row per lesion, using the columns timing and shape, lesion, where it is best heard, and the one feature that gives it away. Cover aortic stenosis, pulmonic stenosis, hypertrophic cardiomyopathy, mitral regurgitation, tricuspid regurgitation, ventricular septal defect, mitral valve prolapse, aortic regurgitation, mitral stenosis, tricuspid stenosis, and patent ductus arteriosus. Above the table write the rule that places any murmur in systole or diastole from the valve involved, and write the grading scale from 1 to 6 with the grade at which a thrill becomes palpable. Underneath, write why an ejection murmur starts after S1 rather than with it, why a holosystolic murmur runs the whole of systole, and why the murmur of mitral valve prolapse comes late with a click in front of it. Finish by writing what happens to the peak of the aortic stenosis murmur and to A2 as the stenosis worsens, and what the interval from S2 to the opening snap says about the severity of mitral stenosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-04-01-C07-BD2

Separate two systolic murmurs with the bedside maneuvers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-4 Heart Valves and the Cardiac Cycle
Competency: NPLEX-P1-CARD-04-01-C07 Activity: NPLEX-P1-CARD-04-01-C07-BD2 Date: _____

Write what each of these changes at the bedside: inspiration, squatting, standing from a squat, handgrip, and Valsalva strain. For each one write whether preload, afterload, or both are altered and in which direction. Then write what most murmurs do with more preload and what regurgitant murmurs do with more afterload, and the reason in each case. Two patients each have a systolic murmur at the left sternal border that grows louder when they stand. Write which two conditions behave this way, what is happening inside the left ventricle as the cavity gets smaller, and what each finding does on squatting and on handgrip. Finish with a third patient who has an ejection murmur radiating to the carotids that gets softer on standing and softer on handgrip, and write why a fixed obstruction behaves the opposite way to the first two.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-04-01-C07-PR01

Predict what standing does to a click and a late systolic murmur

A 29-year-old woman has a midsystolic click followed by a late systolic murmur at the apex. She stands up quickly from a squat. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return and to left ventricular end-diastolic volume		
What happens to the size of the left ventricular cavity at the start of systole		
What happens to the point in systole at which the mitral cusps give way		
What happens to the timing of the click		
What happens to the length and loudness of the murmur		
What happens to the click and the murmur when she squats again and when she performs handgrip		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-05

CV-5 Cardiac Muscle Contraction

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-05-01-C01				
NPLEX-P1-CARD-05-01-C02				
NPLEX-P1-CARD-05-01-C03				
NPLEX-P1-CARD-05-01-C04				
NPLEX-P1-CARD-05-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C01-BD1

Draw the ventricular action potential

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C01 Activity: NPLEX-P1-CARD-05-01-C01-BD1 Date: _____

Without looking at your notes, draw the action potential of a ventricular myocyte with voltage on the vertical axis and time on the horizontal axis. Label phase 4, phase 0, phase 1, phase 2, and phase 3. Beside each phase write the current and the named channel that dominates it, the direction that current carries charge, and the part of the ECG that phase corresponds to. Write the resting voltage, the threshold voltage, the peak voltage, how long the plateau lasts, and how long the whole action potential lasts. Then write what inward rectification means for the IK1 channel and give two things that property does for a working myocyte. Finish by writing how current reaches this cell in the first place and which structure in the intercalated disc carries it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C01-BD2

Refractoriness and a change in plasma potassium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C01 Activity: NPLEX-P1-CARD-05-01-C01-BD2 Date: _____

Write how long the effective refractory period lasts, the voltage at which it ends, what keeps the cell unexcitable during it, and how long the relative refractory period lasts and what a stimulus during it produces. Use that to explain in three or four sentences why the ventricle cannot be tetanized when skeletal muscle can. Next, a patient has a plasma K+ of 8.0 mmol/L (reference range 3.5 to 5.0 mmol/L). Work through the prediction in steps: calculate the new K+ equilibrium potential using 140 mmol/L inside the cell, write where the resting potential moves and why IK1 makes it move there, write what that does to the fast Na+ channels between beats, and write what happens to phase 0, to conduction velocity, to the QRS, and to phase 3 and the T wave. Say which ECG change appears first. Then write what a plasma K+ of 2.6 mmol/L does to phase 3 instead, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C02-BD1

Draw the dyad and the path from voltage to force

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-BD1 Date: _____

Draw one ventricular myocyte in cross section with a T tubule meeting a terminal cisterna of the sarcoplasmic reticulum. Label the T tubule, the dyad, the width of the cleft, the L-type Ca2+ channel with its protein name, the RyR2 cluster, the sarcoplasmic reticulum, and calsequestrin. Write where along the sarcomere the T tubule enters. Beside the drawing write the diastolic cytosolic free Ca2+, the peak cytosolic Ca2+ during a beat, the extracellular ionized Ca2+, and the share of activator Ca2+ that comes from the sarcoplasmic reticulum in human ventricle. Then write out calcium-induced calcium release in your own words, including what a Ca2+ spark is, what the Ca2+ transient is, and what ends release. Finish by writing the cross-bridge cycle from Ca2+ binding troponin C to the head detaching, naming each step and the point at which ATP is needed.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C02-BD2

Remove the trigger, then let the store leak

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-BD2 Date: _____

An isolated heart and an isolated skeletal muscle are both moved into a Ca2+ free solution and stimulated. Write what happens to contraction in each over the next minute, how long each takes to change, and the structural reason for the difference. Say which transporter runs the cardiac store down once trigger entry stops. Then write what a drug that blocks cardiac L-type Ca2+ channels does to force and why skeletal twitches are spared. Next, a young person has a RyR2 mutation that lets the sarcoplasmic reticulum leak Ca2+ in diastole, and collapses during exercise. Write the sequence from adrenergic stress to the arrhythmia, naming what the Na+/Ca2+ exchanger does with the leaked Ca2+, the charge it moves per cycle, the resulting membrane event, and the name of that condition. Finish by naming two other situations in which the notes describe the same mechanism.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-05-01-C02-RT01

Rebuild excitation-contraction coupling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-RT01 Date: _____

Rebuild the six-step sequence from the arriving action potential to a shortened sarcomere, one step per line and in order. Beside step two write which channel opens and where it sits. Beside step three write what opens RyR2 and from which side of the channel. Beside step four write the starting and peak cytosolic Ca2+ values. Beside step five write what troponin I and tropomyosin each do. Then write the sequence backwards as the relaxation sequence, naming what has to happen at troponin C and at each myosin head.

Rebuild from memory: Excitation-contraction coupling in a ventricular myocyte

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-05-01-C02-CP01

Skeletal against cardiac excitation-contraction coupling

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Skeletal muscle	Cardiac (ventricular) muscle
What excites the cell		
Where the T tubule sits along the sarcomere, and the name of the junction		
How the voltage sensor opens the sarcoplasmic reticulum release channel		
Whether extracellular Ca2+ is needed for a twitch		
Duration of the action potential		
Ca2+ sensor protein and how many Ca2+ binding sites it has		
What a Ca2+ free solution does to contraction within a minute		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C03-BD1

Build the calcium removal table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-BD1 Date: _____

On blank paper build a table of the pathways that lower cytosolic Ca2+ in a ventricular myocyte, one row each for SERCA2a, the Na+/Ca2+ exchanger, and the plasma membrane Ca2+-ATPase with mitochondrial uptake. Use the columns location, energy source with the stoichiometry, approximate share of Ca2+ removed, and main regulator, and fill every cell. Underneath write the two values cytosolic Ca2+ moves between during relaxation. Then write three sentences on why relaxation costs ATP, naming what each myosin head needs in order to let go of actin. Finish by writing what unphosphorylated phospholamban does to SERCA2a, what phosphorylation of phospholamban does, and which two kinases phosphorylate it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C03-BD2

Partial inhibition of the sodium pump, beat by beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-BD2 Date: _____

The Na+/K+-ATPase of a ventricular myocyte is partially inhibited. Work out what follows in six numbered steps, from the change in cytosolic Na+ through to the force of each beat, naming at every step which transporter is affected and why. Say which beat the force change appears on and why it takes several beats to settle. Then write what happens if the inhibition goes further: name the state the sarcoplasmic reticulum reaches, what it does in diastole, what the Na+/Ca2+ exchanger makes of that release, and the membrane event that results. Finish by writing why ischemia drives cytosolic Ca2+ up by the same route, naming the exchanger that adds Na+ during ischemia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-05-01-C03-RT01

Rebuild calcium cycling in one heartbeat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-RT01 Date: _____

Rebuild the six-step Ca2+ cycle of a single heartbeat, one step per line and in order, starting with trigger entry and ending with the relaxed cell. Beside each step write the structure doing the work and, where the notes gave one, the share of Ca2+ it handles. Then write the two balances that must hold for the cell to stay in a steady state, one line each. Underneath, write what happens to sarcoplasmic reticulum load and to the force of later beats if extrusion falls behind entry, and what happens if reuptake falls behind release.

Rebuild from memory: Ca2+ cycling in one heartbeat

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-05-01-C03-PR01

Predict what happens when coronary flow stops

A coronary artery is suddenly occluded and the myocytes it supplies lose oxidative phosphorylation. Predict each item below, write the direction of the change, give the time course where the notes gave one, and name in one line the molecule or transporter that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to creatine phosphate and to the products of anaerobic glycolysis		
What happens to the Ca2+ sensitivity of troponin C and to force per cross-bridge, and how soon		
What happens to cytosolic Na+ and to the driving force for the Na+/Ca2+ exchanger		
What happens to the rate of SERCA2a pumping and to diastolic cytosolic Ca2+		
What happens to ventricular stiffness and to end-diastolic pressure		
Which of contraction and relaxation is impaired first, and where that sits in the order of findings in the ischemic cascade		
What happens inside the cell if the occlusion lasts 20 to 40 minutes, and what Ca2+ overload does on reperfusion		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C04-BD1

Draw the beta-1 cascade and its targets

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C04 Activity: NPLEX-P1-CARD-05-01-C04-BD1 Date: _____

Draw the beta-1 adrenergic cascade in a ventricular myocyte from the catecholamine arriving at the receptor to the change in the Ca2+ transient, naming the G protein and its alpha subunit, the enzyme it stimulates, the second messenger, the kinase, and the protein that holds the kinase near its targets. Name the two catecholamines and where each comes from. Beside the drawing build a table of protein kinase A targets with one row each for the L-type Ca2+ channel, phospholamban, troponin I, myosin-binding protein C, and the IKs channel, with columns for what phosphorylation does to the protein and what that means for the beat. Finish by writing the three things that end the signal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C04-BD2

Take one target out of the cascade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C04 Activity: NPLEX-P1-CARD-05-01-C04-BD2 Date: _____

A mouse expresses a mutant phospholamban that cannot be phosphorylated but still binds SERCA2a. It is given a beta-1 agonist. Write what still happens to heart rate, to the L-type Ca2+ current, and to the force of contraction, then write what does not happen, naming the two separate jobs the normal phospholamban response does within one beat and across successive beats. Write what filling looks like at a fast heart rate in this animal and why. Next, explain in three sentences why troponin I phosphorylation does not weaken the beat in a normal heart. Then write why a person who has lost IKs develops arrhythmias during exercise rather than at rest. Finish with chronic heart failure: list the four changes the notes give in the beta-1 pathway and its downstream proteins, and write the consequence of each for force, for relaxation, and for the response to exercise.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-05-01-C04-CA01

Breathlessness on exertion in a man with a dilated ventricle

A 61-year-old man comes to the physician because of 6 months of increasing breathlessness on exertion. He stops after one flight of stairs, where a year ago he walked up three. Temperature is 36.8 °C (98.2 °F), pulse is 104/min and regular, respirations are 20/min, and blood pressure is 112/78 mm Hg. Physical examination shows a displaced apical impulse, crackles at both lung bases, and jugular venous distension. Echocardiography shows a dilated left ventricle with reduced wall motion, a low ejection fraction, and slowed early diastolic filling. During a treadmill test his pulse rises only to 118/min and his stroke volume does not increase. Plasma norepinephrine is elevated. Myocardial biopsy from a research protocol shows reduced beta-1 receptor density, increased G protein-coupled receptor kinase 2, reduced SERCA2a, and phospholamban that is largely unphosphorylated.

1. Recognize. Name the signaling pathway that is failing in this myocardium and list the four biopsy findings that place the fault in it.

2. Explain. Explain how each of those four findings changes the Ca2+ transient, the force of contraction, and the speed of relaxation, naming the protein kinase A target involved in each.

3. Differentiate. Write how this picture differs from a heart whose contractility is normal but whose filling is simply low, naming what the Ca2+ transient is doing in each and what the ventricular function curve does in each.

4. Investigate. Write which measurements on the exercise test and the echocardiogram you would use to show that the fault is in the response to catecholamines rather than in filling, and what each measurement would show.

5. Decide. State which single cellular lever is most responsible for the slowed early diastolic filling, and what that does to filling once the heart rate rises.

6. Justify. Justify that answer from the share of activator Ca2+ that SERCA2a normally removes, what unphosphorylated phospholamban does to the pump, and what a smaller sarcoplasmic reticulum store then does to the force of later beats.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C05-BD1

Draw the length-tension curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C05 Activity: NPLEX-P1-CARD-05-01-C05-BD1 Date: _____

Draw sarcomere length on the horizontal axis and tension on the vertical axis, and plot both an active tension curve and a passive tension curve for cardiac muscle. Mark the thick filament length, the thin filament length, the sarcomere length at which overlap is optimal, the length at which active force reaches zero on the descending limb, and the range ventricular sarcomeres actually work over from low filling to high filling. Write what limits force on the ascending limb and what limits it on the descending limb, one line each. Then write what length-dependent activation is, which way it shifts the force versus Ca2+ curve, and the three mechanisms the notes propose for it. Finish by writing which two tissue components make passive tension climb steeply above 2.2 µm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C05-BD2

Tell a preload change from an inotropic change

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C05 Activity: NPLEX-P1-CARD-05-01-C05-BD2 Date: _____

Write the two ways skeletal muscle grades its force, then write why the heart can use neither, naming the property of the myocyte network that rules out one and the property of the action potential that rules out the other. List the three levers the heart uses instead and, for each, give one thing that raises it and one thing that lowers it. Write the one condition about troponin C that has to hold for these levers to work at all. Next, two hearts each raise their stroke volume. In the first the sarcomeres lengthen with almost no change in the Ca2+ transient; in the second the Ca2+ transient grows. For each write what happens on the ventricular function curve, whether force rises at a given sarcomere length, and which of the two a beta-1 agonist produces. Finish by writing why a dilated failing ventricle is weak even though it is stretched, naming the three reasons the notes give and the structural change that keeps its sarcomeres off the descending limb.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-05-01-C05-CP01

Length-tension and force grading in skeletal against cardiac muscle

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Skeletal muscle	Cardiac muscle
Sarcomere length the muscle normally works at		
What the active length-tension curve rests on		
Passive tension at long sarcomere lengths		
Whether recruitment and summation are available, and why		
How large a part Ca2+ delivery and myofilament Ca2+ sensitivity play in grading force		
How much starting length matters in normal function, and the named mechanism in the heart		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-05-01-C01-Q01

A widened QRS with a high plasma potassium

A 58-year-old woman with chronic kidney disease is brought to the emergency department because of weakness. Temperature is 36.9 °C (98.4 °F), pulse is 58/min, respirations are 16/min, and blood pressure is 104/68 mm Hg. Serum studies show a K+ of 8.1 mmol/L (reference range 3.5 to 5.0 mmol/L). The electrocardiogram shows peaked T waves and a QRS duration of 0.16 s. Which of the following best explains the widened QRS complex?

A. Increased conductance of the rapid delayed rectifier K+ current

B. Prolonged opening of L-type Ca2+ channels during the plateau

C. A less negative resting potential leaving fast Na+ channels inactivated

D. Reduced Ca2+ release from the sarcoplasmic reticulum through RyR2

Why did you pick it?

Board-style question NPLEX-P1-CARD-05-01-C02-Q02

Two muscles in a calcium free bath

In an experiment, an isolated perfused mammalian heart and an isolated skeletal muscle preparation are switched to a Ca2+ free solution and stimulated at a constant rate. Within 1 minute the heart develops almost no force, while the skeletal muscle continues to twitch with little change. Which of the following best explains the difference between the two preparations?

A. Cardiac release channels open only in response to Ca2+ entering through L-type channels, while the skeletal voltage sensor opens its release channel mechanically

B. Cardiac myocytes use calmodulin and myosin light chain kinase, while skeletal fibers use troponin C

C. Cardiac sarcoplasmic reticulum holds no Ca2+ store, so all activator Ca2+ enters from outside the cell

D. Cardiac T tubules lie at the A-I junction, which places L-type channels too far from the release channels

Why did you pick it?

Board-style question NPLEX-P1-CARD-05-01-C05-Q03

Force doubles with a small change in overlap

A strip of ventricular muscle is stretched from a sarcomere length of 1.9 µm to 2.2 µm and stimulated at a fixed interval. Active force roughly doubles, while the calculated change in thick and thin filament overlap over this range is modest and the peak cytosolic Ca2+ of each twitch is unchanged. Which of the following best accounts for most of the increase in force?

A. Recruitment of additional myocytes that were not contracting at the shorter length

B. Summation of successive twitches as the refractory period shortens with stretch

C. A larger trigger Ca2+ influx through L-type Ca2+ channels at the longer length

D. Greater myofilament Ca2+ sensitivity at the longer sarcomere length

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-06

CV-6 Cardiac Output and Its Regulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-06-01-C01				
NPLEX-P1-CARD-06-01-C02				
NPLEX-P1-CARD-06-01-C03				
NPLEX-P1-CARD-06-01-C04				
NPLEX-P1-CARD-06-01-C05				
NPLEX-P1-CARD-06-01-C06				
NPLEX-P1-CARD-06-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C01-BD1

Write out the output and pressure equations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C01 Activity: NPLEX-P1-CARD-06-01-C01-BD1 Date: _____

Without looking at your notes, write out the equations for cardiac output, stroke volume, ejection fraction, cardiac index, mean arterial pressure from cardiac output and resistance, mean arterial pressure from a cuff reading, and cardiac output by the Fick principle. Beside each equation write the normal resting value of every quantity in it, with units. Then write the full formula for the oxygen content of blood, label which part is carried by hemoglobin and which is dissolved, and write how many millilitres per decilitre the dissolved part contributes. Write why the cuff rule for mean arterial pressure uses one third of the pulse pressure and when that rule underestimates the true value. Finish by writing where arterial and mixed venous samples must be drawn and why mixed venous blood has to come from that site.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C01-BD2

Work one patient two ways and read the venous saturation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C01 Activity: NPLEX-P1-CARD-06-01-C01-BD2 Date: _____

A resting patient has a heart rate of 60 bpm, an end-diastolic volume of 150 mL and an end-systolic volume of 90 mL. Calculate stroke volume, cardiac output and ejection fraction, and state whether each one is normal. The same patient has an oxygen consumption of 240 mL/min, hemoglobin of 12 g/dL, arterial saturation of 97 percent with an arterial PO2 of 90 mmHg, and pulmonary artery saturation of 55 percent with a PO2 of 30 mmHg. Calculate cardiac output by the Fick principle, showing every step including the conversion from millilitres per decilitre to millilitres per litre. Then write what the low pulmonary artery saturation tells you about oxygen delivery relative to demand, and explain from the Fick equation why the arteriovenous oxygen difference widened. Finish by writing what a sample taken from a forearm vein instead of the pulmonary artery would do to your calculated output, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C02-BD1

Draw the Frank-Starling curve and define the three determinants

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C02 Activity: NPLEX-P1-CARD-06-01-C02-BD1 Date: _____

Draw a set of axes with end-diastolic volume on the horizontal axis and stroke volume on the vertical axis, and draw the normal Frank-Starling curve on it. Mark the part of the curve the normal heart works on and write what limits further stretch. Then draw a second curve for raised contractility and a third for reduced contractility, and label which direction each one shifted. Separately, define preload, afterload and contractility, and for each one write the strict definition, the measurement used in practice as an index of it, and what the ventricle is doing physically. Write why a stiff ventricle can show a high filling pressure at a normal end-diastolic volume. Finish by writing the sarcomere length that gives maximum force and the two cellular reasons stretch increases force, naming which of the two matters more.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C02-BD2

Follow three single changes through to stroke volume

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C02 Activity: NPLEX-P1-CARD-06-01-C02-BD2 Date: _____

For each of the three changes below, write which of the three determinants of stroke volume it acts on first, then what happens to end-diastolic volume, end-systolic volume, stroke volume and ejection fraction, then how the Frank-Starling curve moves, then how the pressure-volume loop changes shape. Write each chain as a sequence of steps rather than single words. The three changes are a sudden rise in aortic pressure with venous return unchanged, a rapid infusion of one litre of saline, and an acute coronary occlusion that leaves part of the ventricle weakly contracting. For the aortic pressure change, explain over several beats why stroke volume partly recovers and what that recovery costs. Then write why ejection fraction is not a clean measure of contractility, and why a heart rate of 200 bpm lowers cardiac output rather than raising it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C03-BD1

Draw the cardiac function and venous return curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-BD1 Date: _____

Draw one set of axes with right atrial pressure on the horizontal axis and flow in litres per minute on the vertical axis. On it draw the cardiac function curve and the venous return curve, and mark the point where they cross. Write what that crossing point represents and why the two flows must be equal. Label the right atrial pressure at which the venous return curve meets the horizontal axis, and write what that pressure is called and its normal value. Write the equation for venous return, name what sets the slope of the curve, and write why the curve flattens at low right atrial pressures. Finish by writing what stressed volume means, roughly what fraction of the blood the veins hold, and what venoconstriction does to mean systemic filling pressure and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C03-BD2

Name the problem from cardiac output and right atrial pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-BD2 Date: _____

Four patients each have a low cardiac output. Patient A has a right atrial pressure of 1 mmHg, patient B has 14 mmHg, patient C has 2 mmHg after three days of vomiting, and patient D has 15 mmHg two days after a large anterior infarction. For each patient write whether the primary problem lies in the heart or in the vascular compartment, give the reasoning from whether the two numbers moved in the same direction or in opposite directions, name which curve shifted and in which direction, and give a cause from the notes that fits. Then write what happens on the diagram when a dilated exercising muscle bed or an arteriovenous fistula lowers total peripheral resistance, and say why cardiac output changes in that case while right atrial pressure stays about the same.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-06-01-C03-RT01

Rebuild the hemorrhage sequence on the curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C03 Activity: NPLEX-P1-CARD-06-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from blood loss to a partly recovered cardiac output, one step per line and in order. Beside the step where mean systemic filling pressure falls, write why losing blood lowers it and which part of the blood volume is responsible. Beside the reflex steps, write which receptors sense the change and which effector action restores each of the two curves. Then draw one set of axes and show three things on it: the starting operating point, the point immediately after the blood loss, and the point after the reflex response. Beside the final point write why right atrial pressure stays low even though cardiac output has improved.

Rebuild from memory: Hemorrhage on the cardiac function and venous return diagram

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C04-BD1

Write the law of Laplace and the determinants of oxygen demand

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C04 Activity: NPLEX-P1-CARD-06-01-C04-BD1 Date: _____

Write the law of Laplace in the form used for the ventricle, name each symbol, and write what the quantity on the left side means physically. Write which phase of the cardiac cycle makes that quantity the afterload. Then write, in one line each, what happens to wall stress when pressure rises, when the chamber radius grows, and when the wall thickens, and explain the thickness effect in terms of how the load is shared. Separately, list the three main determinants of myocardial oxygen demand, write which kind of work costs the most oxygen, and write the simple bedside index of demand and the two numbers it multiplies. Finish by writing the percentage of delivered oxygen the resting heart already extracts, what that leaves as the only way to meet higher demand, and which part of the cardiac cycle carries most left ventricular coronary flow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C04-BD2

Remodeling and the ischemic cost of a faster rate

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C04 Activity: NPLEX-P1-CARD-06-01-C04-BD2 Date: _____

A patient's systolic pressure has run at 200 mmHg for years and the ventricular wall has thickened from 1.0 cm to about 1.7 cm with a normal cavity size. Write which overload this is, which measurement rose to signal it, how the myocytes added sarcomeres, and what the thickening did to wall stress. Then write the two costs of that adaptation, including what happens to filling pressure and which extra heart sound appears, and explain why the subendocardium becomes vulnerable to ischemia even when the coronary arteries are normal. Next, a second patient has severe mitral regurgitation and a large dilated ventricle. Write which overload this is, which wall stress rose, how the sarcomeres were added, and what the thickness to radius ratio does while compensation holds and when it fails. Finish with a third patient who has a fixed coronary stenosis and a heart rate that rises from 70 to 140 bpm: write the two separate reasons the faster rate causes ischemia, one on the demand side and one on the supply side.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-06-01-C04-CP01

Concentric against eccentric remodeling

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Concentric hypertrophy	Eccentric hypertrophy and dilation
Typical cause		
Which wall stress signals it		
How sarcomeres are added and what the cell shape does		
Thickness to radius ratio		
What happens to wall stress		
Effect on chamber stiffness and filling pressure		
Typical consequence and the extra heart sound		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C05-BD1

Draw the baroreceptor reflex arc

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-BD1 Date: _____

Draw the baroreceptor reflex as a labelled pathway from sensor to effector. Name the two receptor sites and the layer of the vessel wall the endings sit in, name the nerve each afferent travels in with its cranial nerve number, and name the medullary nucleus they all reach. Then draw the central connections: write which output rises and which falls when pressure rises, and name the two medullary regions that carry the sympathetic side. List every effector the reflex acts on and write what the reflex changes at each one. Finish by writing the pressure range over which carotid receptors fire, how the aortic receptors differ, what the receptors do within one to two days, and what that resetting means for long-term pressure control.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C05-BD2

Run the reflex forwards and backwards in four situations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-BD2 Date: _____

For each situation below write what happens to baroreceptor firing, to vagal output, to sympathetic output, and then to heart rate, contractility, arteriolar tone and venous tone, and finally to mean arterial pressure. The four situations are a patient who has lost about 10 percent of blood volume, a patient given a drug that constricts arterioles through alpha-1 receptors, a patient whose carotid sinus is massaged during a fast rhythm, and an older man who faints when a tight collar presses on his neck. For the blood loss patient also write why the skin is cool and pale, why urine output falls, why the pulse pressure narrows, and what arterial pressure itself is doing at that point. For the carotid massage patient write what happens at the atrioventricular node and why that can stop the rhythm. Finish by writing what is added below a mean arterial pressure of about 80 mmHg and what happens below about 60 mmHg.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-06-01-C05-RT01

Rebuild the reflex response to a fall in pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C05 Activity: NPLEX-P1-CARD-06-01-C05-RT01 Date: _____

Rebuild the six-step cycle that runs from a fall in arterial pressure back to pressure returning toward its set point, one step per line and in order, and draw it as a loop with arrows so the feedback closes. Beside the afferent step write the two cranial nerves and the nucleus they reach. Beside the central step write what happens to vagal output and to sympathetic output. Beside the effector step list the four things that rise. Then write beside the loop what makes this negative feedback rather than positive feedback. Underneath, write the same six steps again for a rise in arterial pressure, with every direction reversed.

Rebuild from memory: Baroreceptor reflex response to a fall in arterial pressure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C06-BD1

Build the volume hormone table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C06 Activity: NPLEX-P1-CARD-06-01-C06-BD1 Date: _____

On blank paper build a table with one row each for angiotensin II, aldosterone, antidiuretic hormone, atrial natriuretic peptide and B-type natriuretic peptide. Use the columns source, main stimulus for release, receptor and second messenger, and main actions, and fill every cell. Then write out the renin-angiotensin-aldosterone sequence as seven numbered steps from the stimulus to the rise in volume and pressure, naming the three stimuli for renin release, the cell that releases it, which step is rate limiting, where the converting enzyme is most abundant, and the other peptide that enzyme breaks down. Finish by writing which arteriole angiotensin II constricts more and what that preserves, and how long each of the three control systems takes to act: the reflex, angiotensin II, and aldosterone.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C06-BD2

Three patients, three hormone patterns

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C06 Activity: NPLEX-P1-CARD-06-01-C06-BD2 Date: _____

Patient A has lost a litre of blood. Patient B has a high plasma osmolality from poor water intake with a normal blood volume. Patient C has an aldosterone-secreting adrenal adenoma. For each patient write what happens to renin, angiotensin II, aldosterone, antidiuretic hormone and the natriuretic peptides, and give the stimulus that moved each one. Then write which patient retains mainly sodium with water and which retains mainly water, and explain that difference from what each hormone acts on in the nephron. For patient C, write why the blood pressure is high and the plasma potassium is low but the patient has no edema, and name the process that halts the volume expansion. Finish by writing why high natriuretic peptide levels in heart failure fail to clear the retained fluid, and why inappropriate antidiuretic hormone secretion causes a low plasma sodium rather than hypertension.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-06-01-C06-PR01

Predict the response to renal artery narrowing

A patient has a narrowed renal artery, so perfusion pressure reaching the juxtaglomerular cells of that kidney is low while systemic arterial pressure is normal. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to renin release from the juxtaglomerular cells of the affected kidney, and name the two stimuli at that cell other than sympathetic input		
What happens to circulating angiotensin II		
What happens to the tone of the efferent arteriole compared with the afferent arteriole, and what that preserves		
What happens to aldosterone secretion and to the number of epithelial sodium channels in the principal cells		
What happens to urinary potassium and hydrogen ion secretion as sodium reabsorption rises		
What happens to plasma volume and to systemic arterial pressure over the next days		
What happens to renin release once systemic pressure has risen, and why the affected kidney keeps releasing it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-06-01-C07-BD1

What standing does and how the body answers it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C07 Activity: NPLEX-P1-CARD-06-01-C07-BD1 Date: _____

Write the numerical definition of orthostatic hypotension, including both pressure criteria, the time limit, and the tilt angle that counts. Then write what standing does to the circulation as a sequence: the venous pressure at the foot in a motionless adult, how much blood pools and where, what filters into the leg tissue over minutes, and what happens in turn to venous return, stroke volume, cardiac output and mean arterial pressure. Next write the four things that normally defend pressure on standing, naming the reflex, the expected heart rate rise in beats per minute, the vessels that constrict, the mechanical pump and the structures that make it work in one direction, and the two hormones that add support over minutes. Finish by writing the mean arterial pressure at head level below which symptoms appear, listing the symptoms, and naming the symptom pattern specific to the neurogenic forms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-06-01-C07-BD2

Separate the mechanisms from two sets of vital signs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-6 Cardiac Output and Its Regulation
Competency: NPLEX-P1-CARD-06-01-C07 Activity: NPLEX-P1-CARD-06-01-C07-BD2 Date: _____

Patient A goes from 112/70 mmHg with a heart rate of 92 bpm lying down to 88/62 mmHg with a heart rate of 128 bpm at three minutes standing. Patient B goes from 162/88 mmHg with a heart rate of 96 bpm to 118/70 mmHg with a heart rate of 100 bpm. For each patient calculate the systolic fall and the heart rate rise, state whether the definition is met, write which limb of the reflex arc is working and which is not, and name the mechanism. Write which single number separates the two mechanisms and why that number behaves the way it does in each. Then write why patient B is hypertensive lying down, and in a patient with long-standing diabetes write the three nerve injury mechanisms from the notes and why a fixed fast resting heart rate can appear before the orthostatic drop. Finish by writing how postural tachycardia syndrome and vasovagal syncope each differ from both of these patients, using the heart rate and the pressure in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-06-01-C07-CA01

Lightheadedness on standing after two weeks in bed

A 74-year-old woman is seen on the ward on the fifteenth day of an admission for a hip fracture, during which she has been in bed almost continuously. She reports that each time she is helped to her feet she feels lightheaded, her vision dims, and her neck and shoulders ache, and twice she has had to be lowered back onto the bed. She takes no medication that blocks adrenergic receptors, has no diabetes, and has had no vomiting, diarrhoea or bleeding. Temperature is 36.8 °C (98.2 °F) and respirations are 16/min. Lying down, blood pressure is 128/76 mm Hg and pulse is 78/min. At three minutes standing, blood pressure is 96/60 mm Hg and pulse is 104/min. Her weight is 3 kg lower than on admission, her plasma sodium is 139 mEq/L (reference range 135 to 145 mEq/L), and her ratio of blood urea nitrogen to creatinine is 14 (reference range 10 to 20).

1. Recognize. State whether the pressure readings meet the definition of orthostatic hypotension, showing the systolic and diastolic falls and the heart rate rise you calculated.

2. Explain. Explain the mechanism by which two weeks in bed lowers plasma volume, naming the receptors stretched, the two hormones that change and in which direction, and write what happens to the size and stiffness of the ventricle and to the strength of the leg muscle pump.

3. Differentiate. Write how this presentation differs from orthostatic hypotension caused by volume loss through vomiting or bleeding and from orthostatic hypotension caused by failure of the sympathetic efferent pathway, naming the feature in these readings that separates it from the second of those.

4. Investigate. Write which measurements and bedside findings you would use to confirm the mechanism and what value or appearance each one would have, including what the heart rate response, the supine pressure and the urea to creatinine ratio should look like.

5. Decide. State which of the four normal defences against standing have been weakened in this patient, and which single one of them remains intact.

6. Justify. Justify your answer from the venous return side of the circulation: explain the fall in mean systemic filling pressure, what that does to the venous return curve and to the operating point, and why the intact reflex produces a brisk heart rate rise yet cannot hold the pressure up.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-06-01-C01-Q01

Reading the arteriovenous oxygen difference

A 62-year-old man is evaluated for shortness of breath. His oxygen consumption is 250 mL/min and is unchanged from a study done one year earlier. Arterial oxygen content is 20 mL/dL on both studies. Mixed venous oxygen content, sampled from the pulmonary artery, has fallen from 15 mL/dL last year to 12 mL/dL today. Hemoglobin concentration and arterial saturation are unchanged. Which of the following best describes this patient's cardiac output today compared with last year?

A. It has risen from about 5.0 L/min to about 6.3 L/min, because tissues are extracting less oxygen

B. It is unchanged at about 5.0 L/min, because oxygen consumption and arterial content are unchanged

C. It has fallen from about 5.0 L/min to about 3.1 L/min, because each litre of blood now gives up more oxygen

D. It cannot be estimated, because mixed venous content does not appear in the Fick relationship

Why did you pick it?

Board-style question NPLEX-P1-CARD-06-01-C04-Q02

Wall stress in a dilating ventricle

A 58-year-old woman with a dilated cardiomyopathy has a left ventricular end-diastolic radius that has increased by 40 percent over two years, while wall thickness and systolic pressure have not changed. Which of the following best explains why the oxygen cost of each beat has risen?

A. Systolic wall stress has risen in proportion to the radius, because wall stress varies with pressure times radius divided by twice the thickness

B. Oxygen extraction by the myocardium has fallen below its resting value of about 25 percent

C. The larger chamber lowers systolic wall stress but raises the volume of blood that must be ejected, and volume work costs more oxygen than pressure work

D. Coronary flow during systole has increased to match the larger chamber, raising the oxygen cost of the beat itself

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-07

CV-7 Conduction System and the Electrocardiogram

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-07-01-C01				
NPLEX-P1-CARD-07-01-C02				
NPLEX-P1-CARD-07-01-C03				
NPLEX-P1-CARD-07-01-C04				
NPLEX-P1-CARD-07-01-C05				
NPLEX-P1-CARD-07-01-C06				
NPLEX-P1-CARD-07-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C01-BD1

Draw the conduction pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C01 Activity: NPLEX-P1-CARD-07-01-C01-BD1 Date: _____

Without looking at your notes, draw the conduction pathway from the SA node to the ventricular myocardium. Name every structure in order and write beside each one where it sits in the heart, its conduction velocity, its intrinsic rate, and which ion carries its upstroke. Mark the one place where conduction slows, and write two sentences on why the cells there conduct slowly. Then mark the structure that insulates the atria from the ventricles and write why that makes the AV node and His bundle the only normal electrical route into the ventricles.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C01-BD2

What the AV delay buys and what happens without it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C01 Activity: NPLEX-P1-CARD-07-01-C01-BD2 Date: _____

The AV delay is about 0.13 s. Write what the ventricle gains during that delay and how much of resting filling it accounts for, then name the situations in which that contribution matters more than at rest. Next, the atria begin firing 400 times per minute. Write what the AV node does with those impulses and which property of nodal tissue lets it do that, then write what the ventricular rate and rhythm look like. Finish by writing what changes when an accessory pathway carries the same atrial impulses instead of the node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C02-BD1

Draw the SA nodal action potential

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-BD1 Date: _____

Draw the SA nodal action potential with voltage on the vertical axis and time on the horizontal axis. Mark the maximum diastolic potential and the voltage at which the upstroke begins. Label phase 4, phase 0, and phase 3, and write beside each phase every current that moves during it and the direction each current carries charge. Write why nodal cells have no stable resting potential and which channel working myocytes have that nodal cells lack. Then write the two features this tracing does not have that a ventricular myocyte action potential does.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C02-BD2

Vagal and sympathetic drive at the node

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-BD2 Date: _____

List the three features of the pacemaker action potential that set firing rate. For acetylcholine at M2 receptors and for norepinephrine at beta-1 receptors, write which of those three features each one changes, in which direction, and through which second messenger and channel. Then write what each does to the PR interval and why. Finish with two patients: one receives a drug that blocks cardiac L-type Ca2+ channels, the other a drug that blocks HCN channels only. Write what happens to sinus rate, AV conduction, and ventricular contractility in each, and explain the difference from the currents involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-07-01-C02-RT01

Rebuild the sympathetic acceleration sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-RT01 Date: _____

Rebuild the six-step sequence that runs from norepinephrine arriving at the SA node to a faster heart rate, one step per line and in order. Beside the cAMP step, write which channel cAMP binds directly and what that binding does to the voltage at which the channel opens. Beside the protein kinase A step, write which channel it phosphorylates and what that does to the upstroke. Then draw the phase 4 slope before and after on one set of axes.

Rebuild from memory: Sympathetic acceleration of the SA node

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C03-BD1

Blood supply of the conduction system

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C03 Activity: NPLEX-P1-CARD-07-01-C03-BD1 Date: _____

Write out the blood supply of the conduction system from memory, one line per structure: SA node, AV node, bundle of His, right bundle branch, left anterior fascicle, left posterior fascicle. For each line name the artery or arteries and, where the notes gave one, the approximate percentage of hearts. Then write which two structures have more than one source and why that protects them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C03-BD2

Why inferior and anterior infarcts block differently

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C03 Activity: NPLEX-P1-CARD-07-01-C03-BD2 Date: _____

Two patients develop complete heart block. One has ST elevation in II, III, and aVF; the other has ST elevation in V1 to V4. For each patient write the occluded artery, the conduction tissue lost, the level of the block, the escape rhythm with its rate and QRS width, and whether the block is likely to recover. Then write the three mechanisms other than cell death that produce the bradycardia in the inferior infarct, and name the reflex that arises from the inferoposterior wall. Finish with the one anatomical variant that lets a circumflex occlusion cause AV nodal block.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C04-BD1

Draw and label one normal beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C04 Activity: NPLEX-P1-CARD-07-01-C04-BD1 Date: _____

Draw one normal ECG beat and label the P wave, PR interval, PR segment, QRS complex, ST segment, T wave, QT interval, and U wave. Beside each label write the electrical event it records and its normal value or appearance. Write why atrial repolarization and the activity of the AV node and His bundle do not appear as waves. Then write why the ST segment is isoelectric and why the T wave points the same way as the QRS. Finish with the paper rules: the duration of a small box and a large box at 25 mm/s, and the voltage of 10 mm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C04-BD2

Work a strip from the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C04 Activity: NPLEX-P1-CARD-07-01-C04-BD2 Date: _____

A tracing at 25 mm/s shows an upright P wave in leads I and II before every QRS. R waves are 3 large boxes apart. The PR interval is 6 small boxes, the QRS is 3 small boxes, and the QT is 8 small boxes. Lead I is positive, lead aVF is negative, and lead II is positive. Calculate the rate, the PR, the QRS, and the QT in seconds, show your working, and correct the QT for rate using the Bazett formula. State whether each value is normal. Then place the mean QRS axis within a range of degrees using the three limb leads, and write the reasoning lead by lead. Finish by naming the rhythm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C05-BD1

Build the AV block table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C05 Activity: NPLEX-P1-CARD-07-01-C05-BD1 Date: _____

On blank paper build a table of the four AV blocks with one row each for first degree, Mobitz I, Mobitz II, and third degree. Use the columns PR interval, site of block, QRS width, and risk of progression, and fill every cell. Beside the Mobitz I row, sketch three beats and the dropped beat and write what happens to the RR intervals before the drop and why. Beside the third degree row, write what the P waves and QRS complexes are doing relative to each other and what that pattern is called. Finish by listing the causes of AV block from the notes, including the infectious one and the antibody that injures the fetal AV node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C05-BD2

Gradual failure against all-or-none failure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C05 Activity: NPLEX-P1-CARD-07-01-C05-BD2 Date: _____

Write the ion channel that carries the upstroke in AV nodal cells and the one that carries it in His-Purkinje cells, then explain why one tissue fails gradually and the other fails suddenly. Use that difference to explain, in two or three sentences each: why the PR lengthens beat to beat in Mobitz I, why P waves drop without warning in Mobitz II, and why the escape rhythm after infranodal block is slower and less reliable than the escape after nodal block. Then a patient has 2:1 block with a QRS of 0.14 s. Write why the PR interval cannot settle the site of block here and which feature points to where it is. Finish by writing what a bifascicular block becomes when the last fascicle fails, with the rate and QRS width you expect.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-07-01-C05-CP01

Right against left bundle branch block

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	RBBB	LBBB
Direction of septal depolarization		
QRS shape in V1		
QRS shape in I and V6		
Septal q wave in I, aVL, V5, V6		
Splitting of the second heart sound		
QRS duration in complete block and in incomplete block		
Usual causes and what the finding implies about the heart		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C06-BD1

Mechanisms of tachyarrhythmia, and the premature beats

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-BD1 Date: _____

Write the three mechanisms that generate tachyarrhythmias. Under each one, write what is happening at the membrane and give the triggers listed in the notes. For reentry, write the three conditions a circuit needs and the formula for wavelength, then write which two changes shorten the wavelength and make reentry easier to sustain. Next, draw a premature atrial contraction and a premature ventricular contraction as they look on a strip. Beside each one write the P wave, the QRS width, the T wave direction, and whether the following pause is compensatory, and explain each pause from what the premature impulse does to the SA node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C06-BD2

Read four strips back to their mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-BD2 Date: _____

Four patients present with a fast rhythm. Patient 1 has an irregularly irregular narrow QRS at 130 bpm with no visible P waves. Patient 2 has a regular narrow QRS at 148 bpm with sawtooth atrial waves at about 300 per minute. Patient 3 has a regular narrow QRS at 190 bpm that began and stopped abruptly, with no P wave visible before the QRS. Patient 4 has a regular wide QRS at 170 bpm with P waves marching through it independently. For each patient name the rhythm and its mechanism, write where in the heart the circuit or focus sits, and explain from the atrial rate and nodal refractoriness why the ventricular rate is what it is. For patient 1, write what is lost from ventricular filling and where stasis forms. For patient 4, write the other two findings that confirm the origin.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-07-01-C06-RT01

Rebuild the reentrant circuit

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-RT01 Date: _____

Draw a branch point with two limbs and rebuild the reentrant circuit in six steps, numbering each step on the drawing and writing it out in words. Mark on the drawing where the unidirectional block sits and where conduction must be slow. Then write the two changes that stop the circuit. Underneath, redraw the circuit as the AV node in nodal reentry: label the fast pathway and the slow pathway with their refractory periods, show which one the premature beat goes down and which one it comes back up, and write where the retrograde P wave ends up on the tracing.

Rebuild from memory: A reentrant circuit

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-07-01-C06-PR01

Predict the effect of a low plasma potassium

A woman taking a diuretic has a plasma K+ of 2.6 mEq/L (reference range 3.5 to 5.0 mEq/L) and is started on a macrolide antibiotic. Predict each item below, write the direction of the change, and give the current or channel that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the repolarizing K+ current IKr		
What happens to the duration of the ventricular action potential and to the QT interval		
What happens to the U wave		
What happens to L-type Ca2+ channels late in the prolonged plateau		
What happens to the rhythm once those reopened channels drive a depolarization to threshold		
What the resulting rhythm looks like on the tracing and what it can degenerate into		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C07-BD1

Preexcitation and the failing sinus node

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C07 Activity: NPLEX-P1-CARD-07-01-C07-BD1 Date: _____

Draw a heart in outline with the AV node and an accessory pathway crossing the fibrous ring. Write the embryological reason the pathway exists and the tissue it is made of, then rebuild in six steps how an impulse crossing it produces the ECG pattern. Draw the resulting complex and label the short PR interval with its value, the delta wave, and the wide fusion QRS, and write which part of the complex each pathway produced. Separately, write what sick sinus syndrome is, its usual cause, and the two findings that follow from having fewer pacemaker cells.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C07-BD2

The dangerous pathway and the long pause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C07 Activity: NPLEX-P1-CARD-07-01-C07-BD2 Date: _____

An accessory pathway with a refractory period of about 0.25 s carries atrial fibrillation to the ventricles. Write how many impulses per minute it can pass, why the AV node would not pass that many, and what the tracing looks like in rate, regularity, QRS width, and QRS shape. Write why this rhythm can turn into ventricular fibrillation, and why slowing the AV node does not slow the ventricular rate. Then compare orthodromic and antidromic reentry using the pathway: for each, write the direction of travel, the QRS width, and whether the delta wave is visible. Finally, two patients each have a pause after a run of atrial fibrillation. One pause is an exact multiple of the P to P interval and one is not. Write what each pause tells you about where the sinus node failed, and explain from overdrive suppression why the pause follows the tachycardia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-07-01-C07-CA01

Palpitations in a young man with a wide, irregular tachycardia

A 24-year-old man is brought to the emergency department because of 40 minutes of palpitations and lightheadedness that began during a soccer match. He has had brief episodes of rapid regular palpitations since adolescence, each starting and stopping abruptly. Temperature is 37.0 °C (98.6 °F), pulse is 230/min and irregular, respirations are 22/min, and blood pressure is 88/54 mm Hg. The electrocardiogram shows a wide-complex tachycardia that is irregularly irregular, with QRS complexes that vary in width and shape from beat to beat and no identifiable P waves. An electrocardiogram taken two years earlier, during sinus rhythm, showed a PR interval of 0.10 s, a slurred upstroke at the start of a QRS of 0.13 s, and ST-T changes directed opposite to the QRS.

1. Recognize. Name the atrial rhythm and the pathway carrying it to the ventricles, and name the finding on the earlier electrocardiogram that identifies that pathway.

2. Explain. Explain why the ventricular rate is this fast and why the QRS complexes vary in width and shape from beat to beat.

3. Differentiate. Write how this tracing differs from atrial fibrillation conducted through the AV node and from ventricular tachycardia, naming the features on the strip that separate them.

4. Investigate. Write which findings on the earlier sinus rhythm electrocardiogram you would measure to confirm the pathway, and what value or appearance each one would have.

5. Decide. State which property of the accessory pathway makes this rhythm dangerous, and what this rhythm can turn into.

6. Justify. Justify that answer from the refractory periods of the accessory pathway and the AV node, and explain why slowing conduction through the AV node would not reduce the ventricular rate.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-07-01-C03-Q01

Complete block after an inferior infarction

A 68-year-old man comes to the emergency department because of 3 hours of chest pain. Pulse is 42/min and blood pressure is 86/58 mm Hg. The electrocardiogram shows ST elevation in leads II, III, and aVF, P waves at a rate of 84/min, and QRS complexes at a rate of 44/min with a duration of 0.08 s, unrelated to the P waves. Which of the following best explains the bradycardia in this patient?

A. Necrosis of the His bundle and the proximal bundle branches

B. Ischemia of the atrioventricular node with adenosine release and reflex vagal discharge

C. Conduction of atrial impulses through an accessory pathway

D. Degenerative fibrosis of the sinoatrial node with overdrive suppression

Why did you pick it?

Board-style question NPLEX-P1-CARD-07-01-C05-Q02

Locating the level of block

A 72-year-old woman comes to the physician because of two episodes of syncope. The electrocardiogram shows a PR interval of 0.18 s that is identical in every conducted beat, with occasional P waves that are followed by no QRS complex. The conducted QRS complexes have a duration of 0.14 s with an rSR prime pattern in lead V1. Which of the following is the most likely site of the conduction failure?

A. Sinoatrial node

B. Atrial muscle

C. Atrioventricular node

D. His bundle and bundle branches

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-08

CV-8 Coronary Circulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-08-01-C01				
NPLEX-P1-CARD-08-01-C02				
NPLEX-P1-CARD-08-01-C03				
NPLEX-P1-CARD-08-01-C04				
NPLEX-P1-CARD-08-01-C05				
NPLEX-P1-CARD-08-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C01-BD1

Draw the coronary arteries and their territories

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C01 Activity: NPLEX-P1-CARD-08-01-C01-BD1 Date: _____

Without looking at your notes, draw the heart from the front and from below. Mark the coronary sulcus, the anterior and posterior interventricular grooves, and the crux. Draw both coronary arteries from their origins in the aortic sinuses, name every branch in order along each artery, and write beside each branch the muscle and the part of the septum it supplies. Give the approximate percentage of hearts in which the right coronary artery gives the sinoatrial nodal artery and the atrioventricular nodal artery. Shade the territory of each artery on your drawing and write the electrocardiogram lead group that looks at each territory. Finish by writing which artery is occluded most often in infarction and why that follows from how much muscle it feeds.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C01-BD2

Work backward from the tracing to the artery

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C01 Activity: NPLEX-P1-CARD-08-01-C01-BD2 Date: _____

Three patients each have an acute coronary occlusion. Patient 1 has ST elevation in V1 to V4, patient 2 in II, III, and aVF, and patient 3 in I, aVL, V5, and V6. For each patient name the occluded artery, the wall and the part of the septum that is losing its supply, and the conduction structures at risk, then write the conduction abnormality you expect and whether the block sits at the atrioventricular node or below it. Next, write why a stenosis that developed over two years can leave the muscle beyond it alive while a sudden thrombus in the same artery infarcts it, and name the two places in the heart where the anastomoses that allow this are best developed. Finish by writing why the right ventricle is damaged in some right coronary occlusions but almost never in left anterior descending occlusions.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C02-BD1

Dominance and the supply of the papillary muscles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-BD1 Date: _____

Write the definition of coronary dominance in one sentence, naming the single artery that the definition turns on. Then build a short table with one row for right dominance, left dominance, and codominance, giving the source of the posterior descending artery and the approximate frequency of each pattern. Beside the table write the three structures the posterior descending artery supplies and the other nodal branch that usually comes from the same artery. Underneath, write what the papillary muscles do during systole, then write for each papillary muscle which arteries supply it and whether that supply is single or dual.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C02-BD2

Sudden decompensation four days after an inferior infarct

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-BD2 Date: _____

A patient had an inferior infarct four days ago and now becomes acutely breathless and hypotensive with a new systolic murmur. Write which structure has most likely failed, which artery supplied it, and why this complication appears on day four rather than on day one, naming the cells that act on the dead muscle and what they do to its strength. Then explain, step by step from pressures and volumes, why pulmonary edema comes on this abruptly and why forward stroke volume falls, and write why the murmur can be short and soft even though the regurgitation is severe. Next, write the other post-infarct rupture that causes sudden decompensation with a new murmur, and name the one measurement that separates the two. Finish by writing how the same event would present in a patient whose heart is left dominant, and what you would expect on the tracing in that patient.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-08-01-C02-RT01

Rebuild the chain from inferior infarct to mitral regurgitation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C02 Activity: NPLEX-P1-CARD-08-01-C02-RT01 Date: _____

Rebuild the seven-step chain that runs from a thrombus in the artery giving the posterior descending artery to acute severe mitral regurgitation, one step per line and in order. Beside the step where the muscle loses its supply, write which papillary muscle it is and why only that one is at risk. Beside the softening step, write the days after infarction when it happens and the cells responsible. Beside the final step, write what happens to left atrial pressure and to forward stroke volume. Then write the one change in the chain if the heart is left dominant.

Rebuild from memory: From inferior infarct to papillary muscle rupture

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-08-01-C02-CP01

Anterolateral against posteromedial papillary muscle

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Anterolateral papillary muscle	Posteromedial papillary muscle
Wall of the left ventricle it sits on		
Arteries that supply it, and whether the supply is single or dual		
Infarct location that puts it at risk		
Relative frequency of rupture		
Effect of a left anterior descending occlusion on its perfusion		
What changes about its supply in a left dominant heart		
Consequence for the mitral valve if it ruptures		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-08-01-C02-CA01

Breathlessness and a new murmur five days after chest pain

A 64-year-old man is readmitted five days after an inferior myocardial infarction. He woke an hour ago with severe breathlessness and cannot lie flat. Temperature is 36.8 °C (98.2 °F), pulse is 118/min, respirations are 32/min, and blood pressure is 84/58 mm Hg. He has coarse crackles throughout both lung fields and a new short systolic murmur at the apex that is soft at the left sternal border. The electrocardiogram taken at the first admission showed ST elevation in leads II, III, and aVF with a brief period of a ventricular rate of 44/min and P waves at 80/min. Oxygen saturation measured in the right atrium is 62% and in the right ventricle is 63%.

1. Recognize. Name the structure that has failed, the artery that supplied it, and the valve lesion that has resulted.

2. Explain. Explain from the blood supply of that structure why it was the one at risk in this infarct, and explain why the failure happened on day five rather than during the first day.

3. Differentiate. Write how this presentation differs from post-infarct ventricular septal rupture and from ischemic papillary muscle dysfunction without rupture, naming the feature that separates each from this patient.

4. Investigate. State what the right atrial and right ventricular saturations in this patient tell you, and write what they would have shown in the lesion you ruled out.

5. Decide. State which layer and which wall of the left ventricle became ischemic first in the original infarct, and name the conduction structure that explains the ventricular rate of 44/min on the first tracing.

6. Justify. Justify the abrupt pulmonary edema and the hypotension from the pressures and volumes involved, and explain why the murmur is short and soft despite severe regurgitation.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C03-BD1

Build the cardiac venous drainage table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C03 Activity: NPLEX-P1-CARD-08-01-C03-BD1 Date: _____

On blank paper build a table of cardiac veins with one row each for the great cardiac vein, the middle cardiac vein, the small cardiac vein, the anterior cardiac veins, and the smallest cardiac veins. Use the columns course and companion artery, region drained, and termination, and fill every cell. Underneath, write where the coronary sinus lies, where it opens into the right atrium relative to the inferior vena cava and the tricuspid orifice, the structure that partly guards its opening, and the triangle whose border its ostium forms. Then write the approximate fraction of total coronary flow that returns through the sinus and which ventricle contributes most of it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C03-BD2

Read the venous side as a measurement

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C03 Activity: NPLEX-P1-CARD-08-01-C03-BD2 Date: _____

Blood sampled from the coronary sinus of a healthy adult is about 25% to 30% saturated while mixed venous blood is about 75% saturated. Write why the coronary figure is so much lower, give the oxygen content of arterial and coronary sinus blood in mL O2/dL, and write what the coronary sinus sample tells you about left ventricular metabolism. Next, an echocardiogram in another patient shows a dilated coronary sinus. Write the embryological structure whose persistence explains this, where the sinus itself comes from in development, and why the finding is usually harmless. Finish by writing which two sources of venous blood make up the normal physiological shunt of about 2% of cardiac output, and explain in two sentences why that shunt keeps the alveolar to arterial oxygen difference above zero in a healthy person.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C04-BD1

Draw coronary flow across the cardiac cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-BD1 Date: _____

Draw one cardiac cycle with aortic pressure, left ventricular pressure, and left coronary flow on the same time axis. Mark where coronary flow falls, where it surges, and where it is highest, and write beside each mark what is happening to the vessels inside the wall. Write the approximate percentage of left ventricular flow that occurs in diastole, the resting coronary flow in mL/min, and that flow as a percentage of cardiac output alongside the heart as a percentage of body weight. Write the equation for left ventricular coronary perfusion pressure and work out a normal value from a blood pressure of 120/80 mm Hg and a left ventricular end-diastolic pressure of 8 mm Hg. Then write why the right ventricle is perfused throughout the cycle instead, and the one condition that changes that. Finish with the oxygen content of arterial and coronary venous blood, the resting extraction percentage for the heart and for the whole body, and the four features of the myocyte that force extraction to be this high.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C04-BD2

Four patients with ischemia and normal arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-BD2 Date: _____

Four patients have chest pain on exertion with no coronary stenosis: one has severe aortic stenosis, one has severe aortic regurgitation with a blood pressure of 150/40 mm Hg and a left ventricular end-diastolic pressure of 25 mm Hg, one has a sustained heart rate of 160/min, and one has a hemoglobin of 6 g/dL (reference range 12 to 16 g/dL). For each patient calculate or state what happens to coronary perfusion pressure, to diastolic perfusion time per minute, to wall stress, and to oxygen demand, and write which term in the supply and demand balance has failed. Write the Laplace relationship you are using and show the aortic regurgitation calculation. Then write which layer of the left ventricle becomes ischemic first in all four patients and give the four reasons that layer loses out, and write why hypertrophy lowers wall stress and still leaves the inner wall vulnerable. Finish by writing what the electrocardiogram shows in these patients during exertion and why that pattern rather than ST elevation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-08-01-C04-RT01

Rebuild the tachycardia sequence with the arithmetic

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C04 Activity: NPLEX-P1-CARD-08-01-C04-RT01 Date: _____

Rebuild the six-step sequence that runs from a rising heart rate to inner wall ischemia, one step per line and in order. Beside the step about shortening diastole, work the arithmetic for 70/min and for 150/min: give the cycle length, the duration of systole, the duration of diastole, and the total diastolic seconds per minute at each rate, then give the percentage fall in perfusion time. Beside the step about demand, write why each extra beat costs oxygen. Beside the last step, write the layer involved and what appears on the tracing. Finish by writing the two things happening at once that make tachycardia so poorly tolerated when a stenosis is present.

Rebuild from memory: How tachycardia produces subendocardial ischemia

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-08-01-C04-PR01

Predict the effect of a fast atrial rhythm in aortic stenosis

A patient with severe aortic stenosis and a hypertrophied, stiff left ventricle develops a sustained ventricular rate of 150/min. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the duration of diastole in each cycle and to total diastolic seconds per minute		
What happens to left ventricular end-diastolic pressure in this stiff ventricle		
What happens to coronary perfusion pressure once end-diastolic pressure has changed		
What happens to myocardial oxygen demand, using cavity pressure, radius, wall thickness, and rate		
What happens to the vasodilator reserve still available in the subendocardium		
Which layer of the wall becomes ischemic first and what appears on the electrocardiogram		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C05-BD1

Rebuild metabolic control of coronary flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C05 Activity: NPLEX-P1-CARD-08-01-C05-BD1 Date: _____

Write the Fick principle as it applies to myocardial oxygen consumption, then explain in three sentences why the heart cannot meet higher demand the way skeletal muscle does, using the resting and maximal extraction percentages. Next, rebuild the adenosine pathway in seven steps from an oxygen deficit to a rise in flow, naming the enzyme that makes adenosine, the receptor and G protein it acts through, the second messenger and kinase, the channels that open, the effect on membrane voltage, the channel that then closes, the ion whose cytosolic level falls, and the enzyme whose activity drops before the muscle relaxes. Write how adenosine is cleared and how fast. Then list every other mediator that contributes in parallel, and write what sympathetic alpha-1 and beta-1 activity each do to coronary flow and which effect wins. Finish by writing the perfusion pressure range over which flow is autoregulated and the definition and normal value of coronary flow reserve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C05-BD2

Work the numbers on a stenosis, then explain steal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C05 Activity: NPLEX-P1-CARD-08-01-C05-BD2 Date: _____

Take resting coronary flow as 2.25 dL/min, arterial oxygen content as 20 mL/dL, and coronary venous content as 6 mL/dL. Calculate the arteriovenous difference, resting myocardial oxygen consumption, and resting extraction, showing your working. Now demand doubles. Calculate the consumption that maximal extraction alone would allow at unchanged flow, then calculate the flow needed to meet the new demand at the resting extraction, and write one sentence on what this proves about a lesion that caps flow. Next, a patient has a 75% diameter stenosis of the left anterior descending artery. Write what resting flow, maximal flow, and arteriolar tone are doing in that territory, give the degree of narrowing at which maximal flow begins to fall and the degree at which resting flow begins to fall, and explain why the patient is comfortable sitting and has chest pain climbing stairs. Finish by writing what happens to flow in the stenosed bed and in the normal beds when a vasodilator is given, name the two drugs the notes use as examples and how the second one raises adenosine levels, and write why this effect is useful during stress perfusion imaging.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-08-01-C06-BD1

Rebuild why ALCAPA fails only after birth

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C06 Activity: NPLEX-P1-CARD-08-01-C06-BD1 Date: _____

Write what is anatomically abnormal in anomalous origin of the left coronary artery from the pulmonary artery, and which coronary artery arises normally. Then rebuild in six steps the sequence from the fetal circulation to infant myocardial ischemia, one step per line. Beside the fetal steps, write what keeps pulmonary artery pressure about equal to aortic pressure, the direction of flow in the anomalous artery, and the two reasons the fetal heart tolerates the lesion. Beside the postnatal steps, write the pulmonary artery pressure and the oxygen saturation the left coronary artery now receives, the time course over which this develops, and why collateral flow from the right coronary artery ends up draining into the pulmonary trunk rather than perfusing muscle. Name that last phenomenon. Finish by drawing the self-reinforcing cycle in six steps, from the ischemic ventricle through dilation, wall stress, mitral regurgitation, and end-diastolic pressure, and write beside each step whether it raises demand or lowers supply.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-08-01-C06-BD2

The infant who looks well at birth and sick at ten weeks

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-8 Coronary Circulation Competency: NPLEX-P1-CARD-08-01-C06 Activity: NPLEX-P1-CARD-08-01-C06-BD2 Date: _____

A 10-week-old infant is pale, sweaty, and breathless during feeds and has gained little weight. Write why symptoms begin at this age rather than in the first week, and why feeding is the exertion that brings them on. List the findings you expect on examination and on the chest film, then write the electrocardiogram pattern with the exact leads involved and why that finding in an infant is so unusual, and write the two mechanisms that make this infant regurgitate through the mitral valve. Next, write the two features that separate this diagnosis from dilated cardiomyopathy and myocarditis. Then write why a minority of these patients instead reach childhood or adulthood, listing what you expect to find in them and the one saturation measurement that can be slightly unexpected and why. Finish by writing why a large ventricular septal defect in the same infant can delay or hide the ischemia, and state in one sentence the variable that actually decides whether the left ventricle is perfused.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-08-01-C01-Q01

Locating the occlusion from the complication

A 59-year-old man comes to the emergency department because of 2 hours of chest pain. Pulse is 48/min and blood pressure is 92/60 mm Hg. The electrocardiogram shows ST elevation in leads II, III, and aVF, with P waves at 82/min and QRS complexes at 46/min of 0.08 s duration that are unrelated to the P waves. Which of the following arteries was most likely occluded, and which structure accounts for the slow ventricular rate?

A. Left anterior descending artery supplying the bundle branches

B. Circumflex artery supplying the lateral wall

C. Right coronary artery supplying the atrioventricular node

D. Left main coronary artery supplying most of the left ventricle

Why did you pick it?

Board-style question NPLEX-P1-CARD-08-01-C04-Q02

Ischemia with unobstructed arteries

A 71-year-old woman with severe aortic regurgitation has chest pain on exertion. Blood pressure is 160/40 mm Hg and left ventricular end-diastolic pressure measured at catheterization is 26 mm Hg. Coronary angiography shows no stenosis. Which of the following best explains her exertional chest pain?

A. A coronary perfusion pressure of about 14 mm Hg with high wall stress in a dilated ventricle, which exhausts subendocardial reserve

B. Failure of myocardial oxygen extraction to rise above 25% during exertion

C. Loss of systolic coronary flow to the subepicardium of the left ventricle

D. Diversion of flow from the left coronary artery into the pulmonary trunk

Why did you pick it?

Board-style question NPLEX-P1-CARD-08-01-C06-Q03

Why the infant deteriorates at two months

A 9-week-old infant has episodes of pallor, sweating, and irritability during feeds, with poor weight gain. The chest film shows cardiomegaly and the electrocardiogram shows abnormal Q waves with ST and T changes in leads I, aVL, V5, and V6. Echocardiography shows that the left coronary artery arises from the pulmonary trunk. Which of the following best explains why the infant was well at birth and is ischemic now?

A. Progressive atherosclerotic narrowing of the anomalous left coronary artery

B. Closure of the foramen ovale, which removed a right to left shunt that had been perfusing the left ventricle

C. A rise in myocardial oxygen extraction above the fetal level, which exceeded the capacity of the anomalous artery

D. A fall in pulmonary vascular resistance and pulmonary artery pressure, which left the anomalous artery with low-pressure desaturated blood and reversed collateral flow into the pulmonary trunk

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-09

CV-9 Blood Distribution to Body Regions

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for one competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-09-01-C01				
NPLEX-P1-CARD-09-01-C02				
NPLEX-P1-CARD-09-01-C03				
NPLEX-P1-CARD-09-01-C04				
NPLEX-P1-CARD-09-01-C05				
NPLEX-P1-CARD-09-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C01-BD1

Draw the aorta and label every branch with its level

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-BD1 Date: _____

Without looking at your notes, draw the aorta from the aortic valve to the bifurcation as one long vessel and label every branch in order. Start with the ascending aorta and write what it gives and the level where it ends. For the arch, name the three branches from right to left, write where the brachiocephalic trunk divides and into what, and mark the level the arch sits at. For the descending thoracic aorta write its upper and lower levels and list the branches it gives, including which intercostal spaces it supplies and where the first two come from instead. For the abdominal aorta write each branch with its vertebral level and what it supplies, in order from the diaphragm down to the bifurcation, and mark which branches are paired. Finish by writing the level at which the aorta passes the diaphragm and the level at which it divides.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C01-BD2

Three patients with a lesion on the aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-BD2 Date: _____

Use the levels you just drew on three new patients. Patient 1 is a 54-year-old with a hoarse voice and a widened mediastinum: write which nerve is involved, what it is lying against, and why the aorta can injure it. Patient 2 has an aortic dissection repaired with a clamp on the descending thoracic aorta and wakes with weak legs, no pain or temperature sensation below the lesion, and intact vibration and position sense: write which artery was interrupted, which part of the cord lost its supply, and why vibration and position sense were spared. Patient 3 is a 28-year-old with high arm pressure, weak and delayed femoral pulses, and notching of the lower borders of ribs 3 to 8: write where the narrowing sits relative to the arch branches, why the arm pressure is high while the legs are underperfused, and why ribs 1 and 2 show no notching. For each patient write one line naming the vertebral level or landmark that places the lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-09-01-C01-RT01

Rebuild the collateral route around a coarctation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-RT01 Date: _____

Rebuild the six-step route blood takes around an adult coarctation, one step per line and in order, starting from the high pressure above the narrowing and ending with the change seen on the ribs. Draw the chest wall in outline and trace the route on your drawing, labelling the subclavian artery, the internal thoracic artery, the anterior and posterior intercostal arteries, and the descending aorta below the narrowing. Mark with an arrow the one vessel in which flow runs backwards compared with normal, and write one sentence on what drives it that way. Then write which ribs are notched and which two are spared, and name the trunk that supplies the spared spaces.

Rebuild from memory: Collateral route around an adult coarctation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C02-BD1

Draw the circle of Willis and its feeding vessels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C02 Activity: NPLEX-P1-CARD-09-01-C02-BD1 Date: _____

Draw the base of the brain from below and build the circle of Willis from memory. Start with the two inflow systems: name the pair of vessels that forms the anterior circulation and the pair that forms the posterior circulation, and write where the second pair runs in the neck and what it joins to become. Label every vessel that forms the ring itself, then add the branches that leave it, including the artery that supplies the eye, the thin deep branches that leave the middle cerebral trunk at right angles, and the three cerebellar arteries with the vessel each arises from. Beside each labelled vessel write the territory it supplies. Mark the point where the oculomotor nerve emerges and the two vessels it lies between. Finish by writing why the external carotid artery supplies no brain tissue and naming the one external carotid branch that still matters inside the skull, with where it runs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C02-BD2

Place four deficits on the vascular map

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C02 Activity: NPLEX-P1-CARD-09-01-C02-BD2 Date: _____

Four patients lose brain blood supply. Patient 1 has weakness of the contralateral leg more than the arm with new incontinence and a change in behavior. Patient 2 has vertigo, difficulty swallowing, hoarseness, a drooping eyelid and small pupil on one side with loss of pain and temperature sensation on that side of the face, and loss of pain and temperature sensation on the other side of the body. Patient 3 has a contralateral homonymous hemianopia with the central field spared. Patient 4 has a pure contralateral hemiparesis with no sensory or language loss and a long history of untreated hypertension. For each patient name the occluded artery, write which circulation it belongs to, and write one sentence tying the deficit to the territory. For patient 2 write what the combination of findings on opposite sides of the face and body tells you about the level of the lesion. For patient 4 write what chronic hypertension does to those particular vessels and the two other lesions it produces in them. Finish by writing why an embolus lodged in a cortical middle cerebral branch still infarcts tissue even when the circle of Willis is complete.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C03-BD1

Limb arteries, their landmarks and their pulse points

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C03 Activity: NPLEX-P1-CARD-09-01-C03-BD1 Date: _____

Write out the arterial trunk of the upper limb and then of the lower limb as a chain of named segments from the aorta to the hand and to the foot. Beside each segment write the landmark at which the name changes and, where a pulse can be felt, exactly where you would place your fingers and against what. Mark the two pulses that are often faint and the one that is absent in a small number of healthy adults. Then write where each trunk ends and name the vessels that rejoin in the hand and in the foot. Add the three nerve and artery pairs from the notes, naming the bony or soft tissue landmark each pair travels with. Finish by writing the order of the structures in the femoral triangle from lateral to medial.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C03-BD2

Localize four limb lesions from the examination

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C03 Activity: NPLEX-P1-CARD-09-01-C03-BD2 Date: _____

Work each of these from the pulse sequence and the landmarks. Patient 1 is a 66-year-old whose left femoral pulse is normal, popliteal pulse is faint, and dorsalis pedis and posterior tibial pulses are absent: write where the obstruction lies and which muscle group aches first on walking. Patient 2 has an ankle systolic pressure of 70 mm Hg with an arm systolic pressure of 140 mm Hg: calculate the ankle-brachial index, write what the normal relationship between ankle and arm pressure is and why, and state what this value means. Patient 3 is a child with a supracondylar humeral fracture whose radial pulse is absent and whose forearm is tense and painful: write which artery is at risk, which muscles are ischemic, and what the healed result is called. Patient 4 has a displaced intracapsular femoral neck fracture: write which artery supplies the femoral head and what the head is at risk of. Finish by writing where emboli most often lodge in the lower limb and why a block in the axillary artery above the subscapular origin is better tolerated than one below it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C04-BD1

Draw venous return and the portal system

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-BD1 Date: _____

Draw the two venae cavae and the hepatic portal system on one page. For the superior vena cava write which veins unite to form it, where they unite, and which vein joins it and at what level. For the inferior vena cava write the level at which it forms, which side of the aorta it ascends on, and the level at which it passes the diaphragm. Draw both renal veins and both gonadal veins and write where each gonadal vein drains, then mark the two structures the left renal vein passes between and write what else crosses in that same angle. For the portal system, name the two veins that unite to form the portal vein and where they meet, then write the share of cardiac output the liver receives, how much of that is portal, and how much of the liver's oxygen each source supplies. Write the normal portal pressure and the normal pressure in the hepatic veins, and note what the absence of valves in the portal system means for any rise in resistance. Finish by listing the four portosystemic anastomoses with the portal tributary and the systemic vein at each site.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C04-BD2

Four patients with an obstructed vein

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-BD2 Date: _____

Patient 1 has a mediastinal tumor, a swollen face, neck and arms, and distended veins over the chest wall. Write which vein is obstructed, then write the two routes blood can take back to the heart and which route is used when the block sits above the entry of the azygos vein and which when it sits below it, naming the vein in which flow reverses. Patient 2 is a 22-year-old man with flank pain, blood in the urine, and a new left-sided varicocele. Write which vein is compressed and between which two structures, and explain from the course of the two gonadal veins why varicocele is usually on the left. Patient 3 is a 58-year-old man with a new isolated right varicocele: write what that should make you look for and why. Patient 4 has cirrhosis with dilated veins radiating from the umbilicus, an enlarged spleen and a low platelet count. Write the site and the two veins involved in the umbilical finding, write why the spleen enlarges when the resistance lies in the liver, and write what the enlarged spleen does to the platelet count and why. Finish by writing the three sites at which raised portal pressure can originate, with one example for each, and the pressure gradient at which varices appear.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-09-01-C04-RT01

Rebuild the cycle that keeps portal pressure high

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-RT01 Date: _____

Rebuild the six-step cycle that holds portal pressure high in cirrhosis, one step per line and in order, then draw it as a closed loop with arrows so the last step feeds back into the first. Beside the first step write the three changes inside the liver that raise resistance. Beside the second step write which mediator dilates the splanchnic arterioles. Beside the fourth step write what the body is sensing when it activates the renin angiotensin aldosterone system and the sympathetic nerves, even though total blood volume is not low. Underneath the loop write the relationship between pressure, resistance and inflow in one line, and state which of the two cirrhosis raises. Finish by writing which step also produces ascites.

Rebuild from memory: Why portal pressure stays high in cirrhosis

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-09-01-C04-CA01

Vomiting blood in a man with long-standing liver disease

A 57-year-old man is brought to the emergency department after vomiting a large volume of bright red blood twice in one hour. He has had alcohol use disorder for 20 years and was told four years ago that he had cirrhosis. Temperature is 36.6 degrees C (97.9 degrees F), pulse is 116/min, respirations are 20/min, and blood pressure is 92/58 mm Hg. The abdomen is distended with shifting dullness, the spleen is palpable 4 cm below the left costal margin, and dilated veins radiate from the umbilicus. Platelet count is 68,000/mm3. Portal pressure measured during the admission gives a hepatic venous pressure gradient of 14 mm Hg.

1. Recognize. Name the vessels that bled, the portal tributary and the systemic vein that meet at that site, and the vein into which that systemic channel drains.

2. Explain. Explain why raised resistance inside the liver sends portal blood through the lower esophagus rather than across the liver to the inferior vena cava, and why the two other findings on this abdomen appear at the same time.

3. Differentiate. Write how this bleeding site differs from the anorectal anastomosis and from internal hemorrhoids, and what separates a bleed caused by raised portal pressure from one caused by a lesion in the stomach wall.

4. Investigate. Write which measurements and venous sites you would use to confirm where the raised resistance lies, and what value or finding each one would give in cirrhosis.

5. Decide. State where the pressure gradient in this patient sits relative to the normal gradient and the varix threshold, and name the site at which the resistance in this man is raised.

6. Justify. Justify that location from the pressures on either side of the liver, the absence of valves in the portal system, and the way the gradient behaves in prehepatic and posthepatic causes.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C05-BD1

Map the watershed zones and explain why they fail first

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C05 Activity: NPLEX-P1-CARD-09-01-C05-BD1 Date: _____

Define a watershed zone in one sentence, then explain in three or four sentences why pressure at that border is lower than anywhere else in either arterial tree and why the collateral channels there cannot rescue it. Write the range of mean arterial pressure over which cerebral blood flow stays nearly constant, what happens to flow below that range, and what chronic hypertension does to the range. Then draw a brain in lateral view and mark the cortical border zones, naming the two arteries that meet at each one and the region of the hemisphere each strip occupies, and mark the deep border zone. Draw the colon in outline and mark the two sites with weak marginal connections, naming the arteries that meet at each and writing which part of the large bowel has an extra supply and from where. Finish by listing the five regions that fail first in low flow, with the reason each one is vulnerable and the injury that results.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C05-BD2

Read a pattern of injury back to a mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C05 Activity: NPLEX-P1-CARD-09-01-C05-BD2 Date: _____

A 71-year-old man is resuscitated after 50 minutes of severe hypotension during surgery. Over the next two days he passes bloody stools with left-sided abdominal pain, is found to have weakness of both shoulders and upper arms with normal grip strength, has rising aminotransferases, and has a rising creatinine with granular casts in the urine. For each of the four findings write the region injured, the arteries or the position on the supply path that makes it vulnerable, and the name of the injury. Write what the four findings have in common and the single mechanism that links them. Then write how this pattern differs from the pattern a cardiac embolus produces, covering the number of territories involved, whether findings are one-sided or two-sided, and the depth of bowel injury. Finish by writing why a man with a severe carotid stenosis can infarct a border zone after a fall in pressure that another patient would tolerate.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-09-01-C05-CP01

Global hypoperfusion against occlusion of one artery

Fill in every cell from memory, then check your table against the notes and correct what you missed.

Feature	Global hypoperfusion	Occlusion of a single artery
Mechanism of the fall in perfusion		
Where the infarct sits within an arterial territory		
One side or both sides		
Number of organs involved at once		
Shape of the cerebral lesion and the arm weakness it produces		
Depth of bowel wall injury		
Typical cause		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C06-BD1

Build the flow and extraction table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C06 Activity: NPLEX-P1-CARD-09-01-C06-BD1 Date: _____

On blank paper build a table with one row for the gut and liver, the kidneys, resting muscle, the brain, the heart and the skin. Use the columns share of resting cardiac output, oxygen extraction, and what the flow is used for, and fill every cell. Rank the rows by share of output and then rank them again by extraction, and write one sentence on why the two rankings differ. Above the table write why each organ can set its own share of cardiac output while every organ sees the same aortic pressure. Below the table write the oxygen content of arterial blood per decilitre, the whole body figures for cardiac output, arteriovenous oxygen difference, oxygen consumption, extraction and mixed venous saturation, and the Fick principle as an equation. Finish with one line on what a low extraction tells you about an organ's reserve and one line on what a high extraction means for how that organ must meet extra demand.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C06-BD2

Work the numbers, then explain two organs under strain

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C06 Activity: NPLEX-P1-CARD-09-01-C06-BD2 Date: _____

Arterial oxygen content is 20 mL/dL in every part of this activity. Kidney blood flow is 11 dL/min and renal venous content is 18.5 mL/dL. Coronary blood flow is 2.25 dL/min and coronary sinus content is 6 mL/dL. Calculate the arteriovenous oxygen difference, the oxygen consumption and the extraction for each organ, showing your working. Write which organ receives more blood and which uses more oxygen, and explain the mismatch from what each organ uses its flow for. Then let demand double in each organ and write how each one can meet it, giving the extraction the kidney would need and the extraction the heart would need, and say which figure is impossible and why. Write what the heart must do instead and name the two mediators that do it. Finish with the brain: write its share of body weight, its share of resting oxygen use, how long consciousness lasts without flow, and what rising arterial carbon dioxide and hyperventilation each do to its arterioles.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-09-01-C06-PR01

Predict the redistribution of cardiac output in hemorrhage

A 30-year-old man loses 1.5 L of blood from a stab wound to the thigh. Predict each item below, write the direction of the change, and give the receptor, nerve or local mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to arterial pressure and to baroreceptor firing		
What happens to sympathetic outflow and which receptor on the arterioles it acts through		
What happens to flow through the skin, the splanchnic bed, the kidneys and resting muscle		
What happens to the blood held in the splanchnic veins and what that does to venous return		
What happens to the diameter of cerebral and coronary arterioles while the other beds constrict		
What happens to the share of cardiac output reaching the brain and heart		
What happens to the oxygen extraction of the constricted beds, and what happens to those organs if the low flow continues		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-09-01-C01-Q01

Weight loss and vomiting with an aortic landmark

A 24-year-old woman comes to the physician because of 3 months of vomiting after meals and a 9 kg weight loss. She describes relief when she lies on her left side. Imaging shows dilatation of the first and second parts of the duodenum with narrowing of the third part where it crosses in front of the aorta, and a reduced angle between the aorta and the artery that passes in front of the duodenum. Which of the following arteries is responsible for the obstruction?

A. Celiac trunk at the level of the twelfth thoracic vertebra

B. Inferior mesenteric artery at the level of the third lumbar vertebra

C. Superior mesenteric artery at the level of the first lumbar vertebra

D. Right renal artery at the level of the second lumbar vertebra

Why did you pick it?

Board-style question NPLEX-P1-CARD-09-01-C02-Q02

Dizziness while working with one arm raised

A 64-year-old man who has smoked for 40 years becomes dizzy and unsteady whenever he paints a ceiling with his left arm. The left radial pulse is weak and delayed, and left arm systolic pressure is 30 mm Hg lower than the right. Neurologic examination between episodes is normal. Which of the following best explains his dizziness during use of the arm?

A. Reversal of flow down the left vertebral artery into the dilated arteries of the exercising arm

B. Embolization from an ulcerated plaque in the left internal carotid artery to the middle cerebral artery

C. Compression of the left subclavian artery between the anterior and middle scalene muscles by a cervical rib

D. Occlusion of the left posterior inferior cerebellar artery supplying the lateral medulla

Why did you pick it?

Board-style question NPLEX-P1-CARD-09-01-C05-Q03

Weak arms after a prolonged fall in blood pressure

A 68-year-old woman is resuscitated after a cardiac arrest lasting 40 minutes. On the second day she is awake and has weakness of both shoulders and both proximal arms, while grip strength, facial movement and leg strength are normal. Imaging shows narrow strips of infarction in both hemispheres along the upper convexity. Which of the following best explains the location of these infarcts?

A. Embolic occlusion of both anterior cerebral arteries beyond the anterior communicating artery

B. Thrombosis of the superior sagittal sinus with venous congestion of the parasagittal cortex

C. Hyaline arteriolosclerosis of the lenticulostriate arteries supplying the internal capsule

D. Perfusion pressure at the distal ends of the anterior and middle cerebral territories falling below the autoregulatory range

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-10

CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Spread the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-10-01-C01				
NPLEX-P1-CARD-10-01-C02				
NPLEX-P1-CARD-10-01-C03				
NPLEX-P1-CARD-10-01-C04				
NPLEX-P1-CARD-10-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C01-BD1

Draw the lymphatic capillary and how lymph forms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-BD1 Date: _____

Without looking at your notes, draw a lymphatic capillary sitting in the interstitium beside a blood capillary. Label the blind end, the overlapping endothelial cells, the flap that acts as a valve, the basal lamina, and the anchoring filaments, and write beside each label what it does. Then write the four Starling forces, mark which two push fluid out and which two hold it in, and write one sentence on why a small net filtration leaves the interstitium with both fluid and protein in it. Finish by writing in order the five steps that take fluid from net capillary filtration to lymph moving on into a valved collecting vessel, and write beside the first and last steps what opens the vessel and what closes it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C01-BD2

What stops working when protein cannot get back

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-BD2 Date: _____

A patient's lymphatic return from one leg is blocked while capillary pressure, plasma albumin, and capillary permeability all stay normal. Write what happens to the protein concentration of the interstitial fluid in that leg and why no other route can clear it, then write what that does to interstitial oncotic pressure, to the gradient that normally holds fluid inside vessels, and to net filtration. Next write the protein concentration of lymph from most tissues, from the liver, and from the intestine, and explain from the structure of hepatic sinusoids why the liver figure is the highest. Then a second patient has a thoracic duct that has been draining externally for ten days. Write which fat-soluble components, which plasma component, and which cells are being lost, and explain from the size of chylomicrons why dietary long-chain fat leaves by this route while medium-chain fatty acids do not. Finish by naming the tissues that have no lymphatic capillaries at all.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-10-01-C01-RT01

Rebuild the steps that form lymph

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-RT01 Date: _____

Rebuild from memory the five-step sequence that runs from net capillary filtration to lymph entering a valved collecting lymphatic, one step per line and in order. Beside the step where the vessel opens, draw the anchoring filaments and show with arrows which way the tissue pulls. Beside the step where the vessel closes, draw the endothelial flaps and show which side the pressure comes from. Underneath, write the resting lymph flow per hour, the share of it carried by the thoracic duct, and the daily volume, and write one sentence saying why that daily volume matches net filtration.

Rebuild from memory: Formation of lymph

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C02-BD1

Draw the thoracic duct from end to end

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C02 Activity: NPLEX-P1-CARD-10-01-C02-BD1 Date: _____

Draw the vertebral column in outline from L2 to C7 and lay the thoracic duct on it from memory. Mark and label the cisterna chyli with its vertebral level, the two lumbar trunks and the intestinal trunk that feed it, the aortic hiatus with its level, the level at which the duct crosses from right to left, and the vein junction where it ends. Write beside the posterior mediastinal part which structure lies in front of the duct, which vessel lies on each side of it, and what the terminal valve does. Then draw the right lymphatic duct, name the three trunks that form it, and shade the four body regions it drains. Finish by writing which lung lobe drains against the pattern you have just shaded and where its lymph goes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C02-BD2

Work backward from the node to the primary site

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C02 Activity: NPLEX-P1-CARD-10-01-C02-BD2 Date: _____

Five patients each present with one enlarged node group: left supraclavicular, para-aortic near the renal vessels, superficial inguinal, axillary, and popliteal. For each one write which regions drain there and name one primary site that would explain the finding. Then explain three of those patterns from development rather than from memory: why a testicular tumour reaches para-aortic nodes while a scrotal skin lesion reaches superficial inguinal nodes, why the anal canal drains to two different node groups depending on which side of the pectinate line a lesion sits, and why an abdominal carcinoma can surface first in a node at the root of the left side of the neck. Finish by writing which single node group drains most of the breast, which group drains its medial part, and why a lesion of the medial hand can enlarge a node at the elbow before any node in the axilla.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-10-01-C02-PR01

Predict the effect of a block low in the thoracic duct

A retroperitoneal lymphoma compresses the thoracic duct just above the cisterna chyli. Predict each item below, state the direction of the change or name the structures involved, and give the anatomical reason in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to lymph outflow from both lower limbs and the pelvis		
What happens to lymph outflow from the abdominal viscera, including intestinal chyle		
Which regions of the left side of the body still drain normally, and through which vessels they reach the veins		
Which regions of the right side of the body still drain normally, and through which vessel they reach the veins		
What happens to the pressure inside the duct below the obstruction		
What happens to the valve leaflets of the dilated collecting vessels below the obstruction, and what that does to forward flow		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C03-BD1

Lay out the whole lymphatic pump

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-BD1 Date: _____

Lymph starts in an interstitium below atmospheric pressure and ends in veins above it, in the legs against gravity. Write the pressure at each end, then write why valves are what make any compression useful. Draw two lymphangions in series with their valves labelled, and write out the six steps of one pump cycle from filling to refilling. Beside the cycle write what stretch from filling does to the rate and force of contraction, and name the cardiac property this resembles. Then list the three influences on that smooth muscle from the notes and the direction each one pushes its activity. Finish by listing the extrinsic pumps, writing the fold rise in flow with exercise, and explaining step by step how one inspiration moves lymph up the duct.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C03-BD2

Why the pump runs out of room

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-BD2 Date: _____

Draw interstitial fluid pressure on the horizontal axis from -6 mmHg to +4 mmHg and lymph flow on the vertical axis, and sketch the relationship. Mark the normal subcutaneous value, write why it is negative, mark where flow has risen about twentyfold, and mark where the curve flattens. Write two sentences on why flow stops rising past that point even though the tissue keeps swelling, and write what accumulates once filtration exceeds that ceiling. Next write the three safety factors that oppose edema with the pressure each is worth and the total, and write what each one depends on. Then remove the regional nodes and vessels from a limb: write which safety factors are lost immediately, write what happens to interstitial protein and therefore to interstitial oncotic pressure, and explain why that limb swells even with a normal capillary pressure. Finish with two patients whose lymph outflow is slowed at the far end rather than the near end, one with right heart failure and one with superior vena cava obstruction, and write for each why filtration rises at the same time as outflow falls.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-10-01-C03-RT01

Rebuild the lymphangion pump cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-RT01 Date: _____

Rebuild from memory the six-step cycle of one lymphangion, one step per line and in order, starting with lymph entering through the open upstream valve. Draw the segment at each of the six steps as a small sketch and shade which valve is open and which is closed. Beside the contraction step write what triggered it and what the trigger was before that. Beside the ejection step write where the ejected lymph goes and what it does to the next segment. Underneath, write the pressure an obstructed thoracic duct can generate, and write what happens to the leaflets and to forward flow if a chronically obstructed vessel dilates.

Rebuild from memory: The lymphangion pump cycle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C04-BD1

Build the lymphoid organs from their compartments

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C04 Activity: NPLEX-P1-CARD-10-01-C04-BD1 Date: _____

Separate the primary from the secondary lymphoid organs, write which lymphocyte matures in each primary organ, and write what each secondary organ samples. Then draw four diagrams from memory and label every compartment named below. For a lymph node: capsule, subcapsular sinus, cortical follicle with germinal center, paracortex, medullary cords, hilum, and the afferent and efferent vessels, with arrows for the direction lymph travels. For the spleen: central arteriole, periarteriolar lymphoid sheath, follicle, marginal zone, and red pulp with its cords and sinusoidal slits. For a thymic lobule: cortex, medulla, Hassall corpuscle, and the corticomedullary junction, and write the two things the thymus does not have. For a Peyer patch: epithelium with its M cell, follicle, and the direction antigen travels. Beside each compartment write the main cell type it holds and the one job it does. Finish by writing the eight-step sequence of T-cell maturation in the thymus from marrow precursor to a naive T cell leaving, and mark which steps happen in the cortex and which in the medulla.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C04-BD2

Read a biopsy and a blood film back to a compartment

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C04 Activity: NPLEX-P1-CARD-10-01-C04-BD2 Date: _____

Three children have node biopsies. The first has follicles and germinal centers present but an empty paracortex. The second has a normal paracortex but no follicles, no germinal centers, and no plasma cells in the medullary cords. The third has a greatly enlarged paracortex with normal follicles. For each child write which cell line is missing or responding, write what the same defect would do to the spleen's periarteriolar sheath, marginal zone, and follicles, and name the condition or the kind of infection that fits. Explain for the second child why serum antibody of every class is low. Next write what a T-cell receptor must do in the thymic cortex to survive and what it must avoid doing in the medulla, and explain which of the three children has lost that process and in which organ. Then a fourth patient has had a splenectomy. Write the three findings from the notes that follow, name the three bacteria that put this patient at risk, and explain from red pulp structure and from marginal zone function why both the red cell findings and the infection risk appear. Finish by writing which single transcription factor defect in thymic medulla produces an autoimmune syndrome and name its three features.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C05-BD1

Causes of lymphedema and how it hardens

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C05 Activity: NPLEX-P1-CARD-10-01-C05-BD1 Date: _____

Write the three mechanisms that produce edema by too much filtration, then write what is different about the mechanism of lymphedema and why the fluid that collects is rich in protein. Build a table of the five causes of lymphedema from the notes with one row each, using the columns cause, how drainage fails, and typical setting, and fill every cell. For the parasitic cause, name the organisms, write where the adult worms live and where the microfilariae are found, and write the three things that damage the vessels. For the developmental cause, name the two gene mutations and the chromosomal condition. Then write out the six-step cycle that takes lymphatic outflow falling to further loss of outflow, and beside the fibrosis step name the cells and the growth factor involved. Finish by writing why early lymphedema pits and improves with elevation while late lymphedema does neither, and name the bedside sign at the base of the second toe.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C05-BD2

Reason out a milky effusion

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C05 Activity: NPLEX-P1-CARD-10-01-C05-BD2 Date: _____

A patient develops a large left pleural effusion after chest surgery. The fluid stays turbid after centrifugation. Write which structure has been injured and, from where the duct lies relative to the midline above and below its crossing, write whether the injury sits above or below that crossing. Then write what the fluid triglyceride would have to be to support this diagnosis, what value makes it unlikely, and which particle confirms it. Write why the fluid was thin and clear while the patient was fasting and turned milky once feeding resumed. Next write how you would separate this fluid from the fluid of an infected pleural space and from a chronic cholesterol effusion, giving the one measurement or observation that settles each. Finish by writing the four things a continuing leak depletes and explaining each loss from what the duct normally carries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-10-01-C05-CP01

Three kinds of swelling that look alike

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Lymphedema	Edema of heart failure or low plasma albumin	Myxedema
What accumulates in the tissue			
Protein content of the fluid			
Which Starling force or drainage step has gone wrong			
Pitting or nonpitting, and why			
Response to elevating the limb			
What the skin looks like over time			
Bedside sign or finding that points to it			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-10-01-C05-CA01

A firm, swollen arm four years after breast surgery

A 58-year-old woman comes to the clinic because of swelling of the right arm that began eight months after a right mastectomy with axillary lymph node dissection four years ago. She also had radiation to the right axilla. The swelling was soft at first and went down overnight, but over the past two years it has become firm and no longer changes with elevation. She has had three episodes of a hot, red, tender right arm with fever in the past year. Temperature is 37.1 C (98.8 F), pulse is 78/min, respirations are 14/min, and blood pressure is 128/76 mm Hg. The right arm is larger than the left from the shoulder to the fingers, the skin is thickened and rough, and a fold of skin at the base of the second finger cannot be pinched up. The jugular venous pressure is normal, the heart sounds are normal, serum albumin is 4.1 g/dL (reference range 3.5 to 5.0 g/dL), and the left arm is normal.

1. Recognize. Name the type of edema in this arm and name the two events in her history that caused it, writing for each one how it removed or disabled lymphatic pathways.

2. Explain. Explain why the fluid in this arm is rich in protein, and explain why the arm swells at all when her capillary pressure, plasma albumin, and capillary permeability are normal.

3. Differentiate. Write how this swelling differs from the leg swelling of heart failure and from the swelling of low plasma albumin, naming the findings in this patient that separate them.

4. Investigate. Write which bedside findings and which laboratory values in the case you would use to confirm your answer, and say what each one shows.

5. Decide. State why the arm changed from soft and pitting to firm and nonpitting over two years, and name what is now occupying the tissue.

6. Justify. Justify from the safety factors against edema why losing the nodes and vessels makes this arm swell, and explain from the self-perpetuating cycle in the notes why her repeated episodes of hot, red, tender skin make the swelling worse each time.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-10-01-C04-Q01

Which compartment the organ lacks

A 7-month-old boy has had three episodes of pneumococcal pneumonia and persistent diarrhea. Serum IgG, IgA, and IgM concentrations are all markedly decreased, and the number of circulating T cells is normal. A lymph node biopsy specimen shows a normal paracortex with high endothelial venules but no follicles, no germinal centers, and no plasma cells in the medullary cords. Which of the following best explains these node findings?

A. Failure of thymocytes to be positively selected in the thymic cortex

B. Failure of naive lymphocytes to enter the node through high endothelial venules

C. Arrest of B-cell maturation in the bone marrow

D. Absence of M cells over mucosal lymphoid follicles

Why did you pick it?

Board-style question NPLEX-P1-CARD-10-01-C02-Q02

Where a testicular tumour spreads first

A 29-year-old man has a firm, painless 2-cm mass in the right testis. The overlying scrotal skin is normal. Imaging is done to look for nodal spread. Which group of lymph nodes is most likely to be involved first?

A. Lumbar (para-aortic) nodes near the renal vessels

B. Superficial inguinal nodes

C. Internal iliac nodes

D. Popliteal nodes

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-11

CV-11 Hemodynamics and Blood Flow Regulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-11-01-C01				
NPLEX-P1-CARD-11-01-C02				
NPLEX-P1-CARD-11-01-C03				
NPLEX-P1-CARD-11-01-C04				
NPLEX-P1-CARD-11-01-C05				
NPLEX-P1-CARD-11-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C01-BD1

Write the flow equations and Poiseuille's law from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C01 Activity: NPLEX-P1-CARD-11-01-C01-BD1 Date: _____

Without looking at your notes, write the equation that relates flow, pressure difference and resistance, then rearrange it for resistance and write why resistance is calculated rather than measured. Write the whole-body version of the same equation using mean arterial pressure, cardiac output and total peripheral resistance, and put in the resting values the notes gave for each one. Next write Poiseuille's law in full, name every symbol, and state for each variable whether resistance changes in direct or inverse proportion and by what power. Then write two or three sentences explaining why radius is raised to the fourth power, using the shape of the velocity profile in laminar flow. Finish by listing the three assumptions Poiseuille's law makes and what a living vessel does instead in each case.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C01-BD2

Work out what single changes do to resistance and flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C01 Activity: NPLEX-P1-CARD-11-01-C01-BD2 Date: _____

Hold the pressure gradient constant and work out each of the following, writing the factor of change for resistance and for flow and one line of reasoning for each: the radius is halved, the radius is doubled, the radius widens by about 19%, viscosity doubles, and vessel length doubles. Then write a clinical example for each from the notes. Next, a man has a hematocrit of 68% (reference range about 40 to 45%). Write what happens to viscosity, to resistance, to flow and to oxygen delivery, and explain why oxygen delivery can fall even though the blood carries more oxygen per unit volume. Name three other changes in blood composition that raise viscosity. Finish by writing what rouleaux formation and a cell-poor layer at the wall each do to apparent viscosity and in which vessels each one matters.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-11-01-C01-PR01

Predict the effect of a focal arterial stenosis

A short segment of a muscular artery narrows so that its internal radius falls to half of normal, while the pressure gradient across the limb stays the same at first. Predict each item below, write the direction and size of the change where the notes let you give a number, and give the relationship or equation that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the resistance of the narrowed segment		
What happens to flow through that segment if the pressure gradient does not change		
What happens to the cross-sectional area of the lumen and to blood velocity inside the narrowed segment		
What happens to the Reynolds number in and just beyond the stenosis		
What is heard over the vessel and why		
What happens to that sound as the stenosis becomes critical and flow falls		
What happens to the resistance of the same stenosis if the diseased segment is long and diffuse instead of short		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-11-01-C01-Q01

Arteriolar dilation and flow

A 34-year-old woman begins moderate exercise. Arterioles in the quadriceps dilate so that their internal radius increases by approximately 19%, while mean arterial pressure is held constant by the baroreceptor reflex. Blood viscosity and vessel length do not change. Which of the following best describes the change in blood flow through those arterioles?

A. Flow increases by about 19%

B. Flow increases by about 50%

C. Flow approximately doubles

D. Flow increases approximately 16-fold

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C02-BD1

Series and parallel resistance, and the pressure profile

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C02 Activity: NPLEX-P1-CARD-11-01-C02-BD1 Date: _____

Write the rule for adding resistances in series and the rule for adding them in parallel, each as an equation. Under the series rule write which quantity is the same in every segment and which segment takes the largest pressure drop, and explain why from the relationship between pressure drop, flow and resistance. Under the parallel rule write which quantity is the same in every branch, what happens to total resistance as branches are added, and what n identical branches of resistance R give. Work both rules once with numbers: three segments of 10, 20 and 20 units in series, and then the same three as parallel branches. Next build a table from memory with one row for the aorta and large arteries, one for small arteries and arterioles, one for capillaries and one for venules and veins, and columns for mean pressure, share of total resistance, total cross-sectional area and share of blood volume. Finish by writing the three or four features of arterioles that make them hold the largest adjustable share of resistance, and why billions of capillaries in parallel do not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C02-BD2

Adding and removing parallel paths, and beds placed in series

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C02 Activity: NPLEX-P1-CARD-11-01-C02-BD2 Date: _____

A surgeon creates a large arteriovenous fistula in the forearm. Write what the fistula does to total peripheral resistance and why, then write in order what happens to mean arterial pressure, what the baroreceptors do, what happens to cardiac output and what happens to diastolic pressure and pulse pressure. Write what a very large fistula does to the heart over time. Next, a patient undergoes a mid-thigh amputation. Write what happens to total peripheral resistance and explain it from the parallel rule. Then write the six-step sequence from a large bed dilating to flow being redirected while mean arterial pressure is defended, one step per line. Finish with the capillary beds that lie in series rather than parallel: name three from the notes, write what raised downstream resistance does to upstream capillary pressure, and use that to explain how portal hypertension develops.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C03-BD1

Velocity against total cross-sectional area

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-BD1 Date: _____

Write the equation that links flow, velocity and cross-sectional area, then write why velocity at each level of the systemic circuit varies inversely with the total cross-sectional area at that level. Draw one set of axes with vessel segments from aorta to vena cava along the horizontal axis, and sketch total cross-sectional area and mean velocity as two curves on it. Mark the aortic and capillary velocities with the values from the notes and write the approximate ratio of capillary area to aortic area. Write why capillary velocity has to be lowest and how long a red cell spends in a systemic capillary, then write what that transit time is for. Finish by writing why velocity rises again in the veins but stays below the aortic value.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C03-BD2

Turbulence at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-BD2 Date: _____

Write the Reynolds number as an equation, name each variable, and state which direction each one moves the number. Write which of those variables barely changes in the body and which three are the clinical ones. Then explain in two or three sentences why turbulent flow is audible and why it needs a larger pressure gradient for the same flow. Next, a woman has a hemoglobin of 7.2 g/dL (reference range about 12 to 16 g/dL) and a soft systolic murmur across structurally normal valves. Rebuild the sequence from the low hemoglobin to the murmur, one step per line, and write why anemia is the classic cause of a flow murmur while polycythemia tends to abolish one. Then write what makes each of these audible or silent and why: a carotid stenosis, a critical carotid stenosis, an arteriovenous fistula, and the brachial artery under a blood pressure cuff at a pressure just below systolic and then below diastolic. Finish by writing where in the arterial tree shear stress is low and oscillating, what that does to endothelial nitric oxide, and what tends to form there.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-11-01-C03-RT01

Rebuild the sequence from anemia to a flow murmur

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from a falling hemoglobin and hematocrit to a systolic flow murmur across normal valves, one step per line and in order. Beside the viscosity step write what viscosity does to resistance and to the Reynolds number, and note that these two effects pull the murmur in opposite directions through different variables. Beside the arteriolar step write which stimulus dilates the arterioles. Beside the cardiac output step write which component of cardiac output rises most. Then write the three other states from the notes that produce a flow murmur by the same reasoning.

Rebuild from memory: From anemia to a flow murmur

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C04-BD1

Compliance, the arterial reservoir and pulse pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C04 Activity: NPLEX-P1-CARD-11-01-C04-BD1 Date: _____

Write the equation for compliance, name its reciprocal, and write how much more compliant systemic veins are than the matching arteries and what that explains about where blood sits. Then describe the reservoir function of the elastic arteries in your own words: what the wall does during ejection, what it does during diastole, and the three things that function achieves. Write the definition of pulse pressure with its normal value, then write what pulse pressure is roughly proportional to. Write what governs how fast stored volume runs off during diastole, give the time constant as a product of two variables, and list the three conditions that let diastolic pressure fall further. Finish with the formula that estimates mean arterial pressure from diastolic and pulse pressure, and write at which heart rates that estimate becomes less accurate and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C04-BD2

Predict the pressure pattern in a stiff aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C04 Activity: NPLEX-P1-CARD-11-01-C04-BD2 Date: _____

Two people have the same stroke volume of 70 mL, the same heart rate and the same total peripheral resistance. One is 25 years old with a pressure of 120/80 mm Hg. The other is 80 years old with aortic compliance halved. Work out the second person's pulse pressure, systolic pressure, diastolic pressure and mean arterial pressure, show your reasoning for each, and write why mean arterial pressure barely moves while systolic and diastolic pressures both change. Then write the wall changes that lower aortic compliance with age, name the three diseases that speed them up, and explain what faster pulse wave velocity does to the timing of the reflected wave and to the systolic peak. Write the two downstream consequences for the left ventricle and for coronary perfusion, and say which part of the ventricular wall is most exposed. Finish by sorting each of the following into widens or narrows pulse pressure, with the mechanism in one line: aortic regurgitation, hemorrhage, aortic stenosis, hyperthyroidism, cardiac tamponade, arteriovenous fistula, exercise.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C05-BD1

Draw the capillary and its Starling forces

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C05 Activity: NPLEX-P1-CARD-11-01-C05-BD1 Date: _____

Draw a capillary with its arterial end on the left and its venous end on the right, and add a lymphatic vessel in the interstitium beside it. Write the Starling equation for net filtration in full and name every term, including the filtration coefficient and the reflection coefficient. On your drawing mark the capillary hydrostatic pressure at each end, the interstitial hydrostatic pressure, the plasma oncotic pressure and the interstitial oncotic pressure, using the teaching values from the notes, and draw an arrow for each force in the direction it moves fluid. Calculate the net pressure at the arterial end and at the venous end, show the working, and say which way fluid moves at each end. Write how much fluid the lymphatics return in a day and what else they carry back. Finish by writing which protein supplies most of the plasma oncotic pressure and roughly what share, and where the reflection coefficient and the filtration coefficient are unusually high or low.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C05-BD2

Work out the edema in four patients

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C05 Activity: NPLEX-P1-CARD-11-01-C05-BD2 Date: _____

For each patient below, write which Starling term changes, which direction it moves, whether the fluid is a transudate or an exudate, whether the swelling pits, and where in the body the swelling appears first and why. Patient 1 has nephrotic syndrome with a plasma albumin of 1.8 g/dL (reference range about 3.5 to 5.0 g/dL). Patient 2 has right heart failure with raised jugular venous pressure. Patient 3 has a deep burn to one forearm. Patient 4 had axillary nodes removed and radiated two years ago. For patient 1, take plasma oncotic pressure from 28 mm Hg down to 16 mm Hg and recalculate the net pressure at both ends of the capillary, showing the working, then write why the edema is generalized. For patient 4, write why the swelling stops pitting as the years pass. Finish by writing the three safety factors that delay edema, how much the notes say capillary hydrostatic pressure can rise before edema becomes obvious, and what changes in the interstitium once its pressure passes about 0 mm Hg.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-11-01-C05-CP01

The four mechanisms of edema side by side

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Raised capillary hydrostatic pressure	Low plasma oncotic pressure	Increased capillary permeability	Lymphatic obstruction
Which Starling term changes and in which direction				
Two or three causes from the notes				
Transudate or exudate				
Protein content of the fluid and specific gravity				
Does the swelling pit, and does that change over time				
Generalized or localized, and where it appears first				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-11-01-C05-CA01

Abdominal swelling and ankle edema in a man with liver disease

A 57-year-old man comes to the clinic because of six weeks of increasing abdominal girth and swelling of both ankles. He has a ten-year history of heavy alcohol use. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 92/min, respirations are 16/min, and blood pressure is 104/62 mm Hg. The abdomen is distended with shifting dullness, and there are dilated veins over the anterior abdominal wall. Pressing over each medial malleolus leaves an indentation that takes several seconds to fill. The jugular venous pressure is not raised and the lungs are clear. Serum albumin is 2.1 g/dL (reference range 3.5 to 5.0 g/dL), total protein is 5.4 g/dL, and the fluid removed at paracentesis has a specific gravity of 1.009.

1. Recognize. Name the two Starling terms that are abnormal in this patient and say which vascular bed each one acts on.

2. Explain. Explain how portal hypertension develops in this patient, using the arrangement of the gut and splenic capillaries relative to the hepatic sinusoids, and explain how the low albumin acts on the same capillaries.

3. Differentiate. Write how the Starling picture here differs from the edema of right heart failure and from the edema of lymphatic obstruction, naming the findings in this case that separate them.

4. Investigate. Write which measurements and physical findings you would use to confirm the mechanism, and what value or appearance each one would have.

5. Decide. State which single mechanism explains why the fluid collects in the abdomen first rather than only in the legs, and state what the specific gravity of the fluid tells you about the capillary wall.

6. Justify. Justify your answer from the Starling equation, naming the term that each finding reflects, and explain why the safety factors that normally delay edema are overwhelmed in this patient.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C06-BD1

The four layers of control over arteriolar tone

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-BD1 Date: _____

Start with the smooth muscle itself: write the sequence from a rise in cytosolic calcium to a contracted arteriole, naming the calcium binding protein and the two enzymes that set the level of myosin light chain phosphorylation, then write the two general routes by which a dilator relaxes the same cell. Now organise the four layers of control on one page, with a heading for each. Under metabolic control, list the substances that accumulate and dilate, name the channel many of them open, and define active hyperemia and reactive hyperemia. Under myogenic control, write the sequence from stretch to constriction naming the two channel types involved. Under endothelial control, list the stimuli that activate the nitric oxide pathway, name the other dilator and the constrictor the endothelium releases, and write what acetylcholine does when the endothelium is damaged and why. Under neural and hormonal control, write which receptor and G protein sympathetic norepinephrine uses, what the main neural dilator mechanism is and why, and then list the hormones that constrict and those that dilate with the receptor and second messenger for each. Finish by writing the approximate mean arterial pressure range over which brain and kidney autoregulate, and what chronic hypertension does to that range.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C06-BD2

Which control wins in which bed

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-BD2 Date: _____

Build a table from memory with one row each for the coronary, cerebral, skeletal muscle, skin, pulmonary and renal beds, and columns for dominant control, response to hypoxia and the effect of sympathetic stimulation. Then answer each of the following in two or three sentences, using the mechanism rather than the table entry. Why does the heart have to raise flow rather than extract more oxygen when demand rises, and what does that do to the alpha-1 effect on coronary arterioles. Why does a patient hyperventilating during a panic attack feel lightheaded, and which molecule crosses the blood-brain barrier to produce that effect while another one does not. Why does flow to exercising muscle rise about 20-fold despite high sympathetic outflow, and what is that override called. Why is the skin pale and cool after a hemorrhage while coronary and cerebral flow are preserved. Why do pulmonary vessels constrict to low alveolar oxygen when every systemic bed dilates, what does that achieve locally, and what happens when the hypoxia involves the whole lung. Finish with a patient whose blood pressure has run at 170/100 mm Hg for years and is lowered quickly to 110/70 mm Hg: write why that person can become hypoperfused at a pressure a normotensive person tolerates.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-11-01-C06-RT01

Rebuild the nitric oxide pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-RT01 Date: _____

Rebuild the seven-step nitric oxide pathway from the stimulus at the endothelium to the dilated arteriole, one step per line and in order, and draw it as two cells side by side with the endothelial cell on the left and the smooth muscle cell on the right so that each step sits in the cell where it happens. Name the enzyme in the endothelium, its substrate and both of its products. Name the enzyme in the smooth muscle that nitric oxide activates, the second messenger it makes and the kinase that messenger activates. At the last two steps write what happens to cytosolic calcium and to myosin light chain phosphorylation. Then write where a nitrate drug enters this pathway and why that matters when the endothelium is damaged.

Rebuild from memory: Nitric oxide pathway from endothelium to smooth muscle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-11-01-C06-Q02

Regional flow during hyperventilation

A 22-year-old woman is brought to the emergency department during a panic attack. Respirations are 34/min, pulse is 108/min, and blood pressure is 128/76 mm Hg. She reports lightheadedness and tingling in both hands. Arterial blood gas analysis shows a PaCO2 of 22 mm Hg and a PaO2 of 108 mm Hg. Which of the following best explains her lightheadedness?

A. A rise in perivascular pH that constricts cerebral arterioles and reduces cerebral blood flow

B. A fall in mean arterial pressure below the lower limit of cerebral autoregulation

C. Inhibition of oxygen-sensitive potassium channels in cerebral arteriolar smooth muscle

D. Loss of myogenic tone in cerebral arterioles with a rise in cerebral blood flow

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-12

CV-12 Exercise and the Cardiovascular System

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for one competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-12-01-C01				
NPLEX-P1-CARD-12-01-C02				
NPLEX-P1-CARD-12-01-C03				
NPLEX-P1-CARD-12-01-C04				
NPLEX-P1-CARD-12-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C01-BD1

Draw the two inputs that start the exercise response

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-BD1 Date: _____

Without looking at your notes, draw the contracting muscle, the medullary cardiovascular centers, and the heart, and show the two signals that raise heart rate at the onset of exercise. Label the feedforward signal and write where it comes from, what its output scales with, and what it does to the arterial baroreflex set point. Label the feedback reflex, then split it into its two arms: for each arm write the afferent fiber class, the stimulus that excites it, when during the contraction it is strongest, and its main cardiovascular effect. Finish by writing the two experimental observations from the notes that show central command and the metaboreflex are separate drivers.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C01-BD2

Why the rate rise is vagal first and sympathetic second

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-BD2 Date: _____

Write the receptor, the G protein, the second messenger or subunit, and the channel for vagal slowing and for sympathetic acceleration of the SA node, then explain from how each transmitter is cleared why heart rate rises within one or two beats when vagal firing stops but sympathetic effects take seconds. Give the approximate heart rate up to which vagal withdrawal accounts for most of the rise, and write what sympathetic drive adds above that rate. Then write the three mechanisms that raise venous return so cardiac output can actually rise, and name the relationship that converts greater filling into a larger stroke volume. Finish with three people starting the same exercise: one has a denervated transplanted heart, one is taking a beta-1 blocker, and one has heart failure. For each, write what happens to heart rate at the onset, how cardiac output rises instead, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-12-01-C01-RT01

Rebuild the onset sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-RT01 Date: _____

Rebuild the seven-step sequence that runs from the recruitment of motor units to a matched rise in muscle blood flow, one step per line and in order. Beside the step where heart rate first rises, write the channel that closes and how long the change takes. Beside the sympathetic step, write the three things sympathetic outflow is doing at once. Beside the venous return step, write the two pumps involved and what each one does mechanically. Beside the final step, write which afferent fibers are responsible and what condition in the muscle brings them in.

Rebuild from memory: Cardiovascular response at the onset of exercise

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C02-BD1

Write the Fick principle and fill in the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C02 Activity: NPLEX-P1-CARD-12-01-C02-BD1 Date: _____

Write the Fick principle as an equation for the whole body and define each term. Write the equation for arterial O2 content and work it out for a hemoglobin of 15 g/dL at a saturation of 98%. Then build a table from memory with rows for heart rate, stroke volume, cardiac output, mixed venous O2 content, the arteriovenous O2 difference, and VO2, and columns for rest, maximal exercise in an untrained young adult, and maximal exercise in an elite endurance athlete. Fill every cell with the approximate value from the notes. Underneath, write the three reasons O2 extraction rises during exercise, and name the effect that shifts the oxyhemoglobin curve in active muscle.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C02-BD2

Work VO2max from the numbers, then take the hemoglobin away

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C02 Activity: NPLEX-P1-CARD-12-01-C02-BD2 Date: _____

A 25-year-old untrained man weighing 75 kg has a resting heart rate of 70/min, a stroke volume of 70 mL, an arterial O2 content of 20 mL/dL, and a mixed venous O2 content of 15 mL/dL. At maximal effort his heart rate is 195/min, his stroke volume is 110 mL, and his mixed venous content falls to 5 mL/dL. Calculate resting VO2 and VO2max in mL/min and in mL/kg/min, showing your working, and state by what factor flow rose and by what factor extraction rose. Now repeat the calculation with a hemoglobin of 9 g/dL and a mixed venous content of 4 mL/dL at maximum, and write the cardiac output that would be needed to hold the original VO2max. Explain from your two sets of numbers what happens to VO2max and to heart rate at a fixed submaximal workload, and name the variable that limits VO2max in a healthy person at sea level. Finish by writing why stroke volume plateaus at about 40% to 60% of VO2max in an untrained person and what carries cardiac output higher after that point.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C03-BD1

Rebuild the distribution of cardiac output

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-BD1 Date: _____

From memory, build a table of where cardiac output goes at rest and at maximal exercise, with one row for skeletal muscle, coronary, brain, skin, splanchnic, and kidney. Use columns for resting flow with its percentage of a total of about 5.8 L/min, maximal exercise flow against a total of about 25 L/min, and the control that dominates that bed. Fill every cell. Underneath, write the resting and peak flow per 100 g of skeletal muscle and the fold increase. Then list every local mediator the notes give for dilation in active muscle and write in one line what each one does, including which channel interstitial K+ acts through and what makes red cells release ATP.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C03-BD2

Why total resistance falls while some beds constrict

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-BD2 Date: _____

Sympathetic outflow rises to every vascular bed during exercise, yet flow to active muscle increases up to 25 fold. Explain how both are true, naming the process that blunts alpha-adrenergic constriction in contracting muscle, the receptor subtype most involved, and why some residual constriction has to remain. Then explain from the arrangement of the vascular beds why total peripheral resistance falls even though gut, kidney, and inactive muscle arterioles are constricting. Next write what happens to coronary flow and why extraction cannot supply the extra demand, then write what that predicts for a person with a coronary stenosis that causes no symptoms at rest. Finish by writing what happens to renal blood flow, to GFR, and therefore to filtration fraction, and why cerebral flow is the exception.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-12-01-C03-RT01

Rebuild how flow concentrates in working muscle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-RT01 Date: _____

Rebuild the six-step sequence that explains how flow is concentrated in working muscle, one step per line and in order. Beside the step on inactive beds, name the receptor that mediates the constriction and list the beds involved. Beside the step on active muscle, list the local mediators. Beside the step on sympatholysis, write which signals do the blunting. Beside the final step, write the one variable all the parallel beds share and the equation that converts a bed's resistance into its share of flow.

Rebuild from memory: How flow concentrates in working muscle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-12-01-C03-PR01

Predict what heat does to a steady run

A healthy adult runs at the same speed for 60 minutes on a cool day and then on a hot, humid day. Core temperature on the hot day rises to 39 °C (102.2 °F). For each item below, write the direction of the change on the hot day and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to skin blood flow at a heavy submaximal workload, and which nerve fibers produce it		
What happens to the volume of blood available to fill the heart		
What happens to stroke volume over the 60 minutes		
What happens to heart rate over the 60 minutes, and what this pattern is called		
What happens to the flow available to the leg muscles		
What happens to skin blood flow if the runner pushes on to maximal effort, and what that costs		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C04-BD1

Pressure in dynamic exercise against pressure in static exercise

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C04 Activity: NPLEX-P1-CARD-12-01-C04-BD1 Date: _____

Write the equation that relates mean arterial pressure to cardiac output and total peripheral resistance. Then, for dynamic exercise and for static exercise in turn, write what happens to heart rate, stroke volume, cardiac output, total peripheral resistance, systolic pressure, diastolic pressure, and pulse pressure, and name the main pressor drive in each. Give the usual peak systolic pressure range in healthy adults during dynamic exercise. For static exercise, explain what rising intramuscular pressure does to the muscle's own blood vessels and how that produces the pressor response. Finish by naming the type of load each form of exercise puts on the left ventricle and writing what happens to the shape of the pressure-volume loop in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C04-BD2

Read pressures and work out resistance

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C04 Activity: NPLEX-P1-CARD-12-01-C04-BD2 Date: _____

Write the formula for mean arterial pressure from systolic and diastolic pressure, and the formula for total peripheral resistance from mean arterial pressure and cardiac output. Then calculate mean arterial pressure and total peripheral resistance in mmHg per L/min for three states, showing your working: at rest with 120/80 mmHg and a cardiac output of 5 L/min; while running with 180/76 mmHg and a cardiac output of 20 L/min; and during a sustained handgrip with 165/110 mmHg and a cardiac output of 6.5 L/min. Write one sentence for each state explaining why pressure moved the way it did, and state which calculation is least accurate at high heart rates and why. Then write the index of myocardial O2 demand given in the notes and use it to explain why a static effort can provoke ischemia at a lower heart rate than running. Finish by writing what a fall in systolic pressure during a rising dynamic workload means and naming three conditions that produce it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-12-01-C04-CP01

Dynamic against static exercise

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Dynamic exercise such as running or cycling	Static exercise such as a handgrip or heavy hold
Heart rate		
Stroke volume		
Cardiac output and how it rises		
Total peripheral resistance		
Systolic pressure		
Diastolic pressure and pulse pressure		
Main pressor drive		
Load on the left ventricle and shape of the pressure-volume loop		
Effect of a sustained handgrip on the murmurs of mitral regurgitation and of hypertrophic cardiomyopathy		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C05-BD1

Eccentric and concentric remodeling, and the trained heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C05 Activity: NPLEX-P1-CARD-12-01-C05-BD1 Date: _____

Write how adult myocytes respond to a chronic load and why. Define eccentric and concentric hypertrophy by where the sarcomeres are added, what happens to cavity size and wall thickness, and which kind of load produces each, then name the relationship that explains why both bring wall stress back toward normal. Next describe the endurance-trained left ventricle: wall thickness, wall-to-cavity ratio, right ventricle and atria, and diastolic filling. Write what happens to plasma volume and red cell mass with training and why hemoglobin concentration can fall. Then explain, using a resting cardiac output of about 5 L/min and a stroke volume of about 100 mL, why the resting heart rate is about 50/min, name the main mechanism of the slowing, and list the three benign conduction findings that come from the same mechanism and what happens to them with exercise. Finish by writing what training changes and what it does not change at a fixed submaximal workload and at maximal effort.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C05-BD2

Separate physiologic growth from sarcomere disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C05 Activity: NPLEX-P1-CARD-12-01-C05-BD2 Date: _____

Write the growth factor and the intracellular pathway that drive physiologic hypertrophy, and the neurohormones, receptor class, and signaling pathway that drive pathologic hypertrophy, then write what pathologic growth does to the fetal gene program, the capillary supply, the interstitium, and ventricular function. Next write the inheritance pattern of hypertrophic cardiomyopathy and the two genes most often involved, then explain the mechanism behind each of these three features: exertional dyspnea, angina with normal epicardial coronary arteries, and sudden death in a young athlete. Then write what causes dynamic outflow obstruction and list the four changes in loading or contractility that worsen it, with the bedside maneuvers that make the murmur louder and softer and why. Finish by writing what happens to each heart after about three months without training and why that result separates the two conditions.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-12-01-C05-CP01

Athlete's heart against hypertrophic cardiomyopathy

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Endurance athlete's heart	Hypertrophic cardiomyopathy
Wall thickness and whether it is symmetric		
Cavity dimension		
Diastolic function		
Histology		
Outflow tract		
Electrocardiogram		
Peak VO2		
Result after about three months of detraining		
Family history		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-12-01-C05-CA01

Near syncope in a young athlete during a hot race

An 18-year-old cross-country runner is seen after nearly fainting in the final kilometer of a race run at 32 °C (89.6 °F). He had a tight sensation in the chest for the last several minutes and felt his heart pounding. He has run competitively for four years and trains daily. A maternal uncle died suddenly at 31 years of age. Temperature is 38.6 °C (101.5 °F), pulse is 58/min and regular, respirations are 16/min, and blood pressure is 112/74 mm Hg. A systolic murmur is heard at the left sternal border; it becomes louder when he stands up from squatting and softer during a sustained handgrip. The electrocardiogram shows sinus bradycardia with deep T wave inversions in leads V4 through V6 and Q waves in leads I and aVL. Echocardiography shows a septal wall thickness of 18 mm, a posterior wall thickness of 10 mm, and a left ventricular cavity dimension of 42 mm, with impaired relaxation on diastolic filling.

1. Recognize. Name the structural diagnosis and the type of protein the underlying mutations affect, and write its usual inheritance pattern.

2. Explain. Explain from the stiff ventricle and the muscle mass why he had chest tightness and breathlessness during the race, and why the epicardial coronary arteries can be normal.

3. Differentiate. Write the features in this case that separate it from the heart of a trained endurance runner, taking wall thickness, symmetry, cavity size, diastolic filling, electrocardiogram, and family history in turn.

4. Investigate. Write which measurements and observations you would use to confirm the diagnosis, and what value or appearance each one would have in this condition rather than in a trained heart.

5. Decide. State which physiologic change during a hot race made the outflow gradient worse, and which mechanism accounts for the risk of sudden death.

6. Justify. Justify your answer from the effect of preload, afterload, contractility, and heart rate on ventricular size, and explain why the murmur got louder on standing and softer during handgrip.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-12-01-C02-Q01

Accounting for the rise in oxygen uptake

A 24-year-old untrained man undergoes a graded cycling test. At rest his cardiac output is 5 L/min, his arterial O2 content is 20 mL/dL, and his mixed venous O2 content is 15 mL/dL. At peak effort his cardiac output is 21 L/min, his arterial O2 content is 20 mL/dL, and his mixed venous O2 content is 5 mL/dL. Which of the following best accounts for the increase in O2 uptake from rest to peak effort?

A. A rise in arterial O2 content produced by more efficient pulmonary gas exchange

B. An approximately fourfold rise in cardiac output together with an approximately threefold rise in O2 extraction

C. A rise in cardiac output with no change in the arteriovenous O2 difference

D. A widening of the arteriovenous O2 difference while cardiac output stays at its resting value

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C03-Q02

Regional flow during submaximal cycling

A healthy 30-year-old woman cycles at 70% of her maximal O2 uptake in a thermoneutral room. Which of the following changes in regional blood flow is expected during the exercise?

A. Coronary blood flow rises about three- to fourfold, because myocardial O2 extraction is already near maximal at rest

B. Renal blood flow rises in proportion to the rise in cardiac output

C. Cerebral blood flow rises to about three times its resting value

D. Splanchnic blood flow is unchanged, because local metabolic control overrides sympathetic constriction in the gut

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C04-Q03

Hemodynamics of a sustained handgrip

A healthy 28-year-old man holds a handgrip at 40% of his maximal voluntary force for two minutes. Which of the following sets of changes is expected at the end of the two minutes, compared with his resting values?

A. Total peripheral resistance falls, diastolic pressure falls slightly, and stroke volume rises markedly

B. Cardiac output rises about fourfold and pulse pressure widens while diastolic pressure is unchanged

C. Heart rate falls as the arterial baroreflex buffers the rise in pressure, and cardiac output is unchanged

D. Total peripheral resistance is unchanged or higher, and systolic and diastolic pressure both rise sharply

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C05-Q04

Reading an echocardiogram in a trained cyclist

A 19-year-old competitive cyclist is evaluated after a resting pulse of 46/min is found at a preparticipation examination. He has no symptoms and no family history of sudden death. The electrocardiogram shows sinus bradycardia with isolated high QRS voltage and upright T waves. Echocardiography shows a symmetric left ventricular wall thickness of 11 mm, a left ventricular cavity dimension of 58 mm, an enlarged right ventricle, and enhanced early diastolic filling. Which of the following findings most strongly supports physiologic adaptation to endurance training rather than hypertrophic cardiomyopathy?

A. The resting heart rate of 46/min

B. The isolated high QRS voltage on the electrocardiogram

C. The enlarged cavity with symmetric wall thickening and enhanced diastolic filling

D. The symmetric wall thickness of 11 mm taken on its own

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-13

CV-13 Cardiovascular Biochemistry and Nutrition

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-13-01-C01				
NPLEX-P1-CARD-13-01-C02				
NPLEX-P1-CARD-13-01-C03				
NPLEX-P1-CARD-13-01-C04				
NPLEX-P1-CARD-13-01-C05				
NPLEX-P1-CARD-13-01-C06				
NPLEX-P1-CARD-13-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C01-BD1

Build the lipoprotein table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-BD1 Date: _____

Without looking at your notes, build a table of the five lipoproteins in order of decreasing size, with one row each for chylomicron, VLDL, IDL, LDL, and HDL. Use the columns made by, key apolipoproteins, and main role, and fill every cell. Beside the table write what happens to a particle's size, density, and cholesterol content as triglyceride is removed, and explain why density runs in the order you wrote. Then write one line each for apo B-48, apo B-100, apo C-II, apo E, and apo A-I saying what that apolipoprotein does. Finish by naming the two enzymes that esterify cholesterol, and write where each one works and which particle or compartment it acts in.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C01-BD2

Three broken steps in lipid transport

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-BD2 Date: _____

Three patients each lack one component. Patient 1 lacks apo C-II. Patient 2 lacks ABCA1. Patient 3 lacks apo E function because both apo E alleles are E2. For each patient write which particle accumulates in plasma, which step stopped, and why the particles upstream cannot move on. Then write which plasma lipid rises most in each one and whether plasma left overnight in the cold would show a creamy top layer, uniform turbidity, or neither, and explain each appearance from the particle involved. Finish by explaining why a high fasting triglyceride usually comes with a low HDL, naming the transfer protein and the enzyme that produce that pairing and what each one does to the HDL particle.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C01-RT01

Rebuild the exogenous lipoprotein pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-RT01 Date: _____

Rebuild the six-step exogenous pathway from the enterocyte to the liver, one step per line and in order. Beside the packaging step write the apolipoprotein used and the transfer protein that loads the core. Beside the transport step write the route the particle takes to reach the blood and why it cannot enter a blood capillary directly. Beside the enzyme step write which apolipoprotein activates the enzyme, where the enzyme sits, and what it releases. Mark on your drawing the step that fails in lipoprotein lipase deficiency, and write how long after a meal a normal plasma sample should be free of these particles. Underneath, rebuild the endogenous pathway in the same way from VLDL secretion to LDL clearance, naming the enzyme that makes IDL, the enzyme that makes LDL, and the apolipoprotein LDL depends on.

Rebuild from memory: Exogenous lipoprotein pathway

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C01-Q01

Reverse cholesterol transport fails at its first step

A 28-year-old man is evaluated for numbness and weakness of both hands. Examination shows enlarged orange-yellow tonsils and reduced sensation in a glove distribution. Laboratory studies show total cholesterol of 110 mg/dL, HDL cholesterol of 3 mg/dL, and triglyceride of 210 mg/dL. Macrophages in his bone marrow are filled with cholesteryl ester. Which of the following steps is most likely defective in this patient?

A. Transfer of free cholesterol and phospholipid from the cell membrane to lipid-poor apolipoprotein A-I

B. Esterification of free cholesterol on the surface of HDL by lecithin-cholesterol acyltransferase

C. Hydrolysis of chylomicron triglyceride by lipoprotein lipase on capillary endothelium

D. Hepatic uptake of remnant particles through apolipoprotein E

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C02-BD1

Build the Fredrickson table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C02 Activity: NPLEX-P1-CARD-13-01-C02-BD1 Date: _____

On blank paper build a table of the primary dyslipidemias with one row each for type I, type IIa, type IIb, type III, type IV, and type V. Use the columns defect, inheritance where the notes gave one, raised particle, which lipids are raised, and consequences, and fill every cell. Then write the four desirable adult lipid values from the notes with their units: total cholesterol, LDL cholesterol, HDL cholesterol in men and in women, and fasting triglyceride. Finish by listing the four secondary causes that must be excluded before any pattern is called familial.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C02-BD2

Name the type from the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C02 Activity: NPLEX-P1-CARD-13-01-C02-BD2 Date: _____

Four patients have fasting lipid studies. Patient 1 is a 9-year-old with abdominal pain, triglyceride of 2,800 mg/dL, cholesterol of 260 mg/dL, and a creamy layer over clear plasma after refrigeration. Patient 2 is a 41-year-old with a body mass index of 34, triglyceride of 450 mg/dL, cholesterol of 210 mg/dL, HDL of 31 mg/dL, and uniformly cloudy plasma. Patient 3 is a 38-year-old with cholesterol of 360 mg/dL, triglyceride of 340 mg/dL, and yellow palmar creases. Patient 4 is a 12-year-old with cholesterol of 390 mg/dL, triglyceride of 120 mg/dL, thickening of both Achilles tendons, and a father who had a myocardial infarction at 40. For each patient write the Fredrickson type, the particle that is accumulating, the molecular defect, and the single finding that settled it. Then explain the mechanism behind the abdominal pain in patient 1 and why that patient's atherosclerotic risk is not greatly raised, and explain why the liver in patient 4 keeps making cholesterol and secreting VLDL even though plasma cholesterol is high.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-13-01-C02-CP01

Type I against type IIa against type III

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Type I	Type IIa	Type III
Molecular defect and inheritance			
Particle that accumulates			
Cholesterol and triglyceride pattern			
Appearance of plasma left overnight in the cold			
Skin or eye findings			
Atherosclerotic risk and why it is high or low			
What else must be present for the pattern to appear			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C03-BD1

Cholesterol synthesis and the four controls on the reductase

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-BD1 Date: _____

Write out cholesterol synthesis from acetyl-CoA to cholesterol, naming every intermediate the notes gave in order. Mark the rate-limiting step, name its enzyme, say which membrane that enzyme sits in, and write how many NADPH the reaction uses. Name the branch point intermediate and list the three non-sterol products made from it. Then write the four levels at which the rate-limiting enzyme is controlled, and for each one name the signal and say whether activity goes up or down. Finish by writing which hormones raise its activity or expression and which lower them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C03-BD2

Follow one LDL particle into the cell and out again

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-BD2 Date: _____

Follow one LDL particle from the plasma into a hepatocyte. Write the apolipoprotein that binds, the receptor tail motif that gathers the receptor into a coated pit, the coat protein, the protein that pinches the vesicle off, what the falling pH in the endosome does to the receptor and the particle, where each one goes next, the enzyme that frees cholesterol in the lysosome, and the two proteins that carry that cholesterol out. Beside each of three steps write the disorder named in the notes that blocks it. Then write the three things the released cholesterol does as a feedback signal. Finish by explaining why hypothyroidism raises plasma LDL, and why macrophage scavenger receptor uptake of oxidized LDL keeps going until the cell is full of cholesteryl ester.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C03-RT01

Rebuild SREBP activation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from falling membrane cholesterol to restored cell sterol, one step per line and in order. Name the escort protein, the protein it lets go of, the compartment the complex travels to, and the two proteases in the order they cut. Write which part of the transcription factor enters the nucleus and the two genes it turns on. Then draw the same system when cell sterol is high and write what holds the complex in place and what happens to the two genes. Underneath, write which SREBP controls fatty acid synthesis genes and what signal activates it.

Rebuild from memory: SREBP activation when cell sterol is low

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C03-Q02

Why one child responds poorly

A 10-year-old girl has a total cholesterol of 680 mg/dL, a normal triglyceride, thickened Achilles tendons, and yellow plaques on both eyelids. Genetic testing shows two null alleles of the LDL receptor gene. She is given an inhibitor of HMG-CoA reductase, and her plasma LDL falls far less than it does in her heterozygous father. Which of the following best explains the small response in this patient?

A. Hepatic cholesterol synthesis is already maximally suppressed by the high plasma LDL concentration

B. Apolipoprotein B-100 on her LDL particles is structurally abnormal and cannot be recognized

C. Most of the plasma effect of the drug comes from SREBP-2 driven production of new LDL receptors, which her cells cannot make

D. The drug blocks the branch of the pathway that makes ubiquinone rather than the branch that makes sterols

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C04-BD1

Cardiac fuels and the carnitine shuttle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-BD1 Date: _____

Write the four fuels the heart uses. For each one write how it enters oxidation, naming the enzyme or pathway, and the state in which its share of ATP production rises. Write what fraction of resting ATP comes from fatty acids, how they arrive at the myocyte, and which transporter carries them in. Then draw the carnitine shuttle across both mitochondrial membranes, labelling CPT-I, the translocase, and CPT-II, and write what each one does and which chain lengths of fatty acid do not need it. Write the molecule that inhibits CPT-I, the enzyme that makes it, and the two signals that raise and lower that enzyme. Finish by writing why the liver makes ketone bodies but cannot use them, and what phosphocreatine and creatine kinase do for the myocyte.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C04-BD2

Work out the oxygen cost of each fuel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-BD2 Date: _____

Using 2.5 ATP per NADH and 1.5 per FADH2, calculate the ATP yield and the O2 consumed for one glucose and for one palmitate, show your working line by line, and convert each result to ATP per oxygen atom. State which fuel gives more ATP per molecule and which gives more ATP per oxygen atom, and write the approximate percentage difference. Then explain in three or four sentences why that difference matters to a heart whose coronary flow is limited but not absent. Finish by writing the glucose-fatty acid cycle in both directions: how high glucose and insulin reduce fatty acid entry into mitochondria, naming the molecule and the transport step involved, and how active fatty acid oxidation shuts down glucose use, naming the three metabolites and the two enzymes they act on.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C04-RT01

Rebuild energy failure in ischemia

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-RT01 Date: _____

Rebuild the eight-step sequence from coronary occlusion to irreversible injury, one step per line and in order. Beside the relevant steps write the ATP yield of anaerobic glycolysis per glucose and per glucose unit from glycogen, the enzyme that regenerates the NAD+ it needs, the approximate time at which contraction fails, and the approximate time at which injury becomes irreversible. Mark the step at which acidosis begins to limit glycolysis and name the enzyme it inhibits. Then write, in order, what happens to the Na+/K+-ATPase, to extracellular K+, to intracellular Na+, to the Na+/Ca2+ exchanger, to cytosolic Ca2+, and to relaxation. Finish by naming the protein released once the membrane is irreversibly damaged and the substance whose appearance in coronary sinus blood marks ischemia.

Rebuild from memory: From coronary occlusion to energy failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-13-01-C04-PR01

Predict what happens when the carnitine shuttle fails

An infant has carnitine palmitoyltransferase II deficiency and has been fasting for 14 hours. Predict each item below, write the direction of the change, and give the step or molecule that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to long-chain fatty acyl-CoA in the cytosol of the cardiac myocyte		
What happens to the rate of beta-oxidation in the mitochondrial matrix		
What happens to acetyl-CoA and NADH inside the mitochondrion, and therefore to pyruvate dehydrogenase activity		
What happens to the myocyte's ability to make ATP during the fast, and what the ventricular wall looks like over time		
What happens if the same infant is given a fat source made of medium-chain fatty acids, and why		
What happens to cardiac fuel use instead if the infant is given glucose, naming the molecule that rises and the transport step it blocks		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C05-BD1

Thiamine pyrophosphate and the enzymes that need it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C05 Activity: NPLEX-P1-CARD-13-01-C05-BD1 Date: _____

Write where thiamine is absorbed, the enzyme that converts it to its coenzyme form, and why deficiency can appear within weeks. Write which part of the coenzyme attacks the substrate, which carbon it attacks, and what is released, then say in one sentence why that chemistry is needed for both oxidative decarboxylation and transketolase. Build a table of the four enzymes that require thiamine pyrophosphate with the columns reaction catalysed and effect of deficiency, and fill every cell. Beside the dehydrogenase rows write the four other coenzymes those complexes need and the vitamin each comes from. Finish by writing the laboratory test the notes describe and what result indicates deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C05-BD2

Explain every finding in wet beriberi

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C05 Activity: NPLEX-P1-CARD-13-01-C05-BD2 Date: _____

Rebuild the six-step mechanism of wet beriberi in order, from the enzyme block to the dilated failing myocardium. Then write each of these findings and beside it the step that produces it: warm extremities, bounding pulse with a wide pulse pressure, tachycardia, raised jugular venous pressure with leg edema, an enlarged heart on radiograph, and a raised lactate with normal oxygen saturation. Explain in two or three sentences why lactate is high when oxygenation is normal. Write the name of the fulminant form. Then write the four clinical settings in which deficiency develops, and explain why giving a glucose load to someone with marginal stores can bring on symptoms. Finish by writing the one laboratory or examination clue that separates each of severe anemia, thyrotoxicosis, and an arteriovenous fistula from wet beriberi, and the two examination findings that separate ordinary low-output failure from all of them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-13-01-C05-CA01

Breathlessness and warm hands in a malnourished man

A 52-year-old man is brought to the emergency department because of 3 weeks of worsening breathlessness and swelling of both legs. He has drunk about a litre of vodka daily for 12 years and has eaten very little in that time. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 118/min, respirations are 24/min, and blood pressure is 134/48 mm Hg. His hands and feet are warm, the pulse is bounding, the jugular venous pressure is raised, and there is pitting edema to both knees with crackles at both lung bases. Laboratory studies show hemoglobin of 13.1 g/dL, thyroid-stimulating hormone within the reference range, lactate of 6.2 mmol/L (reference below about 2), arterial oxygen saturation of 97% on room air, and normal troponin. A chest radiograph shows an enlarged cardiac silhouette.

1. Recognize. Name the nutritional deficiency and the form of heart failure it has produced, and write the two findings in this presentation that tell you cardiac output is high rather than low.

2. Explain. Write the enzymes that lose their coenzyme, and explain how one biochemical block produces both the low systemic vascular resistance and the failing myocardium.

3. Differentiate. Write how you would separate this from severe anemia, thyrotoxicosis, an arteriovenous fistula, and ordinary low-output heart failure, naming the finding in this patient that argues against each one.

4. Investigate. Write which laboratory test from the notes would confirm the deficiency, what it measures, and what result indicates deficiency, and name the other test that explains his raised lactate.

5. Decide. State why his lactate is raised when his oxygen saturation is normal, and name the fulminant form of this condition that his lactate and blood pressure should make you watch for.

6. Justify. Justify your answer from the chemistry of the coenzyme: write which reaction type it makes possible, which substrate accumulates behind the block, and why the nervous system and the heart are affected before other tissues.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C05-Q03

A high lactate with normal oxygenation

A 46-year-old woman who has had persistent vomiting for 5 weeks after bariatric surgery is admitted with breathlessness and leg edema. Pulse is 116/min and blood pressure is 128/46 mm Hg. Her extremities are warm and the pulse is bounding. Arterial oxygen saturation is 98% on room air, and venous lactate is 5.8 mmol/L (reference below about 2). Addition of the active coenzyme to a sample of her red cells raises transketolase activity sharply. Which of the following enzyme pairs has lost its coenzyme in this patient?

A. Carnitine palmitoyltransferase I and carnitine palmitoyltransferase II

B. Methionine synthase and cystathionine beta-synthase

C. Lysyl oxidase and prolyl hydroxylase

D. Pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C06-BD1

Draw the methionine cycle and its two exits

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C06 Activity: NPLEX-P1-CARD-13-01-C06-BD1 Date: _____

Draw the methionine cycle from memory with methionine, S-adenosylmethionine, S-adenosylhomocysteine, and homocysteine in a loop. Label the step that uses ATP, the step that donates a methyl group, and the hydrolysis step. Draw both exits for homocysteine: name the remethylating enzyme with its vitamin cofactor and its methyl donor, name the enzyme that makes that methyl donor, name the liver enzyme that remethylates by a second route, and name the two enzymes of transsulfuration with the vitamin they share and the amino acid they produce. Mark which exit removes homocysteine permanently. Then write what the methylfolate trap is, which deficiency causes it, and the one laboratory value that separates vitamin B12 deficiency from folate deficiency, naming the other enzyme that explains it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C06-BD2

Two ways micronutrients let a vessel fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C06 Activity: NPLEX-P1-CARD-13-01-C06-BD2 Date: _____

Write the four mechanisms by which raised homocysteine injures a blood vessel, naming the pathway it impairs in the anticoagulant system and what it does to nitric oxide. Then write the normal fasting range for homocysteine and say how a mildly raised value should be read and why. Next, a 7-month-old boy has kinky pale hair, hypotonia, low serum copper and ceruloplasmin, and tortuous elongated arteries. A separate 60-year-old man has bleeding gums, perifollicular hemorrhages, bruising, and poor wound healing. For each one write the enzyme that has failed, the metal or vitamin it needs, which structural protein cannot be finished, which bond or residue is missing, and how that explains the vascular finding. Write where in the cell each of those two enzymes acts. Finish by writing the gene and inheritance of the infant's disease, two acquired causes of the same deficiency, and the disease caused by the opposite transport defect.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-13-01-C06-CP01

Sorting out a raised homocysteine

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Cause	Step in the pathway that is affected	Methylmalonic acid	Other findings that point to it
Folate deficiency			
Vitamin B12 deficiency			
Vitamin B6 deficiency			
Cystathionine beta-synthase deficiency			
MTHFR C677T variant			
Chronic kidney disease			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C07-BD1

Map each electrolyte onto the action potential and the tracing

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C07 Activity: NPLEX-P1-CARD-13-01-C07-BD1 Date: _____

Draw a ventricular myocyte action potential and label phase 4, phase 0, phase 2, and phase 3. Beside each phase write the ion and current that produce it and the part of the ECG it corresponds to. Then build a table with one row each for hyperkalemia, hypokalemia, hypocalcemia, hypercalcemia, and hypomagnesemia, using the columns change in the action potential, ECG findings, and main rhythm risk, and fill every cell. Beside the table write the adult reference range for potassium, total calcium, ionized calcium, and magnesium with their units, the correction to total calcium for a low albumin, the artefact that falsely raises a potassium result, and the reason a normal plasma magnesium can hide a real deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C07-BD2

Why one potassium goes high and another will not come up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C07 Activity: NPLEX-P1-CARD-13-01-C07-BD2 Date: _____

Take a potassium that is rising from 4.5 to 8.5 mEq/L. Write the first change on the tracing, the current responsible, and why that current responds before anything else. Then write what has to happen to the resting membrane before the QRS widens, which channels are inactivated, and the order in which the P wave, the PR interval, and the QRS change. Separately, a woman on a loop diuretic has a potassium of 2.8 mEq/L that does not rise despite replacement, and her QT interval is long. Write which other electrolyte you would measure and why its loss keeps the potassium low, naming the renal channel involved, and write the two other consequences of that same deficiency given in the notes. Finish by explaining how a long QT becomes torsades de pointes, naming the depolarization type, the channel that drives it, and the beat sequence that often precedes it, then write why a low potassium increases cardiac glycoside toxicity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-13-01-C07-PR01

Predict the tracing in kidney failure

A man with advanced kidney failure has a potassium of 7.4 mEq/L (reference 3.5 to 5.0 mEq/L), a total calcium of 6.8 mg/dL (reference 8.5 to 10.5 mg/dL), and an albumin of 4.0 g/dL. Predict each item below, write the direction of the change, and give the current, phase, or channel that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to phase 3 of the ventricular action potential and to the shape of the T wave		
What happens to the length of phase 2 and to the ST segment		
What happens to the QT interval, and which of the two electrolyte changes is driving that		
What happens to the resting membrane potential and to the fraction of fast Na+ channels available		
What happens to the P wave, the PR interval, and the QRS duration as potassium rises further		
What the tracing looks like immediately before cardiac arrest and which two rhythms this patient is at risk of		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C07-Q04

A potassium that will not correct

A 58-year-old woman taking a thiazide diuretic and a proton pump inhibitor is admitted after an episode of collapse. Her plasma potassium is 2.7 mEq/L (reference 3.5 to 5.0 mEq/L) and remains at 2.8 mEq/L after two days of potassium replacement. The electrocardiogram shows flat T waves, prominent U waves, and a corrected QT interval of 0.52 s, and telemetry captures a short run of polymorphic ventricular tachycardia. Which of the following best explains why her potassium has not corrected?

A. Reduced parathyroid hormone release is holding her ionized calcium down and prolonging phase 2

B. Magnesium depletion has removed its normal block of renal ROMK channels, so potassium continues to be lost in the urine

C. A raised extracellular potassium concentration is increasing IKr and speeding phase 3 repolarization

D. Alkalosis has increased albumin binding of calcium, which raises the threshold of excitable membranes

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-14

CV-14 Cardiovascular Genetics

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-14-01-C01				
NPLEX-P1-CARD-14-01-C02				
NPLEX-P1-CARD-14-01-C03				
NPLEX-P1-CARD-14-01-C04				
NPLEX-P1-CARD-14-01-C05				
NPLEX-P1-CARD-14-01-C06				
NPLEX-P1-CARD-14-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C01-BD1

Draw the LDL receptor pathway and its feedback

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-BD1 Date: _____

Without looking at your notes, draw the path of one LDL particle from plasma to free cholesterol inside a hepatocyte. Label the apolipoprotein on LDL that the receptor binds, the coated pit, the acidified endosome, the recycling receptor, and the lysosome, and write what percentage of plasma LDL the liver clears. Beside the drawing, add the feedback limb: show where SCAP, SREBP-2, and INSIG sit when endoplasmic reticulum cholesterol is low and when it is high, and list the three genes the active SREBP-2 fragment switches on. Then write two sentences on what PCSK9 does to a receptor that has just delivered its cargo, and what that means for how many trips each receptor makes. Finish by naming the other apolipoprotein the receptor binds and the lipoprotein that carries it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C01-BD2

Why plasma LDL stays high while the liver cell starves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-BD2 Date: _____

A man has untreated LDL cholesterol of 310 mg/dL (8.0 mmol/L) with normal triglycerides, tendon xanthomas at the Achilles, and corneal arcus at age 40. His father had a myocardial infarction at 45. Write the inheritance pattern and the three genes that can produce this picture, and for each gene write whether the variant is loss of function or gain of function. Then explain in four or five sentences why his hepatocytes behave as though they are short of cholesterol while his plasma is overloaded, and why that keeps new cholesterol synthesis and VLDL secretion going. Next, write what happens to his plasma LDL if his cells are given cholesterol by a route that does not use the receptor, and what that result proves about which step is broken. Then write why lowering hepatic synthesis helps him more than it would help a child with no working receptors at all. Finish by writing which receptor on macrophages keeps taking up oxidized LDL no matter how much cholesterol the cell already holds, and what that produces in skin, tendon, and artery wall.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C01-RT01

Rebuild the familial hypercholesterolemia sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-RT01 Date: _____

Rebuild the six-step sequence that runs from a variant in one of the three genes to xanthomas and premature atherosclerosis, one step per line and in order. Beside the step about receptor trips, write why a shortage of receptors raises LDL production as well as lowering LDL clearance. Beside the step about the endoplasmic reticulum, write which transcription factor stays active and name the rate-limiting enzyme it keeps switching on. Then add a line under the sequence naming the five classes of LDL receptor variant by the step in the receptor's life that fails, and mark which class is the most common.

Rebuild from memory: Pathogenesis of familial hypercholesterolemia

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C02-BD1

Sarcomere genes to the hypertrophic cardiomyopathy phenotype

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C02 Activity: NPLEX-P1-CARD-14-01-C02-BD1 Date: _____

Write the definition of hypertrophic cardiomyopathy, its approximate prevalence in adults, and its inheritance pattern. Name the two genes that account for most genotype-positive cases, write the protein each one encodes, and write which one acts through a dominant negative mechanism and which through haploinsufficiency, with the type of variant behind each. List the less common sarcomere genes from the notes. Then write the share of patients in whom testing finds a variant, and one sentence on what a negative test does and does not mean. Next, rebuild in six steps how a faulty sarcomere protein leads to a stiff, nondilated, arrhythmia-prone ventricle, and beside the contractility step write the two changes at the level of the myosin head and the thin filament that make the heart hypercontractile and wasteful of ATP. Finish with the morphology: wall thickness that is diagnostic against the normal value, the pattern of hypertrophy and cavity size, the hallmark microscopic finding, and the two other tissue changes that accompany it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C02-BD2

Two relatives, one variant, and a murmur that moves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C02 Activity: NPLEX-P1-CARD-14-01-C02-BD2 Date: _____

A woman with a pathogenic MYH7 variant has a septal wall thickness of 19 mm and exertional dyspnea. Her 16-year-old son carries the same variant and has a normal echocardiogram. Write the recurrence risk for each of her children, then explain in three or four sentences why the son can carry the variant and look normal, naming the two genetic terms involved and saying what makes penetrance age dependent. Write what you would do with his evaluation over time and why, and write which gene is most likely to delay hypertrophy into middle age. Next, write why the mother's symptoms and her fourth heart sound follow from the way the ventricle fills, and why atrial fibrillation would make her much worse. Then take her systolic murmur: write what happens to it on standing and during the strain of a Valsalva maneuver, what happens on squatting, leg raising, and handgrip, and give the mechanism in terms of ventricular volume, the septum, and the anterior mitral leaflet. Write how the murmur of aortic stenosis responds to the same maneuvers. Finish by writing why she can have angina with normal coronary arteries, and by listing the conditions from the notes that can imitate this picture with a line each on what separates them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-14-01-C02-CA01

Collapse during a basketball game in a 17-year-old

A 17-year-old boy collapses during a basketball game and recovers consciousness within a minute. He has had exertional breathlessness for two years that he attributed to being out of condition. A paternal uncle died suddenly at age 29. Temperature is 37.1 C (98.8 F), pulse is 76/min and regular, respirations are 16/min, and blood pressure is 118/72 mm Hg. A fourth heart sound is present. A systolic murmur is heard at the left sternal border; it becomes louder when he stands up from a squat and softer when he squats again or grips a handgrip dynamometer. The electrocardiogram shows high voltage with deep narrow Q waves in the lateral leads. Echocardiography shows asymmetric septal thickening of 20 mm, a small left ventricular cavity, systolic anterior motion of the mitral valve with mitral regurgitation, and normal left ventricular emptying. He trains year round.

1. Recognize. Name the condition, its inheritance pattern, and the two genes that most often cause it, writing the protein each one encodes.

2. Explain. Explain how a variant in one of those proteins produces a thick, nondilated ventricle when loading conditions are normal, taking the mechanism from the protein through contractility and energy use to hypertrophy and fibrosis.

3. Differentiate. Write how this picture differs from athlete's heart, from hypertensive hypertrophy, and from the other phenocopies in the notes, naming the finding that separates each one.

4. Investigate. Write which features of the murmur, the electrocardiogram, and the tissue you would examine to support this diagnosis, and what value or appearance each would have.

5. Decide. State which tissue changes in this ventricle create the substrate for the arrhythmia that caused his collapse, and state what determines the risk to his first-degree relatives.

6. Justify. Justify the behavior of his murmur from ventricular volume, the position of the septum, and the anterior mitral leaflet, and explain why his uncle's variant could be present in relatives who have a normal echocardiogram.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C03-BD1

Marfan syndrome from gene to dissection

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C03 Activity: NPLEX-P1-CARD-14-01-C03-BD1 Date: _____

Write what the aortic media is built from, naming the elastic lamellae, the cells between them, and the two matrix components that hold them. Then write the gene, chromosome, and protein in Marfan syndrome, its prevalence, the proportion of cases that are new variants, and the inheritance pattern. Write the two normal jobs of that protein, including what it anchors in the eye and the growth factor it holds in a latent form. Rebuild in six steps the sequence from the variant to aortic regurgitation and dissection, and beside the dilation step write Laplace's law in words and explain why dilation feeds further dilation. Name the lesion seen in the wall, list its three microscopic features, and write the older term for it and why that term is misleading. Finish by listing the features outside the vessels that come from pleiotropy, including the direction the lens is displaced, and write how homocystinuria differs on three points.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C03-BD2

Three families, three failing wall components

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C03 Activity: NPLEX-P1-CARD-14-01-C03-BD2 Date: _____

Three unrelated patients each have arterial disease. Patient 1 has a dilated aortic root at the sinuses of Valsalva, an upward displaced lens, and long fingers. Patient 2 at age 22 has aneurysms at several sites with tortuous arteries, widely spaced eyes, and a split uvula, and his aorta dissected at a diameter smaller than usually expected. Patient 3 has thin translucent skin, easy bruising, and a dissection of a medium-sized artery with no preceding aneurysm, and a sister died of sigmoid colon rupture. For each patient write the gene, the protein or receptor involved, and the inheritance pattern, then explain in two or three sentences how the failing wall component produces that particular arterial pattern. For patient 2, write the paradox between what the variant does to receptor function in cells and what happens to signaling in the aortic wall. For patient 3, write which amino acid in the Gly-X-Y repeat is usually replaced, why no other amino acid fits that position, and what happens to the triple helix. Then work the arithmetic: if mutant and normal chains are made in equal amounts and assemble at random, calculate the fraction of fully normal molecules for a three-chain collagen and for a four-subunit potassium channel, and compare each with the fraction left by a null allele. Use those numbers to explain why a missense variant is often more severe than a complete loss of one allele.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-14-01-C03-CP01

The three heritable disorders of the arterial wall

Fill in every cell of the table from memory, then open the notes and correct anything you missed.

Feature	Marfan syndrome	Loeys-Dietz syndrome	Vascular Ehlers-Danlos syndrome
Gene or genes and the protein affected			
Inheritance pattern			
Where in the arterial tree the disease appears, and at what age and diameter			
Whether an aneurysm precedes rupture or dissection			
What goes wrong in the wall at tissue level			
Findings in the eye			
Findings in the face, mouth, and skull			
Findings in skin, bowel, and uterus			
Skeletal and joint findings			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C04-BD1

Repolarizing currents and the long QT genotypes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-BD1 Date: _____

Draw the ventricular action potential with voltage on the vertical axis and time on the horizontal axis, and label phases 0 to 4. Write beside phase 3 the two currents that do most of the work there, giving the gene for each and the accessory subunit that partners one of them, and name the current that finishes repolarization. Explain in three or four sentences what repolarization reserve means and why a carrier can have a resting QT interval that is barely long, then list the four situations from the notes that can unmask the problem. Next, build the table of the three main long QT genotypes from memory with one row each, using the columns gene, direction of the functional change, the current affected, the typical trigger, and the appearance of the T wave. Mark which genotype is the most common. Finish by writing the one sentence that captures the pattern across the three genes, separating what happens to the potassium channel genes from what happens to the sodium channel gene.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C04-BD2

Why the trigger, the ear, and the fever all follow from the current

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-BD2 Date: _____

Three young people have events. One faints while swimming, one collapses when a fire alarm sounds, and one is found dead in bed. For each, write the likely genotype and explain from the behavior of the affected current why the event happened in that setting rather than another. Then write what deranges rhythm at the cellular level in all three, naming the afterdepolarization and the arrhythmia it produces. Next, a deaf child has a QT interval above 500 ms and two affected alleles of the same gene. Write the syndrome name and its inheritance pattern, name the common dominant form with normal hearing, and explain why the same channel that repolarizes the ventricle is also needed for hearing, naming the structure in the ear and the ion it secretes. Write why both parents hear normally and what their electrocardiograms may still show. Finish with one gene that produces two different diseases: write the gene, which functional change causes which disease, and what the term for this is.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C04-RT01

Rebuild the Brugada sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-RT01 Date: _____

Rebuild the six-step sequence that runs from a loss of sodium current to ventricular fibrillation in Brugada syndrome, one step per line and in order. Beside the first step name the gene, the inheritance pattern, and the approximate share of patients in whom the variant is found. Beside the notch step name the current that deepens it and write why it matters that this current is large in right ventricular outflow epicardium and smaller in endocardium. Under the sequence, draw two action potentials on one set of axes, one epicardial and one endocardial, and show the lost dome; then draw the surface tracing in V1 and V2 and label the shape of the ST segment, the minimum elevation that counts, and the direction of the T wave. Finish by listing the three things from the notes that can unmask the pattern, with one line on why warming does it, and write the two features of the patient population in which this condition is most common.

Rebuild from memory: Brugada syndrome from SCN5A loss to ventricular fibrillation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-14-01-C04-PR01

Predict the effect of a low plasma potassium in LQT2

A woman with a KCNH2 variant and a baseline QT interval at the upper limit has been vomiting for three days. Her plasma K+ is 2.7 mEq/L (reference range 3.5 to 5.0 mEq/L). Predict each item below, write the direction of the change, and give the current, channel, or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the rapid delayed rectifier current IKr carried by her remaining normal channels		
What happens to the margin left by repolarization reserve		
What happens to the duration of the ventricular action potential and to the QT interval		
What happens to L-type Ca2+ channels late in the prolonged plateau		
What kind of afterdepolarization appears, and at which phase of the action potential		
What the resulting rhythm is called and what it can degenerate into		
What happens to her risk if she is also given a drug that blocks IKr		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C05-BD1

Embryology to the characteristic heart defect

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C05 Activity: NPLEX-P1-CARD-14-01-C05-BD1 Date: _____

Write what the endocardial cushions do during development, naming the canal they divide, the opening they close, and the valves they help build. Then write what the neural crest cells contribute and which vessel their septum divides. Next, take each of these in turn: Down syndrome, Turner syndrome, the 22q11.2 deletion, Williams syndrome, and Noonan syndrome. For each one write the genetic basis, the characteristic cardiovascular lesion, and the embryonic process or protein that links the two, then add the findings outside the heart from the notes. Include for Down syndrome the three mechanisms by which the trisomy arises with their approximate shares and which parental meiotic division is usually involved, and the proportion of patients who have a heart defect. Include for Turner syndrome the karyotypes, the frequency of each left-sided lesion, and the hypothesis about fetal lymphedema. Include for the 22q11.2 deletion the size of the deletion, the gene whose single copy matters, and the two pouch derivatives that fail with the consequence of each. Include for Williams syndrome the gene and what elastin tells smooth muscle to do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C05-BD2

Read the findings back to the mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C05 Activity: NPLEX-P1-CARD-14-01-C05-BD2 Date: _____

Five infants and children present. Child 1 has hypotonia, a flat face, duodenal atresia, and a single murmur with a common atrioventricular valve on echocardiography. Child 2 has hypocalcemic seizures, recurrent infections, a cleft palate, and an interrupted aortic arch. Child 3 is a girl with short stature, a webbed neck, blood pressure higher in the arms than the legs, and weak delayed femoral pulses. Child 4 has elfin facies, a high calcium level, and a systolic murmur with narrowing just above the aortic valve. Child 5 has a normal karyotype, features that resemble Turner syndrome, a dysplastic stenotic pulmonary valve, and a thick ventricle. For each child name the condition, the genetic basis, and the embryonic or cellular mechanism that produced the cardiac lesion. For child 1, write why a large shunt in this syndrome reaches pulmonary hypertension sooner than the same shunt in another child. For child 2, write why the immune and parathyroid findings travel with the cardiac findings. For child 3, write the three physical and radiographic consequences of her aortic lesion and what her aorta is at risk of later. For child 4, write why a variant in that one gene alone gives a narrower picture than the deletion does. Then write the difference between a syndrome, an association, and a sequence, give the example of each from the notes, state how many features are needed for the association and what its recurrence risk is, and name the two single-gene conditions that imitate it with the feature that gives each one away.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-14-01-C05-CP01

Genetic syndromes and their cardiovascular lesions

Fill in every cell of the table from memory, then open the notes and correct anything you missed.

Syndrome	Genetic basis	Characteristic cardiovascular lesion	Developmental process disturbed	Other findings that point to it
Down syndrome				
Turner syndrome				
22q11.2 deletion syndrome				
Williams syndrome				
VACTERL				
Noonan syndrome				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-14-01-C05-Q01

A newborn with hypocalcemia and an arch anomaly

A newborn boy has jitteriness and a seizure on the second day of life. Serum calcium is low. Examination shows a cleft palate and absent thymic tissue on imaging. Echocardiography shows interruption of the aortic arch distal to the left subclavian artery. Which of the following best explains the combination of findings in this infant?

A. Trisomy of chromosome 21 causing failure of endocardial cushion fusion

B. Haploinsufficiency of elastin causing smooth muscle proliferation in the arterial media

C. Loss of one copy of TBX1 disturbing the pharyngeal arches and the neural crest

D. An expansion of a trinucleotide repeat silencing a mitochondrial iron handling protein

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C06-BD1

Dystrophin, frataxin, and mitochondrial DNA

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C06 Activity: NPLEX-P1-CARD-14-01-C06-BD1 Date: _____

Draw the link from actin inside the myocyte to laminin in the basement membrane, labelling dystrophin and the two dystroglycans, and write what that link protects during contraction. Write the gene, its chromosome, why so many cases are new variants, and the inheritance pattern, then give the prevalence of Duchenne dystrophy in males and the proportion of variants that are deletions. State the reading frame rule in full: what an in-frame deletion produces and which disease it gives, what a frameshift produces and which disease it gives, and how often the rule holds. Then write the pathogenesis from a torn membrane to myocyte necrosis, the laboratory finding, the two physical signs from the notes, and the cardiac picture including where fibrosis starts, what cardiomyopathy results and by what age, and the three electrocardiographic findings. Next, write the gene, the repeat, its position in the gene, the normal and disease repeat ranges, and the inheritance pattern for Friedreich ataxia, then write what the missing protein is needed for and which complexes and enzyme depend on it, which tissues are hit and why, and what the cardiomyopathy looks like. Finish with mitochondrial DNA: its approximate size, what it encodes, how it is inherited, and the three mechanisms that make the mutant fraction differ between tissues and siblings, then write the syndrome from the notes with its genetic lesion, its age of onset, its three defining features, and its conduction consequence with the muscle biopsy finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C06-BD2

Work the frame, the carrier, and the threshold

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C06 Activity: NPLEX-P1-CARD-14-01-C06-BD2 Date: _____

Two boys have DMD deletions. Boy A has lost exons totaling 522 coding nucleotides and boy B has lost 197. For each boy, divide and show your working, state whether the downstream reading frame survives, state what protein is made, and name the disease, then write which boy has the larger deletion and why that does not predict the milder disease. Write what happens to the transcript when a premature stop codon appears. Next, a 32-year-old woman whose son has Duchenne dystrophy has no limb weakness but has a dilated, poorly contracting left ventricle. Explain in three or four sentences how a carrier of an X-linked recessive disease develops cardiomyopathy, naming the process in her myocytes and what happens to the cells that express only the mutant allele. Write why her heart cannot replace them. Then two sisters carry the same mitochondrial DNA variant and only one has cardiac disease. Explain the result using heteroplasmy, random segregation, the oocyte bottleneck, and tissue threshold, and write which tissues have the lowest threshold and why. Finish by writing why membrane fragility tends to give a dilated ventricle, why a failure of energy supply tends to give a thick one, and why a large mitochondrial DNA deletion shows itself first in the conduction system.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-14-01-C06-Q02

Progressive block in a young adult

A 19-year-old woman has had drooping eyelids and limited eye movements since childhood. Funduscopy shows pigmentary changes in the retina. The electrocardiogram shows a widening QRS complex with progressive atrioventricular block, and muscle biopsy shows ragged red fibers on modified Gomori trichrome staining. Her mother and two brothers are unaffected. Which of the following best accounts for her findings?

A. A large deletion of mitochondrial DNA removing respiratory chain subunits and transfer RNAs

B. A frameshift deletion in DMD on the X chromosome abolishing dystrophin

C. A GAA repeat expansion in the first intron of FXN silencing transcription

D. A missense variant in MYH7 acting as a dominant negative in the thick filament

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C07-BD1

Why the adult heart grows by hypertrophy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-BD1 Date: _____

Write what happens to cardiomyocyte division around birth and what many of the cells become instead, then list the four molecular brakes that hold the cells out of the cell cycle. Write the measured renewal rate in young adults, the method that produced that figure, and the term for a tissue that heals by scar, then contrast cardiac muscle with skeletal muscle and smooth muscle on their capacity to replace lost cells. Next write Laplace's law for ventricular wall stress as a relationship between pressure, radius, and wall thickness, then write for pressure overload and for volume overload the load that causes it, where the new sarcomeres are added, the shape of hypertrophy that results, and what each one does to wall stress. Give normal heart weight for women and for men. Then write how stretch is sensed, naming three sensors, and trace the pathologic signaling route from angiotensin II, endothelin-1, and norepinephrine through the receptor family, the enzyme, the second messengers, the phosphatase, and the transcription factors that reach the nucleus. Write the separate route used by exercise hypertrophy. Finish with the fetal gene program: name the genes switched on and the two switched down, and write the functional consequence of each of those changes, including what the natriuretic peptides do and the value of BNP below which heart failure is an unlikely cause of acute breathlessness.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C07-BD2

Two thick hearts and one scar

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-BD2 Date: _____

A 28-year-old rower and a 58-year-old man with untreated hypertension both have increased left ventricular wall thickness. For each, write the signaling route that produced it, whether the fetal gene program is switched on, what the capillary supply and the interstitium look like, and what happens to the hypertrophy if the stimulus is removed. Then explain in three or four sentences why one of them is at risk of progressing to heart failure and the other is not, and why the ventricle that keeps losing myocytes eventually dilates even though it started by thickening. Next, take a man six weeks after a myocardial infarction. Write the sequence of repair from macrophage clearance through the growth factor released, the cell type that appears, the first collagen laid down, and the collagen that replaces it. Write the two things the finished scar cannot do and the two consequences of each for pump function and for rhythm. Write which forms of injured myocardium can recover and which cannot. Finish by writing why every disease in this chapter that kills myocytes ends in the same three findings, and name two diseases from earlier in the chapter where that repair follows myocyte loss.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C07-RT01

Rebuild the maladaptive hypertrophy cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-RT01 Date: _____

Draw the six-step cycle of pathologic hypertrophy as a closed loop, numbering each step and writing it out in words, starting from a sustained load that raises wall stress. Beside the step on gene expression, name the genes switched on. Beside the step on the interstitium and the calcium pump, write what each change does to relaxation and to oxygen supply. Beside the step on myocyte death, write why the survivors have more work to do. Beside the closing step, write Laplace's law in words and explain why dilation sends the loop round again. Underneath the loop, write the one sentence about cardiomyocyte renewal that explains why the loop cannot be broken by replacing the lost cells.

Rebuild from memory: Maladaptive cycle of pathologic hypertrophy

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-15

CV-15 Heart and Lung Interactions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-15-01-C01				
NPLEX-P1-CARD-15-01-C02				
NPLEX-P1-CARD-15-01-C03				
NPLEX-P1-CARD-15-01-C04				
NPLEX-P1-CARD-15-01-C05				
NPLEX-P1-CARD-15-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C01-BD1

Build the two circuits side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C01 Activity: NPLEX-P1-CARD-15-01-C01-BD1 Date: _____

Without looking at your notes, draw the pulmonary and systemic circuits in series and write on the drawing the resting mean arterial pressure, the receiving atrial pressure, the capillary hydrostatic pressure, and the resistance in Wood units for each circuit. Beside the drawing write the flow each ventricle carries at rest, and show from pressure and flow how the resistance figures follow. Name the main site of resistance in each circuit and write two sentences on the wall thickness and smooth muscle differences that explain it. Then write the volume of blood the pulmonary circuit holds, what supplies the airways themselves and at what pressure and what share of cardiac output, and the two consequences of running the lung at low pressure. Finish with the mean pulmonary artery pressure that defines pulmonary hypertension.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C01-BD2

Work the pulmonary resistance numbers through exercise and disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C01 Activity: NPLEX-P1-CARD-15-01-C01-BD2 Date: _____

A person goes from rest, with a cardiac output of 5 L/min, a mean pulmonary artery pressure of 15 mmHg, and a left atrial pressure of 5 mmHg, to heavy exercise, with a cardiac output of 15 L/min, a mean pulmonary artery pressure of 22 mmHg, and a left atrial pressure of 10 mmHg. Calculate pulmonary vascular resistance at rest and in exercise, show your working, and write what the mean pulmonary artery pressure would have been in exercise had resistance not changed. Name the two passive mechanisms that lowered it, say which one dominates at low pressure and which at higher pressure, and explain why a small change in vessel radius matters so much. Write what happens to capillary transit time and why gas exchange still completes. Then write what pulmonary vascular resistance does at high lung volume, at low lung volume, and near functional residual capacity, naming the class of vessel responsible in each case. Finish with two patients, one after removal of a healthy lung and one with half the vascular bed lost to emphysema and old emboli: write whose resting pulmonary artery pressure rises, whose rises steeply on exercise, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C02-BD1

Draw hypoxic pulmonary vasoconstriction from sensor to constriction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-BD1 Date: _____

Without your notes, draw a small pulmonary artery supplying a poorly ventilated alveolus and write out the chain of events from low alveolar oxygen to contraction of that vessel, one step per line. Name the channel family that is inhibited first, what that does to the membrane potential, which calcium channel opens next, and the two routes by which calcium reaches the contractile proteins. Add the enzyme that raises calcium sensitivity and name the two endothelial mediators that shift in this setting and in which direction each moves. Write which gas tension is the stimulus, the value below which the response becomes strong, and the evidence from an isolated denervated lung that the response is local. Then write the purpose this response serves and what the leading proposal is for the upstream oxygen sensor.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C02-BD2

Regional against global hypoxia, and what chronic hypoxia leaves behind

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-BD2 Date: _____

Three people are hypoxic: one with right lower lobe pneumonia, one who has climbed rapidly to 4,000 m, and one with twenty years of COPD and chronic hypoventilation. For each, write whether the hypoxia is regional or global, what hypoxic pulmonary vasoconstriction achieves, what happens to mean pulmonary artery pressure, and what happens to arterial oxygen. Explain why the same response helps in one case and harms in another. Write what hypercapnia and acidosis do to the constriction. For the climber, explain the accepted mechanism of high-altitude pulmonary edema from uneven constriction. For the patient with COPD, write the transcription factors stabilised by sustained hypoxia, the three structural changes they drive in the vessel wall, the hormone they raise and what the resulting blood change does to resistance, and which part of the pulmonary hypertension no longer reverses with oxygen. Finish with the fetus: write what keeps pulmonary vascular resistance high before birth, where right ventricular output goes instead, the three changes at birth that drop resistance, and the name for failure of that drop.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-15-01-C02-RT01

Rebuild the hypoxic pulmonary vasoconstriction sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-RT01 Date: _____

Rebuild the six-step sequence that runs from a fall in regional alveolar oxygen to blood being diverted toward ventilated lung, one step per line and in order. Beside the first step write the alveolar oxygen tension at which the response becomes strong. Beside the channel step write which channels are involved and what holds the resting potential negative when they are open. Beside the calcium step write the two sources of calcium and the enzyme that raises calcium sensitivity. Underneath, draw the same vessel in a systemic tissue bed made hypoxic and write beside it the channel that opens there, the direction the membrane potential moves, and the resulting change in vessel diameter.

Rebuild from memory: Hypoxic pulmonary vasoconstriction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-15-01-C02-CP01

Pulmonary against systemic arterioles

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Pulmonary arterioles	Systemic arterioles
Response to local hypoxia		
Key ion channel event and the direction the membrane potential moves		
Other mediators that contribute to the response		
What the response accomplishes for the body		
Response to acidosis and a high carbon dioxide tension		
Result when hypoxia involves the whole bed at once		
Amount of smooth muscle in the wall and how much of the circuit's resistance sits there		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C03-BD1

One breath, four chambers, and the second heart sound

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-BD1 Date: _____

Write the definition of transmural pressure and why a change in pleural pressure changes the filling of every cardiac chamber at once. Give pleural pressure at functional residual capacity and at the end of a quiet inspiration. Write the two pressures whose difference drives venous return and the normal value of the upstream one. Then take one spontaneous inspiration and write in order what happens to right atrial pressure, abdominal pressure, venous return, and right ventricular stroke volume. List the three reasons left ventricular stroke volume falls slightly in the same breath, and write when the extra right ventricular output reaches the left heart. Draw one normal second heart sound in inspiration and one in expiration, label A2 and P2, and write beside the inspiratory pair the two reasons P2 is delayed and the reason A2 is slightly early. Give the normal limit for the inspiratory fall in systolic pressure and where on the chest the split is best heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C03-BD2

Exaggerated and reversed breathing effects

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-BD2 Date: _____

Four patients are examined. The first has cardiac tamponade, the second a severe asthma attack, the third constrictive pericarditis, and the fourth is on a ventilator with a positive end-expiratory pressure of 15 cmH2O after major blood loss. For the first two, write the mechanism by which the inspiratory fall in systolic pressure exceeds 10 mmHg, naming for each whether the pericardium or the pleural pressure swing is doing the work. For the third, write why this finding is less common and what happens to the jugular venous pressure on inspiration instead. For the fourth, write what happens to right atrial pressure, to the gradient for venous return, to cardiac output, and to left ventricular afterload, and explain why the fall in output is larger because of the blood loss. Then write what happens in the same ventilated patient if the left ventricle is instead failing with high filling pressures, and why stroke volume may hold up. Finish by writing which phase of a Valsalva manoeuvre reproduces this pattern and what the heart rate does and through which reflex. Give the four S2 splitting patterns with the timing fault behind each and one example condition for each, including why the split in an atrial septal defect does not move with breathing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-15-01-C03-RT01

Rebuild the inspiration sequence through to S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-RT01 Date: _____

Rebuild the six-step sequence that runs from a fall in pleural pressure to a split second heart sound and a small dip in arterial pressure, one step per line and in order. Beside the venous return step write the two forces that push blood toward the thorax. Beside the step where left stroke volume falls, write the three reasons it falls. Beside the final step write the normal limit for the fall in systolic pressure. Then write the same six steps as they run during expiration, showing that each one reverses, and mark on which phase of the breath the extra right ventricular output finally reaches the aorta.

Rebuild from memory: Inspiration, ventricular output, and S2

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C04-BD1

Starling forces at the pulmonary capillary and the safety margin

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C04 Activity: NPLEX-P1-CARD-15-01-C04-BD1 Date: _____

Draw a pulmonary capillary and its alveolus and mark every Starling force with its value and its direction: capillary hydrostatic pressure, interstitial hydrostatic pressure, interstitial oncotic pressure, and plasma oncotic pressure. Total the outward forces and the inward force, write the net filtration pressure, and write what carries that filtrate away. Then write the three things that hold off edema as capillary pressure climbs, the capillary pressure at which alveolar flooding begins, and the size of the safety margin in mmHg. Write in order what fills first and what floods second. Finish with three modifiers: what a low plasma albumin does to the margin, what years of raised pressure do to it and by how much the lymphatics can enlarge, and how permeability injury produces edema at a normal capillary pressure and how the fluid itself differs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C04-BD2

Which failing chamber, and which side of the circuit caused it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C04 Activity: NPLEX-P1-CARD-15-01-C04-BD2 Date: _____

Two patients have raised pulmonary artery pressure and a failing right ventricle. One has twenty years of COPD and the other has long-standing mitral stenosis. For each, write the pulmonary hypertension group, the mechanism raising pulmonary artery pressure, whether the term cor pulmonale applies, and your reason. Write the five groups with their causes and say which ones load the right ventricle from the lung side. Then for the patient with COPD write how the right ventricle first adapts to slowly rising afterload and why a thicker wall helps, what happens to the tricuspid valve as it later dilates, the three systemic signs of the raised venous pressure, the two findings on examination that point to pulmonary hypertension, and the three ECG changes. For the patient with mitral stenosis write which measured pressure rises and its normal range, what shortens diastole and makes her worse, why the measured wedge pressure sits above left ventricular diastolic pressure here, the receptor whose stimulation causes her dyspnoea, why lying flat is worse, and the two chest film findings. Finish with a third patient who has a massive pulmonary embolus: write the two mechanisms obstructing the bed, the highest mean pressure a previously normal right ventricle can generate, and the share of the circulation whose obstruction brings acute failure or sudden death.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-15-01-C04-PR01

Predict the course of a massive pulmonary embolus

A large clot lodges in both main pulmonary arteries of a patient whose right ventricle was previously normal. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to right ventricular afterload, and the two things that raise it besides the clot itself		
What happens to right ventricular size and to right ventricular output		
What happens to the position of the interventricular septum		
What happens to left ventricular filling, left ventricular output, and systemic arterial pressure		
What happens to right coronary perfusion while right ventricular wall stress stays high		
What happens to right ventricular contraction once that perfusion falls, and what this means for a patient who is already hypotensive		
What happens to the arterial oxygen tension and to the arterial carbon dioxide tension, and which regions of lung explain each		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C05-BD1

Draw the upright lung and its three zones

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C05 Activity: NPLEX-P1-CARD-15-01-C05-BD1 Date: _____

Draw an upright lung with the pulmonic valve marked, and write the vertical distance from the valve to the apex and to the base. Write how much pressure a centimetre of blood column adds and how many centimetres of water make 1 mmHg. With a pulmonary artery pressure of 25/8 mmHg, calculate and write the systolic and diastolic pressure at the apex and at the base, and write what this does to flow per unit volume from apex to base. Then write the third pressure that acts on the alveolar wall capillaries and what it stays near in quiet breathing. Build a table of the three zones with one row each and the columns pressure order, blood flow, driving pressure, and usual location, and fill every cell. Beside the zone 2 row write why venous pressure does not affect flow there. Beside the zone 1 row write why it is normally absent.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C05-BD2

Make zone 1 appear, then work the ventilation and perfusion ratios

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C05 Activity: NPLEX-P1-CARD-15-01-C05-BD2 Date: _____

Write the two ways zone 1 can appear and give one clinical cause of each, then show with numbers why a positive end-expiratory pressure of 15 cmH2O creates zone 1 at the apex. Write what zone 1 alveoli do to carbon dioxide elimination and what has to happen to prevent the arterial carbon dioxide tension from rising, and write what exercise does to the zones instead. Next, write the apical and basal pleural pressures at functional residual capacity, explain from the pressure-volume curve why apical alveoli are more expanded but ventilate less per breath, and write whether ventilation or perfusion rises more steeply toward the base. Give the whole lung average ventilation-perfusion ratio with the two numbers it comes from, then give the ratio, the alveolar oxygen tension, and the alveolar carbon dioxide tension at the apex and at the base. Use the apical value to explain where reactivation tuberculosis sits and why. Then explain, using the shape of the oxyhaemoglobin dissociation curve, why high ratio regions cannot compensate for low ratio regions in oxygen but can for carbon dioxide, and write the blood gas pattern that follows, the normal alveolar to arterial oxygen difference in a young adult, and the two things that produce it. Finish by writing what happens to the apex to base differences when the patient lies down and which regions then receive most of the flow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C06-BD1

Shunt, and the work the pulmonary endothelium does

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C06 Activity: NPLEX-P1-CARD-15-01-C06-BD1 Date: _____

Write the definition of a shunt and the ventilation-perfusion ratio that goes with it. List the anatomical shunts, including the normal one and the share of cardiac output it carries, then list the intrapulmonary shunts with the conditions that cause them. Separately, write what the pulmonary endothelium does to angiotensin I, name the enzyme, the metal it needs, and the second name it carries, and write which peptide it inactivates and what accumulates when the enzyme is blocked and which two drug effects that explains. Write which mediators the lung clears in one passage and roughly how much norepinephrine it removes, and which three hormones pass through largely unchanged. Finish with the filter function: what the capillaries trap, the two reasons small emboli rarely infarct the lung, the condition that makes infarction more likely and why, and the name for a clot that crosses a patent foramen ovale along with how often that opening is present.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C06-BD2

Separate the causes of hypoxaemia with oxygen and arithmetic

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C06 Activity: NPLEX-P1-CARD-15-01-C06-BD2 Date: _____

Five hypoxaemic patients are breathing room air: one at 4,000 m, one after an opioid overdose, one with interstitial fibrosis, one with a COPD exacerbation, and one with dense right lower lobe pneumonia. For each write whether the alveolar to arterial oxygen difference is normal or raised, what the arterial carbon dioxide tension is doing, and whether 100% oxygen corrects the hypoxaemia. Then explain in full why oxygen barely helps the last patient: write what shunted blood never meets, the saturation the blood leaving ventilated alveoli already carries, how much oxygen dissolves per mmHg, and why mixing has to be done on content rather than on partial pressure before the dissociation curve turns that content into a low oxygen tension. Work the numbers for a 30% shunt with a haemoglobin of 15 g/dL, a saturated content of 20.1 mL/dL, and an arteriovenous difference of 5 mL/dL: give the arterial oxygen content, saturation, and oxygen tension on room air, then on 100% oxygen with an alveolar oxygen tension of 660 mmHg, and compare the final oxygen tension with a normal lung on 100% oxygen. Write why the arterial carbon dioxide tension stays normal or low. Finish with a child who has a ventricular septal defect and no cyanosis: write why there is no cyanosis at first, what the extra flow and pressure do to the pulmonary vessels, the six steps that end in reversal of the shunt, the name of that end state, the three clinical findings it brings, and why the vessels no longer relax by then.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-15-01-C06-CP02

Mechanisms of arterial hypoxaemia

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Mechanism	Alveolar to arterial oxygen difference	Arterial carbon dioxide tension	Response to 100% oxygen	Example condition
Low inspired oxygen tension				
Hypoventilation				
Diffusion limitation				
Low ventilation-perfusion regions				
Shunt with a ventilation-perfusion ratio of zero				
Alveolar dead space with a very high ventilation-perfusion ratio				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-15-01-C06-CA01

Hypoxaemia that does not lift on oxygen

A 64-year-old man is admitted with 3 days of fever, a productive cough, and worsening breathlessness. Temperature is 39.1 °C (102.4 °F), pulse is 112/min, respirations are 30/min, and blood pressure is 118/72 mm Hg. Breath sounds are reduced over the right lower chest with bronchial breathing and crackles, and the chest film shows dense consolidation of the right lower lobe with a clear left lung and a normal heart size. On room air, arterial blood gas analysis shows a PaO2 of 54 mm Hg, a PaCO2 of 31 mm Hg, and a pH of 7.47. He is placed on a non-rebreather mask delivering close to 100% oxygen, and a repeat sample 20 minutes later shows a PaO2 of 71 mm Hg. Mean pulmonary artery pressure measured during the admission is 18 mm Hg and the pulmonary capillary wedge pressure is 9 mm Hg.

1. Recognize. Name the mechanism of this man's hypoxaemia, give the ventilation-perfusion ratio of the affected lung units, and name the two features of his course that point to it.

2. Explain. Explain why oxygen content rather than oxygen tension has to be used when the two streams of blood mix, and why his arterial oxygen tension stays low once they have mixed.

3. Differentiate. Write how his blood gases and his response to oxygen differ from those of a patient with an opioid overdose and from those of a patient with a COPD exacerbation, naming the measurement that separates each one.

4. Investigate. Write which measurements you would use to show that the consolidated lobe rather than the left heart is the source, and the value or direction you would expect for each.

5. Decide. State what the local pulmonary arterioles are doing to the blood flow of the consolidated lobe and what his arterial oxygen tension would be without that response.

6. Justify. Justify that answer from the stimulus for the response, the channel events behind it, and the reason his mean pulmonary artery pressure stays near normal even though a whole lobe is constricted.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-15-01-C01-Q01

Why pulmonary pressure barely rises in exercise

A healthy 26-year-old woman is studied during heavy exercise. Her cardiac output rises from 5 L/min to 15 L/min, while mean pulmonary artery pressure rises only from 15 mm Hg to 22 mm Hg and left atrial pressure rises from 5 mm Hg to 10 mm Hg. Which of the following best explains the small rise in pulmonary artery pressure?

A. Sympathetic withdrawal relaxing the smooth muscle of the pulmonary arterioles

B. Diversion of part of the cardiac output into the bronchial circulation

C. Opening of previously unperfused capillaries together with widening of those already open

D. A rise in alveolar pressure that compresses the capillaries of the alveolar walls

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C04-Q02

Alveolar flooding at a normal wedge pressure

A 58-year-old man develops severe breathlessness 2 days after an episode of gram-negative sepsis. Respirations are 34/min and oxygen saturation is 84% on high-flow oxygen. The chest film shows diffuse bilateral alveolar opacities with a normal heart size. Pulmonary capillary wedge pressure is 8 mm Hg, and fluid suctioned from the airway has a protein concentration close to that of plasma. Which of the following best explains the pulmonary edema in this patient?

A. Injury to the alveolar capillary barrier that allows protein to cross at a normal capillary pressure

B. Backward transmission of a raised left atrial pressure through the pulmonary veins

C. A fall in plasma oncotic pressure that narrows the usual safety margin

D. Obstruction of pulmonary lymphatic drainage by the septic process

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C05-Q03

Dead space created at the apex

A 45-year-old woman is receiving mechanical ventilation after major blood loss. Positive end-expiratory pressure is increased to 15 cmH2O, which is about 11 mm Hg. Her pulmonary artery pressure is 18/6 mm Hg, and the apex of her lung sits about 20 cm above the pulmonic valve. Minute ventilation is held constant, and her arterial PCO2 rises over the next hour. Which of the following best explains the rise in arterial PCO2?

A. Hypoxic constriction of the arterioles supplying the apical alveoli

B. Loss of radial traction on the extra-alveolar vessels of the apex

C. Collapse of apical alveoli with continued perfusion of their capillaries

D. Alveolar pressure at the apex now exceeding arterial pressure, so those ventilated alveoli carry no blood flow

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C03-Q04

A large inspiratory fall in systolic pressure

A 51-year-old woman with metastatic breast cancer comes to the emergency department because of 2 days of breathlessness. Pulse is 118/min, blood pressure is 96/70 mm Hg, and the jugular veins are distended. During quiet breathing the systolic pressure falls by 22 mm Hg with each inspiration. Heart sounds are distant, the lungs are clear, and echocardiography shows a large circumferential pericardial effusion with equal diastolic pressures in the two ventricles. Which of the following best explains the inspiratory fall in systolic pressure?

A. Large swings in pleural pressure from obstructed expiratory airflow raise left ventricular afterload

B. Pericardial fluid fixes the combined ventricular volume, so inspiratory right ventricular filling displaces the septum into the left ventricle

C. Inspiratory constriction of the pulmonary arterioles reduces the volume returning to the left atrium

D. A rigid pericardium shields the heart from the fall in pleural pressure

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-16

CV-16 Immune Responses in Cardiovascular Disease

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-16-01-C01				
NPLEX-P1-CARD-16-01-C02				
NPLEX-P1-CARD-16-01-C03				
NPLEX-P1-CARD-16-01-C04				
NPLEX-P1-CARD-16-01-C05				
NPLEX-P1-CARD-16-01-C06				
NPLEX-P1-CARD-16-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C01-BD1

Rebuild atheroma formation as an inflammatory response

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-BD1 Date: _____

Without looking at your notes, draw the sequence that turns a healthy artery into a fibrofatty plaque, from endothelial dysfunction through to a cap over a necrotic core. Beside each step write the molecules doing the work: the transcription factor the injured endothelium activates, the two adhesion molecules and the chemokine it induces, the cytokine that turns monocytes into macrophages, and the growth factors that build the cap. Write which white cells arrive first and why the adhesion molecule involved selects them. Then write what happens to LDL once it is retained in the intima, which receptors carry it into the macrophage, why those receptors do not switch off as cell cholesterol rises, which enzyme stores the cholesterol, and which transporters move it back out. Finish by naming the earliest visible lesion and the age group it already appears in, and list the sites in the arterial tree where plaques form first with the flow reason.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C01-BD2

Why the smaller plaque is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-BD2 Date: _____

Two coronary lesions sit in the same artery. Lesion A narrows the lumen by about 40%, has a large lipid core, a thin cap, and dense macrophages at its edges. Lesion B narrows it by about 85% under a thick collagenous cap with many smooth muscle cells and few inflammatory cells. For each lesion write what the patient is likely to experience and which one is more likely to cause an infarction, then explain the difference from the enzymes and cytokines acting on cap collagen. Name the enzymes that degrade the collagen, the cytokine that stops smooth muscle cells replacing it, and what happens to those smooth muscle cells. Write where on the plaque rupture usually begins and why, what is exposed when it tears, and how quickly thrombus follows. Then write the other surface event that can cause coronary thrombosis without a tear in a cap. Finish by explaining why a patient with rheumatoid arthritis or lupus develops coronary disease faster, naming the cytokines that connect the two, and state what a high sensitivity CRP of 4 mg/L tells you about this patient and the cytokine route that produced that number.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-16-01-C01-RT01

Rebuild the atheroma sequence and the inflammasome branch

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-RT01 Date: _____

Rebuild the eight-step sequence of atheroma formation from memory, one step per line and in order, from the first insult to the formation of thrombus. Beside the step where monocytes enter, write the two molecules that made them stick and the one that drew them in. Beside the foam cell step, write the receptor family involved and why uptake has no brake. Then, underneath the sequence, draw a second short branch coming off the foam cell step: start with oxidized LDL on Toll-like receptors, add what the macrophage makes ready, then what cholesterol crystals do inside the cell, which complex assembles, which enzyme it activates, which cytokine is released in its active form, which second cytokine follows, and what the liver then produces. Write the three risk bands for that liver protein with their values.

Rebuild from memory: Atheroma formation as a chronic inflammatory response

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C02-BD1

Rheumatic fever from the throat to the valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C02 Activity: NPLEX-P1-CARD-16-01-C02-BD1 Date: _____

Write out acute rheumatic fever from memory as a sequence of six steps, beginning with the infection and ending with the chronic valve lesion. Name the organism, the site of infection that leads to this disease, the usual age range, and the length of the gap between infection and carditis. Name the two bacterial antigens involved and, for each, the host protein it resembles or binds, then write in two sentences why antibodies and T cells raised against the bacterium attack the heart. Write which hypersensitivity type the disease is assigned to and which cell type actually drives the long-term valve damage. Then describe the diagnostic microscopic lesion: name it, name the altered macrophage inside it, and write which layers of the heart wall contain it and what that means for the extent of the carditis. Finish by writing what the acute valve lesions look like, their size, where on the leaflet they sit, which valve is affected most, and what repeated attacks turn the mitral valve into.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C02-BD2

Work a suspected case from timing and serology

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C02 Activity: NPLEX-P1-CARD-16-01-C02-BD2 Date: _____

An 11-year-old girl has fever of 38.7 degrees C (101.7 degrees F), migratory pain and swelling of both knees and one elbow, a new murmur, an ESR of 72 mm/h, and a PR interval of 0.24 s. Her throat culture is negative. Using the Jones criteria, write which of her findings are major and which are minor, state whether the criteria are met for an initial attack and show the counting, and write which of her findings cannot be counted here and why. Write how you would prove the recent streptococcal infection now that the culture is negative, naming both antibody tests, when each rises and peaks, which one stays up longer, and why a rising value is better evidence than one measurement. Then write why her joints will recover completely while her valve may not. Finally, a different child presents months later with involuntary writhing movements and no fever, and another child develops skin sores rather than a sore throat and later has dark urine and red cell casts. For each child write which disease fits, the immune mechanism behind it, and why the site of the original infection decides the answer.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C03-BD1

Organize the vasculitides by immune effector

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C03 Activity: NPLEX-P1-CARD-16-01-C03-BD1 Date: _____

On blank paper sort the vasculitides from this chapter into three groups by the immune effector that injures the vessel: immune complex disease, ANCA disease, and granulomatous T-cell disease. List the diseases under each heading. For the immune complex group write how the complexes form, where they deposit and why those sites, which complement pathway they use, which complement fragment brings in neutrophils, what the biopsy shows on immunofluorescence, what the dying neutrophils leave behind and what that appearance is called, and what the lesion looks like in the skin. For the ANCA group write the two target enzymes with their staining patterns, why one of them stains the nucleus, what primes the neutrophil, the two ways the antibody engages the same cell, what the neutrophil then releases, and why the biopsy shows almost no deposits. For the granulomatous group write which cells start the reaction, which cytokine recruits the macrophages, which layer of the artery they gather in, what happens to the internal elastic lamina and the intima, and which cytokine drives the very high ESR.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C03-BD2

Name the vasculitis from complement and serology

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C03 Activity: NPLEX-P1-CARD-16-01-C03-BD2 Date: _____

Four patients have palpable purpura. Patient 1 is 52, has a C4 that is very low with a near normal C3, rheumatoid factor, and hepatitis C antibody. Patient 2 is 61, has normal complement, hemoptysis, red cell casts, and MPO-ANCA. Patient 3 is 8 years old, had a cold two weeks ago, has abdominal pain, arthralgia, normal complement, and mesangial IgA. Patient 4 is 47, has hypertension, abdominal pain after eating, a foot drop with a wrist drop on the other side, hepatitis B surface antigen, and negative ANCA with clear lungs. For each patient name the disease, write the immune mechanism, and write the one laboratory or biopsy finding that settles it. Then explain in one or two lines each why C4 is very low in patient 1, why complement is normal in patient 3 even though complexes are deposited, why patient 2 has no deposits on immunofluorescence, and why patient 4 has renal infarcts and renin-dependent hypertension but no glomerulonephritis. Finish with two more patients: a 74-year-old with a temporal headache, pain on chewing, sudden vision loss in one eye, and an ESR of 105 mm/h, and a 29-year-old woman with a weak left arm pulse and a 30 mm Hg difference in arm blood pressures. Write what each has, the shared mechanism, the vessels involved in each, and which artery occlusion causes the vision loss.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-16-01-C03-CP01

Immune complex, ANCA, and granulomatous vasculitis side by side

Fill in every cell of the table from memory, then check it against the notes and correct what you missed.

Feature	Immune complex vasculitis	ANCA vasculitis	Granulomatous large vessel arteritis
Hypersensitivity type and immune effector cell or molecule			
What immunofluorescence of the biopsy shows			
What serum complement does and why			
Serum marker or antibody to order			
Vessel sizes and tissues usually involved			
Example diseases from this chapter			
Typical skin, kidney, and lung findings			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C04-BD1

Three immune routes into the heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-BD1 Date: _____

Write the three conditions in this section as three short blocks on one page: delayed pericarditis after infarction, the sterile valve lesion of lupus, and heart block in a newborn. For each one write how the antigen or antibody reaches the heart, the immune mechanism, and the tissue injured. For the pericarditis, write how long after the infarct it appears and how that differs in timing and mechanism from the pericarditis of the first few days, then list the symptoms, the examination finding, the blood results, and what the ECG shows and how that pattern differs from the ECG of the infarct itself. Name the operation that produces the same syndrome. For the valve lesion, write what the vegetations are made of, which antibody promotes the thrombosis, which surfaces of the leaflet they sit on, which valve is involved most often, what they leave behind as they heal, and where fragments travel. For the newborn, name the maternal antibody and the second one often found with it, the receptor that carries it across the placenta, the trimester it starts, the maternal diseases to ask about, and the structure destroyed.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C04-BD2

Explain the newborn and sort four sets of vegetations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-BD2 Date: _____

A newborn has a ventricular rate of 50 per minute, P waves at 142 per minute that bear no relation to the QRS complexes, and annular pink lesions on the face. The mother has no symptoms. Write what the atria and ventricles are each doing and what the ventricular rhythm is called, then explain the two separate ways the maternal antibody slows conduction, one immediate and one permanent, and name the cytokines released by the macrophages. Write the window in gestation when the node is injured and why the ion channel the antibody blocks matters at this particular site. Then write what has happened by 8 months to the rash and to the bradycardia, and explain each outcome. Write the risk of this complication in an antibody positive pregnancy and the risk after one affected child. Finish by taking four patients with vegetations on the mitral valve: one had streptococcal pharyngitis as a child, one has lupus with antiphospholipid antibodies, one has fever with positive blood cultures, and one has advanced pancreatic cancer. For each write the cause, the size and position of the vegetations on the leaflet, whether organisms are present, and what the valve looks like years later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-16-01-C04-RT01

Rebuild the pathway to congenital heart block

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-RT01 Date: _____

Rebuild the six-step pathway of congenital heart block from memory, one step per line and in order, from the maternal antibody to the permanent rhythm. Beside the placental step name the receptor that carries the antibody and the trimester transfer begins. Beside the binding step write what the antibody binds on the fetal myocyte and which ion channels it also inhibits. Beside the macrophage step name the two mediators released. Beside the final step write the ventricular rate you would expect against the normal fetal rate range. Then write one line under the sequence saying which step makes the block permanent and why that tissue cannot recover.

Rebuild from memory: Pathogenesis of congenital heart block

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C05-BD1

Build the rejection table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C05 Activity: NPLEX-P1-CARD-16-01-C05-BD1 Date: _____

On blank paper build a table of the four types of heart transplant rejection with one row each, using the columns timing after transplant, immune effector, hypersensitivity type, and findings in the graft, and fill every cell. Above the table write the two ways recipient T cells see donor HLA: for each, write whose antigen presenting cells are involved, where the HLA peptides sit, which one drives the strong early response and why, and which one keeps working after the donor cells carried in the graft are gone and gives B cells the help to make antibody. Beneath the table write, for each type, the one feature that identifies it: what the preformed antibody does to small vessels, what the biopsy shows when T cells are the effector, which complement fragment marks the capillaries in antibody-mediated rejection and why it stays bound, and what the coronary arteries look like in the chronic form. Finish with the two reasons hyperacute rejection is now rare.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C05-BD2

Three recipients and a denervated heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C05 Activity: NPLEX-P1-CARD-16-01-C05-BD2 Date: _____

Three heart transplant recipients come for review. Recipient 1, at 3 weeks, feels well; biopsy shows lymphocytes around vessels and between damaged myocytes with no C4d. Recipient 2, at 5 months, is hypotensive and breathless; biopsy shows swollen capillary endothelium with macrophages inside capillaries, strong C4d, few interstitial lymphocytes, and a new donor-specific anti-HLA antibody has appeared. Recipient 3, at 6 years, has breathlessness with no chest pain, and every coronary branch is smoothly and evenly narrowed. For each recipient name the type of rejection, write the effector, and explain why the timing fits that effector. Write why recipient 2 is more unwell than recipient 1 despite a similar biopsy grade. For recipient 3, write how the artery wall differs from ordinary atherosclerosis in distribution, in the elastic lamina, and in lipid and calcium content, name the immune and the nonimmune drivers, and write why angiography understates the narrowing. Then write the resting heart rate you expect in all three and why, how the rate responds to exercise and through what route, and why recipient 3 has no angina and what he presents with instead. Finish by writing what a graft that thromboses within minutes of reperfusion would mean and which antibodies cause it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C06-BD1

Viral myocarditis in three phases

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C06 Activity: NPLEX-P1-CARD-16-01-C06-BD1 Date: _____

Write the definition of myocarditis in one sentence. Then rebuild viral myocarditis from memory as three numbered phases. For phase 1 name the classic virus, the receptor and co-receptor it uses to enter the myocyte, the viral enzyme that cuts a cytoskeletal protein and which protein that is, the time frame, and the innate defences that limit spread. For phase 2 write the time frame, which T cells kill infected myocytes and with which molecules, which cytokines the helper cells release, why this phase causes most of the muscle loss even though it clears the virus, and which released cardiac protein can set off autoimmunity and by what mechanism. For phase 3 write what happens to most patients and what a minority develop, with the two reasons it happens. Then write how the diagnosis is made: name the histological criteria and their weakness, and write where the MRI signal sits in the wall and how that differs from infarction. Finish with the clinical findings and explain each from the pathology: the blood marker, the chest pain with ST changes, the heart failure, and the arrhythmias that can kill a young athlete.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C06-BD2

Separate four patients with myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C06 Activity: NPLEX-P1-CARD-16-01-C06-BD2 Date: _____

Four patients have myocardial inflammation. Patient 1 is 22 and had fever and sore throat ten days ago, now with chest pain, a troponin of 4.2 ng/mL, and ST changes crossing more than one coronary territory. Patient 2 started a sulfonamide two weeks ago and has a rash, fever, a blood eosinophil count of 2.4 x 10 to the 9 per L, and a falling ejection fraction. Patient 3 has long standing asthma, an eosinophil count of 9 x 10 to the 9 per L, a numb foot, and a thickened endocardium with mural thrombus. Patient 4 is 17, male, and developed chest pain three days after a second dose of an mRNA vaccine, with a mildly raised troponin and subepicardial enhancement on MRI. For each patient write the form of myocarditis, the effector cell or molecule doing the damage, and what you expect the biopsy to show. Write which patient is most likely to end with restrictive physiology and explain the route from eosinophil granule to that physiology. Write what you expect to happen to patient 2 once the drug is stopped and why, and state whether the reaction depends on the dose. For patient 4 write what the notes say about the mechanism and the usual course. Then add a fifth patient whose heart failure worsens over two weeks with wide necrosis and multinucleated cells on biopsy and no granulomas: name the form, its associations, and its outlook.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-16-01-C06-PR01

Predict what the T-cell phase does to the myocardium

A 24-year-old man is in the second week after a coxsackievirus B infection of the myocardium, and the CD8 T-cell response is now at its peak. Predict each item below, write the direction of the change, and give the cell, molecule, or structural event that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the amount of virus in the myocardium		
What happens to the number of surviving myocytes compared with the first few days		
What happens to the serum troponin		
What happens to the integrity of dystrophin and the myocyte membrane		
What happens to circulating exposure to cardiac myosin and to the chance of autoimmunity		
What happens to ventricular chamber size and wall thickness over the following months in the minority who do not recover		
What happens to the risk of reentrant ventricular arrhythmia as fibrosis replaces lost muscle		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C07-BD1

HHV-8 latency and the cytokines of myxoma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C07 Activity: NPLEX-P1-CARD-16-01-C07-BD1 Date: _____

Divide a page in two. On the left, write Kaposi sarcoma from memory: name the virus and its family, how it spreads, which cell it infects latently and what type of endothelium that cell resembles. Then list the viral genes and what each one does, covering the protein that holds the viral genome in the cell and the two tumour suppressors it blocks, the viral cyclin and the partner kinase it drives, the protein that blocks apoptosis and the transcription factor it turns on, the receptor that induces VEGF, and the viral cytokine and the receptor chain it does not need. Write what the lesions look like under the microscope, the three stages they pass through, the tissues involved, and the stain that confirms the virus. Name two other diseases the same virus causes. On the right, write atrial myxoma: its rank among primary cardiac tumours in adults and in children, which chamber and which spot in it, what the mass looks like and what cells and matrix it contains, the usual patient, and the inherited syndrome with its gene and two accompanying features.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C07-BD2

Failed surveillance, and one cytokine imitating three diseases

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C07 Activity: NPLEX-P1-CARD-16-01-C07-BD2 Date: _____

Write which cells normally hold the latent virus of Kaposi sarcoma in check and which helper cells they depend on. Then take three patients: one with untreated HIV and a CD4 count of 90 cells per microlitre, one taking a calcineurin inhibitor after a kidney transplant, and an 68-year-old man of Mediterranean descent with indolent plaques on both shins. For each write why the tumour appeared, how aggressive it is, and what you expect to happen to the lesions if immunity is restored or immunosuppression is reduced, including the transient change that can occur first and which lesions may not respond. Write which viral protein in HIV adds to the proliferation. Then move to a 54-year-old woman with three months of fever, weight loss, fatigue, an ESR of 88 mm/h, a haemoglobin of 9.8 g/dL, raised immunoglobulins across all classes, and breathlessness and dizziness that come on when she stands up, with a sound in early diastole. Write which single cytokine explains the systemic picture and trace it to each finding: the fever, the high ESR, the anaemia, and the raised immunoglobulins, naming the mediator or protein at each step. Explain the positional symptoms and the diastolic sound from where the mass sits, write what else fragments of it can do, and write what happens to all of the systemic findings after the mass is removed and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-16-01-C07-CA01

Fever, weight loss, and a high ESR in a woman with positional breathlessness

A 56-year-old woman is referred after four months of intermittent fever, 7 kg of weight loss, and aching joints. She also describes breathlessness and lightheadedness that come on within seconds of standing and settle when she lies down, and she has had two brief episodes of weakness of the right arm. Temperature is 38.2 degrees C (100.8 degrees F), pulse is 96/min and regular, and blood pressure is 118/72 mm Hg. On auscultation there is a low-pitched sound shortly after the second heart sound, and the intensity of a diastolic murmur changes when she sits up. Haemoglobin is 9.7 g/dL with a mean corpuscular volume of 84 fL, ferritin is 260 ng/mL, ESR is 92 mm/h, and serum immunoglobulins are raised in all classes. Three sets of blood cultures are sterile, antinuclear antibody and ANCA are negative, and complement is normal. Transthoracic echocardiography shows a mobile pedunculated mass attached to the interatrial septum near the fossa ovalis that moves toward the mitral orifice during diastole.

1. Recognize. Name the lesion, the chamber it arises in, and the attachment site that the echocardiographic description identifies.

2. Explain. Name the cytokine the lesion secretes and trace it to the fever, the high ESR, the anaemia, and the raised immunoglobulins, naming the mediator or liver protein at each step.

3. Differentiate. Write how this picture differs from infective endocarditis, from a systemic vasculitis, and from lupus, naming the findings in this patient that separate them.

4. Investigate. Write which features of the mass and the blood work you would examine to support the diagnosis, and what the histology and the inheritance history would show.

5. Decide. State what is causing the positional breathlessness and the episodes of arm weakness, and name the microscopic feature of the mass that permits the second of these.

6. Justify. Justify your answer by explaining why a benign tumour produces a picture resembling chronic infection or malignancy, and what you expect to happen to the fever, anaemia, ESR, and immunoglobulins after the mass is removed.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-16-01-C02-Q01

Carditis three weeks after a sore throat

A 10-year-old boy is brought to the physician because of fever and painful swelling that began in the right knee and moved to the left ankle. He had a sore throat 3 weeks ago that was not treated. Temperature is 38.6 degrees C (101.5 degrees F). A new apical systolic murmur is heard. ESR is 78 mm/h, throat culture is negative, and the antistreptolysin O titer is rising. Which of the following best explains the cardiac findings in this patient?

A. Bacterial invasion of the valve leaflets with destruction of the endocardial surface

B. Antibodies and T cells raised against streptococcal M protein and group A carbohydrate that also recognize cardiac myosin and valve proteins

C. Deposition of circulating streptococcal antigen and antibody complexes in the valve with classical pathway activation

D. Antineutrophil cytoplasmic antibodies activating primed neutrophils on the endocardial surface

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C03-Q02

Purpura, casts, and a very low C4

A 58-year-old woman comes to the physician because of raised purple lesions on both lower legs that do not blanch, aching wrists, and numbness of both feet. Serum C4 is markedly decreased with a near normal C3, rheumatoid factor is present, and hepatitis C antibody is positive. Urinalysis shows red cell casts. Skin biopsy shows granular immunoglobulin and complement in the walls of dermal venules. Which of the following is the mechanism of vascular injury in this patient?

A. IgM rheumatoid factor bound to IgG forming complexes that deposit in vessel walls and consume the classical pathway

B. Antibodies against myeloperoxidase activating primed neutrophils with little immunoglobulin deposition

C. Antibody binding to type IV collagen in the vessel basement membrane

D. Dendritic cell activation of CD4 T cells with macrophage granuloma formation in the vessel media

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C05-Q03

Hypotension five months after transplantation

A 49-year-old man who received a heart transplant 5 months ago is admitted because of 2 days of breathlessness. Blood pressure is 84/56 mm Hg. Endomyocardial biopsy shows swollen capillary endothelial cells with macrophages within capillary lumens, strong capillary staining for C4d, and only scattered interstitial lymphocytes. A donor-specific anti-HLA antibody that was absent before transplantation is now detected. Which of the following best accounts for these findings?

A. CD8 T cells recognizing intact donor HLA on passenger dendritic cells

B. Smooth muscle migration into the intima driven by PDGF and TGF-beta

C. Donor-specific antibody binding capillary endothelium with classical pathway activation

D. Preformed anti-ABO IgM causing immediate small vessel thrombosis

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C06-Q04

A young man two weeks after a viral illness

A 25-year-old man comes to the emergency department because of 2 days of chest pain and breathlessness. He had fever, myalgia, and diarrhoea 12 days ago. Troponin I is 5.6 ng/mL, and the electrocardiogram shows ST segment changes in leads that span more than one coronary territory. Coronary angiography is normal. Cardiac MRI shows late gadolinium enhancement in the subepicardial layer of the lateral wall. Which of the following is most responsible for the myocyte loss at this point in his illness?

A. Cleavage of dystrophin by viral protease 2A

B. Eosinophil major basic protein released into the endocardium

C. Subendocardial necrosis from occlusion of an epicardial coronary artery

D. CD8 T cells killing infected myocytes with perforin and granzymes

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-17

CV-17 Cardiovascular Pathogens

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-17-01-C01				
NPLEX-P1-CARD-17-01-C02				
NPLEX-P1-CARD-17-01-C03				
NPLEX-P1-CARD-17-01-C04				
NPLEX-P1-CARD-17-01-C05				
NPLEX-P1-CARD-17-01-C06				
NPLEX-P1-CARD-17-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C01-BD1

Rebuild the four endocarditis cocci

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C01 Activity: NPLEX-P1-CARD-17-01-C01-BD1 Date: _____

Without looking at your notes, write out the four classic gram-positive cocci of infective endocarditis. For each one write its arrangement on Gram stain, whether it is catalase positive or negative, what the coagulase or hemolysis or growth test shows, and the named virulence factors it uses. Then write one sentence per organism saying how those factors let it attach to a valve and how fast it destroys the leaflet. Finish by writing the three bench results that separate staphylococci from streptococci and enterococci, and the one test that separates Staphylococcus aureus from Staphylococcus epidermidis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C01-BD2

Two patients, two valves, two time courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C01 Activity: NPLEX-P1-CARD-17-01-C01-BD2 Date: _____

Patient one injects drugs and becomes ill over three days with a loud tricuspid murmur and a perforated leaflet found at surgery. Patient two had a dental cleaning six weeks ago, has a known bicuspid aortic valve, and has felt tired and feverish ever since. For each patient write the likely organism, the portal of entry, why that organism can or cannot colonize the valve as it was before the infection, and the virulence factors that explain the speed of leaflet damage. Then write why protein A and bound coagulase make neutrophils and antibody less effective, and write what a sterile platelet-fibrin deposit on a scarred valve does for an organism that invades poorly. Finish by writing which of the two patients you would expect to have an abscess and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-17-01-C01-CP01

The four gram-positive cocci of infective endocarditis

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Staphylococcus aureus	Viridans streptococci	Enterococcus faecalis	Staphylococcus epidermidis
Arrangement on Gram stain				
Catalase result				
Other bench tests that identify it				
Named virulence factors				
How it attaches to the valve surface				
Condition of the valve it needs				
Usual portal of entry				
Speed of valve destruction and course				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C02-BD1

Why blood cultures stay negative

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C02 Activity: NPLEX-P1-CARD-17-01-C02-BD1 Date: _____

Write the definition of culture-negative endocarditis and then list its causes in order, starting with the commonest. For each cause write the organism type, the exposure or body site it comes from, and the reason routine blood cultures fail to grow it. Spell out what the letters of HACEK stand for and write where those organisms normally live. For Coxiella burnetii write the cell compartment it multiplies in, the form that survives drying, the material people inhale, and the test used instead of culture. For the two Bartonella species write the vector or animal source for each. Finish by writing the bench profile of Streptococcus gallolyticus and the one test that separates it from Enterococcus.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C02-BD2

Match the exposure to the organism and read the bench results

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C02 Activity: NPLEX-P1-CARD-17-01-C02-BD2 Date: _____

Four adults have endocarditis with no growth in routine blood cultures. One works on a goat farm and helps at kidding. One lives outdoors and has body lice. One has poor dental health and large friable vegetations that have embolized. One took an antibiotic for a sore throat the week before cultures were drawn. For each person write the likely cause, the structural or metabolic feature of the organism that explains why it did not grow, and the kind of test that would identify it. Then a different patient grows catalase-negative cocci in chains that are nonhemolytic, hydrolyze esculin in bile, and do not grow in 6.5% NaCl. Write the organism, the body site it came from, and what its presence in the blood suggests about that site. Finish by writing what the result would have been instead if the isolate had grown in 6.5% NaCl.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-17-01-C02-Q01

A group D organism from the bench tests

A 71-year-old man comes to the physician because of 5 weeks of fatigue, low-grade fever, and weight loss. Temperature is 38.1 °C (100.6 °F). A new murmur is heard at the cardiac apex, and echocardiography shows a vegetation on the mitral valve. Blood cultures grow gram-positive cocci in chains. The isolate is catalase negative and nonhemolytic, hydrolyzes esculin in the presence of bile, and fails to grow in 6.5% sodium chloride. Which of the following organisms is most consistent with these findings?

A. Enterococcus faecalis

B. Streptococcus mitis

C. Streptococcus gallolyticus

D. Staphylococcus epidermidis

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C03-BD1

The two human forms and every route of transmission

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-BD1 Date: _____

Write the name of the organism that causes Chagas disease and the kind of organism it is. Describe its two human forms side by side: for each one write whether it is inside or outside cells, whether it has an external flagellum, and whether it divides. Write where in the body the dividing form multiplies and name the four cell types it favors. Write the insect vector, where the vector lives, when it feeds, and the part of the insect that carries the infective parasite, then write why the bite itself does not transmit it. List every other route of transmission named in the notes. Finish by writing what the acute phase looks like, how long it lasts, what is seen on a stained blood smear during it, and the two local signs of entry with the site each one follows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C03-BD2

Predict chronic Chagas disease from the cells the parasite lives in

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-BD2 Date: _____

A woman who grew up in rural Bolivia and moved away 30 years ago has had no symptoms since. Write the name of the phase she has been in, why serology rather than a blood smear is used to find infection in that phase, and the approximate share of infected people who go on to chronic disease. Then, working only from the cell types the parasite infects, predict what three decades of parasite persistence and inflammation do to the heart: write what replaces the lost myocytes, what happens to chamber size and contractility, where a thin-walled aneurysm forms and what can collect in it, and which conduction pattern is typical. Predict what the loss of enteric plexus neurons does to the esophagus and colon, and explain in two sentences why those organs dilate. Finish by writing why a symptom-free carrier living far from any triatomine bug still matters to a blood bank and a transplant program.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C03-RT01

Rebuild the Trypanosoma cruzi life cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-RT01 Date: _____

Rebuild the eight-step life cycle of Trypanosoma cruzi from memory, one step per line and in order, starting with the bug taking a blood meal. Beside each step write which parasite form is present at that point. Draw a line across the page to separate the steps that happen in the insect from the steps that happen in the human. Then circle the two steps where the parasite multiplies, and write one sentence saying why the number of parasites in human blood can rise even though the circulating form does not divide.

Rebuild from memory: Life cycle of Trypanosoma cruzi

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C04-BD1

Two tick-borne bacteria from the ground up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-BD1 Date: _____

Draw Borrelia burgdorferi and Rickettsia rickettsii side by side and label the structural features of each. For Borrelia write where its flagella sit, what their rotation does for the cell, what its outer membrane is rich in and what it lacks, and how its genome is arranged. For Rickettsia write its shape and Gram character and the two metabolic reasons it must live inside a host cell. For both write the stains used to see them and why Gram stain is not useful. Then write the tick genus that carries each one, the reservoir for each, and which tick stage transmits most human Lyme disease. Finish by writing the region of the United States where most Rocky Mountain spotted fever cases occur.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C04-BD2

Explain the block and explain the rash

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-BD2 Date: _____

Patient one was camping in Connecticut, has an expanding red skin lesion, and now has a heart rate of 44 per minute with a varying degree of atrioventricular block that changes within hours. Patient two was in Oklahoma, has had fever for 4 days, and now has macules on the wrists and ankles spreading to the trunk, palms, and soles, with falling blood pressure and falling platelets. For each patient name the organism, the tick that carried it, and the tissue the organism is damaging, then explain the cardiovascular finding from the way that organism moves and survives. For patient one write where in the conduction system the block usually sits and why it can change degree hour by hour. For patient two write how the organism gets from one endothelial cell to the next and explain how damaged endothelium produces edema, hypotension, and a petechial rash. Finish by writing why a tick removed within the first day rarely transmits Lyme disease, naming the surface proteins involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C04-RT01

Rebuild the Borrelia surface switch from tick to host

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-RT01 Date: _____

Rebuild the seven-step sequence that takes Borrelia burgdorferi from an unfed tick gut into human skin, one step per line and in order. Beside the first step write which surface protein anchors the spirochete in the gut and what it binds. Beside the step where the proteins change, write which protein falls and which rises, and what triggers the change. Beside the salivary gland step write the tick protein that the new surface protein binds and what that does for the spirochete. Write the number of hours of tick attachment the whole sequence usually needs. Then write the last step separately: name the surface lipoprotein that keeps changing in the host, write the genetic mechanism that changes it, and write in one sentence what that does to antibody made against the earlier version.

Rebuild from memory: Borrelia surface switching from tick to host

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C05-BD1

Flaviviruses and filoviruses side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C05 Activity: NPLEX-P1-CARD-17-01-C05-BD1 Date: _____

Write the two virus families that cause viral hemorrhagic fever in this chapter and name the example viruses in each. For each family write the shape of the virion, whether it is enveloped, the genome type and strand sense, the reservoir, the route of transmission, and the first cells it infects. Then write what a positive-sense genome does on its own inside a cell and what a negative-sense virion must carry with it, and write which of the two naked genomes would be infectious if injected alone. For yellow fever write the organ and the zone of the liver lobule that dies, the name of the dying cell on the slide, and the two consequences of lost hepatocyte function. For dengue write the number of serotypes, the cells it infects, when in the illness the plasma leak peaks, and the secreted viral protein that damages the endothelial glycocalyx.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C05-BD2

Why the second dengue infection is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C05 Activity: NPLEX-P1-CARD-17-01-C05-BD2 Date: _____

A man had dengue 4 years ago and recovered. He now has a second infection with a different serotype and is much sicker than the first time. Write why immunity from the first serotype does not protect him, then rebuild the mechanism in order: what the old antibody does to the new virus, which receptor on which cells takes the coated virus in, what that route of entry does to viral load, and what it does to the interferon response. Write what cross-reactive memory T cells add. Then explain why an infant with waning maternal antibody can have severe disease at a first infection. Separately, write what a rising hematocrit with falling platelets tells you about where the plasma has gone, and why the patient gets worse as the fever breaks. Finish by writing how the mechanism of bleeding in Ebola differs from this one, naming the two interferon-blocking proteins and the macrophage product that starts disseminated intravascular coagulation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-17-01-C05-PR01

Predict the course of a second dengue infection

A traveler who had mild DENV-1 infection 4 years ago is now on day 5 of a febrile illness caused by DENV-2, and her fever is falling. Predict each item below, write the direction of the change, and give the mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What the DENV-1 antibody does when it meets DENV-2		
What happens to the number of virus particles entering monocytes and macrophages		
What happens to the type I interferon response in those cells		
What happens to the amount of fluid in the vessels and to the hematocrit		
What happens to the platelet count		
What happens to her blood pressure and alertness as the fever falls		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C06-BD1

Coxsackie B from gut to dilated heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C06 Activity: NPLEX-P1-CARD-17-01-C06-BD1 Date: _____

Write the definition of myocarditis and the commonest identified cause of it in high-income countries. For coxsackie B virus write the family and genus, the genome type and strand sense, the shape of the capsid, and whether it has an envelope, then write the three things that missing envelope lets the virus survive and the route of spread that follows from it. Rebuild the seven-step sequence that runs from entry at the mouth to a dilated ventricle, one step per line and in order. Beside the receptor step write both surface molecules the virus uses, where in the heart muscle one of them concentrates, and why that matters in a newborn. Beside the protease step write the two host proteins protease 2A cuts and what each cut does to the cell. Finish by writing the other two syndromes this virus causes and what often happens to the pericardium alongside the myocardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C06-BD2

Explain the findings and judge a biopsy result

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C06 Activity: NPLEX-P1-CARD-17-01-C06-BD2 Date: _____

A 14-year-old had diarrhea in August and now has chest pain, a rub, a troponin of 3.2 ng/mL (reference under 0.04 ng/mL), and frequent ectopic beats with ST and T wave changes. Explain each finding from the mechanism of injury: write why troponin appears in the blood, why the chest pain is pleuritic with a rub, why the rhythm is unstable, and why contractility falls. Write the structural protein whose loss lets the membrane tear under contractile stress, name the inherited disease with the same structural failure, and explain why the two diseases damage the same link in different ways. Then write what happens over the following weeks when T cells arrive, and why that phase can leave a dilated ventricle even after the virus is cleared. Finish with this result: a heart biopsy from a different patient is positive for parvovirus B19 DNA. Write why that result does not by itself establish the cause, and write what parvovirus B19 binds and which vessels it injures.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-17-01-C06-Q02

The torn sarcolemma in coxsackie myocarditis

A 9-year-old boy is brought to the physician in September because of 4 days of fever, pleuritic chest pain, and shortness of breath that began a week after an episode of watery diarrhea. Temperature is 38.4 °C (101.1 °F) and a pericardial friction rub is heard. Serum troponin is elevated. A nonenveloped, positive-sense single-stranded RNA virus is recovered from a stool specimen. Which of the following best explains the tearing of the myocyte sarcolemma in this illness?

A. Viral protease cleavage of dystrophin, which links sarcomeric actin to the sarcolemma

B. ADP-ribosylation of elongation factor 2 and arrest of myocyte protein synthesis

C. Polymerization of host actin into tails that push the organism between cells

D. Nonneutralizing antibody carrying virus into cells through Fc gamma receptors

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C07-BD1

Diphtheria toxin and the tuberculous pericardium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-BD1 Date: _____

For Corynebacterium diphtheriae write the shape of the rod, how the cells lie beside each other, the granules inside them, and the colony color on tellurite agar. Write which strains can make toxin and what they carry, then write the one nutrient condition that turns toxin production on and the repressor that holds it off. Rebuild the toxin mechanism in order, from the B portion binding its receptor to the molecule that stops working, and write what the cell does next. Write what forms in the throat and what it is made of, then write when myocarditis appears in the illness, what the muscle looks like microscopically, and which conduction problems follow. For Mycobacterium tuberculosis write the staining property and the wall component behind it, the route by which it usually reaches the pericardium, the kind of granuloma and effusion it produces, and what the two pericardial layers become as the inflammation heals.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C07-BD2

Predict the physiology of a fixed pericardium and a starved aortic media

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-BD2 Date: _____

A man has a thick calcified pericardial shell after treated tuberculosis. Working from the fact that the shell fixes the total volume of the heart, predict what happens to early diastolic filling and what stops it, the extra heart sound this produces, the shape of the jugular venous pulse, and what happens to jugular venous pressure during inspiration and why. Write what the diastolic pressures in the four chambers do relative to each other, and list the findings that follow from systemic venous congestion. Then a second man has untreated syphilis of 25 years. Write which vessels the treponemes persist around, what those vessels supply, and what the narrowing of those vessels does to the outer media. Predict which part of the aorta fails and why that part rather than another, the appearance of the intima over the scarred media, and the three consequences of a dilated aortic root and arch. Finish by writing why atherosclerotic aneurysms instead favor the infrarenal abdominal aorta.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C07-RT01

Rebuild the path from vasa vasorum to aneurysm

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-RT01 Date: _____

Rebuild the seven-step sequence that runs from treponemes in the adventitia to an aortic aneurysm, one step per line and in order. Beside the step where the small vessels narrow, write the name of that lesion and the cells that cuff the vessels. Beside the step where the media is affected, write which two structural components of the media are lost and what the wall can no longer do. Then draw a cross section of the thoracic aorta and label the adventitia, the vasa vasorum, the outer media, and the intima, and mark the layer that becomes ischemic. Finish by writing one sentence saying which does the damage, the organism itself or the loss of blood supply to the wall.

Rebuild from memory: From vasa vasorum endarteritis to aortic aneurysm

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-17-01-C07-CA01

Chest discomfort and a widened ascending aorta in a man with a remote untreated infection

A 58-year-old man comes to the physician because of 6 months of chest discomfort on exertion and increasing breathlessness when lying flat. He recalls a painless genital ulcer and a rash on his palms and soles in his early thirties, neither of which was treated. Temperature is 37.1 °C (98.8 °F), pulse is 88/min and collapsing in character, respirations are 18/min, and blood pressure is 160/48 mm Hg. A high-pitched early diastolic murmur is heard at the left sternal border. Chest radiography shows a widened ascending aorta and arch with a thin rim of calcification. At the time of later surgery the ascending aorta is dilated, the intima is wrinkled into a coarse tree-bark pattern, and the aortic valve ring is stretched with structurally normal leaflets.

1. Recognize. Name the organism, the stage of its infection, and the usual interval between untreated infection and this presentation, and name the lesion that produced the tree-bark intima.

2. Explain. Explain how the organism damages the aortic wall, starting with where it persists and ending with the loss of elastic fibers and smooth muscle, and explain why the valve leaflets are normal while the valve leaks.

3. Differentiate. Write how this aneurysm differs from an atherosclerotic aortic aneurysm in location and in mechanism, and name the feature of the aortic wall that explains the difference in site.

4. Investigate. Write which microscopy methods show this organism and why Gram stain does not, and write what you would look for in the aortic wall and in the adventitial vessels on histology.

5. Decide. State which structural change produces the exertional chest discomfort, and name the two other structures a large aneurysm of the arch can compress.

6. Justify. Justify the mechanism from the blood supply of the thoracic aortic media, explaining why ischemia of the outer media rather than direct destruction by the organism is what weakens the wall.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-01-C01				
NPLEX-P1-CARD-18-01-C02				
NPLEX-P1-CARD-18-01-C03				
NPLEX-P1-CARD-18-01-C04				
NPLEX-P1-CARD-18-01-C05				
NPLEX-P1-CARD-18-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C01-BD1

Blood pressure categories and why the kidney sets pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C01 Activity: NPLEX-P1-CARD-18-01-C01-BD1 Date: _____

Without looking at your notes, write out the four blood pressure categories of the 2017 ACC/AHA scheme with the systolic and diastolic values for each, and write the rule that applies when the two values fall in different categories. Beside that, write how many readings and occasions the diagnosis needs. Then write the equation that relates mean arterial pressure to cardiac output and systemic vascular resistance, and state which of the two is raised in established primary hypertension. Underneath, explain in four or five sentences why baroreceptors cannot hold pressure high over months and how pressure natriuresis works, and finish by writing the one change in the kidney that every sustained hypertension must involve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C01-BD2

Office readings, night-time pressure, and two inherited salt channels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C01 Activity: NPLEX-P1-CARD-18-01-C01-BD2 Date: _____

Three patients are measured. The first reads 128/86 mmHg in clinic and 118/74 mmHg at home, the second reads 118/76 mmHg in clinic and 142/90 mmHg at home, and the third reads 150/94 mmHg in clinic with no fall in pressure during sleep. For each one, name what the pattern is called, state whether the ACC/AHA category assigned in clinic is the right one, and write what the pattern suggests about the patient. Then take a 19-year-old with hypertension, a plasma potassium of 2.9 mEq/L (reference range 3.5 to 5.0 mEq/L), metabolic alkalosis, and low renin with low aldosterone. Write the two inherited disorders and the one dietary exposure from the notes that produce this picture, give the protein or enzyme altered in each, and explain step by step why each one raises sodium reabsorption, wastes potassium, and leaves both renin and aldosterone low. Finish by writing which risk factors in the notes act partly by raising sympathetic outflow and which act mainly by expanding volume.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C02-BD1

Build the secondary hypertension table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C02 Activity: NPLEX-P1-CARD-18-01-C02-BD1 Date: _____

On blank paper build a table of the secondary causes of hypertension with one row each for renal artery stenosis, primary aldosteronism, the mineralocorticoid group of Cushing syndrome with licorice and Liddle syndrome, pheochromocytoma, coarctation of the aorta, and obstructive sleep apnea. Use the columns key clinical clue, renin, and aldosterone, and fill every cell. Beside the renal artery stenosis row, write the two diseases that narrow the artery, the age and sex each one affects, and the part of the artery each one involves. Beside the pheochromocytoma row, write which receptors raise resistance and which raise cardiac output. Then write the share of all hypertension that is secondary and list the four features of a patient that should make you look for a cause.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C02-BD2

Reason from the laboratory pattern back to the organ

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C02 Activity: NPLEX-P1-CARD-18-01-C02-BD2 Date: _____

Four patients are hypertensive and hypokalemic. Patient A has high renin and high aldosterone with an abdominal bruit. Patient B has low renin and high aldosterone with no edema. Patient C has low renin and low aldosterone and drinks licorice tea daily. Patient D has normal renin with episodic headache, sweating, and palpitations. For each patient write the organ or vessel at fault and explain in two or three sentences why renin is where it is and why aldosterone is where it is. For patient B, explain the tubular steps that produce the low potassium and the alkalosis, and explain why the patient has no edema despite retaining sodium. For patient C, name the enzyme involved and write which receptor is being occupied and by which hormone. For patient D, write why the plasma metanephrines stay high between attacks and why an epinephrine-secreting tumor is almost always inside the adrenal gland. Finish by writing what happens to the glomerular filtration rate of a stenotic kidney when angiotensin II is blocked, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-01-C02-CP01

Renal artery stenosis against primary aldosteronism

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Renal artery stenosis	Primary aldosteronism
Where the primary abnormality sits		
Plasma renin and why it is at that level		
Plasma aldosterone and why it is at that level		
Plasma potassium and acid-base state		
Physical examination and imaging clues		
Whether edema develops, and the reason		
What happens to the kidney when angiotensin II is blocked		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C03-BD1

Hyaline arterioles and the concentrically hypertrophied ventricle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-BD1 Date: _____

Write the law of Laplace as the notes give it, with each term named, then explain in four or five sentences why a raised afterload makes myocytes add sarcomeres in parallel and what that does to wall stress. Write the thickness of a normal left ventricular wall and the thickness reached in hypertensive heart disease, and write how the sarcomere arrangement and the chamber size differ between pressure overload and volume overload. Underneath, describe hyaline arteriolosclerosis: what leaks into the wall, what the smooth muscle cells add, how the wall looks on routine stains, what happens to the lumen, and the two other conditions in which the same lesion appears. Finish by writing what happens to the myocyte nuclei in a hypertrophied wall.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C03-BD2

Trace chronic pressure into four organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-BD2 Date: _____

A 70-year-old woman has had untreated hypertension for 25 years. Her echocardiogram shows a left ventricular wall of 1.9 cm with an ejection fraction of 60%, and she has just developed atrial fibrillation with breathlessness at rest. Write why a ventricle with a normal ejection fraction can congest the lungs, and write what the loss of atrial contraction and the fast rate each take away from filling. Then write the extra heart sound this ventricle produces, which event in the cardiac cycle makes it, and the ECG findings in V1, V5, and V6 that reflect the increased mass, including the strain pattern. Next write why this thick wall is prone to ischemia even with open coronary arteries, naming three reasons from the notes. Then, for her kidneys, retinas, and brain, write what you expect to find: name the renal lesion with the appearance of the kidney surface and what happens to proteinuria and glomerular filtration rate, name five retinal findings and the mechanism of the one caused by a thick arteriole crossing a vein, and name the two lesions of penetrating cerebral arteries with the size and the usual sites of each. Finish by writing why hypertensive bleeds sit deep while amyloid angiopathy bleeds sit in the lobes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-01-C03-RT01

Rebuild the pressure overload sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-RT01 Date: _____

Rebuild from memory the seven-step sequence that runs from sustained hypertension to atrial fibrillation and heart failure with a preserved ejection fraction, one step per line and in order. Beside the second step, write the law of Laplace and show which term rises first. Beside the third step, write where the new sarcomeres go and what happens to chamber volume. Beside the fifth step, write why a higher left atrial pressure becomes necessary. Then draw the left ventricle and left atrium in cross section at the start of the sequence and at the end, label the wall thickness and the atrial size on each drawing, and write one sentence on what can happen to the chamber and to systolic function very late in the course.

Rebuild from memory: Pressure overload to diastolic failure and atrial fibrillation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C04-BD1

Autoregulation, breakthrough, and the acute arteriolar lesion

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C04 Activity: NPLEX-P1-CARD-18-01-C04-BD1 Date: _____

Write the definition of a hypertensive emergency with the pressure usually quoted, then write how it differs from hypertensive urgency and what the older term malignant hypertension described. Write which matters more, the level reached or the rate of rise. Draw the cerebral autoregulation curve with mean arterial pressure on the horizontal axis and cerebral blood flow on the vertical axis, mark the lower and upper limits in a healthy person, and draw a second curve showing what chronic hypertension does to it. On the drawing, mark the point of breakthrough and write in five or six sentences what happens to the arterioles, the capillaries, and the blood-brain barrier beyond that point, the symptoms this produces, and which part of the brain the edema favors and why. Underneath, describe the acute arteriolar lesion: what fibrinoid necrosis is and how it looks on routine stains, what the surviving arterioles build and what that lesion is called, where in the kidney it is most conspicuous, and one other disease that produces the same lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C04-BD2

Why the lower pressure did more damage

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C04 Activity: NPLEX-P1-CARD-18-01-C04-BD2 Date: _____

A 72-year-old woman with 30 years of hypertension has a pressure of 186/112 mmHg, no symptoms, and a normal fundus. A previously normotensive 24-year-old woman at 36 weeks of pregnancy has a pressure of 168/108 mmHg with headache and confusion. Write which of the two has a hypertensive emergency and which has urgency, then explain from the position of each woman's autoregulation curve why the lower pressure caused acute brain injury. Next, the younger woman's creatinine rises, her urine shows blood, her platelet count falls, and her blood film shows schistocytes with a raised lactate dehydrogenase. Write the arteriolar lesion behind each of those findings and explain how fibrin in a damaged small vessel produces that blood film. Then write out the renal vicious cycle in this setting in six steps and explain why the renin level is very high here while it is variable in chronic hypertension, and why a shrinking plasma volume pushes renin higher still. Finish by listing the triggers from the notes that precipitate an emergency, and write why stopping clonidine abruptly belongs on that list.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-01-C04-PR01

Predict what a rapid severe rise in pressure does

A previously normotensive 30-year-old develops acute glomerulonephritis and his pressure rises from 118/72 mmHg to 196/124 mmHg over three days. Predict each item below, write the direction of the change, and give the mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to cerebral arteriolar constriction once mean arterial pressure passes the upper limit of autoregulation		
What happens to the permeability of the blood-brain barrier and to the water content of the brain		
What the renal arteriolar walls look like on biopsy within days		
What happens to renin release from the juxtaglomerular cells and why		
What happens to the platelet count, the blood film, and the lactate dehydrogenase		
What the kidney surface looks like at this stage and how that differs from the kidney of long-standing moderate hypertension		
What happens to the left ventricle acutely, given that it has had no time to hypertrophy		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C05-BD1

Pulmonary pressures, the resistance equation, and the five groups

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C05 Activity: NPLEX-P1-CARD-18-01-C05-BD1 Date: _____

Write the normal pulmonary artery pressure as systolic over diastolic with the mean value, and write how pulmonary vascular resistance compares with systemic resistance. Write what happens to pulmonary vascular resistance during exercise and the two mechanical reasons for it. Then write the catheter-based definition of pulmonary hypertension with the threshold the notes use and the older value many textbooks still quote, and write which test raises the suspicion first and which measurement it uses. Write the equation for pulmonary vascular resistance, name the units, and give the value of one of those units. Set out the three hemodynamic types with the wedge pressure and resistance values that define each. Finally, build the five-group classification from memory as a table with the main mechanism and two examples in each group, and mark which group is the most common cause and which group is the most common after it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C05-BD2

Separate two catheter studies and explain two mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C05 Activity: NPLEX-P1-CARD-18-01-C05-BD2 Date: _____

Two patients each have a mean pulmonary artery pressure of 36 mmHg and a cardiac output of 5 L/min. Patient A has a wedge pressure of 10 mmHg and patient B has a wedge pressure of 28 mmHg. Calculate the pulmonary vascular resistance for each, show your working, state which hemodynamic type each one has, and write two diseases that would produce each result. Write in two sentences why the same mean pressure points to different organs. Then take a patient with severe chronic obstructive pulmonary disease and write out, step by step, how low alveolar oxygen reaches pulmonary arterial smooth muscle contraction, naming the channels and the ion movements in order. Write why that response is useful when one region is poorly ventilated and harmful when the whole lung is hypoxic, and write the one blood change in chronic hypoxemia that adds to the pressure and how it does so. Finish by writing how long-standing mitral stenosis can add a precapillary component to a problem that began behind the left atrium, and how the perfusion scan of chronic thromboembolic disease differs from that of group 1.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C06-BD1

Rebuild the pathogenesis of pulmonary arterial hypertension

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C06 Activity: NPLEX-P1-CARD-18-01-C06-BD1 Date: _____

Name the gene most often mutated in heritable pulmonary arterial hypertension, the receptor family it belongs to, the signaling pathway it uses, and the two things that signaling normally does to smooth muscle. Write the inheritance pattern and why a second hit is thought to be needed. Then build a table of the three endothelial mediators from the notes with their receptor and second messenger, their effect on vessel diameter and on cell growth, and the direction each one moves in this disease. Underneath, rebuild the seven-step sequence from the initial endothelial injury to right ventricular failure, one step per line and in order. Beside the remodeling step, write the three structural changes in the small arteries in the order they appear and describe the lesion seen in severe disease, where it forms and what it resembles. Finish by listing the acquired triggers and associated diseases from the notes, and write what happens to a large left-to-right shunt once pulmonary resistance exceeds systemic resistance.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C06-BD2

Explain the signs of the pressure-loaded right heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C06 Activity: NPLEX-P1-CARD-18-01-C06-BD2 Date: _____

A patient with long-standing pulmonary arterial hypertension has a loud P2, a left parasternal heave, a right-sided fourth heart sound, a holosystolic murmur that gets louder on inspiration, large v waves in the jugular venous pulse, raised jugular venous pressure, hepatomegaly, and ankle edema. Her ECG shows right axis deviation, a tall R wave in V1, and peaked P waves. Take each finding in turn and write the mechanism that produces it. Then write the two stages the right ventricle passes through under a rising load and explain how dilation produces the murmur and what it does to the interventricular septum and to left ventricular filling. Write why the right ventricle becomes ischemic as its pressure approaches systemic pressure, naming what happens to its coronary perfusion. Write the definition of cor pulmonale, including what it excludes, and contrast the chronic form with the form that follows a massive pulmonary embolism. Finish by explaining why this patient's systemic blood pressure can be low although her left ventricle is structurally normal, and why syncope on exertion is an ominous finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-01-C06-CA01

Progressive breathlessness in a young woman with a loud P2

A 32-year-old woman comes to the physician because of 10 months of worsening breathlessness on exertion and fatigue, and two episodes of fainting while climbing stairs. She has become hoarse over the past month. She takes no medicines, has never smoked, and has no history of liver disease, joint pain, or skin tightening. Her mother died at 38 years of age of an unexplained heart condition. Temperature is 36.8 °C (98.2 °F), pulse is 104/min, respirations are 22/min, and blood pressure is 96/62 mmHg. Jugular venous pressure is raised with large v waves, the second heart sound has a loud pulmonary component, there is a heave to the left of the sternum, and a holosystolic murmur at the lower left sternal border becomes louder on inspiration. The liver is enlarged and there is pitting edema of both ankles. The lungs are clear. The electrocardiogram shows right axis deviation, a tall R wave in V1, and peaked P waves in lead II. Right heart catheterization shows a mean pulmonary artery pressure of 52 mmHg, a wedge pressure of 9 mmHg, and a cardiac output of 3.4 L/min. Ventilation-perfusion scanning shows no mismatched perfusion defects.

1. Recognize. Name the hemodynamic type of pulmonary hypertension shown by the catheter numbers, calculate the pulmonary vascular resistance with your working, and name the classification group this patient belongs to.

2. Explain. Explain the molecular pathogenesis you expect in this patient, from the inherited defect her family history suggests through the endothelial mediator changes to the structural remodeling of the small arteries.

3. Differentiate. Write how the catheter numbers and the clinical picture separate this from pulmonary hypertension caused by left heart disease, by chronic lung disease, and by chronic thromboembolic obstruction, naming the finding that excludes each one.

4. Investigate. Write which further findings you would look for to support this group and to exclude the associated diseases listed in the notes, and what each finding would show.

5. Decide. State which chamber is failing, name the two physical findings on this examination that show it is failing rather than only hypertrophied, and name the term for right ventricular failure caused by pulmonary vascular disease.

6. Justify. Justify why her cardiac output is low and her systemic pressure is 96/62 mmHg although her left ventricle is structurally normal, why she fainted on exertion, and why she has become hoarse.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-01-C04-Q01

Arteriolar lesion in an abrupt severe rise in pressure

A 34-year-old man comes to the emergency department because of headache, blurred vision, and confusion that began 2 days ago. He was told his blood pressure was normal at a check 4 months ago. Blood pressure is 212/128 mmHg. Funduscopic examination shows flame-shaped hemorrhages, exudates, and papilledema. Serum creatinine is 3.2 mg/dL, urinalysis shows blood, the platelet count is 68,000/mm3, and the blood film shows fragmented erythrocytes. A kidney biopsy is performed. Which of the following is most likely to be seen in the renal arterioles?

A. Homogeneous glassy pink thickening of the wall with a narrowed lumen

B. Subendothelial lipid-laden macrophages with a fibrous cap

C. Concentric laminated layers of smooth muscle with fibrin deposition and wall necrosis

D. Nodular mesangial matrix expansion with capillary basement membrane thickening

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-01-C05-Q02

Reading a right heart catheter study

A 68-year-old woman comes to the physician because of 6 months of breathlessness on exertion. Blood pressure is 148/88 mmHg. Right heart catheterization shows a mean pulmonary artery pressure of 34 mmHg, a pulmonary artery wedge pressure of 26 mmHg, and a cardiac output of 4 L/min. Echocardiography shows a left ventricular wall thickness of 1.8 cm, an ejection fraction of 60%, and an enlarged left atrium. Which of the following best accounts for her pulmonary artery pressure?

A. Backward transmission of a raised left atrial pressure

B. Proliferative remodeling of small pulmonary arteries with plexiform lesions

C. Hypoxic pulmonary vasoconstriction with loss of alveolar capillaries

D. Organized thrombus obstructing the proximal pulmonary arteries

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-B Congestive Heart Failure

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-02-C01				
NPLEX-P1-CARD-18-02-C02				
NPLEX-P1-CARD-18-02-C03				
NPLEX-P1-CARD-18-02-C04				
NPLEX-P1-CARD-18-02-C05				
NPLEX-P1-CARD-18-02-C06				
NPLEX-P1-CARD-18-02-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C01-BD1

From a failing left ventricle to alveolar flooding

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-BD1 Date: _____

Without looking at your notes, draw the route that pressure takes from a left ventricle that cannot empty or fill normally to fluid inside the alveoli. Name every chamber and vessel in order and write beside each one what happens to its pressure, and mark the place where the absence of valves lets pressure pass backward. Write the normal pulmonary capillary pressure and the normal plasma oncotic pressure with their units, the normal range for pulmonary capillary wedge pressure, and the point at which filtration outruns lymphatic clearance. Then list the causes of left-sided failure in groups: loss of muscle, pressure load, volume load, and direct muscle damage. Finish by naming the two lesions that produce every pulmonary finding while the left ventricle itself stays normal or underfilled.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C01-BD2

Explain three breathing patterns and one blood test

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-BD2 Date: _____

A patient with left-sided failure is breathless walking to the bathroom, has to sleep on three pillows, and wakes gasping about two hours after falling asleep. For each of those three findings write the mechanism in two or three sentences, naming what moves where and which receptors drive the breathing pattern. Explain why the breathlessness on lying flat starts within a minute or two while the one that wakes the patient takes one to a few hours. Then write why this patient wheezes and why the crackles are heard at the bases first. Finish with the blood findings of forward failure: write what happens to the BUN to creatinine ratio, give the value that marks the change, and explain the tubular handling of urea that produces it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-02-C01-RT01

Rebuild the seven-step congestion sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-RT01 Date: _____

Rebuild the seven-step sequence that runs from a rise in left ventricular end-diastolic pressure to pulmonary edema, one step per line and in order. Beside the step where interstitial fluid appears, write which two forces are now out of balance and which drainage route has been exceeded. Beside the step where lying flat matters, write how much blood shifts into the chest and why a failing ventricle turns that shift into higher capillary pressure rather than more stroke volume. Then write what two things a long history of mitral stenosis changes about where the alveolar flooding step sits on this sequence, and why a patient one day after a large infarct floods at a lower pressure.

Rebuild from memory: From a failing left ventricle to orthopnea and PND

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C02-BD1

Right-sided failure from causes to organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C02 Activity: NPLEX-P1-CARD-18-02-C02-BD1 Date: _____

Write out right-sided heart failure from memory in four blocks. First, list the causes in order, starting with the most common one, and separate the ones that arise in the lungs or pulmonary vessels from the ones that do not, naming the acute and the chronic version of each. Second, write how the neck veins are used as a manometer: the vein used, the patient position in degrees, the normal mean right atrial pressure, the height above the sternal angle that counts as raised, and what a sustained response to abdominal pressure means and how long that pressure is held. Third, write the two requirements for dependent edema, why weight gain comes before visible swelling, where the fluid settles in a walking patient and a bedbound one, and why it pits. Fourth, draw the hepatic acinus and label the zone that atrophies and the zone that becomes fatty, then write what the liver looks like grossly, what low output added to congestion produces, and what years of congestion produce.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C02-BD2

Three patients with a raised JVP

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C02 Activity: NPLEX-P1-CARD-18-02-C02-BD2 Date: _____

Three patients each have a jugular venous pulsation 9 cm above the sternal angle. Patient 1 has basal crackles, an S3, and a displaced apex beat. Patient 2 has clear lungs, long-standing chronic obstructive pulmonary disease, and a loud pulmonary component of the second heart sound. Patient 3 has clear lungs, ST elevation in leads II, III, and aVF, and a blood pressure of 82/56 mm Hg. For each patient write which ventricle is failing, where the pressure load or muscle loss sits, and whether the lungs are congested, and explain the lung finding from the position of the lesion in the two circulations. For patient 2 write the pulmonary mechanism that raised the pressure and what the arterioles have done over time. For patient 3 write what has happened to left ventricular filling and give the two reasons, then explain why giving a large volume load helps this patient while it harms patient 1.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C03-BD1

Two failing ventricles, two sounds, one peptide

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C03 Activity: NPLEX-P1-CARD-18-02-C03-BD1 Date: _____

Write the formula for ejection fraction and its normal range, then the three ejection fraction categories with their cutoffs. For the reduced type, write what has happened to the end-systolic pressure-volume relationship, which direction sarcomeres are added, and use the law of Laplace, stated as a formula, to explain the wall stress, the oxygen demand, and the leaking valve that follow. For the preserved type, write what active relaxation needs and the pump it uses, then list what raises passive stiffness, what happens to the end-diastolic pressure-volume relationship, and which patients have it. Next, describe the S3 and the S4: the timing in diastole, the pitch, where each is heard, what is happening in the ventricle at that moment, which failure type each accompanies, when an S3 is normal, and which rhythm abolishes the S4. Finish with BNP: write the precursor and the two fragments, which receptor and second messenger the active one uses, how each fragment is cleared, the rule-out values for both, and four states that raise the level and one that lowers it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C03-BD2

Why the numbers can look reassuring

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C03 Activity: NPLEX-P1-CARD-18-02-C03-BD2 Date: _____

A woman of 78 with long-standing hypertension and a body mass index of 38 arrives with pulmonary edema that began when her heart rate rose to 150 per minute in atrial fibrillation. Her ejection fraction is 62%, her chest radiograph shows a cardiothoracic ratio of 0.46, and her BNP is modestly raised. Write which failure type this is and defend it from the history and the geometry of her ventricle. Then explain, one at a time, why a normal ejection fraction does not exclude failure here, why her BNP is lower than you would expect for her filling pressure and name both the wall stress reason and the body habitus reason, and why her chest radiograph shows no enlarged heart. Next write why the arrhythmia alone tipped her into edema, naming the two things a fast atrial fibrillation takes away from a stiff ventricle. Finish by writing what her radiograph would show in order as her wedge pressure climbed from normal to above 25 mmHg, with the pressure band beside each sign, and write why those signs would appear late if her failure had been building for years.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-02-C03-CP01

Reduced against preserved ejection fraction

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	HFrEF (systolic)	HFpEF (diastolic)
Ejection fraction cutoff		
Core defect in the muscle		
Pressure-volume loop change		
Ventricular size, wall thickness, and direction of sarcomere addition		
Typical causes		
Extra heart sound and what produces it		
How much the ventricle depends on atrial contraction		
Why the filling pressure rises in each		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C04-BD1

Lay out the three compensatory systems

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-BD1 Date: _____

Write what the baroreceptors and kidney are actually sensing when they switch on their defenses in heart failure, and define the volume term that explains why they stay switched on while total extracellular volume is expanded. Then take the three systems one at a time. For the sympathetic nervous system, write what reduced baroreceptor firing does to outflow, then list the beta-1, alpha-1, and venous effects, and the harms that appear over months including what happens to the receptors themselves and the proteins that do it. For the renin-angiotensin-aldosterone system, write the three stimuli for renin release, then six actions of angiotensin II at its receptor, then what aldosterone does, in which cell, and at the expense of which two ions. For antidiuretic hormone, write which stimulus is driving it here, the receptor, the channel it inserts, and the electrolyte result. Finish by writing why retained volume adds little stroke volume in this setting, naming the curve and the part of it the ventricle is sitting on.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C04-BD2

Why blocking hormones works and squeezing harder does not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-BD2 Date: _____

Two patients with reduced ejection fraction are treated along different lines. One has beta-1 receptors, angiotensin II, and aldosterone blocked, and over months the ventricle becomes smaller and less spherical. The other is given a drug that raises contractility alone, and the ventricle keeps enlarging. Write what remodeling is, listing the changes in the myocytes, the genes they re-express, the cells that die and how, and the hormones that drive the fibrosis. Then explain from those mechanisms why the first course changes the course of the disease and the second does not, and write what a larger more spherical ventricle does to wall stress and to the papillary muscles, and why that makes the process self-sustaining. Next write where the natriuretic peptides fit: which chambers release which peptide, the receptor and second messenger they work through, the three actions that oppose the other systems, and the three reasons they cannot win. Finish by predicting the serum sodium and the urine sodium of a patient with advanced disease and marked edema, with the mechanism for each in one line.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-02-C04-RT01

Rebuild the neurohormonal cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-RT01 Date: _____

Draw the neurohormonal cycle of heart failure as a closed loop with seven steps, numbering each step and writing it out in words so the loop returns to its starting point. Mark on the loop the one step that is genuinely useful in a short crisis and the two steps where the response begins to harm the heart. Beside the step on vasoconstriction and retention, write which variable of ventricular work each one raises. Beside the remodeling step, write the three hormones responsible and the tissue change each one causes. Then write the single sentence that explains why a response built for hemorrhage damages a failing pump, and write where on your loop a drug that blocks an angiotensin II receptor and a drug that blocks a mineralocorticoid receptor each interrupt it.

Rebuild from memory: The neurohormonal cycle of heart failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-02-C04-PR01

Predict what six months of neurohormonal drive does

A man with an ejection fraction of 25% after an infarction has had six months of untreated sympathetic, renin-angiotensin-aldosterone, and antidiuretic hormone activation. Predict each item below, write the direction of the change, and give the hormone, receptor, or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to systemic vascular resistance and so to the work the ventricle does against it		
What happens to preload and to the venous pressure behind both ventricles		
What happens to the number of beta-1 receptors on the myocyte surface and how		
What happens to serum potassium and to serum sodium		
What happens to the interstitial collagen content of the myocardium		
What happens to ventricular size and shape, and what that does to wall stress		
What happens to the ability of natriuretic peptides to offset all of the above		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C05-BD1

Organise the complications by their cause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C05 Activity: NPLEX-P1-CARD-18-02-C05-BD1 Date: _____

Write the three headings that complications of heart failure fall under, then place each complication below under its heading and write its mechanism. Cover acute pulmonary edema with its five triggers and what the sympathetic surge adds, cardiogenic shock with its three numerical criteria and the fraction of ventricle classically lost, the atrial and the ventricular arrhythmias with the membrane changes behind each, mural thrombi with the three parts of the triad named and the sites they form and travel to, cardiorenal syndrome with both the arterial and the venous mechanism, and cardiac cachexia with its three contributors. Then write the three-number haemodynamic profile that separates cardiogenic shock from hypovolaemic shock and from early septic shock, using output, wedge pressure, and systemic vascular resistance for all three.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C05-BD2

Four deteriorations in one patient

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C05 Activity: NPLEX-P1-CARD-18-02-C05-BD2 Date: _____

A man with an ejection fraction of 20% deteriorates on four separate occasions. On the first he develops severe breathlessness with frothy pink sputum and diffuse crackles over two hours. On the second he is found to have a wedge pressure of 10 mmHg with the same lung picture. On the third his creatinine rises while his jugular venous pressure is markedly raised and his blood pressure is 96/60 mm Hg. On the fourth he has a sudden right-sided weakness. For each occasion write what has happened and the mechanism, and for the second write what the normal wedge pressure tells you about where the fluid is coming from. For the third, write the two separate mechanisms by which his kidney is being injured and explain how a raised venous pressure lowers filtration using the pressures inside the capsule. For the fourth, write where the material formed and why, naming the three conditions that allowed it and the appearance it would show on section. Finish by writing why his weight has not risen despite progressive muscle loss.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-02-C05-CA01

Cold, wet, and oliguric three days after an infarction

A 64-year-old man is in hospital three days after an anterior myocardial infarction that was not reperfused. Over six hours he becomes confused and breathless. Temperature is 35.8 C (96.4 F), pulse is 118/min, respirations are 30/min, and blood pressure is 84/58 mm Hg. His hands and feet are cold and mottled, his skin is clammy, the jugular venous pulsation is 7 cm above the sternal angle, and there are crackles over both lower and mid zones with an S3 at the apex. Urine output has been 12 mL per hour for four hours. A pulmonary artery catheter shows a cardiac index of 1.8 L/min/m2, a pulmonary capillary wedge pressure of 26 mmHg, and a systemic vascular resistance well above normal. Mixed venous oxygen saturation is 44%, serum lactate is raised, creatinine has risen from 94 to 168 micromol/L, and serum sodium is 129 mmol/L.

1. Recognize. Name the complication of heart failure this patient has developed and state which of its numerical criteria his measurements meet.

2. Explain. Explain the mechanism of the cold mottled skin, the crackles with the wedge pressure of 26 mmHg, the mixed venous saturation of 44%, and the raised lactate, taking each one back to either low forward output or high filling pressure.

3. Differentiate. Write how his three measured numbers would differ in hypovolaemic shock and in early septic shock, using cardiac output, wedge pressure, and systemic vascular resistance for each.

4. Investigate. Write which further findings you would look for to explain the falling urine output and the serum sodium of 129 mmol/L, and what each finding would show.

5. Decide. State which single mechanism makes this state self-reinforcing once the blood pressure has fallen, and name the organ whose perfusion depends on that pressure.

6. Justify. Justify that answer from coronary perfusion and from the proportion of left ventricle that must be lost for this complication to appear, and explain why the sympathetic surge that accompanies it adds to the problem rather than solving it.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C06-BD1

How a high output still leaves a congested patient

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C06 Activity: NPLEX-P1-CARD-18-02-C06-BD1 Date: _____

Define high-output heart failure and name its defining haemodynamic feature. Write the relationship between mean arterial pressure, cardiac output, and systemic vascular resistance as a formula, then use it to explain in four or five steps why a fall in resistance ends in congestion and edema, naming the receptors and hormones that are activated and what they retain. Then work the fistula numbers from memory: starting from a mean arterial pressure of about 93 mmHg and an output of 5 L/min, calculate the resistance in Wood units, show what the resistance falls to when a large fistula is added, and calculate the output needed to hold the pressure near 90 mmHg. Write why the tissues gain little from that extra flow, what the venous blood looks like, and what happens to the heart rate when the fistula is compressed and why. Finish by listing five causes of high-output failure with the mechanism for each, and for the thiamine one name the three enzymes that lose their cofactor.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C06-BD2

Sort two congested patients by their resistance

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C06 Activity: NPLEX-P1-CARD-18-02-C06-BD2 Date: _____

Two patients have orthopnea, raised jugular venous pressure, ankle edema, and an S3. Patient 1 has cool pale hands, a weak pulse with a narrow pulse pressure, and a mixed venous oxygen saturation of 42%. Patient 2 has warm flushed hands, a bounding pulse with a wide pulse pressure, and a mixed venous oxygen saturation of 78%. Write which has the low output and which the high output, then explain each of the three paired findings from the resistance, naming what the arteriovenous oxygen difference is doing in each patient and why. Write why both patients have the same congestive findings even though their outputs are opposite. Then, for patient 2, write the questions you would ask and the findings you would look for to separate severe anemia, thyrotoxicosis, an arteriovenous fistula, wet beriberi, and extensive Paget disease, giving one discriminating clue for each. Finish by explaining why a patient whose heart was already diseased reaches congestion sooner than a patient whose heart was normal when the resistance fell.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C07-BD1

Rupture to shock, and why the murmur can be quiet

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C07 Activity: NPLEX-P1-CARD-18-02-C07-BD1 Date: _____

Draw the mitral apparatus and label the leaflets, the chordae, and both papillary muscles with the arteries that supply each. Write the causes of chordal rupture and the causes of papillary muscle rupture, and for the infarct-related cause give the number of days after the infarct, the reason the tissue is weakest then, which papillary muscle ruptures and with which infarct, and why that muscle is the vulnerable one. Then rebuild the six-step sequence from rupture to cardiogenic shock in order, and beside the atrial step write what the pressure tracing shows and what makes that atrium unable to absorb the volume. Write why the murmur may be early systolic, soft, or absent, and why the heart looks normal in size on the radiograph. Finish by writing the two findings that separate ventricular septal rupture from this lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C07-BD2

Work the volumes behind a normal ejection fraction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C07 Activity: NPLEX-P1-CARD-18-02-C07-BD2 Date: _____

A patient with acute mitral regurgitation is in shock with an ejection fraction of 75%. Starting from end-diastolic volume 120 mL and end-systolic volume 30 mL, calculate total stroke volume, ejection fraction, forward stroke volume if 50 mL enters the left atrium each beat, the regurgitant fraction, and forward cardiac output at a heart rate of 100 per minute. Show your working and define regurgitant fraction in one line. Then write why the ejection fraction rose after the rupture even though the patient is in shock, and explain what the ejection fraction is measuring that forward output is not. Next write where those 50 mL per beat go and what they do to the wedge pressure. Finish by writing what the same regurgitant volume would do in a patient who had carried mitral regurgitation for fifteen years, naming the two structural adaptations that patient has and this one lacks, and explain from compliance why one gets sudden pulmonary edema and the other does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-02-C02-Q01

Hypotension with a raised JVP and clear lungs

A 59-year-old man comes to the emergency department because of 4 hours of chest pain. Pulse is 54/min and blood pressure is 80/54 mm Hg. The jugular venous pulsation is 10 cm above the sternal angle and rises further when pressure is applied to the abdomen. The lung fields are clear on auscultation and on chest radiograph, and there is no S3. The electrocardiogram shows ST elevation in leads II, III, and aVF. Which of the following best explains the hypotension in this patient?

A. Alveolar flooding from a rapid rise in pulmonary capillary pressure

B. Reduced left ventricular filling because right ventricular output has fallen and the septum has shifted leftward

C. A fall in systemic vascular resistance from arteriolar vasodilation

D. Acute regurgitation of blood into a small, noncompliant left atrium

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-02-C06-Q02

Congestion with warm extremities on dialysis

A 47-year-old woman receiving hemodialysis through a large upper arm arteriovenous fistula comes to the physician because of 3 weeks of breathlessness on lying flat and swelling of both ankles. Temperature is 36.8 C (98.2 F), pulse is 104/min, and blood pressure is 142/58 mm Hg. Her hands are warm, the pulse is bounding, the jugular venous pulsation is raised, and an S3 is present at the apex. Cardiac output is 9 L/min and mixed venous oxygen saturation is 79%. Which of the following best explains her mixed venous oxygen saturation?

A. A large share of the cardiac output returns to the heart without passing through capillaries

B. Tissues are extracting more oxygen per unit of blood delivered

C. Pulmonary edema is limiting oxygen uptake across the alveolar membrane

D. Reduced hemoglobin concentration has lowered the oxygen carrying capacity of blood

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-C Ischemic Heart Disease

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-03-C01				
NPLEX-P1-CARD-18-03-C02				
NPLEX-P1-CARD-18-03-C03				
NPLEX-P1-CARD-18-03-C04				
NPLEX-P1-CARD-18-03-C05				
NPLEX-P1-CARD-18-03-C06				
NPLEX-P1-CARD-18-03-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C01-BD1

Myocardial oxygen supply and demand from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C01 Activity: NPLEX-P1-CARD-18-03-C01-BD1 Date: _____

Without looking at your notes, divide a page into supply and demand. On the supply side write the resting coronary flow in mL/min and as a share of cardiac output, how much of the oxygen the myocardium extracts at rest, how many fold flow can rise in heavy exercise, and the local signals that dilate the resistance arterioles. Write the formula for the driving pressure that perfuses the left ventricle and name the phase of the cardiac cycle in which that perfusion happens. On the demand side list the three determinants and write the relationship that sets wall stress from pressure, radius, and wall thickness. Then write three sentences explaining why the subendocardium becomes ischemic before the subepicardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C01-BD2

Four patients with chest pain and clean coronary arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C01 Activity: NPLEX-P1-CARD-18-03-C01-BD2 Date: _____

Four patients have exertional chest pain, and coronary angiography shows no significant stenosis in any of them. Patient 1 has severe aortic stenosis with left ventricular hypertrophy. Patient 2 has a hemoglobin of 6.5 g/dL (reference range 12.0 to 16.0 g/dL). Patient 3 has untreated hyperthyroidism with a pulse of 136/min. Patient 4 used cocaine two hours before the pain started. For each patient write which determinants of oxygen balance are disturbed, whether the problem is on the supply side, the demand side, or both, and the mechanism in two or three sentences. Then write which layer of the ventricular wall becomes ischemic in patients 1 to 3 and why, and say which patient can instead develop ischemia through the whole thickness of the wall and what makes that possible.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-03-C01-PR01

Predict what happens when demand rises on a fixed stenosis

A man has a fixed 75% stenosis of the left anterior descending artery and has been free of symptoms at rest. He develops pneumonia with a temperature of 39.2 degrees Celsius (102.6 degrees Fahrenheit), a pulse of 134/min, a blood pressure of 86/50 mm Hg, and a hemoglobin of 7.2 g/dL (reference range 13.5 to 17.5 g/dL). For each item below write the direction of the change and the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the oxygen content of each mL of coronary blood		
What happens to the time available in each cardiac cycle for left ventricular perfusion		
What happens to the pressure gradient that drives subendocardial flow		
What happens to the oxygen used per minute by the myocardium		
What happens to the arteriolar reserve distal to the stenosis		
Which layer of the wall becomes ischemic first and what the ST segment does in the leads facing it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C02-BD1

Stable angina and the moving ST segment

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-BD1 Date: _____

Write out what angina pectoris is and why the pain is felt in the chest, left arm, neck, and jaw, naming the chemicals released by ischemic myocytes, the nerves they travel in, and the spinal levels involved. Then write the stenosis severity that produces exertional angina, state how resistance changes when the radius of a vessel is halved and name the relationship that gives that result, and explain in three or four sentences why flow is normal at rest but cannot rise with exertion. Explain why nitroglycerin relieves the pain, naming the second messenger and which vessels dilate most. Finish with the reason the ST segment is normally flat, and write why subendocardial ischemia depresses it while ischemia through the whole wall thickness elevates it, and which of the two localizes the artery.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C02-BD2

Sort three episodes of chest pain by mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-BD2 Date: _____

Three patients come to the emergency department. Patient 1 has 5 minutes of chest pressure that comes on after two blocks of walking, eases with rest, and is accompanied by ST depression in V4 to V6 during the pain. Patient 2 has 25 minutes of chest pressure at rest that is more severe than any previous episode, with ST depression and T wave inversion, and serial troponin stays below the upper reference limit. Patient 3 is a 38-year-old smoker with chest pain at 3 in the morning, transient ST elevation in II, III, and aVF during the pain, and no stenosis on angiography. For each patient name the condition, the lesion in the artery, the layer of the wall that is ischemic, and the expected troponin result, and explain in two or three sentences how the lesion produces that ECG pattern. Then write what single change would make patient 2's diagnosis a non-ST-elevation myocardial infarction, and name the agents other than atherosclerosis that can provoke patient 3's mechanism.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-03-C02-RT01

Rebuild the ischemic cascade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-RT01 Date: _____

Rebuild the six steps of the ischemic cascade from the moment demand exceeds supply to the moment the patient feels pain, one step per line and in order. Beside the step where relaxation slows, write which pump needs ATP and what that pump is doing. Beside the step where the ST segment moves, write what is different about the membrane potential of ischemic cells and what that difference generates between ischemic and normal muscle. Then write one sentence on what the position of pain in this order means for a patient with diabetic autonomic neuropathy, and one sentence on what reverses the whole cascade when no myocytes have died.

Rebuild from memory: The ischemic cascade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-03-C02-CP01

Stable, unstable, and Prinzmetal angina

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Stable angina	Unstable angina	Prinzmetal angina
Lesion in the artery			
Pattern and timing of the pain			
Layer of the ventricular wall that is ischemic			
ECG during the pain			
Troponin result			
Typical patient and main provoking factors			
Main risk if it is left alone			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C03-BD1

From a ruptured cap to a dead myocyte

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-BD1 Date: _____

Write the sequence that runs from rupture of a fibrous cap to transmural necrosis, one step per line, naming the material exposed in the plaque, the protein that binds platelet glycoprotein Ib, the two agonists that recruit more platelets, the receptor that binds fibrinogen, and the factor that drives thrombin and fibrin. Give the time from occlusion to loss of contraction, the time to irreversible injury, and the time for necrosis to reach the epicardium. Then write what is happening inside the myocyte at each stage: which pathway makes ATP once oxygen is gone and what accumulates, which pump fails and what enters the cell, which ion floods in and which enzymes it activates, and the two structural changes that mark the point of no return. Finish by naming the three proteins that leak into blood and the time of day and the two triggers that make rupture more likely.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C03-BD2

Two infarcts and two reperfused hearts

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-BD2 Date: _____

A man has a complete occlusion of the left anterior descending artery that stays occluded for 8 hours. A woman has 30 minutes of partial occlusion of the same artery that then opens. For each one write how far the necrosis extends through the wall, what the ST segment does, whether Q waves are expected, what happens to troponin, and which label fits the electrical picture. Then write which of the two can cause pericarditis and rupture of the free wall and why the other rarely does. Next, consider two hearts reperfused at 3 hours. Write what the infarct looks like on gross examination, name the change seen in the hypercontracted myocytes and what makes them hypercontract, and explain how reactive oxygen species and calcium opening the mitochondrial permeability transition pore kill cells that might have lived. Finish with the one situation in which a subendocardial infarct runs right around the ventricle instead of following one artery, and say why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-03-C03-RT01

Rebuild the path from plaque rupture to transmural infarction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-RT01 Date: _____

Draw a cross section of a coronary artery with a disrupted plaque, then rebuild the seven-step sequence from cap rupture to transmural necrosis beside it, numbering each step on the drawing. Mark on the drawing where von Willebrand factor acts, where the glycoprotein IIb/IIIa receptor acts, and where tissue factor comes from. Underneath, draw the ventricular wall as three layers and shade how far the necrosis has spread at 30 minutes, at 2 hours, and at 6 hours, writing the layer that dies first beside the drawing. Then write the two things that can stop the wavefront before it reaches the epicardium.

Rebuild from memory: From plaque rupture to transmural infarction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C04-BD1

Build the coronary territory table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C04 Activity: NPLEX-P1-CARD-18-03-C04-BD1 Date: _____

On blank paper build a table of infarct territories with one row each for anterior, lateral, inferior, right ventricular, and posterior. Use the columns artery, leads with ST elevation, and associated findings, and fill every cell. Write beside the table which wall and which part of the septum each of the three main arteries supplies, which artery usually supplies the sinoatrial and atrioventricular nodes, where the posterior descending artery arises and in what share of hearts, and which artery is occluded most often with its approximate frequency. Then write what reciprocal ST depression tells you and the condition it helps you rule out. Finish by writing why a pathologic Q wave appears over dead muscle and give the width and height criteria from the notes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C04-BD2

Read the biomarkers across four days

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C04 Activity: NPLEX-P1-CARD-18-03-C04-BD2 Date: _____

Draw one set of axes with time from 0 to 10 days on the horizontal axis and plasma concentration on the vertical axis, and sketch the curves for troponin, the MB isoenzyme of creatine kinase, and myoglobin after an infarction. Label each curve with its first rise, its peak, and the time it returns to normal. Write why troponin stays high so much longer than the MB isoenzyme, naming the two intracellular pools of troponin and where the MB isoenzyme sits in the cell. Then a patient who had an anterior infarct four days ago has chest pain again. Write which marker can show new necrosis and which cannot, and why. Write what you would conclude if that marker is normal and the pain is sharp, pleuritic, and worse lying flat. Finish by listing five conditions other than infarction that raise troponin and write the two things the definition of infarction requires beyond a single high value.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-03-C04-Q01

Hypotension with clear lungs after an inferior infarction

A 61-year-old man comes to the emergency department because of 2 hours of chest pressure and nausea. Temperature is 36.8 degrees Celsius (98.2 degrees Fahrenheit), pulse is 44/min, respirations are 18/min, and blood pressure is 82/54 mm Hg. Jugular venous pressure is increased and the lungs are clear on auscultation. The electrocardiogram shows ST elevation in leads II, III, and aVF with ST depression in I and aVL, and additional ST elevation in lead V4R. Which of the following best explains the hypotension with clear lung fields in this patient?

A. Acute mitral regurgitation from rupture of the posteromedial papillary muscle

B. Blood in the pericardial sac restricting ventricular filling

C. Right ventricular infarction limiting delivery of blood to the left heart

D. Loss of more than 40% of left ventricular muscle with pulmonary congestion

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C05-BD1

Build the healing timeline from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C05 Activity: NPLEX-P1-CARD-18-03-C05-BD1 Date: _____

On blank paper build the morphologic timeline of an infarct with one row for each of these intervals: 0 to 30 minutes, 30 minutes to 4 hours, 4 to 24 hours, 1 to 3 days, 3 to 7 days, 7 to 10 days, 10 to 14 days, 2 to 8 weeks, and over 2 months. Use the columns gross appearance, light microscopy, and the complication linked to that stage, and fill every cell. Write beside the table why dead myocytes keep their outlines for several days, naming what ischemia and acidosis do to structural proteins and to lytic enzymes, and why adult myocardium is replaced by scar instead of new muscle. Then write the direction healing travels through an infarct and what that means for the center of a large one. Finish with three early details: why wavy fibers appear at the border within hours, what triphenyltetrazolium chloride stains and after how long an infarct looks pale with it, and when neutrophils arrive and what draws them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C05-BD2

Date four infarcts from what the tissue shows

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C05 Activity: NPLEX-P1-CARD-18-03-C05-BD2 Date: _____

Four hearts come to autopsy. Heart 1 shows a yellow-tan softened center with a hyperemic border, and microscopy shows disintegrating myocytes with macrophages at the edge. Heart 2 shows no gross change, and microscopy shows complete coagulative necrosis with loss of nuclei and dense neutrophils. Heart 3 shows a depressed red-gray border, and microscopy shows new vessels and collagen among fibroblasts. Heart 4 shows a firm white area, and microscopy shows dense collagen with few cells. For each heart give the age of the infarct as a range, name the stage of healing, and write the complication that heart is most at risk of, with the mechanism in one sentence. Then write which of the four hearts has the weakest wall and explain from the balance between removal of dead muscle and deposition of collagen why that stage is the weakest. Finish by writing what a pale unstained zone on a tetrazolium-stained heart tells you when light microscopy of the same tissue looks normal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C06-BD1

Complications laid out on a time line

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C06 Activity: NPLEX-P1-CARD-18-03-C06-BD1 Date: _____

Draw a horizontal time line from the moment of occlusion out to several months and place every complication from the notes on it at the time it occurs: ventricular fibrillation, bradycardia and atrioventricular block, pump failure and cardiogenic shock, right ventricular infarction, fibrinous pericarditis, free wall rupture, septal rupture, papillary muscle rupture, mural thrombus, true aneurysm, Dressler syndrome, and scar-related ventricular tachycardia. Beside the arrhythmia entries write the three electrical changes in ischemic myocytes that set up reentry and the four other factors that add to the instability. Write how much of the left ventricle must be lost for cardiogenic shock, and write why an inferior infarct slows the heart while bundle branch block from an anterior septal infarct carries a worse outlook. Then rebuild the six-step cycle of cardiogenic shock after a large infarction in order.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C06-BD2

Four patients deteriorate after an infarction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C06 Activity: NPLEX-P1-CARD-18-03-C06-BD2 Date: _____

Four patients deteriorate in hospital. Patient 1, four days after an inferior infarct, develops sudden pulmonary edema and hypotension with a soft apical murmur. Patient 2, five days after an anterior infarct, develops a new loud holosystolic murmur at the left lower sternal border and the oxygen saturation of blood sampled from the right ventricle is higher than expected. Patient 3, four days after an anterolateral infarct, suddenly loses her pulse with distended neck veins and muffled heart sounds while the monitor still shows organized electrical activity. Patient 4, two days after a transmural infarct, has sharp pain that is worse lying flat and a scratching sound over the precordium. For each patient name the complication, the anatomy or tissue that failed, and why that failure happens at that point in the healing sequence. For patient 1 write why one papillary muscle ruptures more often than the other, naming the arteries involved. Then write what separates a pseudoaneurysm from a true aneurysm in wall composition, neck shape, timing, and risk of rupture, and write how Dressler syndrome differs in timing and mechanism from patient 4's problem.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-03-C06-CA01

Sudden breathlessness on the fifth day after an inferior infarction

A 67-year-old woman is in hospital on the fifth day after an inferior myocardial infarction. The admission electrocardiogram showed ST elevation in leads II, III, and aVF, and troponin peaked on the first day and has been falling since. While sitting up for breakfast she becomes abruptly breathless. Temperature is 37.1 degrees Celsius (98.8 degrees Fahrenheit), pulse is 118/min, respirations are 32/min, and blood pressure is 78/46 mm Hg. There are crackles throughout both lung fields, the jugular venous pressure is not raised, and a soft apical systolic murmur is heard. The electrocardiogram shows sinus tachycardia with the inferior Q waves present since admission and no new ST elevation. A repeat troponin is higher than the value from the previous morning but lower than the first day peak.

1. Recognize. Name the structural complication and the valve lesion it has produced, and name the exact structure that has failed.

2. Explain. Explain why this structure fails on the fifth day, using the healing sequence and the strength of the tissue at that point.

3. Differentiate. Write how this differs from free wall rupture, from septal rupture, and from reinfarction, naming the finding in this patient that separates each one.

4. Investigate. Write which findings you would look for to confirm the lesion and what each would show, including why the murmur can be soft rather than loud.

5. Decide. State which papillary muscle has failed and why that one rather than the other.

6. Justify. Justify the answer from the blood supply of the two papillary muscles and from the artery that supplied this infarct, and explain the rising troponin without new ST elevation.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-03-C06-Q02

Timing of free wall rupture

A 58-year-old woman who had her first myocardial infarction 5 days ago suddenly loses consciousness. The monitor shows organized electrical complexes at a rate of 96/min but no pulse is palpable. The jugular veins are distended, the heart sounds are muffled, and she dies despite resuscitation. Autopsy shows a transmural anterolateral infarct with a tear through the free wall and 400 mL of blood in the pericardial sac; the heart is not hypertrophied and no collateral vessels are seen. Which of the following best explains why the wall tore on the fifth day?

A. Macrophages have removed the necrotic myocytes and collagen deposition has not yet strengthened the wall

B. Neutrophil elastase is digesting the fibrous cap of the culprit coronary plaque

C. Granulation tissue has matured into a thin fibrous scar that bulges during systole

D. Dense collagen in the healed infarct conducts slowly and sets up a reentrant circuit

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C07-BD1

The chronically ischemic heart and its remodeling cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C07 Activity: NPLEX-P1-CARD-18-03-C07-BD1 Date: _____

Write what chronic ischemic heart disease is, what usually causes it, and which form of heart failure it is one of the commonest causes of. Then describe the heart as it would appear at autopsy: size and weight, the state of the left ventricular chamber and wall, what the scars look like and where they sit, the state of the coronary arteries, and the two findings on the endocardium. Write the three microscopic findings, including the change in subendocardial myocytes that marks chronic sublethal ischemia. Write the one feature that separates this from other dilated cardiomyopathies and what the coronary arteries look like in those. Then rebuild the six-step remodeling cycle in order, and write beside it which hormone systems are activated, which two of their mediators promote fibrosis, how sarcomeres are added in eccentric hypertrophy, and what happens to the shape of the chamber.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C07-BD2

Work the cost of dilation and sort recoverable muscle from scar

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C07 Activity: NPLEX-P1-CARD-18-03-C07-BD2 Date: _____

Take a ventricle generating 120 mm Hg with a radius of 2.5 cm and a wall 1.0 cm thick, and a remodeled ventricle generating the same 120 mm Hg with a radius of 3.5 cm and an average wall thinned to 0.8 cm. Write the relationship for wall stress, calculate both values, show your working, give the ratio, and write in two sentences what that ratio means for oxygen demand in a heart whose arteries are already narrowed and why hypertrophy only partly helps. Next, a patient with an ejection fraction of 25% has three regions that contract poorly. Write how hibernating myocardium, stunned myocardium, and scar differ in viability, in the flow they are receiving, and in whether function can return, and say which one restoring flow cannot help. Then write how dilation produces mitral regurgitation without any disease of the valve itself and what that regurgitation adds to the ventricle. Finish by writing why slow conduction at scar borders produces monomorphic ventricular tachycardia, and name the two findings on examination and the one on the electrocardiogram that follow from a dilated ventricle with old infarcts.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-D Valvular Heart Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-04-C01				
NPLEX-P1-CARD-18-04-C02				
NPLEX-P1-CARD-18-04-C03				
NPLEX-P1-CARD-18-04-C04				
NPLEX-P1-CARD-18-04-C05				
NPLEX-P1-CARD-18-04-C06				
NPLEX-P1-CARD-18-04-C07				
NPLEX-P1-CARD-18-04-C08				
NPLEX-P1-CARD-18-04-C09				
NPLEX-P1-CARD-18-04-C10				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C01-BD1

Rebuild calcific aortic stenosis and the wall it builds

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C01 Activity: NPLEX-P1-CARD-18-04-C01-BD1 Date: _____

Without looking at your notes, write the normal aortic valve area and the area at which symptoms usually begin. Then write out the steps of age-related calcific aortic stenosis from the first endothelial injury to the heaped calcified masses, naming what enters the cusp, which cell changes its behaviour, and what that cell starts to deposit. State what the serum calcium is in this process and what the commissures look like. Beside that, write the two other causes of aortic stenosis, the typical age of each, and the one feature that identifies the rheumatic valve. Next, write the law of Laplace as a relationship between wall stress, pressure, radius, and wall thickness, and use it to explain why the ventricle thickens around a normal cavity rather than dilating. Name that pattern of hypertrophy, write what happens to ejection fraction for the first years, and write which heart sound appears once the wall is stiff and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C01-BD2

Work a Doppler study back to the symptoms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C01 Activity: NPLEX-P1-CARD-18-04-C01-BD2 Date: _____

A woman of 76 years has an aortic jet velocity of 4.6 m/s, a valve area of 0.8 cm2, and an arterial pressure of 128/76 mm Hg. Convert the jet velocity to a peak gradient using the simplified Bernoulli relationship, show your working, and state the pressure the left ventricle must generate. Using Laplace with an unchanged cavity radius, calculate how thick a wall that began at 1.0 cm must become to hold wall stress steady, and show the ratio you used. State whether this study meets the thresholds for severe disease and give each threshold. Then explain, in two or three sentences each, why she has angina even though her coronary arteries are clean, why her syncope comes only on exertion, and why she is breathless. Write what her carotid upstroke and apical impulse feel like and why. Finish by writing what happens to the loudness of her murmur if the ventricle begins to fail, and why loudness is a poor guide to severity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C01-Q01

Chest pain with clean coronary arteries

A 74-year-old man comes to the physician because of 6 months of chest tightness on climbing stairs and one episode of collapse while running. Temperature is 36.8 °C (98.2 °F), pulse is 72/min, and blood pressure is 126/80 mm Hg. The carotid upstroke is weak and late and the apical impulse is sustained. A crescendo-decrescendo systolic murmur is loudest in the right second intercostal space and radiates to both carotids, A2 is soft, and an S4 is present. Coronary angiography shows no epicardial stenosis. Echocardiography shows a valve area of 0.8 cm2, a mean gradient of 48 mm Hg, and a left ventricular wall thickness of 1.6 cm with an ejection fraction of 60%. Which of the following best explains this patient's chest pain?

A. Thrombotic occlusion of a coronary artery over a ruptured atherosclerotic plaque

B. Embolism of a calcified valve fragment into a coronary artery

C. High oxygen demand in a thick wall whose raised diastolic pressure compresses subendocardial vessels while prolonged systole shortens diastolic perfusion time

D. Reduced oxygen carrying capacity from loss of large von Willebrand factor multimers

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C02-BD1

Causes of aortic regurgitation and the wide pulse pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C02 Activity: NPLEX-P1-CARD-18-04-C02-BD1 Date: _____

Divide the causes of aortic regurgitation into root disease and leaflet disease and list every cause the notes gave under each heading. For Marfan syndrome, name the mutated protein, the growth factor whose activity rises, and the name given to the change in the aortic media. For tertiary syphilis, write which vessels are inflamed, what that does to the media, and what the intima looks like. Then write what the left ventricle receives in diastole in chronic aortic regurgitation, which mechanism raises total stroke volume, and name and describe the pattern of hypertrophy that follows. Write why diastolic pressure in the ventricle stays near normal for years. Next, give a representative blood pressure in chronic aortic regurgitation and write the two reasons diastolic arterial pressure falls. List the named peripheral signs from the notes and write the one shared finding that produces all of them. Finish by writing the murmur: its pitch, its timing, where it starts in the cardiac cycle and why, where and in which position it is heard best, and what handgrip does to it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C02-BD2

Two patients with a leaking aortic valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C02 Activity: NPLEX-P1-CARD-18-04-C02-BD2 Date: _____

Patient 1 has had a dilated aortic root for 15 years, feels well, and has a blood pressure of 172/44 mm Hg with bounding pulses and a long early diastolic murmur. Patient 2 presents within an hour of an ascending aortic dissection with a blood pressure of 92/68 mm Hg, pulmonary edema, and a short soft diastolic murmur. For each patient write the size and stiffness of the left ventricle, what happens to left ventricular diastolic pressure, the length of the murmur and why, and whether the pulse pressure is wide, explaining each answer from the compliance of the chamber receiving the regurgitant volume. Write why patient 2 can have severe regurgitation and almost no murmur, and what the rising ventricular pressure does to the mitral valve. Then write what a fall in heart rate does to the regurgitant volume per beat in each patient and why. Finish by writing how you would separate the apical rumble of an Austin Flint murmur from mitral stenosis at the bedside, naming the two findings mitral stenosis adds.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C03-BD1

Mitral stenosis: the chambers that pay and the sounds you hear

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-BD1 Date: _____

Write the normal mitral valve area, the area at which a gradient first appears, and the area that counts as severe. Name the usual cause and the two uncommon ones. Then write what happens to the size and pressure of the left atrium and to the size and diastolic pressure of the left ventricle, and write the one haemodynamic measurement that identifies mitral stenosis when compared with left ventricular diastolic pressure. Explain why hemoptysis occurs. Next, list from memory the findings of mitral stenosis with the mechanism of each: the character of S1, the extra diastolic sound and what makes it, what happens to the interval between S2 and that sound as the disease worsens and why, the murmur with its pitch, timing, where and in which position it is heard, which feature of the murmur tracks severity, the accentuation before systole and what abolishes it, and the two electrocardiographic patterns. Finish by writing the three structures an enlarged left atrium can press on and the symptom each produces.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C03-BD2

Why a faster heart rate floods the lungs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-BD2 Date: _____

A woman of 32 years with known mitral stenosis has been well for years. At 28 weeks of pregnancy she develops a fever of 38.6 °C (101.5 °F) and a heart rate that rises from 72/min to 124/min, and within hours she has pulmonary edema. Write the relationship between flow across a narrowed orifice, the orifice area, and the pressure difference across it, then explain why a rise in heart rate raises left atrial pressure so sharply, naming what tachycardia does to diastole. Estimate what happens to the gradient at the new heart rate and say what the notes give for a rise of this size. List every factor in this patient that pushed her over, and write what pregnancy itself contributed. Then write her left ventricular end-diastolic pressure and ventricular size and justify each from where the obstruction sits. Next, she develops atrial fibrillation. Write the two ways that worsens filling, where thrombus forms and why, and name four organs the emboli reach. Finish by writing how her picture differs from pulmonary edema caused by left ventricular failure, using the wedge pressure and the ventricular diastolic pressure to make the distinction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-04-C03-RT01

Rebuild the backward transmission of pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C03 Activity: NPLEX-P1-CARD-18-04-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from the narrowed mitral orifice to right heart failure, one step per line and in order. Beside the left atrial step, write what happens to the wall of the atrium and what happens to the left ventricle at the same time. Beside the pulmonary venous step, write the three symptoms that follow. Beside the pulmonary arterial step, write what the arterioles do first and what then changes in their walls. Beside the right ventricular step, name the valve lesion that appears and why. Beside the final step, write the three physical findings you expect. Then draw a vertical bar for every chamber and vessel from the pulmonary veins to the left ventricle and mark with an arrow which pressures are raised and which are not, and write one sentence on why the ventricle is the one chamber spared.

Rebuild from memory: Backward transmission of pressure in mitral stenosis

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C04-BD1

The mitral apparatus and how each part fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C04 Activity: NPLEX-P1-CARD-18-04-C04-BD1 Date: _____

Name the five parts of the apparatus that keep the mitral valve competent. Then separate the causes of mitral regurgitation into primary and secondary, list the causes the notes gave under each, and name the most common primary cause in developed countries. Write which papillary muscle ruptures most often after an infarction and the anatomical reason. For secondary regurgitation, write what a dilated ventricle does to the annulus and to the papillary muscles, and explain in two sentences why this form feeds itself. Next, write why backflow begins at the very onset of systole and what that does to isovolumetric contraction. Write which chambers carry the extra volume and when the regurgitant volume returns to the ventricle. Describe what happens to the size and compliance of the left atrium and the rhythm it often develops, then name the pattern of ventricular hypertrophy. Finish with the murmur and sounds: timing, pitch, where it is loudest, where it radiates, what handgrip and squatting do, what standing and Valsalva do, the character of S1, what happens to the split of S2, and which extra sound the loaded atrium produces and what it does and does not prove.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C04-BD2

Why a normal ejection fraction misleads

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C04 Activity: NPLEX-P1-CARD-18-04-C04-BD2 Date: _____

A man with chronic mitral regurgitation has an end-diastolic volume of 210 mL and an end-systolic volume of 75 mL, and 60% of his total stroke volume returns to the left atrium. Calculate his total stroke volume, his ejection fraction, his regurgitant volume, and his forward stroke volume, showing each step. Then repeat the calculation with contractility reduced so that end-systolic volume rises to 105 mL with the same filling and the same regurgitant fraction, and state the new ejection fraction and forward stroke volume. Write two or three sentences explaining why an ejection fraction a clinician might call low-normal already reports a failing ventricle in this lesion, naming the feature of the outlet that flatters it. Then compare this with aortic stenosis, where the ventricle ejects against a high pressure, and write what that difference does to the usefulness of ejection fraction. Finish by writing which single measurement in this patient would tell you how much blood is actually reaching his tissues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-04-C04-CP01

Chronic against acute regurgitation

Fill in every cell from memory for chronic and for acute regurgitation, covering both the aortic and the mitral lesion where the row allows it. Then check your table against the notes and correct any cell you missed.

Feature	Chronic regurgitation	Acute regurgitation
Size and compliance of the chamber receiving the backflow		
What happens to filling pressure, and which pressure you would name in aortic and in mitral disease		
Length and loudness of the murmur, named for the aortic and the mitral lesion		
Pulse pressure in aortic regurgitation		
How the patient presents and over what time		
Usual causes in aortic and in mitral disease		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-04-C04-CA01

Sudden breathlessness four days after an infarction

A 64-year-old man is on the cardiology ward on the fourth day after an inferior myocardial infarction that was treated late. He becomes acutely breathless over 20 minutes. Temperature is 36.9 °C (98.4 °F), pulse is 118/min, respirations are 32/min, and blood pressure is 84/60 mm Hg. Oxygen saturation is 86% on room air. There are coarse crackles throughout both lung fields. A short, soft systolic murmur is just audible at the apex. Chest radiography shows pulmonary edema with a normal cardiac silhouette. A pulmonary artery catheter shows a wedge pressure of 34 mm Hg with tall v waves. Echocardiography shows a left atrium of normal size, a left ventricle of normal size with inferior wall hypokinesis, and a flail segment of the posterior mitral leaflet.

1. Recognize. Name the valve lesion, state whether it is acute or chronic, and name the structure that has failed and the part of the apparatus it belongs to.

2. Explain. Explain why the pulmonary edema and the tall v waves appeared within minutes rather than over years, using the size and compliance of the receiving chamber.

3. Differentiate. Write how this presentation differs from chronic mitral regurgitation and from acute aortic regurgitation, naming the findings on examination and on the catheter tracing that separate them.

4. Investigate. Write which findings you would look for on the echocardiogram and the catheter tracing to confirm the lesion, and what each one would show.

5. Decide. State why the murmur is soft and short here even though the regurgitant volume is large, and what you would conclude about severity from its loudness.

6. Justify. Justify your reasoning from the vascular supply of the structure that ruptured, the territory of the infarct, and the pressure relationship between the left atrium and the left ventricle late in systole.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C05-BD1

How a vegetation forms and what the organism decides

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C05 Activity: NPLEX-P1-CARD-18-04-C05-BD1 Date: _____

Write what a vegetation is made of and explain in two sentences why organisms growing inside it survive an intact immune system. Then rebuild the six-step pathogenesis of infective endocarditis in order, one step per line, and write beside the first two steps which organism can skip them and what that allows it to do. List the conditions that expose collagen and tissue factor on a valve, and list the everyday events that deliver a transient bacteremia. Name the surface molecules by which viridans streptococci and Staphylococcus aureus attach. Next, build a two-column comparison of acute and subacute disease with a row each for the usual organism, the state of the valve before infection, the time course, the fever and systemic features, the size and destructiveness of the vegetation, and whether embolic or immune signs dominate, and write one sentence on why virulence sets the tempo. Finish by writing which valve is most often involved in native valve disease, which valve is most often involved in people who inject drugs and why, where right-sided emboli travel and what they look like, and four places left-sided emboli lodge.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C05-BD2

Read the peripheral signs and the laboratory panel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C05 Activity: NPLEX-P1-CARD-18-04-C05-BD2 Date: _____

A woman of 58 years has had 7 weeks of low-grade fever, fatigue, and weight loss. She has a repaired bicuspid aortic valve, painful red-purple nodules on three fingertips, painless flat red macules on both palms, retinal hemorrhages with pale centres, hematuria with red cell casts, an erythrocyte sedimentation rate of 86 mm/h, a hemoglobin of 9.4 g/dL, and a positive rheumatoid factor. Two sets of blood cultures drawn 8 hours apart grow the same organism. For each physical finding, name it, state whether it is painful, and write its mechanism, then write the single rule that lets you sort the painful from the painless signs. Explain why her urine has red cell casts and her complement is low, and why her prolonged illness favours this mechanism. Explain why her anemia developed and name the hormone that drives it. Write why both culture sets grew the same organism and what that tells you about the vegetation. Name the organism group you expect from this tempo and this valve. Finish by listing which of her findings count as major and which as minor by the Duke criteria, and give the temperature threshold the criteria use in Celsius with Fahrenheit in brackets.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C05-Q02

A painful nodule on the finger pad

A 61-year-old woman comes to the physician because of 6 weeks of low-grade fever, fatigue, and a 4-kg weight loss. She has rheumatic mitral valve disease. Temperature is 37.9 °C (100.2 °F). Examination shows tender red-purple nodules on the pads of two fingers, splenomegaly, and a holosystolic apical murmur. Urinalysis shows hematuria with red cell casts, serum complement is low, and rheumatoid factor is positive. Blood cultures drawn at separate times grow viridans streptococci. Which of the following best explains the tender nodules on her finger pads?

A. Immune complex deposition with small vessel vasculitis in the digital pad

B. Septic microemboli lodging in dermal capillaries with small abscesses

C. Fibrin and platelet plugs occluding retinal arterioles

D. Dilation of a digital artery wall seeded by circulating organisms

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C06-BD1

The two sterile vegetations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C06 Activity: NPLEX-P1-CARD-18-04-C06-BD1 Date: _____

Write the full name of nonbacterial thrombotic endocarditis and the other name it goes by, then describe its vegetations: size in millimetres, what they are made of, how firmly they are attached, which valves and which part of the leaflet they sit on, and how much inflammation surrounds them. Write the one condition it is most associated with, naming the tumour types, and write the two tumour products that drive the clotting and the vein condition they also cause. List the other settings that produce it. Then write whether a murmur is usual, where the fragments travel, and the presentation in a patient with cancer that should bring it to mind. Write one further thing an existing sterile lesion can become. Next, write the disease in which Libman-Sacks endocarditis occurs and the antibody often found with it, then describe its vegetations: size, appearance, which leaflet surfaces and which other structures they occupy, the histological change they show, and the valve lesion their healing most often produces. Finish by writing the single positional feature that points to Libman-Sacks rather than the other vegetations in this chapter.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C06-BD2

Strokes in three territories with negative cultures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C06 Activity: NPLEX-P1-CARD-18-04-C06-BD2 Date: _____

A man of 67 years with pancreatic adenocarcinoma has three strokes in two weeks, in the territories of the right middle cerebral, left posterior cerebral, and left middle cerebral arteries. He has no fever and no murmur, and three sets of blood cultures are negative. Echocardiography shows small nodules along the closure lines of the mitral and aortic leaflets. Write which lesion this is and the two findings in his history and imaging that point to it. Explain the mechanism that put those nodules there, naming the tumour products involved and why this tumour type in particular. Explain why infarcts appear in unrelated arterial territories and why he has no murmur. Then write why negative cultures do not prove a vegetation is sterile, naming the two situations that produce a culture-negative infection, and list the findings that would push you back toward infection and the findings that keep you with a sterile lesion. Finish by writing what would change in the appearance and the behaviour of these nodules if organisms seeded them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-04-C06-CP02

Four kinds of valve vegetation

Fill in every cell from memory, then check your table against the notes and correct any cell you missed.

Feature	Infective endocarditis	Nonbacterial thrombotic endocarditis	Libman-Sacks endocarditis	Acute rheumatic fever
Size and consistency of the vegetation				
Position on the valve and whether it spreads beyond the leaflet				
Organisms present or absent				
Degree of inflammation and the damage left behind				
Nature of the emboli				
Clinical setting in which it arises				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C07-BD1

Myxomatous degeneration and the mid-systolic click

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C07 Activity: NPLEX-P1-CARD-18-04-C07-BD1 Date: _____

Define mitral valve prolapse in terms of what the leaflets do and where they end up, and give the proportion of adults affected. Name the underlying lesion, then write what accumulates in the spongiosa, what happens to the collagen of the fibrosa, and which cell drives the change and through which products. Describe what the leaflets, the chordae, and the annulus look like, and name the leaflet most often involved. Name the three inherited disorders the notes link to it and write the matrix defect in each. Next, explain in steps why a click appears in mid systole: write what the papillary muscles and annulus do as the ventricle empties, what the redundant leaflets then do, what the chordae do at the end of it, and where the sound is heard. Write what follows the click if the leaflets no longer meet, and the timing of that murmur. Finish by listing the complications the notes give and writing one line on how the weak tissue produces each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C07-BD2

Tell four systolic findings apart at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C07 Activity: NPLEX-P1-CARD-18-04-C07-BD2 Date: _____

Write the rule that links left ventricular size to the timing of the click in mitral valve prolapse and the length of the murmur that follows it. List the manoeuvres that make the ventricle smaller and those that make it larger, and state what each group does to the click and to the murmur. Then four patients each have a systolic finding at the apex or the base: one has mitral valve prolapse, one has hypertrophic cardiomyopathy, one has aortic stenosis, and one has chronic mitral regurgitation. For each, write what standing and the strain phase of Valsalva do to the murmur, what squatting and sustained handgrip do, and the reason in one sentence drawn from preload, afterload, or chamber size. Write which two of the four behave the same way and the finding that still separates them. Then write the two differences between the mid-systolic click of prolapse and the opening snap of mitral stenosis, covering when in the cycle each occurs and which variable moves each one. Finish by writing which of these patients is at risk of sudden severe regurgitation and what has to happen for it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-04-C07-PR01

Predict what happens when a patient stands up

A patient with mitral valve prolapse and mild regurgitation moves from squatting to standing. Predict each item below, write the direction of the change, and give the reason in one line drawn from preload, ventricular size, or afterload.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return and to left ventricular end-diastolic volume		
What happens to the size the ventricle reaches during systole, and how soon the leaflets lose support		
What happens to the timing of the mid-systolic click relative to S1		
What happens to the length of the late systolic murmur		
What would happen instead to the murmur of hypertrophic cardiomyopathy, and why		
What would happen instead to the murmur of aortic stenosis, and why		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C08-BD1

The chronic rheumatic valve and which valves it takes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-BD1 Date: _____

Write the interval between group A streptococcal pharyngitis and acute rheumatic fever. Name the streptococcal antigen most involved, the two host tissues the immune response cross-reacts with, the name of that process, and the inflammatory lesion found in the heart during the acute illness. Then write why a single attack may leave little damage and what each new streptococcal throat infection does to an already scarred valve, naming the two tissue changes that let damage accumulate and explaining why the disease becomes evident only years to decades later. Next, write the four changes that define the chronic lesion. Describe what the stenotic mitral valve looks like seen from the atrium, and write why Aschoff bodies are rarely found at this stage. Write which of the four changes causes stenosis and which two prevent closure, and explain in two sentences why stenosis and regurgitation so often occur together here when degenerative disease usually causes one lesion alone. Finish with the valve distribution: the proportion with mitral disease alone, the proportion with mitral and aortic disease, what happens to the tricuspid and pulmonic valves, the usual explanation for the left-sided preference, and the one thing isolated aortic disease in an older adult should make you think of instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C08-BD2

Two narrowed valves with different histories

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-BD2 Date: _____

Patient 1 is 34 years old, grew up where streptococcal pharyngitis went untreated, and has a tight mitral valve with a loud S1, an opening snap, and a diastolic rumble, together with an aortic murmur. Patient 2 is 78 years old and has a tight aortic valve with a soft A2, pulsus parvus et tardus, and a normal mitral valve. For each patient write the likely cause of the valve disease and the two features of the valve itself, visible to a pathologist, that would confirm it. Write why the finding of an unaffected mitral valve in patient 2 effectively rules out rheumatic disease. Write what the commissures look like in each. Then write why patient 1 may have both narrowing and leakage of the same valve, naming the structural change behind each. Write which infection patient 1 is now at particular risk of, which organism group causes it, and why her valve invites it. Finish by writing why patient 1 is likely to decompensate during pregnancy, naming the three changes pregnancy brings and the one feature of her lesion that makes each of them dangerous.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-04-C08-RT02

Rebuild the path from a sore throat to a fused valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C08 Activity: NPLEX-P1-CARD-18-04-C08-RT02 Date: _____

Rebuild the seven-step sequence that runs from group A streptococcal pharyngitis to the chronic rheumatic valve, one step per line and in order. Beside the second step, write the antigen and the cross-reacting host tissues. Beside the third step, write where on the leaflet the vegetations sit and their size. Beside the fourth step, write the two healing changes and why one of them matters in a tissue that is normally avascular. Beside the fifth step, write what each recurrence adds. Beside the sixth step, write what happens to the commissures and to the chordae. Beside the final step, write what the orifice looks like and which other valve is commonly involved. Then write one sentence stating what the chronic disease actually is, so that it is clear the valve is not still infected.

Rebuild from memory: From streptococcal pharyngitis to chronic rheumatic valve disease

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C09-BD1

Serotonin, two filters, and the right-sided plaques

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C09 Activity: NPLEX-P1-CARD-18-04-C09-BD1 Date: _____

Write what kind of tumour a carcinoid is, the cell it arises from, and its most common site. Write the amino acid serotonin is made from and name the two enzymes in that pathway, then name the enzyme that breaks serotonin down, the metabolite produced, and where that metabolite is measured. List the other mediators these tumours release and name the peptide that one of them generates. Next, rebuild in seven steps the sequence from a serotonin-secreting bowel tumour to fibrous plaques on the right-sided valves. Using that sequence, explain why a tumour confined to the bowel usually causes no valve disease, why liver metastases change that, and how the lung protects the left heart, naming the two things pulmonary endothelium does to serotonin. Write the three situations in which left-sided lesions appear and the one tumour site that causes the syndrome without liver metastases, with the reason. Then describe the lesions: their colour, the cells and matrix they contain, which surfaces they lie on, and whether the valve beneath is destroyed. Name the receptor involved, the cell it acts on, the growth factor that helps, and two drug classes that produce similar plaques. Finish by writing what happens to the tricuspid valve and to the pulmonic valve, naming the commonest lesion of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C09-BD2

A flushing patient with a murmur that changes with breathing

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C09 Activity: NPLEX-P1-CARD-18-04-C09-BD2 Date: _____

A man of 59 years has 8 months of episodic flushing, wheezing, and watery diarrhea with cramps, and a scaly rash with poor memory. He has a holosystolic murmur at the left lower sternal border that gets louder when he breathes in, large jugular v waves, a pulsatile liver, ankle edema, and ascites, plus a systolic ejection murmur at the left upper sternal border. For each symptom and sign write the mediator or the lesion that produces it, and name the valve lesion behind each of the two murmurs. Explain why the first murmur gets louder on inspiration. Explain the rash and the memory change, naming the vitamin that is short, the amino acid diverted away from making it, and the three features of that deficiency. Write the single urine test you expect to be abnormal and what is measured. Then write where this tumour must have spread for the heart to be involved and the reason, and write what you would predict about his left-sided valves and why. Finish with a second patient who has a bronchial carcinoid, no liver metastases, and a new valve lesion: write which side of the heart is affected and justify it from the route his tumour products take.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C09-Q03

Valve disease in a patient with liver metastases

A 56-year-old woman comes to the physician because of 9 months of episodic flushing, wheezing, and watery diarrhea, and 3 months of ankle swelling. Examination shows a pulsatile liver, large jugular venous v waves, ascites, and a holosystolic murmur at the left lower sternal border that increases on inspiration. Urinary excretion of 5-hydroxyindoleacetic acid over 24 hours is markedly raised. Imaging shows an ileal mass with multiple liver metastases. Echocardiography shows pearly white plaques on the endocardial surfaces of the right-sided valves with normal left-sided valves. Which of the following best describes the valve lesions expected in this patient?

A. Mitral regurgitation from warty vegetations on both surfaces of the leaflets

B. Aortic stenosis from heaped calcified cusps with unfused commissures

C. Mitral stenosis from commissural fusion with a rigid fish-mouth orifice

D. Tricuspid regurgitation with pulmonic stenosis from fibrous plaques that fix the leaflets

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-04-C10-BD1

Two kinds of prosthesis and the ways each one fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C10 Activity: NPLEX-P1-CARD-18-04-C10-BD1 Date: _____

Write what a mechanical valve is usually made of, how long it lasts, and the property of its surface that causes most of its trouble. Write what a bioprosthetic valve is made of, what glutaraldehyde does to the tissue and why that is done, and the consequence of the tissue being nonviable, giving the usual time to structural failure and the group in whom it comes sooner. Write what a transcatheter valve is. Then set out Virchow's triad and write what satisfies each of the three parts in a patient with a mechanical valve. Write which position carries the greater thrombotic risk and give two reasons. Write the two things a thrombus on the valve can do and the findings that follow when a leaflet is immobilised, then name the slower fibrous process that narrows the orifice and say where it grows from and over what time. Next, write what high-velocity turbulent flow does to red cells, name the defect that makes it worst and define it, and list every laboratory finding of that hemolysis with the reason for each. Write why chronic iron deficiency can follow. Finish by writing what happens to bioprosthetic tissue over time, the chemistry behind it, the group in whom it is faster, and the two ways the cusps then fail.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-04-C10-BD2

Read the panel and name the complication

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-D Valvular Heart Diseases Competency: NPLEX-P1-CARD-18-04-C10 Activity: NPLEX-P1-CARD-18-04-C10-BD2 Date: _____

A woman with a mechanical mitral valve has 3 months of fatigue and dark urine. Hemoglobin is 8.8 g/dL, the blood film shows schistocytes including helmet cells, lactate dehydrogenase is high, unconjugated bilirubin is raised, haptoglobin is undetectable, reticulocytes are high, the platelet count is normal, the direct antiglobulin test is negative, and the urine contains hemoglobin and hemosiderin. Write the complication, the structural defect most likely behind it, and the mechanism in one sentence. Then take each laboratory abnormality in turn and write which step in the destruction of red cells it marks. Write why the platelet count and the direct antiglobulin test matter here, naming the three conditions that also produce schistocytes and the one that would also destroy red cells immunologically. Write why she may become iron deficient despite ongoing hemolysis, naming where the iron is lost. Then a second patient develops fever 6 weeks after valve replacement, with a new paravalvular leak and heart block. Write where this infection starts and where it spreads, the organism you expect, the structure on the prosthesis it depends on and what that structure does to host defences, and the four laboratory features that identify that organism. Finish by writing how late disease after valve replacement differs in its organisms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-04-C10-Q04

Anemia three years after valve replacement

A 58-year-old woman comes to the physician because of 4 months of fatigue and intermittently dark urine. She underwent mechanical mitral valve replacement 3 years ago. Temperature is 36.7 °C (98.1 °F). Hemoglobin is 8.6 g/dL, the platelet count is 240,000/mm3, and the blood film shows fragmented red cells including helmet cells. Lactate dehydrogenase is high, unconjugated bilirubin is raised, haptoglobin is undetectable, and the direct antiglobulin test is negative. Urinalysis shows hemoglobin and hemosiderin with no red cells. Transesophageal echocardiography shows a gap between the sewing ring and the annulus with a high-velocity jet through it. Which of the following best explains her anemia?

A. Destruction of red cells by warm autoantibodies directed against the red cell membrane

B. Shear fragmentation of red cells by a high-velocity jet through a paravalvular leak

C. Consumption of platelets and fibrin strand formation in the microvasculature

D. Hepcidin-driven iron restriction from chronic inflammation on the prosthesis

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-E Primary Myocardial Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-05-C01				
NPLEX-P1-CARD-18-05-C02				
NPLEX-P1-CARD-18-05-C03				
NPLEX-P1-CARD-18-05-C04				
NPLEX-P1-CARD-18-05-C05				
NPLEX-P1-CARD-18-05-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C01-BD1

Rebuild dilated cardiomyopathy from its central defect

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-BD1 Date: _____

Without looking at your notes, write the central defect in dilated cardiomyopathy in one sentence, then draw the ventricle in cross section beside a normal ventricle and label cavity size and wall thickness in each. Write which direction sarcomeres are added, what that pattern of hypertrophy is called, and why total heart weight rises while the wall does not thicken. Write the law of Laplace as a relationship between wall stress, pressure, radius, and wall thickness, then use it to explain in three or four sentences why dilation keeps worsening once it starts. Finish by listing at least seven causes, and beside each one write how it injures myocytes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C01-BD2

Explain every finding in a large, weak ventricle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-BD2 Date: _____

A 46-year-old man has an echocardiogram showing all four chambers enlarged, a left ventricular wall of 8 mm, and an ejection fraction of 22%. He has a holosystolic murmur at the apex, a second holosystolic murmur at the left lower sternal border, an extra sound in early diastole, raised jugular venous pressure, and ankle edema. Explain each of those findings from the size and weakness of the ventricle, naming for the two murmurs why the valves leak even though the leaflets are structurally normal. Write what the extra diastolic sound is and the mechanical event that produces it. Then write the two complications that follow from blood sitting in a poorly contracting chamber and from stretched, fibrotic myocardium, and say why each one happens. Finish by writing what you would look for on imaging and in the history to separate this from ischemic cardiomyopathy, and name the feature of wall motion that differs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-05-C01-RT01

Rebuild the remodeling spiral

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C01 Activity: NPLEX-P1-CARD-18-05-C01-RT01 Date: _____

Draw the remodeling spiral of dilated cardiomyopathy as a closed loop with five steps, writing each step in order around the loop. Beside the step where the cavity enlarges, write the two separate processes that enlarge it and which neurohormonal change drives one of them. Beside the step on wall stress, write the Laplace relationship and mark which two variables are moving and in which direction. Beside the final step, write why higher wall stress costs the myocardium oxygen. Then write one sentence saying what happens to the ejection fraction with each turn of the loop.

Rebuild from memory: The remodeling spiral of dilated cardiomyopathy

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C02-BD1

Rebuild hypertrophic cardiomyopathy from gene to tissue to filling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C02 Activity: NPLEX-P1-CARD-18-05-C02-BD1 Date: _____

Write the inheritance pattern of hypertrophic cardiomyopathy, its approximate prevalence, and the two genes most often involved with the protein each one encodes. Write what the mutant sarcomeres do differently and why the muscle thickens with no pressure load on it. Draw a short axis view of the left ventricle showing where the hypertrophy is thickest and what has happened to the cavity, then list the three histological findings. Next write which half of the cardiac cycle is primarily impaired and why a thick, fibrotic wall produces that impairment, then write what happens to the ejection fraction and why. Finish by writing which extra heart sound appears, which part of filling the patient becomes dependent on, why atrial fibrillation is poorly tolerated, and why exertional angina occurs with normal epicardial coronary arteries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C02-BD2

Reason through obstruction, syncope, and sudden death

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C02 Activity: NPLEX-P1-CARD-18-05-C02-BD2 Date: _____

Rebuild the five-step sequence that turns septal thickening into dynamic outflow obstruction, then write in two or three sentences the single rule that decides whether the obstruction gets better or worse, stated in terms of ventricular cavity size and contractility. Apply that rule to a 17-year-old who has a louder murmur when he stands up quickly and a softer one when he squats, and explain each change. Describe the murmur itself, where it is loudest, whether it radiates to the carotids, and what the carotid pulse contour feels like. Then write why this disease is a leading cause of sudden death in young athletes, naming the tissue features that create the arrhythmia substrate and the three things exertion adds. Write the two separate mechanisms that can produce exertional syncope here. Finish with the four features that separate a young athlete with physiologic hypertrophy from this disease.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-05-C02-CP01

Hypertrophic cardiomyopathy against the athlete's heart

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Hypertrophic cardiomyopathy	Athlete's heart
What drives the hypertrophy		
Symmetry and site of wall thickening		
Left ventricular cavity size		
Diastolic filling		
What happens after a period of rest from training		
Tissue architecture of the myocardium		
Risk of exertional arrhythmia and sudden death		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-05-C02-PR01

Predict what the strain phase of Valsalva does in hypertrophic cardiomyopathy

A 24-year-old woman with hypertrophic cardiomyopathy and a resting outflow murmur is asked to bear down and hold it. Predict each item below for the strain phase, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to venous return to the right heart		
What happens to left ventricular end-diastolic volume		
What happens to the distance between the anterior mitral leaflet and the septum		
What happens to the timing and extent of systolic anterior motion		
What happens to the loudness of the outflow murmur		
What happens to the loudness of a murmur of aortic stenosis under the same maneuver		
What happens to the murmur instead when she squats or her legs are raised passively		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C03-BD1

Rebuild restrictive cardiomyopathy and its causes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C03 Activity: NPLEX-P1-CARD-18-05-C03-BD1 Date: _____

Write what makes the ventricular wall stiff in restrictive cardiomyopathy, naming the two broad ways it happens. Define compliance in your own words, then draw a pressure against volume curve for filling in a normal ventricle and in a restrictive one on the same axes and label which is steeper. Draw a ventricular pressure tracing through diastole showing rapid early filling that stops abruptly, and write the name of that pattern. Write what happens to cavity size and ejection fraction and why. Then build a table of six causes with one row each, using the columns cause, what stiffens the wall, and clues, and fill every cell. Finish by writing which cause is most common and, for the eosinophilic cause, the three stages that run from the initial injury to the obliterated apex.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C03-BD2

Trace high filling pressures to the bedside and read the ECG mismatch

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C03 Activity: NPLEX-P1-CARD-18-05-C03-BD2 Date: _____

A 71-year-old man has a normal left ventricular cavity, a wall of 16 mm, an ejection fraction of 55%, and both atria markedly enlarged. Write why the atria are this large and the two problems that follow from atrial dilation. Write which side of the circulation tends to dominate his examination and list the findings you expect there, including what his jugular venous pressure does on inspiration and the name of that sign. Explain in two or three sentences why his cardiac output cannot rise much with exercise, referring to how stroke volume would normally be increased. His electrocardiogram shows low voltage in the limb leads. Rebuild the five-step sequence that explains low voltage despite a thick wall, and write what the same wall thickness would do to voltage if it were true muscular hypertrophy. Finish by writing which stain is used on the biopsy, what it shows under polarized light, and which other condition produces nearly this whole picture from outside the myocardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C04-BD1

Sort the three cardiomyopathies by which half of the cycle fails

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C04 Activity: NPLEX-P1-CARD-18-05-C04-BD1 Date: _____

Write the single question you ask first to separate the three cardiomyopathies, then answer it for dilated, hypertrophic, and restrictive disease in one sentence each, naming whether the dysfunction is systolic or diastolic and whether the ejection fraction is reduced or preserved. For the two diastolic diseases, write what makes the wall stiff in each. Then draw pressure against volume loops on one set of axes for a normal ventricle, a dilated one, a hypertrophic one, and a restrictive one. Label the end-systolic pressure-volume relationship and the end-diastolic pressure-volume relationship on each, mark which line has moved in which direction, and write beside each loop which property of the ventricle that movement represents. Finish by writing the one consequence all three diseases share and the tissue change that explains it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C04-BD2

Classify four sets of measurements

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C04 Activity: NPLEX-P1-CARD-18-05-C04-BD2 Date: _____

Four patients have echocardiography and an electrocardiogram. Patient A: left ventricular end-diastolic diameter enlarged, wall 7 mm, ejection fraction 20%, extra sound in early diastole. Patient B: septum 21 mm, posterior wall 10 mm, small cavity, ejection fraction 72%, anterior mitral leaflet touching the septum in systole. Patient C: normal cavity, wall 16 mm, ejection fraction 55%, both atria very large, low voltage in the limb leads. Patient D: wall 19 mm, tall QRS complexes, small cavity, ejection fraction 70%, deep narrow Q waves in the septal leads. For each patient write in order whether emptying or filling has failed, which cardiomyopathy it is, and the one measurement that settled it. For patients C and D explain why the same wall thickness gives opposite voltages. Then write what you expect each patient's pressure-volume loop to look like, and which extra heart sound each is most likely to have.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-05-C04-CP02

The three cardiomyopathies side by side

Fill in the table from memory on blank paper, then check it against the notes and correct any cell you missed.

Feature	Dilated	Hypertrophic	Restrictive
Ventricular cavity size			
Wall thickness			
Pattern of hypertrophy, if any			
Ejection fraction			
Primary dysfunction			
Position and shape of the pressure-volume loop			
Typical extra heart sound			
Characteristic electrocardiogram finding			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C05-BD1

Rebuild the causes and histology of myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-BD1 Date: _____

Define myocarditis in one sentence, including the word that excludes the commonest other cause of myocyte death. Write which group of organisms causes most cases in developed countries and name at least five of them. For the classic virus, write the receptor it uses to enter myocytes, which other virus shares that receptor, and which structural protein a viral protease cleaves. Write the two separate processes that injure myocytes in viral disease. Then build a table of seven causes with one row each, using the columns cause and mechanism with distinguishing features, and fill every cell, including the vector or route where the notes gave one. Finish by writing the defining histological finding and the name of those criteria, then describe the infiltrate in viral disease, in hypersensitivity disease, and in giant cell disease, and write why a negative biopsy does not exclude the diagnosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C05-BD2

Explain the findings of myocarditis and its late outcome

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-BD2 Date: _____

A 23-year-old man has chest pain and palpitations ten days after a febrile illness. Troponin is raised, the coronary arteries are normal, the electrocardiogram shows diffuse ST elevation, and the ejection fraction is 35%. Explain each of those four findings from the lesion in the myocardium, saying for troponin why it rises without coronary occlusion, for the rhythm disturbance which two electrical mechanisms the inflammation creates, and for the low ejection fraction what has been lost. Write why his chest pain could be either ischemic in character or pleuritic, and what the second pattern implies about which layer is involved. Then rebuild the six-step sequence from viral entry to dilated cardiomyopathy and write which step makes the damage permanent. Finish with three patients: one has many eosinophils with rash and fever, one has multinucleated giant cells in widespread necrosis, and one has parasites inside myocytes. For each, write the cause and the one feature of the history or tissue that named it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-05-C05-RT02

Rebuild the path from viral infection to dilated cardiomyopathy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C05 Activity: NPLEX-P1-CARD-18-05-C05-RT02 Date: _____

Rebuild the six-step sequence from viral entry into the myocyte to established dilated cardiomyopathy, one step per line and in order. Beside step one write the receptor and the two viruses that use it. Beside the immune steps write which cell does the killing and how exposed cardiac proteins can keep the process going after the virus is cleared. Beside the final two steps write what replaces the lost myocytes and the one mechanical property that tissue does not have. Then write one sentence saying why the ejection fraction falls as a result.

Rebuild from memory: From viral infection to dilated cardiomyopathy

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-05-C05-CA01

A young man with chest pain, a raised troponin, and clean coronary arteries

A 26-year-old man comes to the emergency department because of 2 days of chest pain, breathlessness on climbing stairs, and the sensation of his heart skipping. Twelve days earlier he had fever, sore throat, and loose stools that settled without treatment. Temperature is 37.8 degrees Celsius (100.0 degrees Fahrenheit), pulse is 108/min, respirations are 22/min, and blood pressure is 102/64 mm Hg. Jugular venous pressure is raised and there is a soft extra sound in early diastole. The electrocardiogram shows ST elevation in the anterior, lateral, and inferior leads with occasional wide early beats, and intermittent delay between the atria and the ventricles. Serum troponin is markedly raised and C-reactive protein is high. Echocardiography shows a left ventricle of normal cavity size with diffusely reduced wall motion and an ejection fraction of 34%. Coronary angiography shows no obstructive disease. An endomyocardial biopsy shows a patchy interstitial infiltrate of lymphocytes surrounding injured myocytes, with no eosinophils, no giant cells, and no granulomas.

1. Recognize. Name the condition, the most likely class of cause given the preceding illness, and the single organism most classically responsible.

2. Explain. Explain the raised troponin, the diffusely reduced wall motion, the atrioventricular delay, and the wide early beats from what the inflammation is doing to myocytes and to conduction.

3. Differentiate. Write how this differs from acute myocardial infarction, from stress cardiomyopathy, and from giant cell and hypersensitivity myocarditis, naming for each the finding on this patient's results that separates it.

4. Investigate. Write which two features on the biopsy you would name to satisfy the defining histological criteria, and explain why a biopsy without them would not exclude the diagnosis.

5. Decide. State what this illness can progress to over months to years if myocytes are lost, and what the ventricle will look like if it does.

6. Justify. Justify that answer by writing the sequence from viral entry to that end state, naming the receptor involved, the immune mechanism, what replaces lost myocytes, and why the law of Laplace keeps the process going once the ventricle dilates.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-05-C06-BD1

Rebuild stress and tachycardia-induced cardiomyopathy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C06 Activity: NPLEX-P1-CARD-18-05-C06-BD1 Date: _____

Write the two other names for stress cardiomyopathy, the patient group it mostly affects, and the kinds of trigger that set it off. Draw the left ventricle in systole in this condition, marking which segments are akinetic and which contract vigorously, and write the name of the pattern and what the shape is named after. Then rebuild the six-step proposed mechanism in order, including which catecholamine matters most, where its receptors are dense, which signaling change occurs at those receptors, and what happens to calcium in the myocyte. Separately, write what tachycardia-induced cardiomyopathy is and list at least three rhythms that cause it. Write what a sustained fast rate does to diastole and name the two things diastole is needed for, then write the four myocardial changes that follow and what the ventricle ends up looking like. Finish by writing the one feature both of these conditions share that sets them apart from the three classic cardiomyopathies.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-05-C06-BD2

Separate a stunned ventricle from an infarcted one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-E Primary Myocardial Diseases
Competency: NPLEX-P1-CARD-18-05-C06 Activity: NPLEX-P1-CARD-18-05-C06-BD2 Date: _____

A 70-year-old woman has chest pain and breathlessness an hour after being told of her sister's death. The electrocardiogram shows anterior ST elevation with deep T wave inversion and a long QT interval, troponin is mildly raised, the apex and mid ventricle are akinetic with a vigorously contracting base, and the coronary arteries are normal. Write the four features of this presentation that argue against infarction as the cause, including how the size of the troponin rise compares with the amount of muscle not moving and why the akinetic area does not fit a coronary territory. Write what a biopsy would show and what that pattern tells you about the injury. Predict what her ventricular function will be in six weeks and explain the cellular reason, then write what the same sized infarct would have left behind. List the four complications that follow from an acutely weak apex and a hypercontracting base, saying for each how the geometry or the weakness produces it. Finish with a second patient whose ejection fraction is 30% during six months of a fast irregular rhythm and 55% two months after the rate is controlled. Write what that recovery tells you about the cause and why rate alone can lower the ejection fraction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-05-C02-Q01

A systolic murmur that changes with position

A 19-year-old man is evaluated after collapsing during a basketball game. His father died suddenly at age 34. Temperature is 37.0 degrees Celsius (98.6 degrees Fahrenheit), pulse is 78/min, and blood pressure is 118/72 mm Hg. A harsh crescendo-decrescendo systolic murmur is heard at the left lower sternal border that does not radiate to the carotids. The murmur becomes louder when he stands up from a squatting position and softer during sustained handgrip. An extra sound is heard just before the first heart sound. Echocardiography shows a septum of 22 mm, a posterior wall of 11 mm, a small left ventricular cavity, and an ejection fraction of 74%. Which of the following best explains why the murmur becomes louder on standing?

A. Reduced afterload increases the pressure gradient across a fixed valvular narrowing

B. Increased venous return raises end-diastolic volume and accelerates flow across the outflow tract

C. Reduced left ventricular volume brings the anterior mitral leaflet closer to the hypertrophied septum

D. Reduced contractility lengthens ejection time and prolongs turbulent flow

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-05-C03-Q02

A thick wall with small complexes

A 74-year-old man comes to the physician because of 4 months of breathlessness on exertion, abdominal swelling, and leg edema. Jugular venous pressure is raised and rises further on inspiration. The liver is enlarged and there is ascites. Echocardiography shows a left ventricular wall thickness of 17 mm with a speckled appearance, a normal left ventricular cavity, an ejection fraction of 54%, and marked enlargement of both atria. The electrocardiogram shows low QRS voltage in the limb leads. Which of the following best explains the combination of a thick wall with low QRS voltage?

A. Extracellular protein deposits add wall thickness without adding electrically active myocytes

B. Myocyte disarray disperses depolarization wavefronts so that their voltages cancel

C. Series addition of sarcomeres lengthens myocytes without increasing their number

D. Pericardial thickening attenuates the voltages recorded at the body surface

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-05-C06-Q03

Akinesis beyond one coronary territory

A 68-year-old woman comes to the emergency department because of chest pressure and breathlessness that began 2 hours after her husband's funeral. Pulse is 104/min and blood pressure is 96/60 mm Hg. The electrocardiogram shows ST elevation in leads V2 to V6 with a QT interval corrected for rate of 0.49 s. Serum troponin is mildly raised. Echocardiography shows akinesis of the apex and mid ventricle with vigorous contraction of the basal segments and an ejection fraction of 35%. Coronary angiography shows no obstructive disease. Which of the following findings would most likely be present on endomyocardial biopsy?

A. Interstitial infiltrate rich in eosinophils with myocyte necrosis

B. Extracellular material staining with Congo red and showing apple-green birefringence

C. Coagulative necrosis with dense neutrophil infiltration in a wedge-shaped region

D. Contraction band necrosis of myocytes without occlusive vascular disease

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-F Pericardial Disease

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Spread the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-06-C01				
NPLEX-P1-CARD-18-06-C02				
NPLEX-P1-CARD-18-06-C03				
NPLEX-P1-CARD-18-06-C04				
NPLEX-P1-CARD-18-06-C05				
NPLEX-P1-CARD-18-06-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C01-BD1

Pericardial layers and the findings of acute pericarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C01 Activity: NPLEX-P1-CARD-18-06-C01-BD1 Date: _____

Without looking at your notes, draw the pericardium in cross section. Label the fibrous sac, the parietal serous layer, the visceral layer, and the space between them, and write the normal volume of fluid in that space and what that fluid is. Write which layers carry pain fibers, which nerve supplies them, and the spinal levels that nerve comes from. Then list the classic findings of acute pericarditis (the pain and what makes it better or worse, where it refers, the friction rub with its components and where you hear it, fever, the blood tests) and write beside each finding the one sentence of mechanism that produces it. Finish by naming the usual cause in developed countries, the classic virus, and the type of exudate that forms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C01-BD2

Read the tracing back to the inflamed epicardium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C01 Activity: NPLEX-P1-CARD-18-06-C01-BD2 Date: _____

A 25-year-old man has sharp chest pain that eases when he leans forward. The electrocardiogram shows concave ST elevation in I, II, III, aVF, and V2 to V6 with PR depression in those leads, ST depression and PR elevation in aVR, and no Q waves. Write why the pericardium itself produces none of these changes and which tissue does. Explain why the ST elevation covers more than one coronary territory, why there is no reciprocal ST depression except in aVR, what makes aVR the mirror lead, and what the PR shift tells you about which chamber is involved. Then write the four stages of the electrocardiographic evolution in order. Finish by writing what a raised troponin in this patient would mean and what that condition is called.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-06-C01-CP01

Acute pericarditis against ST elevation myocardial infarction

Fill in every cell from memory, then check your table against the notes and correct anything you missed.

Feature	Acute pericarditis	ST elevation myocardial infarction
Distribution of ST elevation		
Shape of the elevated ST segment		
Reciprocal ST depression		
PR segment		
Q waves		
Timing of T wave inversion relative to the ST segment		
Tissue that generates the ST shift and why the distribution follows from it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-06-C01-Q01

Diffuse ST elevation with PR depression

A 24-year-old man comes to the emergency department because of 2 days of sharp retrosternal chest pain that worsens when he lies flat and eases when he sits forward. He had a diarrheal illness 10 days earlier. Temperature is 37.9 °C (100.2 °F), pulse is 96/min, respirations is 18/min, and blood pressure is 118/74 mm Hg. A scratchy sound is heard at the lower left sternal border. The electrocardiogram shows concave upward ST elevation in leads I, II, III, aVF, and V2 through V6 with PR segment depression in the same leads, ST depression with PR elevation in lead aVR, and no Q waves. Which of the following best explains the distribution of the ST segment changes?

A. Occlusion of a dominant right coronary artery supplying both the inferior and lateral walls

B. Electrical current generated by the inflamed fibrous and parietal pericardial layers

C. Current of injury from inflammation of the epicardial myocardium across the whole cardiac surface

D. Beat to beat change in cardiac axis produced by the heart swinging within pericardial fluid

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C02-BD1

Build the secondary pericarditis table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-BD1 Date: _____

On blank paper build a table of the causes of secondary pericarditis with one row for each of these: early post-infarction pericarditis, Dressler syndrome, cardiac surgery, uremia, systemic lupus erythematosus, rheumatoid arthritis, mediastinal radiation, tuberculosis, pyogenic bacteria, and cancer. Use the columns Mechanism of injury and Usual exudate, and fill every cell. Above the table, write the rule that links the kind of injury to the kind of fluid: write what degree and type of injury gives serous fluid, fibrinous fluid, pus, blood, and caseous material. Give the descriptive name used for a heavily fibrinous pericardium. Underneath the table, mark which exudates are the most common and which two are the most likely to organize into constriction.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C02-BD2

Four patients, four exudates

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-BD2 Date: _____

Four patients develop pericarditis. Patient 1 is on day 2 after a transmural anterior infarct. Patient 2 is 4 weeks after the same infarct and now has fever, pleuritic pain, a rub, and an effusion. Patient 3 has end stage renal failure and no diffuse ST elevation on the tracing, with bloody fluid drained from the sac. Patient 4 has pneumonia with an empyema and is acutely toxic. For each patient write the mechanism of pericardial injury, the expected exudate, and the one feature of the history that identifies that mechanism. Then explain why patient 1 would not be expected to have pericarditis if the infarct had been subendocardial, why the timing in patient 2 fits an immune mechanism rather than direct injury and what the same picture after cardiac surgery is called, why the tracing in patient 3 can lack diffuse ST elevation and why the fluid is bloody, and which of these four is most likely to heal by dense fibrosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C02-RT01

Rebuild the Dressler syndrome sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C02 Activity: NPLEX-P1-CARD-18-06-C02-RT01 Date: _____

Rebuild from memory the five step sequence that runs from myocardial injury to the clinical picture of Dressler syndrome, one step per line and in order. Beside the step where antigens are released, write where those antigens normally sit and why the immune system has not met them before. Beside the priming step, write how long it takes and which arms of the immune system are involved. Beside the final step, write the findings the patient has and the exudate in the sac. Underneath, write the single feature that separates this process from early post-infarction pericarditis and the name given to the same process after cardiac surgery.

Rebuild from memory: Dressler syndrome as an immune reaction to injury

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C03-BD1

The rigid shell and what it does to filling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-BD1 Date: _____

Write the five step sequence that turns an acute pericardial exudate into a rigid constricting shell, one step per line. Name the major cause worldwide and the three usual causes in developed countries. Then write what happens to diastolic filling inside that shell: describe early diastole, describe where and why filling stops, and name the sound this makes and how it differs from an S3 in timing and pitch. Draw the ventricular pressure tracing of this filling pattern and name it, and draw the jugular venous pulse and mark the descent that is steep. Write what happens to diastolic pressure in each of the four chambers, what happens to systolic function and to stroke volume, and list the signs that follow from the high venous pressure. Finish with the imaging finding that can be seen on a plain film or CT.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C03-BD2

Kussmaul sign and telling the sac from the muscle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-BD2 Date: _____

Write what the jugular veins normally do during inspiration and the two steps of mechanism behind that. Then write what they do in constrictive pericarditis, name that sign, and explain it from two things the rigid shell does: what it blocks and what the descending diaphragm still does. Use the same isolation of the heart from intrathoracic pressure to explain why the two ventricles fill reciprocally with breathing, naming what happens to the pulmonary veins and to the septum. Write the two other conditions in which this sign appears. Then a 58-year-old woman has raised venous pressure, ascites, edema, clear lung fields, and a normal ejection fraction. Write the findings you would use to decide between a stiff pericardium and a stiff myocardium, and for each finding write the value or appearance you expect in each condition. Name four diseases that infiltrate or scar the myocardium to produce the second picture.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C03-RT01

Rebuild the path from pericarditis to constriction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C03 Activity: NPLEX-P1-CARD-18-06-C03-RT01 Date: _____

Rebuild the five step sequence that runs from an acute pericardial exudate to a shell that fixes total cardiac volume, one step per line and in order. Beside step one, write which three kinds of exudate are most likely to take this path. Beside the organization step, write which tissue does the organizing. Beside the fusion step, write what happens to the pericardial space and to the normal pericardial fluid. Beside the final step, write what is now fixed and what the myocardium inside is still able to do normally. Underneath, draw the pericardium before and after and write one sentence on what the change does to the pericardial pressure-volume relationship.

Rebuild from memory: From pericarditis to constriction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C04-BD1

Effusion: the curve, the causes, and the signs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C04 Activity: NPLEX-P1-CARD-18-06-C04-BD1 Date: _____

Write the normal volume of pericardial fluid, then draw the pericardial pressure-volume curve with volume on the horizontal axis and pressure on the vertical axis. Label the reserve volume and the steep limb, and write what is happening in the collagen of the sac along each part of the curve. Draw a second curve for a sac that has been stretched over months and write which way it shifts and why. Then group the causes of effusion by the composition of the fluid: write which causes give a transudate and the Starling change behind each, which cause gives a large protein-rich slowly forming effusion and through what two mechanisms, and which causes give hemorrhagic fluid. Finish by listing the findings of a large effusion, with one line of mechanism beside each, and write which of those findings are not specific and what else causes them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C04-BD2

Two patients, and why volume is the wrong number

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C04 Activity: NPLEX-P1-CARD-18-06-C04-BD2 Date: _____

Patient A has untreated hypothyroidism and 1,200 mL of pericardial fluid that accumulated over four months, with a blood pressure of 124/78 mm Hg. Patient B has a stab wound to the chest and 200 mL of blood in the sac after 20 minutes, with a blood pressure of 82/60 mm Hg and distended neck veins. Place each patient on the pressure-volume curve you drew and explain, for each one, what the sac had time to do and where pericardial pressure sits relative to right atrial pressure, giving the normal value of right atrial pressure. Write which chamber stops filling first and why that chamber rather than another. Then write what the chest film and the electrocardiogram are likely to show in each patient, and why the lung fields are clear in patient A. Finish with the rule this pair of patients teaches about predicting compression.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-06-C04-PR01

Predict the effect of a slowly accumulating effusion

A 52-year-old woman with untreated hypothyroidism has an effusion that has been building for several months and now measures 1,100 mL. For each item below, predict the finding or the direction of change and write the one line of mechanism that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the shape and position of the pericardial pressure-volume curve as the sac stretches over months		
What happens to pericardial pressure compared with right atrial pressure at this volume		
What happens to diastolic filling and to stroke volume		
What the amplitude of the QRS complexes does in the limb leads		
What the cardiac silhouette looks like on a chest film		
What the lung fields look like and why		
What would change if the same 1,100 mL had collected in one hour instead		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C05-BD1

Tamponade: definition, causes, and the fall in output

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-BD1 Date: _____

Write the definition of cardiac tamponade in one sentence that makes clear it is a pressure state and not a volume. List the causes, and for each of the three classic acute causes write the anatomical reason that cause fills the sac: name the chamber penetrating trauma usually enters and why, the usual timing of free wall rupture after a transmural infarct and what has happened in the wall by then, and why a type A aortic dissection can bleed into the sac. Then write the six step sequence from fluid collecting to hypotension with distended neck veins, one step per line, and write which side of the heart is compressed first and why. Write which pressures become equal, name the triad of findings with the mechanism of each part, and name the compensating response and the electrocardiographic finding that may appear.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C05-BD2

Separate fluid under pressure from a fixed shell

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-BD2 Date: _____

Write when filling occurs in tamponade and what makes it possible at that moment, then write what the jugular x and y descents look like and explain each from the timing of filling. Now build a comparison of tamponade and constriction on blank paper covering onset, pulsus paradoxus, Kussmaul sign, the y descent, which part of diastole filling is impaired in, whether diastolic pressures equalize, and the extra heart sound, and write one sentence explaining why fluid and a shell differ on the part of diastole they affect. Explain why Kussmaul sign is usually absent in tamponade. Then a patient has a pulsus paradoxus of 24 mmHg. Write how that number is obtained at the bedside with the two pressures you read, write the normal cut-off, explain the finding from ventricular interdependence step by step, and name three conditions other than tamponade that produce it and the mechanism in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-06-C05-RT01

Rebuild pulsus paradoxus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C05 Activity: NPLEX-P1-CARD-18-06-C05-RT01 Date: _____

Draw the heart inside a tense fluid-filled sac, then rebuild the seven step sequence from inspiration to a fall in systolic pressure, numbering each step on the drawing and writing it out in words. Mark on the drawing where the septum moves and in which direction, and mark where the pulmonary veins sit and what they do on inspiration. Beside the step where the right ventricle cannot expand, write the property of the sac that prevents it and the name of the principle that links the two ventricles. Beside the last step, write the number of millimetres of mercury that makes the finding abnormal. Underneath, write what happens to total cardiac volume through the respiratory cycle and use that to say in one sentence why right heart gain becomes left heart loss.

Rebuild from memory: Pulsus paradoxus in tamponade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-06-C05-CA01

Sudden hypotension six days after an anterior infarction

A 71-year-old woman is in the hospital on day 6 after a large anterior ST elevation myocardial infarction that was not reperfused. She suddenly becomes short of breath and loses consciousness. Temperature is 36.8 °C (98.2 °F), pulse is 128/min and weak, respirations are 30/min, and blood pressure is 68/50 mm Hg. The jugular veins are distended to the angle of the jaw, the heart sounds are distant, and the lung fields are clear on auscultation. Systolic pressure falls by 22 mm Hg during quiet inspiration. The monitor shows sinus tachycardia with an organized QRS complex and no palpable pulse. Bedside echocardiography shows an echo-free space around the heart with collapse of the right atrium and right ventricle during diastole.

1. Recognize. Name the mechanical complication that has occurred, the structure that has failed, and the hemodynamic state it has produced, and state why day 6 fits this complication.

2. Explain. Explain why cardiac output has fallen, naming which pressures have equalized, which chambers are compressed first and why, and why electrical activity continues while no pulse is felt.

3. Differentiate. Write how this picture differs from acute left ventricular failure after reinfarction and from constrictive pericarditis, naming the findings in this patient that separate them.

4. Investigate. Write which bedside and echocardiographic findings you would look for to confirm the diagnosis and what each one would show in this condition.

5. Decide. State the property of the pericardial sac that made this volume of blood enough to stop filling, and name the point on the pressure-volume curve the sac has reached.

6. Justify. Justify the inspiratory fall in systolic pressure from ventricular interdependence, and explain why the jugular venous pressure does not rise with inspiration in this patient even though it does in constriction.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-06-C05-Q02

Blunted y descent in a tense sac

A 63-year-old man with metastatic lung carcinoma comes to the emergency department because of 3 days of worsening dyspnea. Temperature is 37.1 °C (98.8 °F), pulse is 122/min, respirations are 26/min, and blood pressure is 88/68 mm Hg with a 19 mm Hg fall in systolic pressure during inspiration. The jugular veins are distended, the heart sounds are distant, and no extra heart sound is heard. The jugular venous waveform shows a preserved x descent and an absent y descent. Right atrial, right ventricular diastolic, and pulmonary capillary wedge pressures are all 22 mm Hg. Which of the following best explains the absent y descent in this patient?

A. Pericardial pressure remains above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur

B. A rigid fibrocalcific pericardium permits very rapid filling in early diastole until the limit of the shell is reached

C. Loss of atrial contraction removes the late diastolic contribution to ventricular filling

D. Impaired myocardial relaxation from amyloid infiltration slows the early diastolic pressure fall in the ventricle

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-06-C06-BD1

How tumor reaches the heart and what it does there

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C06 Activity: NPLEX-P1-CARD-18-06-C06-BD1 Date: _____

Write how common cardiac and pericardial metastases are at autopsy in patients dying of cancer, and how that frequency compares with primary cardiac tumors, naming the usual primary tumor. List the three sites of cardiac involvement in order of frequency. Then build a table of the four routes of spread to the heart with the tumors typical of each, and write one line under the table explaining which routes are explained by proximity and which by wide bloodstream spread. Write which two carcinomas account for the largest number of cases and why, and which tumor involves the heart in a high proportion of patients with widespread disease despite being less common. Finish with the two hematologic malignancies that often infiltrate the heart.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-06-C06-BD2

The effusion that keeps coming back

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-F Pericardial Disease Competency: NPLEX-P1-CARD-18-06-C06 Activity: NPLEX-P1-CARD-18-06-C06-BD2 Date: _____

A 66-year-old woman with breast carcinoma has progressive dyspnea, distant heart sounds, and an effusion that reaccumulates two weeks after it is drained. Write the five step sequence from tumor reaching the pericardium to failure of filling, one step per line. Write the two mechanisms by which pericardial deposits produce fluid, name the kind of exudate that results, and write the test on the fluid that confirms tumor as the cause. Explain why a rapidly growing or repeatedly reaccumulating effusion points to tumor, and write why malignancy is a leading cause of nontraumatic tamponade. Then write the two other pericardial problems a patient with cancer can have instead of tumor in the sac, naming the treatment-related one and what tumor encasing the heart can imitate. Finish with what myocardial and intracavitary deposits do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-G Congenital Heart Conditions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-07-C01				
NPLEX-P1-CARD-18-07-C02				
NPLEX-P1-CARD-18-07-C03				
NPLEX-P1-CARD-18-07-C04				
NPLEX-P1-CARD-18-07-C05				
NPLEX-P1-CARD-18-07-C06				
NPLEX-P1-CARD-18-07-C07				
NPLEX-P1-CARD-18-07-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C01-BD1

Draw the three left-to-right shunts and label the pressures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-BD1 Date: _____

Without looking at your notes, draw a four-chamber heart with the aorta and pulmonary trunk, then add a ventricular septal defect, an atrial septal defect, and a patent ductus arteriosus to the same drawing. Write the normal pressure in each ventricle after birth and mark the direction blood crosses each defect with an arrow. Beside each arrow write what makes blood move that way, naming pressure for one defect and chamber compliance for another. Then write what Qp:Qs means, what value it takes with a net left-to-right shunt, and why systemic arterial saturation stays normal in all three. Finish by writing which of the three defects transmit systemic pressure to the pulmonary arteries and which adds only extra volume, and explain the difference from where each defect sits relative to the tricuspid valve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C01-BD2

Explain every Eisenmenger finding from the reversal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-BD2 Date: _____

A 29-year-old woman has an uncorrected ventricular septal defect and is now cyanotic. Write what has happened to her pulmonary arterioles, naming the structural changes in the order they develop and the two endothelial mediators that fall and the one that rises. Then take each of these findings and write in one or two sentences why it is present: cyanosis that appears only now, cyanosis that barely improves on supplemental oxygen, a raised red cell mass, clubbed fingers, a loud P2, and a murmur that is softer than it was ten years ago. Write which chamber has hypertrophied and what raised its pressure. Finish by writing why a child with an atrial septal defect of the same size would be expected to reach this point later, and whether the arteriolar changes at this stage can reverse.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C01-RT01

Rebuild the path from shunt to reversal

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C01 Activity: NPLEX-P1-CARD-18-07-C01-RT01 Date: _____

Rebuild the six-step sequence that runs from a left-to-right shunt to late cyanosis, one step per line and in order. Beside the step where resistance rises, list the three arteriolar changes the notes name and write which one marks irreversible disease. Beside the step where the shunt reverses, write the pressure comparison that must be true for reversal to happen. Underneath the sequence, draw the shunt arrow at three points in time: early, when the shunt is bidirectional, and after reversal, and write the patient's systemic arterial saturation at each point.

Rebuild from memory: From left-to-right shunt to Eisenmenger syndrome

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-07-C01-PR01

Predict what happens as pulmonary vascular resistance falls after birth

A term infant is born with a large ventricular septal defect. Over the first eight weeks pulmonary vascular resistance falls from its high fetal value toward the normal postnatal value. Predict each item below, write the direction of the change, and give the pressure, resistance or volume reason in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the volume of blood crossing the defect from left to right		
What happens to Qp:Qs		
What happens to pulmonary blood flow and to left atrial and left ventricular volume		
What happens to the loudness of the murmur in a defect that is large and nonrestrictive		
What the infant looks like clinically by the second month, listing the findings of heart failure the notes give		
What happens to systemic arterial oxygen saturation during these eight weeks, and why		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C02-BD1

Build the ventricular septal defect from the septum outward

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C02 Activity: NPLEX-P1-CARD-18-07-C02-BD1 Date: _____

Draw the interventricular septum and label its muscular part and its membranous part, marking where the membranous part sits relative to the aortic valve. Write the three structures whose fusion forms the membranous septum and write why that part is the one most often left incomplete. Name the two main locations of ventricular septal defects and say which is more common. Then write the murmur in full: its timing across the cardiac cycle, its quality, where it is loudest by intercostal space, what else may be felt there, and what sustained handgrip does to it and why. Finish by writing which chambers carry the extra volume and enlarge, and explain from where in the cycle the shunted blood leaves the left ventricle why the right ventricle is spared until pulmonary pressure rises. List the associations the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C02-BD2

Two infants, two murmurs, two courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C02 Activity: NPLEX-P1-CARD-18-07-C02-BD2 Date: _____

Two infants each have a ventricular septal defect. Infant A has a loud harsh holosystolic murmur with a thrill and is feeding and growing well. Infant B has a soft short systolic murmur, an apical mid-diastolic rumble, a loud P2, tachypnea and sweating with feeds, and poor weight gain. For each infant write which defect is the larger one and justify it from the pressure gradient across the defect and from right ventricular pressure. Write what sets the size of the shunt in each. Then write the expected course of each over childhood, including which is likely to close on its own and by what mechanism, and which is at risk of irreversible pulmonary vascular disease and roughly how soon. Finish with two complications of a defect sitting just below the aortic valve, and write the mechanism of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C03-BD1

Build the atrial septum, then explain the fixed split S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C03 Activity: NPLEX-P1-CARD-18-07-C03-BD1 Date: _____

Draw the developing atrial septum from memory and label the septum primum, the septum secundum, the ostium primum, the ostium secundum, and the foramen ovale, marking which structure acts as the flap valve. Write what holds the flap open in the fetus and what presses it shut after birth. Then write why an atrial septal defect shunts left to right even though left and right atrial pressures differ by only a few mmHg, naming the property of the right ventricle that is responsible. Write which chambers and vessels receive the extra volume and enlarge. Finish by explaining the two auscultatory findings: say why flow across the defect itself is silent, where the systolic murmur actually comes from and where it is heard, and then explain step by step why the second heart sound is widely split and stays split in both inspiration and expiration.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C03-BD2

A stroke at 34 with no murmur

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C03 Activity: NPLEX-P1-CARD-18-07-C03-BD2 Date: _____

A 34-year-old man with no cardiac history has a stroke two days after a long flight, and a deep vein thrombosis is found in his leg. Write the cardiac finding that would explain the stroke and write why it caused neither a murmur nor any chamber enlargement in the years before this. Write the pressure relationship that normally keeps it closed and list the everyday events the notes name that reverse that relationship for a moment. Then write why the same event can send a clot from the leg into the systemic arteries, and name that process. Separately, write the three adult presentations of an atrial septal defect that was never found in childhood, and for the rhythm problem write which chamber change produces it. Finish by writing which atrial lesion involves missing septal tissue and which involves tissue that simply never sealed, and which of the two is associated with a cleft mitral leaflet.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-07-C03-CP01

The three lesions of the atrial septum

Fill in every cell from memory, then open the notes chapter and correct anything you missed.

Feature	Ostium secundum ASD	Ostium primum ASD	Patent foramen ovale
What went wrong in development			
Is septal tissue missing, or is complete tissue unsealed			
How common it is			
Genetic or exposure associations			
Associated valve abnormality, if any			
Direction of the axis on the ECG, and any conduction finding			
Whether there is a resting shunt large enough to overload the right heart			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C04-BD1

The ductus arteriosus from arch to ligament

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-BD1 Date: _____

Write which aortic arch the ductus arteriosus comes from, then draw the aortic arch and pulmonary trunk and show exactly where the ductus attaches at each end, naming the branch of the arch it lies just distal to. Write what the ductus does in fetal circulation and why so little right ventricular output goes to the lungs. Name the two signals that keep its smooth muscle relaxed before birth and write where the prostaglandin comes from and why the fetus does not clear it. Then rebuild the five-step sequence of closure after birth in order, and write what the closed ductus becomes. Finish by naming the risk factors for a ductus that stays open and writing the mechanism for each, and name the nerve that hooks under the ligamentum arteriosum.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C04-BD2

Pulses, pressures and blue toes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-BD2 Date: _____

A term infant has a continuous machine-like murmur below the left clavicle, a wide pulse pressure and bounding pulses. Write the pressure comparison between the aorta and the pulmonary artery in systole and in diastole, and use it to explain why the murmur never stops and why it peaks where it does. Then explain the pulse findings in two parts: why diastolic pressure falls, and why systolic pressure rises. Write which chambers dilate and why. Next, write why a premature infant with the same ductus may have a murmur only in systole. Finally, this patient reaches adulthood uncorrected and develops Eisenmenger physiology. Write which limbs become cyanotic, which stay pink, and justify the pattern from the attachment point of the ductus on the arch, then name that finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C04-RT01

Rebuild ductal closure after birth

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C04 Activity: NPLEX-P1-CARD-18-07-C04-RT01 Date: _____

Rebuild the five-step sequence of ductal closure after birth, one step per line and in order. Beside the oxygen step write what rising arterial PO2 does to ductal smooth muscle, and beside the prostaglandin step write which prostaglandin falls and what two changes at birth lower it. Beside the last step write the name of the remaining structure and how long anatomic closure takes. Underneath, write what happens to the ductus when cyclooxygenase is inhibited and what happens when prostaglandin E1 is given, and then write which of these two findings explains why a premature infant is at the highest risk.

Rebuild from memory: Closure of the ductus arteriosus after birth

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C05-BD1

Build tetralogy of Fallot from one developmental error

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C05 Activity: NPLEX-P1-CARD-18-07-C05-BD1 Date: _____

Write the single developmental error that produces tetralogy of Fallot, naming the septum involved and the direction it is displaced. Then list the four features and write beside each one how that displacement produces it, making clear which feature is a consequence of the other three rather than a primary defect. Draw the right ventricular outflow tract, the septal defect and the aortic root in one sketch and label all four features on the drawing. Write which cell population builds the outflow septum and name the deletion syndrome that follows from its failure, with the two other organs that syndrome affects and the two other cardiac lesions it is linked to. Finish with the other associations the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C05-BD2

Why one child with tetralogy is blue and another is not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C05 Activity: NPLEX-P1-CARD-18-07-C05-BD2 Date: _____

Two children have tetralogy of Fallot. One is deeply cyanotic from the first weeks and the other has almost normal saturation at age four. Write the one variable that differs between them and explain, using the two exits available to the right ventricle and the resistance of each, how that variable decides the saturation. Write why the ventricular septal defect itself produces no murmur, where the murmur in this lesion actually comes from, and why S2 is single. Write why pulmonary vascular disease and Eisenmenger reversal do not develop here. Then the second child has a hot bath, cries hard, and becomes deeply cyanotic while the murmur fades. Write the five-step loop that produced the spell, write why the murmur softened, and write what squatting does to systemic vascular resistance and to the share of right ventricular output reaching the lungs. Finish by naming two findings of chronic hypoxemia and two consequences of right-to-left flow that the notes give.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C06-BD1

Bicuspid aortic valve and its four complications

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C06 Activity: NPLEX-P1-CARD-18-07-C06-BD1 Date: _____

Write how common a bicuspid aortic valve is, which sex it favors, and which two cusps most often fail to separate, then draw the valve from above showing two unequal leaflets and label the ridge that marks the fused commissure. Name the gene the notes give for some familial cases and the two conditions that cluster with it. Then write the four complications of this valve and for each one write its mechanism in one or two sentences, making clear which complication dominates in younger patients and which in older ones. Write the sequence of tissue changes that leads to calcific stenosis, and write roughly how much earlier significant stenosis appears than with a three-cusp valve. Finish by writing the one auscultatory clue the notes give for a mobile bicuspid valve and why it disappears later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C06-BD2

Why the aorta is part of the same disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C06 Activity: NPLEX-P1-CARD-18-07-C06-BD2 Date: _____

A 38-year-old man with a known bicuspid aortic valve is found to have a dilated ascending aorta even though the valve is working well. Write the two reasons the notes give for the aortic wall changes, saying which one is intrinsic to the wall and which comes from flow, and name the two histologic changes in the media. Write why the finding can appear while the valve still functions normally, and name the two complications the dilated aorta can lead to. Then write which two coexisting conditions raise that risk further. Separately, explain why this patient has a higher risk of infective endocarditis than someone with a three-cusp valve, naming what forms on the leaflet surface first. Finish by writing why a 22-year-old with the same valve is more likely to present with a regurgitant murmur while a 62-year-old presents with a stenotic one, giving a mechanism for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C07-BD1

Coarctation: where it sits and what it does to the pulses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-BD1 Date: _____

Draw the aortic arch with its three branches and mark exactly where a coarctation sits, naming the branch it lies just distal to and the structure its narrowing is tied to. Write the leading explanation for why the narrowing forms there. Name the sex it favors, the chromosomal syndrome it accompanies, and the valve lesion present in most patients. Then write what a fixed high resistance in series does to pressure upstream and to pressure and flow downstream, and from that write the full examination picture: arm blood pressure, leg blood pressure and how the two normally compare, the femoral pulse in strength and in timing relative to the radial pulse, and what the legs feel like with exercise. Write which hormonal system is activated and why. Finish by listing the complications of long-standing hypertension proximal to the narrowing that the notes give, and name which chamber hypertrophies.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C07-BD2

The newborn who collapses and the teenager who does not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-BD2 Date: _____

A newborn with coarctation is well on day one and in shock with absent femoral pulses on day six. A 15-year-old with coarctation is found on a sports physical with hypertension and no symptoms. For each patient write how severe the narrowing is, what the lower body was being perfused by before the problem appeared, and what event or passage of time produced the presentation. For the newborn write why lower body cyanosis can be present, what happens to the kidneys when the ductus closes, and why acidosis follows. For the teenager write what has developed over years that prevents collapse, then rebuild the five-step collateral route in order from the vessel above the narrowing to the aorta below it, naming the one segment in which flow runs backwards. Finish by writing the three chest film findings the notes give, why the first two ribs are spared from notching, why notching is rarely seen in a young child, and where the murmur of collateral and coarctation flow is heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-07-C07-RT01

Rebuild the collateral route around a coarctation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C07 Activity: NPLEX-P1-CARD-18-07-C07-RT01 Date: _____

Draw the aortic arch, the coarctation, and the chest wall, then rebuild the five-step collateral route in order from the vessel above the narrowing to the descending aorta below it, labelling each vessel on your drawing. Mark the one segment in which flow runs in the reverse of its normal direction. Beside the drawing write which vessels enlarge over years, which bony structures they erode and where on those structures, and the radiographic name of that finding. Then write which two ribs are spared and give the arterial reason, and write why this finding is unusual in a young child.

Rebuild from memory: Collateral flow around a coarctation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-07-C07-CA01

Hypertension in a 16-year-old with weak femoral pulses

A 16-year-old girl is seen for a sports physical. She has no symptoms other than aching calves after long runs. She is 147 cm tall with a webbed neck and a broad chest. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 76/min and regular, and respirations are 14/min. Blood pressure is 158/92 mm Hg in the right arm and 104/64 mm Hg in the right leg. The radial pulses are strong and the femoral pulses are weak and arrive slightly after the radial pulse. A systolic murmur is heard over the back between the scapulae, and a separate early systolic click and soft systolic murmur are heard at the right upper sternal border. The chest film shows notching along the inferior margins of the fourth through eighth ribs and an indented aortic contour with dilation above and below it.

1. Recognize. Name the aortic lesion, state where in the aorta it sits, and name the second cardiac lesion the click and upper sternal murmur point to and the chromosomal syndrome her growth and physical features suggest.

2. Explain. Explain from flow, pressure and resistance why her arm pressure is high while her leg pressure is low, and explain why the femoral pulse is both weak and delayed.

3. Differentiate. Write how this pressure and pulse pattern differs from the two other causes of leg desaturation or weak lower body perfusion in this chapter, a patent ductus arteriosus with Eisenmenger physiology and a severe neonatal coarctation, naming the findings that separate them.

4. Investigate. Write what the rib notching and the indented aortic contour tell you, which vessels produce the notching, and which measurements on examination you would record to document the gradient.

5. Decide. State which hormonal system is adding to her hypertension and name the three complications of long-standing hypertension proximal to the narrowing that she is at risk of.

6. Justify. Justify the renal part of your answer from perfusion pressure at the kidney, and justify the dissection risk from both the pressure load and the aortic wall changes that accompany her valve lesion.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-07-C08-BD1

Draw the parallel circulations of transposition

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C08 Activity: NPLEX-P1-CARD-18-07-C08-BD1 Date: _____

Write what the conotruncal ridges normally do as they grow, including how far the aorticopulmonary septum twists and which ventricle each great vessel ends up connected to. Then write what goes wrong in transposition and which vessel arises from which ventricle, with the usual position of the aorta. Draw both circuits as two separate loops and label, for each loop, the chambers in order, the vessel leaving the heart, and whether the blood in it is saturated or desaturated. Write the four-step path through each loop and then write the fifth line: what must be present for the infant to survive and the three communications that can provide it. Name the maternal risk factor and the sex this lesion favors. Finish by writing why the circulations are described as parallel rather than in series, and what that means for the tissues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-07-C08-BD2

Why oxygen barely helps and why day three is worse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-G Congenital Heart Conditions
Competency: NPLEX-P1-CARD-18-07-C08 Activity: NPLEX-P1-CARD-18-07-C08-BD2 Date: _____

A newborn with transposition and an intact ventricular septum is mildly cyanotic at birth and deeply cyanotic on day three, and 100% inspired oxygen raises his saturation only slightly. Write why the fetus had no difficulty before delivery. Write which two communications he was depending on after birth and which one is now closing. Then explain why supplemental oxygen barely changes systemic saturation, naming which loop it oxygenates and where that blood goes. Write why mixing must work in both directions rather than one. Next, compare him with an infant who has transposition plus a large ventricular septal defect: write which one is less cyanotic, which one develops heart failure earlier, and give the reason for each. Finish by writing which congenital lesion is the most common cause of cyanosis on the first day of life and which is the most common cyanotic lesion overall, and write what makes the upper mediastinum look narrow on the chest film and what that appearance is called.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-07-C08-CP01

Tetralogy of Fallot against transposition of the great vessels

Fill in every cell from memory, then open the notes chapter and correct anything you missed.

Feature	Tetralogy of Fallot	Transposition of the great vessels
Developmental error in the outflow septum		
Genetic or maternal associations		
Arrangement of the circulations and the route of desaturated blood into the aorta		
When cyanosis appears and what sets its severity		
Pulmonary blood flow		
Change in saturation with supplemental oxygen		
Chest film appearance		
Whether survival depends on a patent ductus or an atrial communication		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-07-C02-Q01

The loud murmur and the quiet one

A 3-week-old infant is brought to the physician for a well child visit. He feeds well and has gained weight appropriately. Temperature is 36.9 degrees C (98.4 degrees F), pulse is 140/min, and respirations are 38/min. A harsh holosystolic murmur with a palpable thrill is heard at the left lower sternal border, and the second heart sound is normal. A second infant of the same age has a soft short systolic murmur at the same location, a loud pulmonic component of the second heart sound, an apical mid-diastolic rumble, tachypnea with feeds, and poor weight gain. Which of the following best explains why the first infant's murmur is louder?

A. A restrictive defect maintains a large systolic pressure gradient between the ventricles

B. A larger defect allows a greater volume of blood to cross the septum each beat

C. Right ventricular pressure has risen to approach left ventricular pressure

D. Pulmonary vascular resistance has already fallen below systemic vascular resistance

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C05-Q02

Squatting during a spell

A 2-year-old boy with tetralogy of Fallot becomes deeply cyanotic and hyperpneic after a prolonged crying episode, and the systolic murmur at his left upper sternal border becomes much softer. His parents place him in a knee-chest position and the cyanosis improves. Which of the following best explains the improvement?

A. Pulmonary vascular resistance falls, drawing more blood through the lungs

B. Venous return from the legs increases, raising right ventricular preload

C. Systemic vascular resistance rises, so a greater share of right ventricular output enters the pulmonary outflow tract

D. The infundibular obstruction relaxes, widening the right ventricular outflow tract

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C04-Q03

Blue toes and pink fingers

A 31-year-old woman with an uncorrected patent ductus arteriosus has become cyanotic over the past several years. Oxygen saturation is 97% on the right hand and 81% on the right foot. The toes are clubbed and the fingers are not. Which of the following best explains the difference in saturation between her hand and her foot?

A. Oxygen extraction is greater in the legs than in the arms because of their larger muscle mass

B. The left ventricle ejects preferentially into the brachiocephalic vessels during early systole

C. Pulmonary venous blood reaches the arch vessels before it reaches the descending aorta

D. Desaturated pulmonary arterial blood enters the aorta distal to the left subclavian artery

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-07-C03-Q04

A second sound that does not move

A 28-year-old woman is evaluated for 6 months of exertional fatigue. A systolic ejection murmur is heard at the left upper sternal border, and the second heart sound is widely split, with the splitting unchanged between inspiration and expiration. The electrocardiogram shows right axis deviation and an incomplete right bundle branch block pattern. Oxygen saturation on room air is 98%. Which of the following best explains the unchanging split of the second heart sound?

A. Pulmonic valve closure is delayed by elevated pulmonary arterial pressure throughout the cycle

B. Right ventricular stroke volume stays nearly constant because inspiratory venous return is offset by reduced shunting

C. Aortic valve closure occurs early because left ventricular stroke volume is reduced

D. Conduction delay in the right bundle branch separates the two components by a fixed interval

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-08-C01				
NPLEX-P1-CARD-18-08-C02				
NPLEX-P1-CARD-18-08-C03				
NPLEX-P1-CARD-18-08-C04				
NPLEX-P1-CARD-18-08-C05				
NPLEX-P1-CARD-18-08-C06				
NPLEX-P1-CARD-18-08-C07				
NPLEX-P1-CARD-18-08-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C01-BD1

Build the Virchow triad and the two thrombus types

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C01 Activity: NPLEX-P1-CARD-18-08-C01-BD1 Date: _____

Without looking at your notes, write out the three limbs of the Virchow triad. Under each limb write the mechanism in one or two sentences and list the clinical settings that act through it. Then draw an artery and a vein side by side and build a thrombus in each. Beside each one write its color, what it is mostly made of, how firmly it attaches to the wall, the direction it grows relative to blood flow, and the vessels or chambers where it typically forms. Finish by listing the ways normal endothelium keeps blood fluid, naming the molecules involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C01-BD2

Tell a thrombus formed in life from one formed after death

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C01 Activity: NPLEX-P1-CARD-18-08-C01-BD2 Date: _____

A pathologist removes two red masses at autopsy. One lifts out of the femoral vein in one piece, is gelatinous, dark red below and yellow above, and shows no layering. The other is attached to the wall of the left atrium and shows alternating pale and dark bands. For each mass write when it formed, the features that prove it, and why the layering appears in one and not the other. Then write which of the two could have produced the pulmonary embolus found in the lung, and explain the physical reason the bands form only in moving blood. Finish by listing the five fates open to a thrombus formed in life, with the consequence of each, and write which fate explains a fibrous thickening in a vein wall found years later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C02-BD1

The brakes on thrombin and the inherited thrombophilias

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C02 Activity: NPLEX-P1-CARD-18-08-C02-BD1 Date: _____

Draw thrombin in the centre of the page and write around it everything thrombin does to fibrinogen, to platelets, and to other clotting factors. Then draw the three brakes that restrain it: write the name of each brake, the factors it inactivates, and what speeds or assists it. Underneath, build a four row table of the inherited hypercoagulable states with columns for the defect, the mechanism of thrombosis, and one distinguishing note for each. Finish by writing whether these states mainly cause arterial or venous thrombosis and listing the four clinical clues that should make you suspect an inherited thrombophilia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C02-BD2

Three clotting puzzles explained from the mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C02 Activity: NPLEX-P1-CARD-18-08-C02-BD2 Date: _____

Work each of these from the mechanism. First, a woman has a prolonged partial thromboplastin time in the laboratory and a deep vein thrombosis in her leg, and her nontreponemal syphilis test is positive although she has no syphilis. Name the disorder, the antibody target, and explain how one disorder produces a long clotting time in the tube and clotting in the patient. Second, a man with low protein C begins warfarin and within days develops patches of skin necrosis. Write which proteins warfarin lowers, the order in which they fall, and why the first days of treatment make him more likely to clot rather than less. Third, a woman has her first deep vein thrombosis at 26 weeks of pregnancy and is found to carry factor V Leiden. Write what pregnancy contributes through each limb of the Virchow triad and why the inherited trait had caused no thrombosis before. Finish with one sentence on why heparin is less effective in antithrombin deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C03-BD1

Where an embolus lodges, and the nonthrombotic emboli

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-BD1 Date: _____

Draw the circulation in outline with the right heart, the lungs, the left heart, and the systemic arteries. On the drawing trace where material entering a systemic vein ends up, where material from the left heart or aorta ends up, and the one route by which a venous embolus reaches a systemic artery. Name that route and the defects that provide it. Then write the sources of systemic thromboemboli with the approximate share each contributes, and the organs where they lodge with the approximate share each receives. Underneath, build a table of the nonthrombotic emboli with one row each for fat and marrow, amniotic fluid, air, nitrogen, and atheroembolism, and columns for the setting, the mechanism, and the findings.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C03-BD2

Three pulmonary emboli of different size

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-BD2 Date: _____

Three patients each have a pulmonary embolus. The first is small and in a peripheral branch, the second is medium sized and the patient has long standing left heart failure, and the third sits astride the bifurcation of the main pulmonary artery. For each patient write what happens to the lung tissue, whether an infarct forms and why, and what happens to the right ventricle and the circulation. Then explain from ventilation and perfusion why the second patient is hypoxemic, what happens to dead space, and why the arterial carbon dioxide falls rather than rises. Finish by writing what repeated small emboli do to the pulmonary circulation over months and why more than 95% of these emboli come from the legs above the knee rather than the calf.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C03-RT01

Rebuild the route from a leg vein to a blocked pulmonary artery

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-RT01 Date: _____

Rebuild the six step sequence that runs from conditions in a deep leg vein to a blocked pulmonary artery, one step per line and in order. Beside the first step name which limb or limbs of the Virchow triad are usually acting. Beside the step where the fragment detaches, write which part of the thrombus breaks away and why that part is loosely held. Beside the final step write what is now true of ventilation and perfusion in that part of the lung and which one of the two is preserved. Then draw the chambers the fragment passes through, in order, and mark the point at which a right to left shunt would send it somewhere else instead.

Rebuild from memory: From deep vein thrombosis to pulmonary embolism

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C04-BD1

Draw an infarct and date it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C04 Activity: NPLEX-P1-CARD-18-08-C04-BD1 Date: _____

Draw a wedge shaped infarct in a kidney. Mark the occluded vessel, the apex, the base, the organ surface, and the margin, and write what covers the overlying serosa and why. Then write the two separate questions that describe every infarct, and for each question give the two possible answers. Underneath, write out the time course of coagulative necrosis from the moment of occlusion: when cell death first becomes visible under the light microscope, when nuclei disappear and the cytoplasm turns eosinophilic, when neutrophils reach the border, when macrophages and granulation tissue appear, and what the area becomes at the end. Finish by writing what the brain does instead at each of those stages and what it leaves behind.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C04-BD2

Same occlusion, four different outcomes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C04 Activity: NPLEX-P1-CARD-18-08-C04-BD2 Date: _____

The same sized embolus lodges in four patients. Patient one has it in a renal artery branch. Patient two has it in a pulmonary artery branch and a normal heart. Patient three has it in a middle cerebral artery branch and the embolus breaks up and flow returns after a few hours. Patient four has it in a coronary artery that has been narrowing slowly for ten years. For each patient write whether an infarct forms, its color, the necrosis pattern, and what the area looks like weeks later, and explain each answer from the blood supply of the organ and the tissue itself. Then write which single patient would have a larger infarct if his hemoglobin were 7 g/dL and explain why. Finish by writing the four factors that decide whether and how large an infarct forms, with one line on each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-08-C04-CP01

White against red infarcts

Fill in every cell of the table from memory, then check it against the notes and correct what you missed.

Feature	White (pale, anemic) infarct	Red (hemorrhagic) infarct
Type of vascular occlusion behind it		
Kind of tissue it occurs in		
Why blood is absent or present in the dead area		
Organs and situations where you see it		
Usual necrosis pattern in that organ		
What the area looks like weeks later		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C05-BD1

Patterns of bleeding into tissue

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C05 Activity: NPLEX-P1-CARD-18-08-C05-BD1 Date: _____

Build a four row table of hemorrhage into tissues with one row each for petechiae, purpura, ecchymosis, and hematoma. For each row write the size and form and the typical causes. Then write the names given to bleeding into the four body cavities. Underneath, write which pattern points to a platelet or small vessel problem and which points to a coagulation factor deficiency, and explain each from what platelets do and what fibrin does at an injured vessel. Finish by writing the color changes of a bruise in order, naming the pigment responsible for each stage, and what a large resorbing hematoma can do to the serum bilirubin.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C05-BD2

Volume, rate, and site in acute blood loss

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C05 Activity: NPLEX-P1-CARD-18-08-C05-BD2 Date: _____

A man weighing 70 kg is brought in 30 minutes after losing about 1.5 L of blood from a stab wound. He is pale and tachycardic, and his hematocrit is 44%. Calculate his blood volume and the percentage he has lost, then write what percentage of blood volume is tolerated with little change, what percentage brings on hypovolemic shock, and what percentage is immediately life threatening. Explain why his hematocrit is normal now, what two processes will lower it over the next day, and why an early normal hematocrit does not exclude severe blood loss. Then write what the baroreceptor reflex is doing to his heart rate, his arterioles, his veins, and his pulse pressure. Finish by writing why a 50 mL bleed inside the skull, a 150 mL bleed into the pericardium, and a 2 L bleed into the retroperitoneum each do what they do, and why slow recurrent external bleeding causes an anemia that bleeding into a body cavity does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C06-BD1

The mechanisms of edema and the fluid each produces

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-BD1 Date: _____

Write the one sentence definition of edema, then write what normally keeps the interstitium dry, naming the forces at the capillary and the structure that returns the excess. Build a five row table of the mechanisms of edema with columns for the mechanism, two or three examples, and the kind of fluid produced. Underneath, write the features that separate a transudate from an exudate: protein content, cell content, specific gravity, clarity, and the state of the vessel wall in each. Name the mediators that open the wall in an exudate. Finish by writing the three pleural fluid criteria that mark a fluid as an exudate, with the number for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C06-BD2

Four swollen patients, four mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-BD2 Date: _____

Four patients have swelling. The first has nephrotic syndrome with heavy albumin loss in the urine. The second has long standing heart failure. The third has had an axillary lymph node dissection for breast cancer and has a swollen arm. The fourth has untreated hypothyroidism. For each patient write the mechanism, the protein content of the retained fluid, whether the swelling pits, and where on the body it appears first, and explain each answer. For the third patient explain why the swelling is soft at first and hard years later, and name what the skin and subcutaneous tissue become. For the fourth explain why the swelling does not pit even though it is not lymphedema. Finish by writing why ordinary edema collects in the legs when standing and over the sacrum when lying.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C06-RT01

Rebuild the edema cycle in heart failure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-RT01 Date: _____

Rebuild the six step cycle that runs from a fall in cardiac output to a further overloaded heart, one step per line and in order, then draw it as a loop with arrows. Beside the renal step name the hormones involved and what each one retains. Beside the step where edema forms, name the Starling force that has changed and write whether the fluid is a transudate or an exudate. Then write in two sentences why a response meant to restore perfusion makes the problem worse, and mark on your loop the one point at which lowering venous pressure would interrupt it.

Rebuild from memory: Edema in heart failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C07-BD1

Build the hemodynamic table of shock

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C07 Activity: NPLEX-P1-CARD-18-08-C07-BD1 Date: _____

Write the one sentence definition of shock and the equation that relates mean arterial pressure to cardiac output and systemic vascular resistance, then write what cardiac output itself depends on. Name the three measurements used to identify the type of shock and write what the third one approximates. Now build a table from memory with one row each for hypovolemic, cardiogenic, obstructive, septic, anaphylactic, and neurogenic shock, and columns for the primary problem, cardiac output, systemic vascular resistance, pulmonary capillary wedge pressure, and the skin. Give at least two causes in the primary problem cell of each row. Finish by writing which rows show a reflex change in resistance and which show resistance as the primary defect, and write the normal range for each of the three measurements.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C07-BD2

Read four hemodynamic profiles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C07 Activity: NPLEX-P1-CARD-18-08-C07-BD2 Date: _____

Four patients are measured. Patient A has a cardiac output of 2.5 L/min, systemic vascular resistance of 1,800 dyn.s/cm5, and a wedge pressure of 26 mm Hg. Patient B has an output of 2.8 L/min, resistance of 1,700, and a wedge pressure of 4 mm Hg. Patient C has an output of 8.5 L/min, resistance of 450, and a wedge pressure of 8 mm Hg. Patient D has an output of 2.4 L/min, resistance of 1,900, a wedge pressure of 7 mm Hg, and a central venous pressure of 20 mm Hg. For each patient name the type of shock and write, value by value, which measurement is the primary problem and which is the reflex response. For patient D explain how the central venous pressure and the wedge pressure can move in opposite directions and name the two causes of shock that do this. Then write what tamponade does to the diastolic pressures of all four chambers and why. Finish with the one patient among these four whose mixed venous oxygen saturation would be normal or high, and explain why tissues are still hypoxic.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-08-C07-CP01

Cardiogenic against hypovolemic against septic shock

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Cardiogenic shock	Hypovolemic shock	Septic shock early
Primary problem and two causes			
Cardiac output and whether it is the primary change or a response			
Systemic vascular resistance and whether it is primary or reflex			
Pulmonary capillary wedge pressure and what it says about the left ventricle			
Skin temperature and appearance			
Mixed venous oxygen saturation			
Why the tissues are underperfused in this type			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-08-C07-PR01

Predict the profile after a spinal cord injury at T4

A man has a complete spinal cord injury at the fourth thoracic level and becomes hypotensive within minutes, with no evidence of bleeding. Predict each item below, give the direction of the change, and write the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to sympathetic outflow to the arterioles and veins below the injury		
What happens to systemic vascular resistance and whether that change is primary or reflex		
What happens to venous return and to the pulmonary capillary wedge pressure		
What happens to cardiac output		
What happens to heart rate, and how this differs from every other type of shock with a low cardiac output		
What the skin feels and looks like, and why it differs from the skin in hypovolemic shock		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C08-BD1

The three stages of shock and the organs that fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-BD1 Date: _____

Write out the three stages of shock in order. For each stage name what the body is doing or failing to do, the mechanisms and hormones involved, and the findings in the patient, including the blood pressure, the urine output, the mental state, and the skin. For the middle stage write the acid base disturbance by name, the metabolic pathway that causes it, the substance that accumulates, and its normal upper value. Then build a table of multiorgan dysfunction with one row each for kidney, lung, blood, liver, and adrenal, and columns for the lesion and the mechanism with its findings. Finish by writing the one sentence rule that explains why each organ fails where it does.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C08-BD2

From one gram negative bacterium to failure in five organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-BD2 Date: _____

A woman with a gram negative urinary infection becomes hypotensive with warm flushed skin, a rising creatinine, hypoxemia, a falling platelet count, and a rising D-dimer. Starting from the bacterial molecule itself, write the chain that explains each of her findings: name the molecule and its toxic portion, the receptors and binding proteins it uses, the transcription factor activated, the two cytokines released first, and the enzyme and gas that lower her vascular resistance. Then take each finding in turn and write which step in that chain produced it. Explain why her coagulation tests are abnormal in both directions at once, naming the three changes cytokines make to the hemostatic balance. Explain why her kidney injury appears in the outer medulla rather than evenly through the kidney, and why her hypoxemia occurs with a normal left atrial pressure. Finish by writing why her shock would continue to worsen even if the infection were cleared at this point, and name the stage at which restoring her circulation would no longer help.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C08-RT01

Rebuild the endotoxin sequence and the vicious cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-RT01 Date: _____

Rebuild the six step sequence that runs from endotoxin binding its receptors to multiorgan dysfunction, one step per line and in order. Beside the second step name the transcription factor and the two cytokines. Beside the third step name the enzyme, the gas it makes, and the second messenger through which that gas relaxes vascular smooth muscle. Beside the fourth step write what happens to tissue factor and to protein C activation and what that combination produces in the small vessels. Underneath, rebuild the six step positive feedback cycle of progressive shock as a loop, and mark the two points at which falling flow causes a further fall in flow.

Rebuild from memory: From endotoxin to septic shock, and the positive feedback cycle of progressive shock

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-08-C08-CA01

Hypotension and bleeding gums three days after a fracture

A 58-year-old woman is in hospital on the third day after surgical fixation of a femoral fracture. She becomes confused and her urine output falls to 15 mL/h. Temperature is 38.9 C (102.0 F), pulse is 124/min, respirations are 28/min, and blood pressure is 82/44 mm Hg. Her hands and feet are warm and the skin is flushed. Urine obtained before surgery grew Escherichia coli. Laboratory studies show a platelet count of 48,000/mm3, fibrinogen of 95 mg/dL, prolonged prothrombin and partial thromboplastin times, a D-dimer well above the reference range, schistocytes on the smear, serum lactate of 5.8 mmol/L, and creatinine rising from 0.9 to 2.6 mg/dL. The urine sediment contains muddy brown granular casts. Arterial blood shows hypoxemia, and the chest radiograph shows bilateral infiltrates. Blood is oozing from her gums and from both intravenous sites.

1. Recognize. Name the type of shock, the stage it has reached, and each of the three organ lesions her results point to.

2. Explain. Starting from the bacterial molecule, write the chain of mediators that produced her low vascular resistance, her warm flushed skin, and her capillary leak.

3. Differentiate. Write how her hemodynamic profile and her skin differ from the profile and skin of cardiogenic and hypovolemic shock, and name the measurement that would separate the three.

4. Investigate. Write which measurements you would obtain to confirm each of the three organ lesions, and the direction or appearance each result would take.

5. Decide. State which single process explains both the microthrombi and the bleeding, and state whether her shock at this point would respond to restoring her circulation.

6. Justify. Justify that answer from what cytokines do to tissue factor, thrombomodulin, and plasminogen activator inhibitor-1, and from what defines the stage of shock you named.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-08-C03-Q01

Sudden collapse after a flight

A 64-year-old woman collapses in an airport after a 12-hour flight. On arrival she is hypotensive with distended neck veins and clear lung fields. Arterial blood gas analysis shows a low partial pressure of oxygen and a low partial pressure of carbon dioxide. She dies within the hour. At autopsy a large mass of layered thrombus sits astride the bifurcation of the main pulmonary artery, and the right ventricle is dilated. Which of the following best explains the low arterial carbon dioxide in this patient?

A. Hyperventilation driven by hypoxemia and stimulated receptors, with ventilation continuing in lung that is no longer perfused

B. Reduced carbon dioxide production by hypoperfused peripheral tissues

C. Loss of alveolar surface area from collapse of the affected lung segments

D. Metabolic alkalosis from catecholamine release during the collapse

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C07-Q02

A profile with a high filling pressure

A 71-year-old man is hypotensive 6 hours after the onset of chest pain. His skin is cool and mottled. Measurements show a cardiac output of 2.4 L/min, a systemic vascular resistance of 1,900 dyn.s/cm5, and a pulmonary capillary wedge pressure of 28 mm Hg. Crackles are heard over both lung fields. Which of the following best accounts for the combination of these three measurements?

A. Loss of circulating volume with reflex arteriolar constriction

B. Reduced sympathetic tone with venous pooling below the lesion

C. Failure of the left ventricle to empty, with reflex arteriolar constriction

D. Cytokine-driven vasodilation with capillary leak

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C02-Q03

A long clotting time with thrombosis

A 29-year-old woman has a deep vein thrombosis of the left leg and a history of two second-trimester pregnancy losses. Her partial thromboplastin time is prolonged and does not correct fully when her plasma is mixed with normal plasma. A nontreponemal screening test for syphilis is positive, and confirmatory treponemal testing is negative. Which of the following mechanisms best explains her thrombosis?

A. Resistance of factor Va to cleavage by activated protein C

B. Antibodies against beta2-glycoprotein I that activate platelets, endothelium, and complement

C. Deficiency of antithrombin, so thrombin and factor Xa are not inactivated

D. Increased plasma prothrombin from a mutation in the untranslated region of its gene

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C04-Q04

A cavity where an infarct had been

An 80-year-old man had a sudden loss of speech and right-sided weakness 8 months ago. He dies of unrelated causes. At autopsy the left frontal and temporal cortex contains a fluid-filled cavity bordered by reactive astrocytes, with no scar tissue. A similar sized old infarct in his left kidney is a firm pale depressed scar. Which of the following best explains why these two infarcts healed so differently?

A. The kidney infarct was white while the brain infarct was red, and blood in the dead tissue prevents scarring

B. The brain was reperfused when the embolus lysed, while the renal artery remained occluded

C. Neurons tolerate hypoxia for only a few minutes while renal tubular cells tolerate it for hours

D. Brain tissue has little collagen framework and is lipid rich, so lysosomal enzymes liquefy it and microglia clear it away

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-I Vascular Conditions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-09-C01				
NPLEX-P1-CARD-18-09-C02				
NPLEX-P1-CARD-18-09-C03				
NPLEX-P1-CARD-18-09-C04				
NPLEX-P1-CARD-18-09-C05				
NPLEX-P1-CARD-18-09-C06				
NPLEX-P1-CARD-18-09-C07				
NPLEX-P1-CARD-18-09-C08				
NPLEX-P1-CARD-18-09-C09				
NPLEX-P1-CARD-18-09-C10				
NPLEX-P1-CARD-18-09-C11				
NPLEX-P1-CARD-18-09-C12				
NPLEX-P1-CARD-18-09-C13				
NPLEX-P1-CARD-18-09-C14				
NPLEX-P1-CARD-18-09-C15				
NPLEX-P1-CARD-18-09-C16				
NPLEX-P1-CARD-18-09-C17				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C01-BD1

Rebuild the three patterns of arteriosclerosis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C01 Activity: NPLEX-P1-CARD-18-09-C01-BD1 Date: _____

Without looking at your notes, write out the three patterns of arteriosclerosis. For each one write the size and type of vessel involved, the layer of the wall that is damaged, what the lesion looks like down the microscope, and whether the lumen narrows. Under arteriolosclerosis, separate the hyaline form from the hyperplastic form and write the clinical setting each belongs to, what the kidney looks like in each, and which one shows fibrinoid necrosis. Finish with one sentence on why the term arteriosclerosis is not the same as atherosclerosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C01-BD2

Two patients with hardened arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C01 Activity: NPLEX-P1-CARD-18-09-C01-BD2 Date: _____

A 72-year-old man with type 2 diabetes and chronic kidney disease has hard, cord-like leg arteries, pipestem calcification on a foot radiograph, and an ankle-brachial index of 1.52. A 44-year-old woman with a blood pressure of 220/128 mm Hg has pinpoint hemorrhages on the renal cortical surface and arterioles with concentric layers of smooth muscle. For each patient write which pattern of arteriosclerosis is present, which layer of the wall holds the lesion, and whether flow through the vessel is reduced. For the first patient write why the ankle pressure reads high and what that does to the usefulness of the index. For the second patient write what the layered appearance is made of and why the kidney becomes ischemic. Finish by writing what a kidney with a finely granular cortical surface and glassy pink arterioles tells you instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-09-C01-CP01

The patterns of arterial hardening side by side

Fill in every cell from memory, then open the notes and correct anything you missed.

Feature	Atherosclerosis	Hyaline arteriolosclerosis	Hyperplastic arteriolosclerosis	Monckeberg medial calcific sclerosis
Vessels involved				
Layer of the wall affected				
What the lesion is made of				
Effect on the lumen and on flow				
Typical clinical setting				
Effect on a measured ankle pressure				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C02-BD1

Build an atheroma from the first injury

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-BD1 Date: _____

With your notes closed, write the sequence that turns a normal artery into a complicated plaque, one step per line and in order, from the first endothelial change to the final events that make a plaque dangerous. Beside the step where macrophages load with lipid, name the receptor they use and write why that uptake does not switch off. Then draw a cross section of a mature atheroma and label the fibrous cap and the core, writing what each is made of and which cell builds the cap. List the arteries that plaque favors, in order from the most affected, and separate the modifiable risk factors from the fixed ones.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C02-BD2

Why a small plaque can kill and a big one can ache

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-BD2 Date: _____

Patient A has a 40% stenosis of the left anterior descending artery and dies of an infarction in that vessel. Patient B has an 80% stenosis of the same artery and has chest pain only when climbing stairs. Explain each outcome from the structure of the plaque rather than its size. Write the three features of a plaque that make it likely to rupture, and name the cell and the enzyme that thin the cap along with the cytokine that stops smooth muscle replacing the collagen. Write what is exposed when the cap tears and what forms on top of it. Then write the degree of stenosis at which flow can no longer rise with demand, and explain what that means for when patient B feels symptoms. Finish by listing the five complications of a plaque with the mechanism of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C02-RT01

Rebuild the atherogenesis sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C02 Activity: NPLEX-P1-CARD-18-09-C02-RT01 Date: _____

Rebuild the eight-step sequence from endothelial injury to complicated plaque, one step per line and in order, with the steps numbered. Beside the first step, write where in the arterial tree the injury starts and why that location and not another. Beside the step where monocytes enter, name the adhesion molecule the endothelium displays. Beside the step where smooth muscle arrives, name the growth factor that recruits it and the layer it comes from. Mark on your list the first lesion that is visible to the naked eye and write at what age it is already present in most people, and mark the first step at which the lumen is meaningfully narrowed.

Rebuild from memory: From endothelial injury to complicated plaque

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-09-C02-Q01

An infarct from a modest stenosis

A 57-year-old man dies suddenly at home. An angiogram performed four months earlier had shown a 40% stenosis of the proximal left anterior descending artery and no other significant lesions. At autopsy the left anterior descending artery is occluded by a fresh thrombus overlying a plaque with a thin fibrous cap, a large lipid core, and dense macrophage infiltration. Which of the following best explains why this plaque occluded the artery?

A. Calcification of the media stiffened the artery and reduced its luminal diameter

B. Gradual growth of the plaque reduced coronary flow reserve beyond the ischemic threshold

C. Macrophage metalloproteinases degraded cap collagen while interferon gamma suppressed its replacement

D. Plasma proteins leaked into the wall and smooth muscle cells laid down excess matrix

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C03-BD1

Draw LDL clearance and then break it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C03 Activity: NPLEX-P1-CARD-18-09-C03-BD1 Date: _____

Draw a hepatocyte and trace one LDL particle from plasma into the lysosome. Label the apolipoprotein the receptor binds, the mechanism of uptake, and what the released cholesterol does to SREBP-2, to HMG-CoA reductase, and to the number of new receptors. Then write the three genes that cause familial hypercholesterolemia, stating for each whether the mutation is loss of function or gain of function and which part of the pathway it breaks. Write the inheritance pattern. Finish by listing the physical deposits of this disease, writing for each where it sits, what it is made of, and which ones are specific enough to point at the diagnosis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C03-BD2

Read the lipid panel and the tendons

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C03 Activity: NPLEX-P1-CARD-18-09-C03-BD2 Date: _____

A 9-year-old boy has an LDL cholesterol of 620 mg/dL, yellow plaques over the knuckles and elbows, and a father who had a myocardial infarction at 41. His 38-year-old mother has an LDL of 236 mg/dL, normal triglycerides, a firm nodule in the Achilles tendon, and a gray-white ring at the corneal margin. For each of them write the likely receptor activity, whether they are heterozygous or homozygous, and the decade in which coronary disease is expected. Explain why the plasma LDL rises even though the liver is detecting too little cholesterol inside its own cells. Write why normal triglycerides matter for this diagnosis. Then write why the corneal ring is strong evidence in the mother but would be weak evidence in a 70-year-old, and explain in two or three sentences how the same receptor failure that raises plasma LDL also builds the tendon nodule and the arterial plaque.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C04-BD1

Build the aneurysm table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C04 Activity: NPLEX-P1-CARD-18-09-C04-BD1 Date: _____

On blank paper build a table of four aneurysms, one row each for abdominal aortic, thoracic aortic, berry, and mycotic. Use the columns site, what weakens the wall, typical patient or association, and main consequence, and fill every cell. Above the table write the definition of an aneurysm, the difference between a true and a false aneurysm, and the diameter at which an abdominal aorta is called aneurysmal. Below the table write the formula that relates wall stress to pressure, radius, and wall thickness, and write two sentences on why dilation tends to continue once it has started. Finish by writing why the infrarenal aorta is the usual site, naming what supplies that part of the media and what plaque does to it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C04-BD2

Work the numbers on a growing aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C04 Activity: NPLEX-P1-CARD-18-09-C04-BD2 Date: _____

A 69-year-old man who smoked for 40 years has an infrarenal aorta measuring 6.0 cm. A normal aorta at that level measures 2.0 cm. Calculate how many times greater the wall stress is in his aorta at the same blood pressure, show your working, and write what happens to the stress as the stretched wall thins. Explain why his rupture risk is far higher than that of a man with a 4.0 cm aorta, and write what uncontrolled hypertension adds to every step of your reasoning. Then write the three findings expected if this aneurysm ruptures. Separately, a 24-year-old tall man with long fingers has a dilated ascending aorta and a new diastolic murmur: write which layer has degenerated, what that degeneration looks like microscopically, and why the valve leaks when the root widens. Finish by writing how tertiary syphilis produces the same root dilation through a different route.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C05-BD1

Rebuild a dissection and its consequences

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C05 Activity: NPLEX-P1-CARD-18-09-C05-BD1 Date: _____

Write the definition of aortic dissection, naming where the blood enters, which layer it travels in, and what the resulting channel is called. List the conditions that weaken the media, separating the common cause in older patients from the inherited and structural causes seen in younger patients, and name the acute triggers. Then rebuild the seven-step sequence of a proximal dissection from weakened media to external rupture, one step per line, and mark which step is the usual cause of death. Draw the Stanford classification as two labelled diagrams and write beside each what it does and does not involve. Finish by writing why heavy atherosclerosis makes a dissection less likely to spread.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C05-BD2

Explain each sign in a dissecting aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C05 Activity: NPLEX-P1-CARD-18-09-C05-BD2 Date: _____

A 58-year-old man with long-standing hypertension has sudden tearing chest pain that moves to between his shoulder blades. Blood pressure is 176/96 mm Hg in the right arm and 128/74 mm Hg in the left, there is a new diastolic murmur, and the chest radiograph shows a widened mediastinum. Explain each of those four findings from the anatomy of the false lumen, writing one short paragraph per finding. Then he becomes hypotensive with distended neck veins and muffled heart sounds: write where the aorta has ruptured, why filling of the ventricles falls, and which Stanford type this must be. Write the Stanford type you would expect instead if the tear had begun beyond the left subclavian artery, and list the complications that type threatens. Finish by writing which single finding in this man would point you to branch occlusion of the renal, mesenteric, carotid, or spinal arteries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-09-C05-Q02

Tearing chest pain in a tall young man

A 31-year-old man who is 196 cm tall with long slender fingers and a high arched palate comes to the emergency department because of sudden severe tearing chest pain radiating to his back. Blood pressure is 88/54 mm Hg, the jugular veins are distended, and the heart sounds are muffled. Which of the following best explains his hypotension?

A. Blood has ruptured from the aorta into the pericardial sac and is limiting ventricular filling

B. The false lumen has occluded both renal arteries and reduced circulating volume

C. Fibrinoid necrosis of the arterioles has produced widespread loss of vascular tone

D. An intimal plaque has starved the underlying media and allowed the aorta to dilate

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C06-BD1

Claudication, the ankle-brachial index, and the cholesterol shower

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C06 Activity: NPLEX-P1-CARD-18-09-C06-BD1 Date: _____

Write what peripheral arterial disease is and name its two strongest risk factors. Explain in three or four sentences why the leg hurts with walking and not at rest, naming what the arterioles do during exercise and why the stenosis prevents flow from rising. Write which muscle group hurts with superficial femoral disease and which hurts with aortoiliac disease. Then rebuild the ankle-brachial index table from memory with a row for each band, giving the numeric range and what it means, and write the one band in which the result cannot be trusted and why. List the findings of chronic limb ischemia on inspection and palpation, write what rest pain means and why hanging the foot down eases it, and name the three features of Leriche syndrome. Finish with atheroembolic disease: write what showers downstream, what procedures set it off, the size of vessel the crystals lodge in, the skin and kidney findings, why the pedal pulses stay normal, and what the biopsy shows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C06-BD2

Three painful feet, three different vessels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C06 Activity: NPLEX-P1-CARD-18-09-C06-BD2 Date: _____

Patient 1 is a 70-year-old smoker with calf cramping after one block of walking that stops within three minutes of standing still, cool hairless shins, absent pedal pulses, and an ankle-brachial index of 0.58. Patient 2 is a 66-year-old who had coronary angiography four days ago and now has purple toes, a mottled net-like pattern on the thighs, rising creatinine, eosinophilia, and normal pedal pulses. Patient 3 is a 74-year-old with diabetes on dialysis who has a nonhealing great toe ulcer and an ankle-brachial index of 1.46. For each patient name the vascular process, write which size of vessel is obstructed and by what material, and explain how the pulse finding follows from that. Explain why patient 2 keeps normal pulses while patient 1 loses them. Write why patient 3's index does not exclude disease and what you would measure instead to judge perfusion. Finish by writing which patient has an ulcer you would expect to be punched out and dry rather than shallow and moist, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-09-C06-PR01

Predict what happens as a femoral stenosis tightens

A 68-year-old smoker has a single stenosis of the superficial femoral artery that narrows steadily over three years. For each item below write the direction of the change and give the one-line mechanism that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to resting blood flow in the calf while the stenosis is still moderate		
What happens to calf blood flow during walking, and what the patient feels		
What happens to the ankle systolic pressure and therefore to the ankle-brachial index		
What happens to the distance he can walk before the pain starts		
What happens to pain at rest in the forefoot at night once the stenosis is severe, and what hanging the foot over the edge of the bed does to it		
What happens to the skin, nails, and hair of the lower leg, and what appears at the toes and pressure points		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C07-BD1

Rebuild pulmonary embolism from clot to blood gas

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-BD1 Date: _____

Write where more than 95% of pulmonary emboli come from, name the specific veins, and write why those veins and not the calf veins alone. Then rebuild the seven-step sequence from arterial obstruction to respiratory alkalosis, one step per line and in order, and write beside the fourth step what is happening to ventilation and perfusion in the obstructed region and in the open region. Write the proportion of the pulmonary circulation whose obstruction can cause shock or sudden death. List the clinical findings of a large embolus and separately those that suggest a small peripheral one. Finish by writing the findings of acute right ventricular strain, including the most common electrocardiographic finding, and write what D-dimer is and the one limitation of the test.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C07-BD2

Calculate the gradient and explain the lung that lives

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-BD2 Date: _____

A 29-year-old woman taking an estrogen-containing contraceptive breathes room air at sea level. Her PaO2 is 58 mm Hg and her PaCO2 is 28 mm Hg. Calculate the alveolar PO2 and the alveolar-arterial gradient, showing every step of the arithmetic, and state whether the gradient is normal for a young adult. Write why her low PaCO2 rules out hypoventilation as the cause of her hypoxemia, and name the mechanism that explains it instead. Then write why only about one in ten emboli infarcts the lung, naming the two things that keep the tissue alive, and write which coexisting condition makes infarction likely and why. Describe what the infarct looks like and where it sits. Finish by writing what repeated small emboli over years produce.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C07-RT02

Rebuild the pulmonary embolism sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C07 Activity: NPLEX-P1-CARD-18-09-C07-RT02 Date: _____

Rebuild the seven-step sequence from pulmonary arterial obstruction to respiratory alkalosis, numbering each step and writing one step per line. Beside the step where pulmonary vascular resistance rises, write why the right ventricle is the chamber that fails and what that failure is called. Beside the step where blood is diverted, draw two alveolar units side by side, one obstructed and one open, and mark on each what is happening to ventilation and to perfusion. Beside the hypoxemia step, write which measurement widens and what that measurement separates this condition from. Finish by writing which step produces the low PaCO2 and why the patient is breathing faster.

Rebuild from memory: From pulmonary arterial obstruction to signs

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-09-C07-CA01

Sudden breathlessness after a long flight

A 34-year-old woman comes to the emergency department because of 2 hours of sudden breathlessness and sharp right-sided chest pain that is worse on deep inspiration. She returned yesterday from a 14-hour flight and has been taking a combined oral contraceptive for six years. Her left calf has been swollen and tender for five days. Temperature is 37.4 degrees C (99.3 degrees F), pulse is 116/min, respirations are 28/min, blood pressure is 104/68 mm Hg, and oxygen saturation is 89% on room air. The jugular venous pressure is raised and there is a loud pulmonary component of the second heart sound. Arterial blood gas on room air at sea level shows PaO2 56 mm Hg and PaCO2 29 mm Hg. The electrocardiogram shows sinus tachycardia. The chest radiograph is clear.

1. Recognize. Name the vascular event, the vein the material most likely came from, and the three elements of risk present in this woman.

2. Explain. Explain her hypoxemia and her low PaCO2 mechanistically, naming what is happening to ventilation and perfusion in the obstructed and the open lung regions.

3. Differentiate. Write how the blood gas pattern here differs from that of pure hypoventilation and from that of a right-to-left shunt, naming the measurement that separates them.

4. Investigate. Calculate the alveolar-arterial gradient from the values given, show the arithmetic, and state what the result tells you; then name the leg finding you would look for on compression ultrasound and what it would show.

5. Decide. State which chamber of the heart is under acute strain and name the three findings in this case that point to it.

6. Justify. Justify from the dual blood supply of the lung why her lung tissue is likely to survive, and write what would change that conclusion.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C08-BD1

Build Virchow triad and the venous thrombus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C08 Activity: NPLEX-P1-CARD-18-09-C08-BD1 Date: _____

Write where venous thrombi usually begin and why that exact site, and write what they are made of and how that differs from an arterial thrombus. Then build a table of Virchow triad with one row per element, giving the mechanism and at least four examples for each, and mark the two clinical situations that supply all three elements at once. Write the inherited and acquired causes of hypercoagulability from the notes, including the one that works by losing a protein in the urine. Finish by writing what a deep vein thrombosis does to capillary hydrostatic pressure and what the leg looks and feels like, why many are silent, which clinical sign is unreliable, and what the thrombosed vein does under the ultrasound probe.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C08-BD2

A red cord, a swollen leg, and a wandering vein

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C08 Activity: NPLEX-P1-CARD-18-09-C08-BD2 Date: _____

Patient 1 has a tender red palpable cord along a varicose vein on the calf with no limb swelling. Patient 2 has a uniformly swollen, warm, tender left leg with a calf circumference 4 cm greater than the right and dilated superficial veins. Patient 3 is a 61-year-old with 48 kg weight loss and painless jaundice who has had thrombophlebitis in the left arm, then the right calf, then the left thigh over ten weeks. For each patient name the venous condition, write which venous system is involved, and write the embolic risk with the reason. Explain why patient 2 has diffuse swelling and patient 1 does not. For patient 3 name the sign, the tumor type most often responsible, the organ to suspect first, and the substances the tumor releases that activate coagulation, then write which identically named sign belongs to a completely different disorder and what that other sign actually is. Finish by writing which of these three patients you would expect to have a normal platelet count and why that question is not useful here.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-09-C08-Q03

Thrombosis at shifting sites

A 63-year-old woman has had 14 kg of weight loss, painless jaundice, and three episodes of tender palpable venous cords over ten weeks, first in the right forearm, then the left calf, then the right thigh, each resolving as the next appeared. Her platelet count and coagulation studies are normal. Which of the following best explains her venous findings?

A. Antiphospholipid antibodies binding phospholipid-protein complexes on the endothelium

B. Loss of antithrombin in the urine from nephrotic syndrome

C. Compression of the deep veins by an enlarging pelvic mass

D. Tissue factor and mucin released by an adenocarcinoma activating coagulation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C09-BD1

The calf pump, the valves, and the skin that follows

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-BD1 Date: _____

Draw a leg vein in outline with a valve, a perforating vein, and the calf muscle. Write how blood is returned against gravity during walking, naming the pump and the direction competent valves allow through the perforators, and write what happens to ankle venous pressure during movement when the valves work. Then write what happens to that pressure when the valves fail and name the resulting state. Define varicose veins, name the venous system most often involved, and list the risk factors with a one-line reason for pregnancy and for prolonged standing. Write which other sites develop varicosities when pressure rises elsewhere in the body and what raises it. Finish by rebuilding the seven-step sequence from valve incompetence to a venous ulcer and write beside the pigment step what the brown colour is made of.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C09-BD2

Two ulcers above two different problems

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-BD2 Date: _____

A 64-year-old woman who stands all day at work has ankle swelling that is worst in the evening, brown discoloration and itchy scaling skin around both lower legs, indurated tissue above the ankles, and a shallow moist irregular ulcer just above the left medial malleolus. A 71-year-old man who smokes has a dry, pale, punched out ulcer on the tip of the second toe, a cool hairless foot, and absent pedal pulses. For each ulcer write the underlying vascular problem, the usual location and why that location, the appearance, the character of the pain, and the state of the surrounding skin. Explain why elevating the leg helps one patient and harms the other, giving the pressure that changes in each case. Then explain each of the woman's four skin findings from the steps of the venous sequence, one short paragraph each. Finish by writing why her swelling is worst in the evening and better by morning.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-09-C09-RT03

Rebuild the path from a failed valve to a broken skin surface

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C09 Activity: NPLEX-P1-CARD-18-09-C09-RT03 Date: _____

Rebuild the seven-step sequence from valve incompetence to a venous ulcer, numbering each step and writing one step per line. Beside the first step, write the three groups of veins in which the valves can fail and what a previous deep vein thrombosis does to deep valves. Beside the edema step, write which pressure has risen and in which vessel. Beside the pigment step, write what leaks out of the capillary and what the brown pigment is. Beside the oxygenation step, write the two changes that impair delivery of oxygen to the skin. Then mark on the list the step at which the changes become visible as a skin disease and name that disease and the fibrotic stage that follows.

Rebuild from memory: From incompetent valves to stasis ulcer

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C10-BD1

The three colours of a vasospastic attack

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C10 Activity: NPLEX-P1-CARD-18-09-C10-BD1 Date: _____

Write the definition of Raynaud phenomenon, naming the vessels involved, the body parts affected, and the two triggers. Write how digital vessels normally respond to cold, naming the nerve supply and the specific receptor subtype, and write the purpose of that response. Then draw the attack as a sequence of five steps and label the three colours in order, writing beside each colour what the blood in the digit is doing to produce it. Explain the red phase in full, naming at least three local metabolites that accumulate during ischemia and what they do to the arterioles. Finish by writing which colour is the most specific and why not every attack shows all three.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C10-BD2

Which Raynaud is in front of you

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C10 Activity: NPLEX-P1-CARD-18-09-C10-BD2 Date: _____

Patient 1 is a 19-year-old woman whose fingers of both hands turn white then blue in cold weather, with no ulcers, normal nailfold capillaries, and a negative antinuclear antibody. Patient 2 is a 47-year-old woman with attacks confined to three fingers of the right hand, pitted scars on two fingertips, tight shiny skin over the fingers, and dilated and dropped-out nailfold capillaries. For each patient write whether the phenomenon is primary or secondary, what is wrong with the vessel, and whether tissue damage is expected. Explain why one patient can lose tissue and the other cannot, using the state of the lumen between attacks. Name the disease most likely behind patient 2 and write what it does to the digital arteries. Then list the other conditions and exposures that cause secondary disease. Finish by writing the five features that should make you look for an underlying disease in any patient with cold-triggered colour change.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C11-BD1

Thromboangiitis obliterans from histology to toes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C11 Activity: NPLEX-P1-CARD-18-09-C11-BD1 Date: _____

Write the definition of thromboangiitis obliterans, naming the vessels involved, the arteries most often affected, and the two other structures the inflammation spreads to. Write who gets it, the age at onset, and the single exposure that is central to it, with what is thought to be the mechanism and what happens to the disease when that exposure stops. Then describe the microscopic lesion in full, naming what is inside the thrombus, what surrounds those collections, what happens to the internal elastic lamina, and what the segment looks like over time. Write what the word segmental means here and why it separates this disease from atherosclerosis. Finish by listing the clinical features in order from earliest to latest, writing beside the claudication which part of the limb hurts and why, and beside the rest pain why it is out of proportion to the visible lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C11-BD2

A 33-year-old smoker with a black fingertip

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C11 Activity: NPLEX-P1-CARD-18-09-C11-BD2 Date: _____

A 33-year-old man who smokes 30 cigarettes a day has pain in the instep of the right foot when walking, a painful ulcer on the tip of the left index finger, absent radial and posterior tibial pulses with normal brachial and femoral pulses, cold-triggered colour change in both hands, and a tender nodular cord that appeared in one calf vein and then another. Write the diagnosis and then justify it finding by finding, writing one line per finding. Explain why the proximal pulses are normal while the distal ones are absent. Write why an upper limb finding argues against atherosclerotic disease in a man of this age, and write three further features that separate the two conditions. Explain what the vein nodules and the colour change tell you about the extent of the disease beyond the arteries. Finish by writing what determines whether this man keeps his fingers and toes, naming the mechanism behind that answer.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C12-BD1

Sort the vasculitides by vessel size, then build giant cell arteritis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C12 Activity: NPLEX-P1-CARD-18-09-C12-BD1 Date: _____

Write the definition of vasculitis and the principle on which the Chapel Hill classification is built. For each of the three vessel sizes write what the clinical picture looks like and why. Then rebuild the vasculitis table from memory with a row for each disease, using the columns vessel size, main immune mechanism, and key association, and include the large, medium, small ANCA, small immune complex, and variable vessel diseases. Next write giant cell arteritis in full: the age at which it starts, the sex and ancestry most affected, the immune mechanism, and every artery named in the notes. Write the cytokine that drives the acute-phase response, then the three ischemic symptoms with the mechanism of each, the laboratory findings with values, and what the biopsy shows. Write why a biopsy can be negative in a patient who has the disease. Finish with polymyalgia rheumatica: the girdles involved, the laboratory result, why strength and creatine kinase are normal, and how often the two conditions overlap.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C12-BD2

Jaw pain, a headache, and sudden loss of vision

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C12 Activity: NPLEX-P1-CARD-18-09-C12-BD2 Date: _____

A 74-year-old woman of Northern European ancestry has six weeks of new temporal headache with a tender thickened cord over the temple, pain in the jaw after a few minutes of chewing, aching and morning stiffness of both shoulders and hips, and this morning sudden painless loss of vision in the left eye. The erythrocyte sedimentation rate is 86 mm/h, C-reactive protein is high, creatine kinase is normal, and shoulder strength is normal. Explain each of her five problems mechanistically, one short paragraph each, naming the artery involved where there is one. Write why her shoulder and hip symptoms are not a muscle disease and what tissue is actually inflamed. Write why the sedimentation rate is so high, naming the cytokine responsible. Then a temporal artery biopsy is reported as normal: write what that does and does not tell you and the feature of the lesion that explains it. Finish by writing which vessel this disease can involve that would later produce a thoracic aneurysm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C13-BD1

Takayasu arteritis from arch to pulse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C13 Activity: NPLEX-P1-CARD-18-09-C13-BD1 Date: _____

Write what Takayasu arteritis is, naming every vessel group the notes list, the usual age and sex, and the region where it is more common. Describe the histology and write which other vasculitis it resembles and the single feature that separates the two. Then write what chronic inflammation leaves behind in the media, adventitia, and intima, and what that does to the origins of the great vessels. Write the constitutional symptoms that come first and the laboratory finding that accompanies them. Explain why the arm pulses weaken, name the older name of the disease, and write what happens to the blood pressure in the affected arm and what is heard over the subclavian and carotid arteries. Finish by writing the findings produced by carotid and vertebral narrowing, by renal artery stenosis, by aortic root involvement, and by pulmonary artery involvement.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C13-BD2

Two women with a weak arm pulse

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C13 Activity: NPLEX-P1-CARD-18-09-C13-BD2 Date: _____

A 28-year-old woman has three months of fever, fatigue, and weight loss, then arm aching with use, a blood pressure of 94/62 mm Hg in the left arm and 148/92 mm Hg in the right, a bruit over the left subclavian artery, and a sedimentation rate of 72 mm/h. A 61-year-old man with hypertension has sudden tearing chest pain radiating between his shoulder blades and a blood pressure of 182/98 mm Hg in the right arm and 126/76 mm Hg in the left. For each patient write the diagnosis and the mechanism by which the arm pressures became unequal. Write which features of the history separate them, and explain why the time course is the most useful single discriminator. For the woman, write why a blood pressure taken in her left arm would underestimate her true pressure and what that means if she also has renal artery stenosis. Then write what you would expect her blood pressure to do if the renal arteries are involved and name the system that produces it. Finish by writing which disease a granulomatous large-vessel arteritis would be in a 68-year-old instead, and the single feature that decides it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-09-C13-CP02

The two granulomatous large-vessel arteritides

Fill in every cell from memory, then open the notes and correct anything you missed.

Feature	Giant cell arteritis	Takayasu arteritis
Age at onset		
Typical patient and ancestry		
Main vessels involved		
Key clinical findings		
Histology		
Laboratory findings		
Late aortic complication		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C14-BD1

Polyarteritis nodosa, vessel by vessel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C14 Activity: NPLEX-P1-CARD-18-09-C14-BD1 Date: _____

Write the definition of polyarteritis nodosa, naming the size and type of vessel involved, the two organ groups most often affected, and the three vessel types that are spared. Write what that sparing means for the kidney and name the lesion the disease therefore cannot cause. Write the age group affected, the infection associated with a substantial minority of cases, the mechanism in those cases including the hypersensitivity type and the antigen involved, and the antibody result that is characteristically negative. Then describe the lesion: what is happening across the wall, the cells present early, what the necrosis does to the wall, and what that produces on angiography and in the name of the disease. Write the hallmark that concerns the age of the lesions. Finish by listing the organ manifestations, giving the mechanism of each, and name the leading cause of death.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C14-BD2

Explain a patient with fever, foot drop, and bloody stools

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C14 Activity: NPLEX-P1-CARD-18-09-C14-BD2 Date: _____

A 46-year-old man with chronic hepatitis B has eight weeks of fever, malaise, and 9 kg weight loss, new hypertension at 168/104 mm Hg, abdominal pain that begins 30 minutes after eating with dark stools, a right foot that will not dorsiflex, painful nodules on the calves, and a creatinine of 2.1 mg/dL with no red cell casts and no hematuria. Chest imaging is normal and ANCA is negative. Write the diagnosis and then explain each finding mechanistically, one short paragraph per finding, naming the vessel involved. Explain why his urine shows no red cell casts and what that tells you about the size of vessel affected. Write why the hepatitis B infection matters and what would be found in his vessel walls. Write what renal and mesenteric angiography would show and why the wall produces that appearance. Then write which small-vessel vasculitis you would have suspected instead if he had hemoptysis, red cell casts, and a positive ANCA. Finish by writing what the coexistence of acute and healed lesions in the same artery tells you about how the disease injures vessels.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C15-BD1

Kawasaki disease and the coronary arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C15 Activity: NPLEX-P1-CARD-18-09-C15-BD1 Date: _____

Write what Kawasaki disease is, naming the size of vessel involved, the age group and ancestry most affected, and why it matters in children as a group. Write the current view of what triggers it and which cells damage the vessel wall, then describe the arteritis and write how it differs in one respect from the lesion of polyarteritis nodosa. Write the fever requirement for the clinical diagnosis and list the five other principal features with a short description of each, then write how many of them are needed alongside the fever. Write what happens to the fingers and toes in the second to third week and list the laboratory findings, including the one that appears only after the first week. Finish by writing the key complication, how often it occurs in untreated children, the mechanism that produces it, the two things inside an aneurysm that favour clot, what that clot can cause, and the other cardiac problems of the acute phase.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C15-BD2

A red-eyed, red-tongued, miserable toddler

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C15 Activity: NPLEX-P1-CARD-18-09-C15-BD2 Date: _____

A 3-year-old boy has had a temperature of 39.4 degrees C (103 degrees F) for six days, red eyes without discharge, cracked red lips with a strawberry tongue, a rash on the trunk and perineum, swollen red hands and feet, and one enlarged cervical lymph node measuring 2 cm. He is extremely irritable. Laboratory testing shows a raised sedimentation rate and C-reactive protein, leukocytosis, and a platelet count of 610,000/microL. Write the diagnosis and then write which features you are counting towards the clinical criteria and whether the criteria are met. Write what the eye finding is called and the detail that distinguishes it, and write what you expect his fingers to do next week. Explain the high platelet count and when it appears. Then write which vessels you are now worried about, the mechanism by which they dilate, and why thrombosis is likely inside the resulting lesion, and name what that thrombosis can cause in a child of this age. Finish by writing why scarlet fever does not explain this picture, naming the organism, the two features of this boy that scarlet fever would not produce, and the complication it does not cause.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C16-BD1

How ANCA injures a capillary, and the three diseases that follow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C16 Activity: NPLEX-P1-CARD-18-09-C16-BD1 Date: _____

Write what the two ANCA target, which immunofluorescence pattern each one gives, and the abbreviation used for each pattern. Then rebuild the mechanism in steps: what priming does to the neutrophil, where the target enzymes go, what the antibody does when it binds, what the neutrophil then releases, and which vessels are injured. Write what pauci-immune means and why these diseases are pauci-immune, and write what the biopsy of an immune complex vasculitis would show instead. Write which two organs are hit hardest and why those organs, naming the renal lesion with its urinary findings and the pulmonary lesion with its symptom. Write what a pulmonary-renal syndrome is and name the other disease that causes one, with the immunofluorescence finding that identifies it. Finish by building the three-column table of the ANCA-associated diseases from memory, with rows for ANCA type, granulomas, upper airway, lung, kidney, and the distinctive feature of each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C16-BD2

Three patients who all cough blood

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C16 Activity: NPLEX-P1-CARD-18-09-C16-BD2 Date: _____

Patient 1 is a 58-year-old with two years of bloody nasal discharge and crusting, a collapsed nasal bridge, two cavitating lung nodules, hematuria with red cell casts, and PR3-ANCA. Patient 2 is a 63-year-old with hemoptysis, hematuria with red cell casts, palpable purpura on the legs, MPO-ANCA, and a renal biopsy showing crescents with almost no immunoglobulin. Patient 3 is a 41-year-old with twelve years of asthma and nasal polyps, a blood eosinophil count of 18%, patchy lung infiltrates, numbness and weakness in two separate nerve distributions, and new heart failure. For each patient name the disease and write the features that place them there rather than in either of the other two. Write what each renal biopsy would show and why the term pauci-immune applies. Explain why patient 1 has upper airway destruction and patient 2 does not. For patient 3 write the blood finding that drives the tissue injury, the proportion of such patients who are ANCA positive, and the organ involvement that is the main cause of death. Finish by writing what would make you suspect anti-GBM disease instead in a patient with lung hemorrhage and glomerulonephritis, and the single test finding that would show it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-09-C16-Q04

Sinuses, cavities, and casts

A 54-year-old woman has had 18 months of bloody nasal discharge with nasal crusting and a recently flattened nasal bridge. Chest imaging shows two cavitating nodules. Urinalysis shows hematuria with red cell casts, and the serum creatinine is 3.4 mg/dL. Renal biopsy shows necrotizing crescentic glomerulonephritis with almost no immunoglobulin or complement on immunofluorescence. Which of the following is the most likely serologic finding?

A. Linear IgG deposition along the glomerular basement membrane

B. PR3-ANCA with a cytoplasmic immunofluorescence pattern

C. IgA-containing immune complexes in the glomerular mesangium

D. Hepatitis B surface antigen in circulating immune complexes

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-09-C17-BD1

IgA vasculitis and Behcet syndrome from mechanism to sign

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C17 Activity: NPLEX-P1-CARD-18-09-C17-BD1 Date: _____

Write what IgA vasculitis is, the age range, the usual preceding event, and the molecular abnormality that starts it, naming the modified antibody and what recognizes it. Write where the resulting complexes deposit, which complement pathways they activate, the hypersensitivity type, and what immunofluorescence shows. Then write the four organ manifestations, giving for each the mechanism and the typical finding, and write where the skin lesions sit and why that distribution. Write what the platelet count and coagulation tests show and which other cause of purpura that result excludes. Then write what Behcet syndrome is, the geographic distribution, the genetic association, and the cell type driving the inflammation. List its mucosal, eye, skin, vascular, and neurologic features, writing what pathergy is. Finish by writing the one finding in each disease that would make you choose it over the other.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-09-C17-BD2

Purpura in a child and ulcers in a young adult

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-I Vascular Conditions Competency: NPLEX-P1-CARD-18-09-C17 Activity: NPLEX-P1-CARD-18-09-C17-BD2 Date: _____

A 6-year-old boy has raised purple spots that do not blanch over his buttocks and the backs of both legs, two weeks after a sore throat, with swollen painful knees and ankles, colicky abdominal pain, dark stools, and urine showing blood. His platelet count and coagulation tests are normal. Write the diagnosis, then explain each of his five findings mechanistically, one short paragraph each. Write why a normal platelet count is an important part of the picture and what it rules out. Write which abdominal complication he is at risk of and the usual segment involved, and write what a kidney biopsy would show and which separate renal disease it is identical to. Then a 26-year-old man of Turkish ancestry has recurrent painful mouth ulcers, genital ulcers, a painful red eye with a layer of pus in the anterior chamber, tender red shin nodules, and a calf deep vein thrombosis, and he develops a pustule where a needle was inserted. Write that diagnosis, the genetic association, the cell driving the inflammation, the name of the needle finding, and what the layer in the eye is called. Finish by writing why the term variable-vessel vasculitis is used for the second man and not the boy.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-J Vascular Neoplasms

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-10-C01				
NPLEX-P1-CARD-18-10-C02				
NPLEX-P1-CARD-18-10-C03				
NPLEX-P1-CARD-18-10-C04				
NPLEX-P1-CARD-18-10-C05				
NPLEX-P1-CARD-18-10-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C01-BD1

Hemangioma against malformation, and the life of an infantile hemangioma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-BD1 Date: _____

Without looking at your notes, write a definition of a hemangioma and a definition of a vascular malformation that make clear which one involves proliferating endothelial cells. Underneath, draw the life cycle of an infantile hemangioma as a timeline with five stages, from what is visible at birth through to what replaces the tumor at the end, and write beside each stage what the endothelial cells are doing. Add the incidence in infants, the groups in which it occurs more often, the color of a superficial lesion and of a deep one, and the marker its endothelium expresses in every phase. Then write what a capillary hemangioma looks like down the microscope and what a cavernous hemangioma looks like, one line each, and state which of the two regresses and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C01-BD2

Site, growth rate and the harm they cause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-BD2 Date: _____

Four infants have hemangiomas: one on the upper eyelid covering the pupil, one spread across the beard area of the jaw and neck, one large and segmental across half the face, and one large lesion in the liver. For each infant write what harm you would expect and the mechanism that produces it, including the enzyme and the hormone involved in the hepatic lesion. Then write which complication is the most common of all and explain it from the growth rate of the tumor and the blood supply of the skin above it. Next, a newborn has a red facial patch that was fully formed at birth and has since grown only in proportion to the child. Write what this lesion is, whether its endothelium would stain for GLUT1, whether you expect it to regress, and the syndrome and somatic mutation it is linked to. Finish by writing which two lesions cause the platelet-trapping consumptive coagulopathy and why the ordinary infantile hemangioma does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-10-C01-RT01

Rebuild the life cycle of an infantile hemangioma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-RT01 Date: _____

Rebuild the five-step life cycle of an infantile hemangioma from memory, one step per line and in order, from the appearance at birth to the final state. Beside the proliferation step write the marker the endothelium carries and the other tissue in the body known for that marker. Beside the regression step write the form of cell death involved and the tissue that takes the place of the vessels, and add the two changes that can be left behind in the skin. Underneath the steps write the one feature of this sequence that separates a tumor from a malformation.

Rebuild from memory: Life cycle of an infantile hemangioma

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-10-C01-CP01

Infantile hemangioma, cavernous hemangioma and port-wine stain

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Infantile (strawberry) hemangioma	Cavernous hemangioma	Port-wine stain
What is visible at birth			
Pattern and speed of growth			
Does it regress			
GLUT1 staining			
What you see on histology			
Usual sites			
The association most often tested			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C02-BD1

Three red papules that are not hemangiomas of infancy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C02 Activity: NPLEX-P1-CARD-18-10-C02-BD1 Date: _____

For pyogenic granuloma, cherry hemangioma and bacillary angiomatosis, write a block of four lines each: the patient and setting in which it appears, what the lesion looks like and how fast it grows, what the histology shows including every cell type and any organism, and what happens to it over time. Write the alternative histologic name of pyogenic granuloma and explain which two words in the name pyogenic granuloma are inaccurate and why. Then write what the pregnancy form is called, where in the mouth it sits, and the two hormones that amplify the response along with the local factor they amplify it against. For bacillary angiomatosis, name both organisms and how each reaches the patient, and write the transcription factor and growth factor through which the bacteria drive capillary growth, along with the stain that shows the organisms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C02-BD2

A purple papule in a man with a CD4 count of 40

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C02 Activity: NPLEX-P1-CARD-18-10-C02-BD2 Date: _____

A man with untreated HIV infection and a CD4 count of 40 cells per microliter has several violaceous papules on his arms, a fever of 38.4 degrees C (101.1 degrees F), and a cat at home. Write the two diagnoses that fit both the skin finding and the immune state, and write why the color of the lesions does not separate them. Then write what on the biopsy would settle it in favour of each one, listing the shape of the vessels, the lining cells, the inflammatory cells and any nuclear or silver stain result for both. Write which clinical feature in this man leans towards an infection and why. Finish by writing what happens to each lesion over the following months if the organism is cleared in one case and if the CD4 count recovers in the other, and explain the difference from what sustains each proliferation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-10-C02-CP02

Red and purple vascular lesions side by side

Fill in the table from memory for all five lesions, then check it against the notes.

Lesion	Typical patient	Cause	Key histology	Course
Infantile hemangioma				
Pyogenic granuloma				
Cherry hemangioma				
Bacillary angiomatosis				
Kaposi sarcoma				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C03-BD1

How HHV-8 drives the spindle cell

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-BD1 Date: _____

Write what kind of virus HHV-8 is, how it spreads from person to person, and the one sentence that explains why most infected people never develop the tumor. Name the tumor cell, the lineage it comes from, and four markers it carries. Then draw the latent programme: for the viral cyclin, for LANA and for viral FLIP, write which host protein or pathway each one acts on, what that action does to the cell cycle or to cell death, and which checkpoint is lost. Beside the viral cyclin write the host protein it is a homolog of, the CDK it binds, the inhibitors it resists, the protein it causes to be phosphorylated and the transcription factor that is then freed. Separately write what the small lytic fraction of cells secretes or expresses, the growth factor that rises as a result, and what those signals do to neighbouring cells. Finish with one line on why early lesions are polyclonal and advanced nodules may be clonal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C03-BD2

From slit-like spaces to the findings at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-BD2 Date: _____

Build the three stages of a Kaposi sarcoma lesion from memory as three rows, writing for each what the lesion looks like to the eye and what the microscope shows. Then explain each of the following findings from the structure you have just described, two sentences each: the purple color of a new lesion, the brown color of an old one, swelling of the whole limb, and bleeding from a lesion in the gut. Write which nuclear stain identifies the virus in the spindle cells and at which stages it works. Next, a transplant recipient on T-cell immunosuppression develops widespread lesions and the immunosuppression is then lowered. Write what you expect the lesions to do and justify it from both the latent viral genes and the cells that normally police infected cells. Finish by naming the two other diseases HHV-8 causes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-10-C03-RT01

Rebuild the path from infection to a nodule

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from HHV-8 infecting an endothelial-lineage cell to a Kaposi sarcoma nodule, one step per line and in order. Mark which steps belong to the latent programme, which belong to the lytic programme and which belong to the host immune system. Beside the latent steps write the host proteins put out of action. Beside the lytic step write the two signals released and the growth factor that rises. Beside the immune step write the two cell types whose loss of control permits expansion. Underneath, write one line on what each of the three groups contributes, so that removing any one of them would stall the sequence.

Rebuild from memory: From HHV-8 infection to a Kaposi sarcoma nodule

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-10-C03-PR01

Predict what happens when single viral genes are lost

A laboratory studies HHV-8 strains in which one viral gene at a time has been inactivated in infected endothelial-lineage cells. For each item below write the direction of the change you expect in the infected cell and give the host protein or pathway that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
With the viral cyclin inactivated, what happens to Rb phosphorylation and to the G1 to S transition		
With the viral cyclin inactivated, what happens to the braking effect of p16, p21 and p27 on the cell cycle		
With LANA lost, what happens to the viral episome in daughter cells after division		
With LANA lost, what happens to p53 and Rb activity and to the fate of a cell that is dividing without a signal to do so		
With viral FLIP lost, what happens to NF-kappaB signalling and to apoptosis		
With the lytic genes silenced in the whole lesion, what happens to viral IL-6 and VEGF levels and to the growth of neighbouring cells		
With cytotoxic T-cell and NK cell function restored in the host while all viral genes remain intact, what happens to the size of the lesions		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C04-BD1

Build the four forms of Kaposi sarcoma from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C04 Activity: NPLEX-P1-CARD-18-10-C04-BD1 Date: _____

On blank paper build a table of the four clinical forms of Kaposi sarcoma with one row each for classic, endemic African, transplant-associated and AIDS-associated. Use the columns population affected, typical sites and course, and fill every cell, including ages, ancestry, HIV status and the route by which the virus is usually acquired where the notes give it. Above the table write the one feature every form shares, both in terms of the virus and in terms of the histology. Underneath the table write the two host variables that decide how the tumor behaves, and give an example of what mild immune decline produces and what profound suppression produces.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C04-BD2

Four patients, four courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C04 Activity: NPLEX-P1-CARD-18-10-C04-BD2 Date: _____

Four patients have biopsy-proven Kaposi sarcoma: a 74-year-old HIV-negative man of Mediterranean descent with plaques on both feet; a 6-year-old in sub-Saharan Africa with large cervical and axillary lymph nodes and no skin lesions; a woman 14 months after kidney transplantation with lesions on the skin, palate and in the stomach; and a man with HIV infection with lesions on the face, hard palate and in the lungs. For each patient name the form, write the cofactor that permitted the tumor, and predict the pace and extent of the disease over the next year. Then write which patient is most likely to die of something unrelated and why, which patient has the illness that defines AIDS, and which patient's disease is most likely to shrink without any treatment aimed at the tumor itself, explaining each from immune status. Finish by writing why the AIDS form is far more common in men who have sex with men than in other groups with HIV, and what you expect the lesions to do transiently during early immune recovery and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C05-BD1

Myxoma and rhabdomyoma from the ground up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C05 Activity: NPLEX-P1-CARD-18-10-C05-BD1 Date: _____

Write a block on myxoma and a block on rhabdomyoma. In each block give the age group it belongs to, any sex difference, the chamber and exact site it arises from with the proportions the notes give, whether it is a true neoplasm or a hamartoma, the cell of origin, and the histology including the matrix or the stored substance and the name given to the characteristic cell. Add for myxoma the range of gross appearances from firm to gelatinous, and for rhabdomyoma whether the lesions are single or multiple and what happens to them in early childhood. Then write the inherited syndrome linked to each, with the gene, the protein it encodes, the pathway involved and the pattern of inheritance, and for the myxoma syndrome add the two non-cardiac findings. Finish with one line on how common primary cardiac tumors are compared with metastases to the heart.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C05-BD2

Why a benign tumor makes a patient faint

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C05 Activity: NPLEX-P1-CARD-18-10-C05-BD2 Date: _____

Draw a left atrium with a stalked mass on the interatrial septum and rebuild in six steps how it produces symptoms, from where it is attached through to what breaks off, one step per line. Then write the three groups of findings a myxoma produces. For the obstructive group write the valve lesion it imitates, the added sound and when in the cycle it occurs, and why the murmur and the symptoms change with body position. For the embolic group write where the fragments go from a left atrial tumor and where they go from a right atrial tumor, with the consequence in each case. For the constitutional group write the cytokine, the response it drives in the liver, the two laboratory values that rise and the reason for the anemia. Finish by writing the two ways a rhabdomyoma in a child causes trouble, and explain from the position of the tumor in the heart why its problems differ from those of a myxoma.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-10-C05-CA01

Breathlessness, fainting on standing and a high ESR

A 45-year-old woman comes to the clinic because of 4 months of breathlessness on exertion and two fainting spells, both of which happened as she stood up from a chair. She has lost 5 kg (11 lb) over the same period and reports low-grade fevers and aching in her hands. Temperature is 37.8 degrees C (100.0 degrees F), pulse is 96/min and regular, respirations are 20/min, and blood pressure is 112/70 mm Hg. On auscultation a low-pitched sound follows the second heart sound, and a diastolic murmur is heard at the apex whose loudness changes when she lies on her left side. Hemoglobin is 10.4 g/dL, the erythrocyte sedimentation rate is 78 mm/h, and C-reactive protein is raised. Blood cultures are repeatedly negative, and there is no history of rheumatic fever.

1. Recognize. Name the lesion, the chamber it sits in and the structure within that chamber it is usually attached to, and name the sound that follows the second heart sound.

2. Explain. Explain the mechanism that links the lesion to her breathlessness and to fainting that happens only when she changes position, following the pressures from the chamber of origin back into the lungs.

3. Differentiate. Write how this picture differs from rheumatic mitral stenosis and from infective endocarditis, naming for each the findings in this woman that do not fit it.

4. Investigate. Write which single imaging study you would use, what it would show and where, and name the two laboratory findings already present that support the systemic part of the picture.

5. Decide. State which complication of this lesion would be the most damaging if a fragment broke away, and name the organ most likely to be affected from a tumor in this chamber.

6. Justify. Justify your account of the fevers, the weight loss, the raised sedimentation rate and the anemia from a single substance the tumor secretes, and explain why negative blood cultures do not argue against the diagnosis.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C06-BD1

Angiosarcoma: what it is, where it sits, what it looks like

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C06 Activity: NPLEX-P1-CARD-18-10-C06-BD1 Date: _____

Write what cell angiosarcoma arises from, the age group it belongs to, and the sites where it most often appears, including the chamber of the heart and the skin site typical of elderly patients. Describe the histology of a well-differentiated tumor in two or three sentences, naming what the channels do to the surrounding collagen and what the lining cells look like, then describe a poorly differentiated tumor and write the two markers used to prove the lineage when the cells no longer form channels. Underneath, list all six entries from the risk factor table with the setting and the main site for each. Finish by writing the behavior of the tumor: how it spreads at its margin, whether it is single or multifocal in skin, where it metastasizes, the five-year survival figure from the notes and which site does worse.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C06-BD2

Same tumor, two different organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C06 Activity: NPLEX-P1-CARD-18-10-C06-BD2 Date: _____

A 68-year-old man who spent 20 years in a plant making polyvinyl chloride has a liver mass. A 61-year-old woman who had a radical mastectomy with axillary node dissection 12 years ago, and whose arm has been swollen ever since, has a purple mass in that arm. Write the tumor both have, then explain from first principles why the organ is different in each case. For the man, rebuild the five-step sequence from inhaled monomer to malignant proliferation, naming the enzyme, the reactive product and what that product does to DNA, and write why the liver rather than the lung is the target. For the woman, name the syndrome and write the two features of a chronically lymphedematous limb that are thought to allow the tumor to arise there. Then add a third patient who had chest wall irradiation 9 years ago and now has a mass in the treated field: write the genetic change these tumors often carry. Finish by writing the one property these exposures share that explains why all three patients present so long after the exposure, and the two other hepatic exposures from the notes with what each one is and how it reached the liver.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-10-C01-Q01

A red lesion that appeared after birth

A 5-week-old girl born at 33 weeks of gestation is brought to the physician because of a bright red raised lesion on her cheek. Her mother says the skin there looked faintly pale at birth, that a small red spot appeared at 2 weeks of age, and that it has enlarged steadily since. Biopsy of a similar lesion in another infant shows lobules of closely packed thin-walled capillaries whose endothelium stains for GLUT1. Which of the following is the most likely course of this lesion?

A. Rapid growth for several months, then apoptosis of the endothelium with fibrofatty replacement

B. Steady enlargement in proportion to the child, with darkening and thickening in adult life

C. Progressive dilatation of large thin-walled channels with thrombosis and calcification

D. Continued growth with infiltration of deep tissue and metastasis to the lungs

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C02-Q02

Purple nodules in an immunocompromised host

A 36-year-old man with untreated HIV infection and a CD4 count of 35 cells per microliter comes to the clinic because of fever and several red to purple nodules on his trunk that appeared over 3 weeks. He adopted two kittens 2 months ago. Biopsy of a nodule shows a proliferation of rounded capillaries lined by plump epithelioid endothelial cells, a dense neutrophilic infiltrate with nuclear debris, and clumps of granular purple material that resolve into bacilli on Warthin-Starry staining. Which of the following best explains the vascular proliferation in this lesion?

A. A viral cyclin homolog that binds CDK6 and phosphorylates Rb in infected spindle cells

B. A somatic activating mutation in GNAQ in dermal endothelial cells

C. Bacterial induction of hypoxia-inducible factor 1 alpha with a rise in VEGF

D. Endothelial expression of type 3 iodothyronine deiodinase with local hormone inactivation

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C03-Q03

The gene that removes two checkpoints at once

A 52-year-old man who received a kidney transplant 18 months ago and takes T-cell immunosuppression has violaceous plaques on his legs and trunk. Biopsy shows fascicles of spindle cells surrounding slit-like vascular spaces containing extravasated erythrocytes, with scattered lymphocytes, plasma cells and hemosiderin-laden macrophages. Immunostaining shows a nuclear antigen in the spindle cells. The protein detected by that stain tethers the viral episome to host chromosomes during mitosis. Which additional effect does this same protein have on the infected cell?

A. It activates NF-kappaB and thereby blocks apoptosis

B. It binds and inhibits both p53 and Rb, removing the checkpoint response to unscheduled proliferation

C. It acts as a constitutively active G protein-coupled receptor that raises VEGF

D. It resists the cyclin-dependent kinase inhibitors p16, p21 and p27

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C06-Q04

A liver mass decades after an industrial exposure

A 70-year-old man who worked for 22 years in a plant that polymerized vinyl chloride into plastic piping has right upper quadrant pain and weight loss. Imaging shows a large vascular mass in the right lobe of the liver, and biopsy shows irregular anastomosing channels dissecting between collagen bundles and lined by plump, multilayered atypical cells that stain for CD31. Which of the following best explains why the liver is the organ affected in this man?

A. Hepatic macrophages retained the inhaled particles and irradiated adjacent cells with alpha particles for decades

B. Impaired lymphatic drainage of the liver reduced local immune surveillance in a growth factor rich tissue

C. Amplification of MYC in hepatic endothelium followed the prolonged exposure

D. Hepatic cytochrome P450 2E1 oxidized the monomer to a reactive epoxide that formed DNA adducts in sinusoidal cells

Why did you pick it?

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-K Infectious Vascular Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-11-C01				
NPLEX-P1-CARD-18-11-C02				
NPLEX-P1-CARD-18-11-C03				
NPLEX-P1-CARD-18-11-C04				
NPLEX-P1-CARD-18-11-C05				
NPLEX-P1-CARD-18-11-C06				
NPLEX-P1-CARD-18-11-C07				
NPLEX-P1-CARD-18-11-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C01-BD1

Build a vegetation from a normal valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-BD1 Date: _____

Without looking at your notes, draw a heart valve in cross section and build the six-step sequence that turns it into an infected vegetation, one step per line and in order. Beside the first step write what kinds of flow and what foreign material injure valve endothelium, and what is exposed underneath once it is injured. Beside the sterile deposit step write what the deposit is made of and what it is called. Beside the adhesion step write the molecule viridans streptococci use to stick to it and the two host proteins that Staphylococcus aureus surface proteins bind. Then write why neutrophils clear this mass poorly and what that does to the blood cultures. Finish by writing which organism can skip the first three steps and infect a valve that was never damaged.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C01-BD2

Reason from the setting to the organism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-BD2 Date: _____

Six patients have infective endocarditis. Patient 1 has had 6 weeks of low-grade fever and weight loss after a tooth extraction, with known mitral regurgitation. Patient 2 injects drugs and has 4 days of high fever with a new holosystolic murmur at the left lower sternal border. Patient 3 had a prosthetic aortic valve placed 3 weeks ago. Patient 4 is 78 and became febrile after a urologic procedure. Patient 5 has bacteremia and iron deficiency anemia with blood in the stool. Patient 6 has 2 months of fever and a vegetation on echocardiography with three sets of negative blood cultures. For each patient name the organism you expect, write the route by which it entered the blood, and write in one line the property of the organism or the surface that makes it the expected one. Then write which patients you expect to have an acute course and which a subacute course, and say what decides that.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-11-C01-RT01

Rebuild the vegetation sequence and mark where vegetations sit

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-RT01 Date: _____

Rebuild the six-step formation of a vegetation from memory, numbering each step. Then draw the mitral and aortic valves and mark with an arrow the surface of each one where a vegetation forms. Beside each arrow write whether that surface faces the higher or the lower pressure chamber and why the regurgitant jet lands there. Write which two valves are infected most often and why, then write the one setting in which the tricuspid valve is the most common site and why the organisms reach it first.

Rebuild from memory: Formation of a vegetation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C01-Q01

Organism in right-sided endocarditis

A 29-year-old man who injects heroin comes to the emergency department because of 4 days of fever and shortness of breath. Temperature is 39.4 °C (102.9 °F), pulse is 112/min, respirations are 24/min, and blood pressure is 104/62 mm Hg. Cardiac examination shows a holosystolic murmur at the left lower sternal border that increases with inspiration. A chest radiograph shows multiple peripheral nodular opacities, two of which are cavitating. Which of the following organisms is most likely to be isolated from his blood?

A. Streptococcus gallolyticus

B. Enterococcus faecalis

C. Staphylococcus aureus

D. Staphylococcus epidermidis

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C02-BD1

Build the complication table by mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C02 Activity: NPLEX-P1-CARD-18-11-C02-BD1 Date: _____

On blank paper build a table of the complications of bacterial endocarditis with one row for each of these mechanisms: valve destruction, perivalvular abscess, left-sided emboli, right-sided emboli, immune complexes, and arterial wall infection. Use the columns Findings and Clue in a question, and fill every cell. Under the abscess row write which valves it complicates most often and name the two conduction structures that lie next to the aortic annulus. Under the immune complex row write which type of hypersensitivity this is, what the urine shows, and what happens to complement levels. Finish by naming the leading cause of death in endocarditis and writing in two sentences how valve destruction produces it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C02-BD2

Sort four new findings back to their mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C02 Activity: NPLEX-P1-CARD-18-11-C02-BD2 Date: _____

Four patients already known to have endocarditis develop something new. Patient 1 has a splenic infarct and a left hemiparesis. Patient 2 has multiple cavitating lung nodules with pleuritic pain and hemoptysis. Patient 3 has a PR interval that has lengthened from 0.16 s to 0.28 s over two days, with an infected aortic valve. Patient 4 has hematuria with red cell casts, a low serum complement level, and painful nodules on the finger pads. For each patient name the mechanism, name the infected valve or the site of the new lesion, and explain in two or three sentences how the mechanism produces that exact finding. Then write what sudden severe headache in week 6 of the illness suggests and explain how a vegetation fragment weakens an arterial wall. Finish by writing why sudden severe regurgitation causes pulmonary edema while regurgitation that develops over years does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-11-C02-CA01

Six weeks of fever, blood in the urine, and a changing murmur

A 64-year-old woman comes to the physician because of 6 weeks of low-grade fever, night sweats, fatigue, and a 4.5 kg (10 lb) weight loss. She had a tooth extracted 2 months ago. She has had a murmur of mitral regurgitation for 12 years. Temperature is 38.1 °C (100.6 °F), pulse is 96/min, respirations are 16/min, and blood pressure is 128/76 mm Hg. Cardiac examination shows a holosystolic murmur at the apex that is louder than it was recorded 1 year ago. There are two tender nodules on the pads of the left index and ring fingers and painless flat red macules on both palms. Funduscopic examination shows two retinal hemorrhages with pale centers. Laboratory studies show a hemoglobin of 10.1 g/dL (reference range 12.0 to 15.5 g/dL), erythrocyte sedimentation rate of 72 mm/h (reference range 0 to 20 mm/h), serum creatinine of 1.6 mg/dL (reference range 0.6 to 1.1 mg/dL), and serum C3 and C4 below the reference range. Urinalysis shows 3+ blood with dysmorphic erythrocytes and erythrocyte casts, and no bacteria.

1. Recognize. Name the infection, the valve involved, the course (acute or subacute), and the organism you expect from this history, and name each of the four peripheral findings described.

2. Explain. Explain how the organism came to sit on this particular valve, writing the sequence from the dental extraction to a growing vegetation and saying what the regurgitant jet contributed.

3. Differentiate. Write which of her findings come from embolism and which come from immune complexes, and explain what separates the two mechanisms in this patient.

4. Investigate. Write what the urinary findings, the dysmorphic erythrocytes, and the low C3 and C4 together tell you about the kidney lesion, and name the hypersensitivity mechanism and the structure in which the complexes deposit.

5. Decide. State which single new finding over the next week would tell you the infection has extended into the aortic annulus, and state which finding would tell you a leaflet or chord has given way.

6. Justify. Justify both of those predictions from anatomy: name the conduction structures next to the aortic annulus, and explain why a ventricle facing sudden severe regurgitation develops pulmonary edema.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C03-BD1

Trypanosoma cruzi from the bug to the tissue

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C03 Activity: NPLEX-P1-CARD-18-11-C03-BD1 Date: _____

Without looking at your notes, write out how Trypanosoma cruzi gets into a person and what it does once it is inside. Name the organism type, the geographic region, the vector and where it lives, when it feeds, and the exact step by which infective forms enter the body. List the four other routes of transmission. Then write the two forms of the parasite in human tissue, which one lives inside cells and multiplies, and which one travels in the blood. Draw the acute entry findings: label the skin lesion and the eye finding by name and write which entry site produces each. Finish by writing how long the acute phase lasts, how the parasite is found in the blood during it, and who is at risk of acute myocarditis or meningoencephalitis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C03-BD2

Explain each sign of chronic Chagas disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C03 Activity: NPLEX-P1-CARD-18-11-C03-BD2 Date: _____

A 52-year-old man who moved from rural Bolivia 20 years ago has an enlarged heart with a low ejection fraction, right bundle branch block with left anterior fascicular block, a thin bulging left ventricular apex, dysphagia, and severe constipation. For each of those findings write what has been destroyed in the tissue and how that destruction produces the finding. Write where the parasite has persisted and what the heart tissue looks like under the microscope, including how many parasites you expect to see. Explain why sudden death can be the first sign, naming the mechanism in the scarred myocardium, and explain what forms in the apical aneurysm and where it travels. Then write why a blood smear is of no use now while it worked in the acute phase, and name what is used instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-11-C03-CP01

Acute against chronic Chagas disease

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Acute	Chronic
Timing after infection and the share of people affected		
Parasite numbers and where the parasites are		
Main findings outside the heart		
Heart findings		
How the infection is detected		
What the tissue looks like under the microscope		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C04-BD1

Build the three stages of Lyme disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C04 Activity: NPLEX-P1-CARD-18-11-C04-BD1 Date: _____

Name the organism that causes Lyme disease and the kind of bacterium it is, then name the tick genus and the two species, the animal reservoir, the animal that sustains the adult ticks, the regions of the United States where risk is concentrated, and the season when most infections are acquired. Write which stage of the tick transmits most infections and why it is missed. Write how long the tick must stay attached and explain that number from where the spirochete sits in the tick. Then build a table of the three stages with one row each for early localized, early disseminated, and late disease, using the columns Timing, Findings, and Mechanism, and fill every cell. Describe the appearance and the size of the early skin lesion, and name the two other infections the same tick can transmit.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C04-BD2

Lyme carditis and why the testing changes with the stage

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C04 Activity: NPLEX-P1-CARD-18-11-C04-BD2 Date: _____

A 19-year-old from rural Wisconsin had an expanding red rash on the thigh 3 weeks ago and now has lightheadedness, a pulse of 40/min, and complete heart block with a narrow QRS. Write which stage of Lyme disease this is, which tissue is inflamed and what the infiltrate is made of, and why the degree of block can change within hours. Write whether you expect the block to resolve and explain your reasoning from what is happening in the node, inflammation or scarring. Then write why a patient with the early skin lesion alone is recognized clinically rather than by antibody testing, and what testing is used once the infection has disseminated. Finish by writing why bilateral facial palsy points toward Lyme disease, and name two other causes of inflamed or destroyed conduction tissue from this chapter and the feature that separates each one from Lyme carditis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C04-Q02

Heart block after a summer rash

A 19-year-old woman comes to the emergency department in August because of 2 days of lightheadedness. She lives in rural Wisconsin and spends weekends hiking. Three weeks ago she had an expanding red patch on the posterior thigh that was 9 cm across and cleared in the center. Temperature is 37.4 °C (99.3 °F), pulse is 40/min, respirations are 14/min, and blood pressure is 102/64 mm Hg. The electrocardiogram shows P waves at 82/min that are unrelated to QRS complexes at 40/min with a duration of 0.08 s. Examination shows weakness of the right side of the face that includes the forehead. Which of the following best explains the conduction abnormality in this patient?

A. Lymphocytic inflammation of the atrioventricular node around a disseminated spirochete

B. Fibrosis of the conduction system after decades of intracellular protozoal persistence

C. Extension of valvular infection into the aortic annulus as a ring abscess

D. Deposition of circulating immune complexes in the conduction system with complement activation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C05-BD1

Rickettsia rickettsii from the tick to the vessel wall

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C05 Activity: NPLEX-P1-CARD-18-11-C05-BD1 Date: _____

Name the organism that causes Rocky Mountain spotted fever and describe what kind of bacterium it is, including why it stains poorly on Gram stain and where it must live to replicate. Name the tick genus and the two United States tick species, name the states where most cases occur, give the season, and give the number of days between the bite and the first symptoms. Then write the cell the organism infects and rebuild in order what it does to that cell: how it gets in, how it escapes the phagosome, what it polymerizes, and how it reaches the next cell. Write the name of the tissue lesion this produces and list the six organs the notes say it reaches. Finish by drawing the rash: write the day it appears, where it starts, the direction it spreads, what it looks like at first, what it becomes, and which body surfaces are characteristically involved late.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C05-BD2

Derive every laboratory finding from the leaking vessel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C05 Activity: NPLEX-P1-CARD-18-11-C05-BD2 Date: _____

Starting from injured endothelium alone, explain each of these findings in two or three sentences: thrombocytopenia, hypoalbuminemia with edema, hyponatremia, a petechial rash, noncardiogenic pulmonary edema, confusion and seizures, and shock with acute kidney injury. For the low platelet count say whether the marrow is the problem and name where the platelets are going. For the low sodium name the hormone involved and the stimulus that releases it. Then write the classic triad, write how often all three are present at the start, and explain why the diagnosis is made clinically in the first week rather than on antibody testing. Finish with a 44-year-old in North Carolina in May who has fever, headache, vomiting, and abdominal pain and no rash: write which three other causes of a rash on the palms and soles you would keep in mind and the single feature that separates each from this illness.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-11-C05-PR01

Predict what happens as the vasculitis spreads

A 38-year-old man was bitten by a Dermacentor tick in Oklahoma in June and now has his fifth day of fever and headache. Rickettsiae are multiplying in the endothelium of small vessels in his skin, lungs, brain, kidneys, and gut. Predict each item below, write the direction of the change, and give in one line the vascular event that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the permeability of the small vessels and to the fluid in the interstitium		
What happens to the serum albumin and to the effective circulating volume		
What happens to ADH release and to the serum sodium		
What happens to the platelet count, and whether the marrow or the vessel wall accounts for it		
What happens to the skin lesions as red cells escape, and where they appear first		
What happens to oxygenation when the same process occurs in the pulmonary capillaries		
What happens to blood pressure and urine output as the leak continues		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C05-Q03

Low sodium and low platelets in a spotted fever

A 41-year-old woman comes to the emergency department in May because of 5 days of fever, severe headache, and vomiting. She removed a tick from her calf 1 week ago while gardening in North Carolina. Temperature is 39.6 °C (103.3 °F), pulse is 118/min, respirations are 22/min, and blood pressure is 92/58 mm Hg. Examination shows petechiae on the wrists, ankles, and trunk and pitting edema of both legs. Laboratory studies show a platelet count of 68,000/microliter (reference range 150,000 to 400,000/microliter), serum sodium of 127 mEq/L (reference range 135 to 145 mEq/L), and serum albumin of 2.4 g/dL (reference range 3.5 to 5.0 g/dL). Which of the following best explains this combination of findings?

A. Deposition of circulating immune complexes in small vessels with complement activation

B. Replication of the organism within platelets and marrow megakaryocytes

C. Non-neutralizing antibody increasing viral entry into monocytes and macrophages

D. Infection of vascular endothelium causing plasma leak and platelet consumption at the vessel wall

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C06-BD1

Yellow fever from the mosquito to the liver

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C06 Activity: NPLEX-P1-CARD-18-11-C06-BD1 Date: _____

Write the virus family and the genome type shared by yellow fever and dengue, and name the mosquito that carries both and where it breeds. For yellow fever write the regions where it occurs, the two transmission cycles and the animals in the jungle cycle, the incubation period, and the findings of the first febrile illness. Write the share of patients who improve briefly and then enter the toxic phase. Then draw a liver lobule, mark the zone the virus damages, and write the name of that pattern of necrosis and why it is unusual compared with toxic and ischemic injury. Name the dying hepatocyte by its eponym and describe what it looks like, and write what else the notes say the liver shows. Finish by writing how liver failure produces jaundice and bleeding, explaining the dark vomit, and naming the two organ failures that follow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C06-BD2

Why the second dengue infection is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C06 Activity: NPLEX-P1-CARD-18-11-C06-BD2 Date: _____

A woman had dengue three years ago and now has a different serotype. Write how many serotypes exist, then explain step by step why her old antibodies make this infection worse: what the antibodies do to the new virus, which part of the antibody binds which receptor on which cells, and what happens to viral load and cytokine release. Name this mechanism and name the other group of patients at risk through the same route. Then write when in the illness severe disease appears, relative to the fever, and explain what is happening at the capillary wall. Her hematocrit rises from 40% to 48% while her platelets fall to 30,000 per microliter: write what the rising hematocrit means about where her plasma has gone, and explain why it warns of shock before the blood pressure falls. Finish by writing where fluid collects as the leak continues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-11-C06-CP01

Yellow fever against dengue

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Yellow fever	Dengue
Where in the world it occurs		
Main target cell		
Hallmark pathology		
Key clinical findings		
Laboratory clue		
What the severe form is linked to		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C07-BD1

The three shared effects of hemorrhagic fever viruses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-BD1 Date: _____

Write what viral hemorrhagic fever means and why it is a syndrome rather than one disease. List the four virus families named in the notes with the diseases that belong to each and the genome type of each family. Write what the viruses have in common about their envelope. Then write out the three shared effects that produce the syndrome: for the first, name the cells infected first, where those cells carry the virus, what the viruses do to type I interferon signaling, and what happens to lymphocytes; for the second, name four mediators released and what they do to endothelial junctions; for the third, name the molecule that activated macrophages express, the process it starts, what is consumed, and the other name for that process. Add what hepatocyte injury contributes. Finish by writing why patients leak plasma and bleed at the same time and what most deaths come from.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C07-BD2

Work out who is at risk in a filovirus outbreak

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-BD2 Date: _____

Describe the structure and genome of Ebola and Marburg viruses and name the likely reservoir. Write the two ways an outbreak starts in a person and the region where these outbreaks occur, then write exactly how the virus passes from one person to the next and name the two groups at highest risk, explaining what each group contacts. Write when a patient becomes infectious relative to symptoms, what happens to viral load as the illness worsens, and where virus can persist after recovery. State whether ordinary airborne spread occurs. Then give the incubation period and write the findings of the first days and of the second week, and explain which of those findings accounts for most of the fluid and electrolyte loss. Finish by listing the laboratory findings you expect and giving the approximate case fatality.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-11-C07-RT01

Rebuild the common pathway of the hemorrhagic fever viruses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-RT01 Date: _____

Rebuild the seven-step common pathway of the hemorrhagic fever viruses from memory, one step per line and in order, from the first cells infected through to multiorgan failure. Beside the interferon step write what blocking interferon signaling buys the virus. Beside the cytokine step name the mediators. Beside the tissue factor step write which cells express it and name what the consumption of platelets and clotting factors is called. Then mark on your sequence the two points at which the patient loses intravascular volume and the two points at which bleeding becomes likely, and write one sentence on why shock and bleeding appear together rather than one after the other.

Rebuild from memory: Common pathway of hemorrhagic fever viruses

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C08-BD1

Build the course of viral myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C08 Activity: NPLEX-P1-CARD-18-11-C08-BD1 Date: _____

Define myocarditis and say what must be excluded as the cause of the myocyte injury. Name the most common group of causes in developed countries, name the classic virus with its virus family, genome and routes of spread, and list the other viruses the notes name. Describe the typical patient and explain why cardiac symptoms appear one to a few weeks after a respiratory or gastrointestinal illness. Then rebuild the six-step course from the first infection to dilated cardiomyopathy, one step per line. Beside the entry step name the receptor on the myocyte and what kind of protein it is. Beside the immune step name the two cell types that destroy infected myocytes and write what cytokines do to contractility. Write what happens in the minority whose inflammation persists, including the cardiac protein autoimmunity can target. Finish by writing which virus infects coronary microvascular endothelium instead of myocytes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C08-BD2

Separate myocarditis from an infarction and from the other infiltrates

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A 23-year-old man has chest pain, diffuse ST elevation, and a raised troponin one week after a febrile diarrheal illness, and his coronary arteries are normal. Write why the troponin is raised and name the process that is killing the myocytes. Write where the ECG changes, the conduction disturbances, and the arrhythmias come from, and name the population in which this is an important cause of sudden death. Explain why his chest pain is pleuritic and the ST elevation diffuse. Then write what the ejection fraction does as contracting myocytes are lost and the symptoms that follow. Write what an endomyocardial biopsy would show and why a single biopsy can be negative in a patient who has the disease. Write where cardiac MRI shows its abnormality and how that pattern differs from an infarction. Finish by naming the cause suggested by an infiltrate rich in eosinophils and the cause suggested by multinucleated giant cells.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

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Board-style question NPLEX-P1-CARD-18-11-C08-Q04

Biopsy after a viral illness

A 23-year-old man comes to the emergency department because of 2 days of sharp chest pain that is worse when he lies flat. He had 4 days of fever and watery diarrhea 1 week ago. Temperature is 37.8 °C (100.0 °F), pulse is 104/min, respirations are 18/min, and blood pressure is 112/70 mm Hg. The electrocardiogram shows diffuse ST elevation. Serum troponin I is elevated. Coronary angiography shows normal coronary arteries. Echocardiography shows global hypokinesis with an ejection fraction of 38%. Which of the following is most likely to be seen on endomyocardial biopsy?

A. Multinucleated giant cells with extensive myocyte loss

B. A lymphocytic infiltrate with necrosis of adjacent myocytes

C. Intracellular amastigotes with interstitial fibrosis

D. An eosinophil-rich infiltrate with little myocyte necrosis

Why did you pick it?